

Econometrics 3, Sciences Po

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Fall 2025

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Chapter 1

Estimating a mean, with an application to surveys

In this chapter, we review tools of asymptotic statistics. We will learn how to use **estimators** to learn about **estimands**, or how we can use a **sample** to learn about a **population**.

1.1 Independent and identically distributed samples

Throughout this chapter, we assume that the data we observe are realizations of $(Y_1, Y_2, \dots, Y_n)$, n independent and identically distributed (iid) random variables. A more compact way of denoting those random variables is $(Y_i)_{1 \leq i \leq n}$. There are two leading cases when the data we observe are indeed realizations of n iid random variables.

1.1.1 Iid sampling in surveys

The first one is in surveys, when the sample of the survey is randomly drawn with replacement from the target population.

Assume we are on November 1st 2016, and you would like to know the fraction of the Pennsylvania electorate that wants to vote for Hillary Clinton. There are 8,448,674 registered voters in Pennsylvania. For every k included between 1 and 8,448,674, let y_k be a number equal to 1 if on November 1st 2016 voter k wants to vote for Clinton, and to 0 otherwise. The y_k s are NOT random variables: whether on November 1st 2016 person k wants to vote for Clinton or not is something deterministic. The y_k s are just numbers. Let $c = y_1 + y_2 + \dots + y_{8,448,674}$.

What does c represent?

c is the number of registered Pennsylvania voters that want to vote for Clinton, as of November 1st 2016. Let $p = \frac{c}{8,448,674}$. p is the percentage of registered Pennsylvania voters that want to vote for Clinton. p is what you would like to know. Unfortunately, to compute p , you would need to interview the 8,448,674 registered voters in Pennsylvania on November 1st 2016, and ask them who they want to vote for. Obviously, this is going to be difficult to do...

A smarter solution is to randomly draw with replacement a sample from the registered voters population, observe their y_k s, and compute the percentage of the sample that wants to vote for Clinton. In this chapter, we will show that this percentage can give us useful information about p . Drawing samples with replacement from the registered voters population and observing their y_k s is essentially what political survey firms do, up to 3 differences:

1. Surveying firms cannot sample from the universe of all registered voters in Pennsylvania. Instead, they typically use the phonebook.
2. Surveying firms do not observe the y_k s of all the voters they sample: some of them refuse to answer the survey. Non response rate is often very high in surveys.
3. Surveying firms do not observe the y_k s of all the voters they sample: some of them might answer the survey, but might lie about who they want to vote for.

For now, we will leave these issues aside: we will assume we can sample from the universe of all registered voters, that all sampled voters answer the survey, and that all of them answer truthfully. As you will see in the problem set, the failure of one of those assumptions must explain why surveying firms failed to predict Trump's victory in Pennsylvania.

We are going to consider the survey we will conduct from an **ex-ante perspective**, before we actually conduct it, hence all the "will" in the pages that follow.

For now, let's assume our survey will only have one respondent. We will randomly choose one voter from the population of registered voters. Specifically, we will randomly draw a number from the uniform distribution on $\{1, 2, \dots, 8,448,674\}$, and we will ask the voter with that number if they want to vote for Clinton. Let Y_1 be a variable denoting the answer that person will give us (1 if that person says they want to vote for Clinton, 0 otherwise).

Is Y_1 a deterministic number or a random variable?

Y_1 is a random variable. If we randomly draw a voter that wants to vote Clinton, $Y_1 = 1$. If we randomly draw a voter that does not want to vote Clinton, $Y_1 = 0$.

What is the expected value of Y_1 ?

$$E(Y_1) = p.$$

Can someone explain why?

Y_1 is a binary random variable. Therefore, $E(Y_1) = P(Y_1 = 1)$. Y_1 will be determined as follows. We will randomly draw one voter out of the 8,448,674 registered voters. Among them, $p\%$ want to vote for Clinton. Therefore, the probability that the voter we will randomly draw wants to vote for Clinton is p .

Remark: as Y_1 is a binary random variable, $E(Y_1) = P(Y_1 = 1)$ fully characterizes its distribution: if we know $P(Y_1 = 1)$, we know $P(Y_1 = 0)$ and there are no other probabilities we need to know.

Other remark: the expectation is taken over the 8,448,674 possible draws of the voter, which each have a probability $\frac{1}{8,448,674}$ of being realized.

More generally, the following result holds.

Theorem 1.1.1 *Expectation with respect to random sampling is the average in the population.*

Assume you randomly draw one individual from a population of N individuals, which each have a quantity of interest y_k attached to them. Let Y denote the quantity attached to the individual you draw. $E(Y) = \frac{1}{N} \sum_{k=1}^N y_k$.

What is the variance of Y_1 ?

$$V(Y_1) = p(1 - p).$$

Can someone explain why?

The variance of a binary variable is equal to its expectation $\times$ (1-its expectation).

For which values of p is the variance of Y_1 equal to 0? What is the intuition for that?

$V(Y_1) = 0$ if and only if $p = 0$ or $p = 1$. If none or all voters want to vote Clinton, Y_1 is a degenerate random variable (i.e. a real number). For instance, if $p = 0$, irrespective of which voter we draw we will always get $Y_1 = 0$.

Now, let's assume our survey will have n respondents. We will randomly draw with replacement a sample of n voters from the population of registered voters. Here is how we will do it:

1. We will randomly draw a first number from the uniform distribution on $\{1, 2, \dots, 8,448,674\}$. We will interview the voter with that number. Let Y_1 denote the answer that person will give us (1 if that person says they want to vote for Clinton, and 0 otherwise).
2. We will randomly draw a second number from the uniform distribution on $\{1, 2, \dots, 8,448,674\}$. If that second random number is different from the first, we will interview the voter with that number. Let Y_2 denote the answer that person will give us (1 if that person says they want to vote for Clinton, and 0 otherwise). If that second random number is the same as the first, we will not interview the same person a second time (we already know their answer), we will just let $Y_2 = Y_1$.
3. ...
4. We will randomly draw an n th number from the uniform distribution on $\{1, 2, \dots, 8,448,674\}$. If that n th random number is different from all the other random numbers we've drawn so far, we will interview the voter with that number. Let Y_n denote the answer that person will give us (1 if that person says they want to vote for Clinton, and 0 otherwise). If that n th random number is the same as one of the random numbers we drew before (say the i th one), we will not interview the same person a second time (we already know their answer), we will just let $Y_n = Y_i$.

What is the expected value of Y_2 ? Do Y_2 and Y_1 have identical or different probability distributions?

$E(Y_2) = p$. Y_2 is a binary random variable. Therefore, $E(Y_2) = P(Y_2 = 1)$. Y_2 will be

determined as follows. We will randomly draw one voter out of the 8,448,674 registered voters. Among them, $p\%$ want to vote for Clinton. Therefore, the probability that the voter we will randomly draw wants to vote for Clinton is p . $E(Y_2) = E(Y_1)$. As these two random variables are binary, their expectations fully characterize their probability distributions. Therefore, Y_2 and Y_1 are identically distributed.

Is Y_2 independent or dependent of Y_1 ?

Y_2 is independent of Y_1 . To determine Y_2 , we will randomly draw again a number from $\{1, 2, \dots, 8,448,674\}$. The way we will draw this second voter will not depend on whether $Y_1 = 1$ or $Y_1 = 0$: in both cases we will draw this second number from the uniform distribution on $\{1, 2, \dots, 8,448,674\}$. Y_2 denotes whether the 2nd voter we will draw wants to vote Clinton. Because Y_1 has no influence on how we will draw this second voter, Y_2 is independent of Y_1 . Irrespective of whether $Y_1 = 1$ or $Y_1 = 0$, the probability that we draw a second voter that wants to vote for Clinton is p .

In this survey with random sampling and replacement, for every $i \in \{1, \dots, n\}$ $E(Y_i) = p$: the Y_i s are identically distributed. Moreover the Y_i s are independent from each other. **Therefore, the $(Y_i)_{1 \leq i \leq n}$ are n iid random variables, and the data we will eventually get to observe at the end of our survey will be realizations of those n iid random variables.**

In practice, surveying companies draw their samples a bit differently. They randomly draw without replacement a sample of n voters. Here is how they do it:

1. The company will randomly draw a first number from the uniform distribution on $\{1, 2, \dots, 8,448,674\}$. They will interview the voter with that number. Let Y_1 denote the answer that person will give them (1 if that person says they want to vote for Clinton, and 0 otherwise).
2. The company will randomly draw a second number from the uniform distribution on $\{1, 2, \dots, 8,448,674\}$, excluding the first number that has already been drawn. Therefore, this second draw is not independent from the first. They will interview the voter with that number. Let Y_2 denote the answer that person will give (1 if that person says they want to vote for Clinton, and 0 otherwise).
3. ...
4. The company will randomly draw an n th number from the uniform distribution on $\{1, 2, \dots, 8,448,674\}$, excluding the first $n - 1$ numbers that have already been drawn.

Therefore, n th draw will not be independent from the first $n - 1$. They will interview the voter with that number. Let Y_n denote the answer that person will give (1 if that person says they want to vote for Clinton, and 0 otherwise).

With this sampling process, are the $(Y_i)_{1 \leq i \leq n}$ iid?

After a bit of work, one can show that the $(Y_i)_{1 \leq i \leq n}$ are still identically distributed. On the other hand, they are not independent anymore: if $Y_1 = 1$, then the probability that $Y_2 = 1$ is $\frac{c-1}{8,448,674-1}$. If $Y_1 = 0$, then the probability that $Y_2 = 1$ is $\frac{c}{8,448,674-1}$. The probability that $Y_2 = 1$ depends on Y_1 . In the problem set, you will see why surveying firms draw their sample with replacement, and how this changes the theoretical results we will derive assuming iid sampling. You will see that when the sample is small relative to the total population (e.g.: you sample 1000 Pennsylvanian voters out of 8,448,674), drawing the sample with or without replacement does not make a big difference. Intuitively, in the example above with sampling with replacement, whether $Y_1 = 0$ or $Y_1 = 1$ has an impact on the probability that $Y_2 = 1$, but that impact is very small. The difference between the probability that $Y_2 = 1$ in those two cases is $\frac{1}{8,448,674-1}$, a very small number. Therefore, Y_2 and Y_1 are “almost” independent, which is why drawing the sample with or without replacement does not make a big difference.

1.1.2 Iid sampling with random number generators

When you write in Stata

```
set obs 1000
```

```
gen u=uniform(),
```

you generate 1000 draws from the uniform distribution on $[0, 1]$, and the data you observe are realizations of $(Y_i)_{1 \leq i \leq 1000}$, 1000 independent random variables following the uniform distribution on $[0, 1]$.

You can think of draws from random number generators as sampling with replacement. If your first draw from the uniform distribution is 0.5, in your second draw you can still draw 0.50. In this experiment, you draw from replacement values from a continuum of values $([0, 1])$.

When we will run Monte Carlo simulations, we will use a random number generator to create the realizations of the random variables we pretend to observe. Those random variables will be iid, which is why all the theory below describes well what we observe in Monte-Carlo studies.

1.2 Using an estimator to estimate an estimand

Now let's get back to our initial, more abstract setting: we assume that the **data we observe are realizations of $(Y_i)_{1 \leq i \leq n}$, n iid random variables**. This is an interesting setting to consider: it is applicable to surveys, and to Monte-Carlo simulations.

In what follows, we are going to **consider our data from an ex-ante perspective**, before the $(Y_i)_{1 \leq i \leq n}$ get realized. However, I will drop the “will” in the pages that follow because that's not very elegant and quite cumbersome to write everything in the future tense. But bear in mind that we are reasoning from the perspective of someone who has not seen the data yet, and only knows he will get to observe realizations of $(Y_i)_{1 \leq i \leq n}$, n iid random variables.

To alleviate a bit the notation, let me introduce **Y , a random variable with the same distribution as the $(Y_i)_{1 \leq i \leq n}$** . We will assume that **Y** (and therefore all the $(Y_i)_{1 \leq i \leq n}$) has a **finite second moment: $E(Y^2) < +\infty$** . This ensures we can use the central limit theorem, see below. We also assume that $V(Y) > 0$: this rules out the case where the $(Y_i)_{1 \leq i \leq n}$ are degenerate (constant) random variables (in the survey example, this corresponds to $p = 0$ or $p = 1$).

Now assume we want to learn the value of a unidimensional estimand θ .¹

Definition 1.2.1 *Definition of an estimand.*

A unidimensional estimand is a real number which is a function of the probability distribution of the random variables we will get to observe, i.e. of the probability distribution of Y .

In the survey example in Subsection 1.1.1, which unidimensional estimand would we like to learn?

In the survey example in Subsection 1.1.1, we would be interested in learning $E(Y)$, the expectation of the $(Y_i)_{1 \leq i \leq n}$, because we have shown that $E(Y)$ is equal to p , the percentage of Pennsylvania voters that wanted to vote for Clinton as of November 1st 2016. $E(Y)$ is a unidimensional estimand: it is equal to $P(Y = 1)$, and it is therefore a function of the probability distribution of Y . Relating a parameter you are interested in (e.g.: p) to an estimand (e.g.: $E(Y)$) is called **identification**.

To learn about our estimand θ , we use the $(Y_i)_{1 \leq i \leq n}$ to build an **estimator**.

Definition 1.2.2 *Definition of an estimator.*

An estimator is a function of the random variables we will get to observe, here the $(Y_i)_{1 \leq i \leq n}$.

¹All of what follows easily extends to multidimensional estimands, but to keep the notation simple I only consider the unidimensional case.

An estimator for θ is usually denoted $\hat{\theta}_n$. As it is a function of $(Y_i)_{1 \leq i \leq n}$, its value will change if the sample size grows from n to $n+1$. That's the reason why I wrote $\hat{\theta}_n$ instead of $\hat{\theta}$. However, people often forget the n index, and I will sometimes omit it in those notes to alleviate the notations. So $\hat{\theta}$ and $\hat{\theta}_n$ mean the same thing: a function of the random variables we will get to observe that is supposed to proxy the estimand θ . Strictly speaking, $(\hat{\theta}_n)$ defines a sequence of estimators, but I most often call it an estimator. Proposing an estimator for an estimand is called **estimation**.

When the parameter of interest is $E(Y)$, a “natural” estimator is $\hat{E}(Y) = \frac{1}{n} \sum_{i=1}^n Y_i$. This indeed defines an estimator, as $\hat{E}(Y)$ is a function of $(Y_i)_{1 \leq i \leq n}$.

Is $\hat{E}(Y)$ a random variable or a real number?

$\hat{E}(Y)$ is a function of the $(Y_i)_{1 \leq i \leq n}$, which are random variables. $\hat{E}(Y)$ is a random variable.

What is the expected value of $\hat{E}(Y)$?

$$\begin{aligned} E(\hat{E}(Y)) &= E\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) \\ &= \frac{1}{n} \sum_{i=1}^n E(Y_i) \\ &= \frac{1}{n} \sum_{i=1}^n E(Y) \\ &= E(Y). \end{aligned}$$

$E(\hat{E}(Y)) = E(Y)$: $\hat{E}(Y)$ is an unbiased estimator of $E(Y)$. Our estimator does not systematically over- or underestimates $E(Y)$, its distribution is “centered” around $E(Y)$.

However, our estimator might still make mistakes when estimating $E(Y)$. A first measure of whether our estimator makes a lot of mistake or not is $E(|\hat{E}(Y) - E(Y)|)$, the expectation of the distance between our estimator and $E(Y)$. This measure is hard to work with, because $x \mapsto |x|$ is not differentiable at 0. Therefore, we prefer to use $E\left(\left(\hat{E}(Y) - E(Y)\right)^2\right)$, the expectation of the squared distance between our estimator and $E(Y)$.

Definition 1.2.3 Definition of the *quadratic risk of an estimator*.

For any estimator $\hat{\theta}$ supposed to proxy an estimand θ , $E\left(\left(\hat{\theta} - \theta\right)^2\right)$ is called the quadratic risk of $\hat{\theta}$.

Show that the quadratic risk of $\hat{E}(Y)$ is equal to its variance. Hint: remember that $\hat{E}(Y)$ is unbiased.

The quadratic risk of $\hat{E}(Y)$ is

$$\begin{aligned} E\left(\left(\hat{E}(Y) - E(Y)\right)^2\right) &= E\left(\left(\hat{E}(Y) - E\left(\hat{E}(Y)\right)\right)^2\right) \\ &= V\left(\hat{E}(Y)\right). \end{aligned}$$

The first equality follows from the fact $\hat{E}(Y)$ is an unbiased estimator of $E(Y)$. The second equality follows from the definition of the variance: for any random variable X , $V(X) = E\left((X - E(X))^2\right)$. More generally, the quadratic risk of any unbiased estimator is equal to the variance of that estimator.

What is the variance of $\hat{E}(Y)$?

$$\begin{aligned} V\left(\hat{E}(Y)\right) &= V\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n V(Y_i) \\ &= \frac{1}{n^2} \sum_{i=1}^n V(Y) \\ &= \frac{V(Y)}{n} \end{aligned}$$

$V\left(\hat{E}(Y)\right) \rightarrow 0$ when $n \rightarrow +\infty$. Therefore, the quadratic risk of $\hat{E}(Y)$ eventually vanishes when our sample grows. This gives us a first sense that $\hat{E}(Y)$ becomes a better proxy of $E(Y)$ when the sample grows. In the next sections we will explain precisely how one can use $\hat{E}(Y)$

to infer plausible values of $E(Y)$. Using an estimator to infer plausible values of an estimand is called **inference**. **Identification**, **estimation**, and **inference** are the three cornerstones of econometrics.

So far we have focused on the case where your parameter of interest is the expectation of Y . Assume that instead, your parameter of interest is the variance of Y , $V(Y)$.

Is $\frac{1}{n} \sum_{i=1}^n (Y_i - E(Y))^2$ an estimator of $V(Y)$?

$\frac{1}{n} \sum_{i=1}^n (Y_i - E(Y))^2$ is not an estimator of $V(Y)$, as it is not a function of $(Y_i)_{1 \leq i \leq n}$ only, it also involves $E(Y)$. Even when the $(Y_i)_{1 \leq i \leq n}$ get realized, we can still not compute $\frac{1}{n} \sum_{i=1}^n (Y_i - E(Y))^2$, because we do not know $E(Y)$. An estimator is something we should be able to compute when the $(Y_i)_{1 \leq i \leq n}$ get realized.

On the contrary, $\hat{V}(Y) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{E}(Y))^2$ is an estimator of $V(Y)$. One can show that this estimator is biased, while $\frac{1}{n-1} \sum_{i=1}^n (Y_i - \hat{E}(Y))^2$ is an unbiased estimator of $V(Y)$. However, because the difference between the 2 is negligible when n is large, we keep $\hat{V}(Y)$ as our estimator of $V(Y)$. Finally, notice that

$$\begin{aligned} \hat{V}(Y) &= \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{E}(Y))^2 \\ &= \frac{1}{n} \sum_{i=1}^n (Y_i^2 - 2Y_i \hat{E}(Y) + \hat{E}(Y)^2) \\ &= \frac{1}{n} \sum_{i=1}^n Y_i^2 - 2\hat{E}(Y) \frac{1}{n} \sum_{i=1}^n Y_i + \hat{E}(Y)^2 \\ &= \frac{1}{n} \sum_{i=1}^n Y_i^2 - 2\hat{E}(Y)^2 + \hat{E}(Y)^2 \\ &= \frac{1}{n} \sum_{i=1}^n Y_i^2 - \left(\frac{1}{n} \sum_{i=1}^n Y_i \right)^2. \end{aligned} \tag{1.2.1}$$

This mimicks the fact that $V(Y) = E(Y^2) - (E(Y))^2$.

Given our estimator of $V(Y)$, can you propose an estimator of $V(\hat{E}(Y))$?

We have shown that $V(\hat{E}(Y)) = \frac{V(Y)}{n}$. Therefore, a natural estimator of $V(\hat{E}(Y))$ is

$$\widehat{V}(\widehat{E}(Y)) = \frac{\widehat{V}(Y)}{n}.$$

$\widehat{E}(Y)$ and $\widehat{V}(Y)$ are estimators from the method of moments. This method amounts to replacing expectations by sample means in order to construct estimators. For instance, $V(Y) = E((Y - E(Y))^2)$, therefore $\widehat{V}(Y) = \frac{1}{n} \sum_{i=1}^n (Y_i - \widehat{E}(Y))^2$. $\widehat{E}(Y)$, the sample mean, is sometimes denoted $\bar{Y}$. Here I denote it $\widehat{E}(Y)$ to emphasize that it is used as an estimator of $E(Y)$.

1.3 Consistency and $\sqrt{n}$ -consistency

An estimator is a function of $(Y_i)_{1 \leq i \leq n}$ which is supposed to proxy an estimand, a function of the probability distribution of Y . In this course we adopt an asymptotic perspective to assess the quality of an estimator. First, a good estimator should converge to the target parameter when the sample size goes to infinity. If we had an infinite sample at hand, would $\widehat{\theta}_n$ finally uncover θ ? Second, as collecting an infinite sample is infinitely costly, a good estimator should converge to the target parameter at a reasonably fast pace.

I first review two notions of convergence for sequences of random variables and random vectors.

Definition 1.3.1 *Definition of convergence in probability*

1. The sequence of random variables $(U_n)_{n \in \mathbb{N}}$ converges in probability towards the random variable V if for every $\varepsilon > 0$,

$$\lim_{n \rightarrow +\infty} P(|U_n - V| > \varepsilon) = 0.$$

2. Let $(U_n)_{n \in \mathbb{N}}$ be a sequence of $k \times 1$ random vectors. Let V be a $k \times 1$ random vector. $(U_n)_{n \in \mathbb{N}}$ converges in probability towards V if each coordinate of $(U_n)_{n \in \mathbb{N}}$ converges in probability towards the corresponding coordinate of V .

Convergence in probability of U_n towards V is denoted $U_n \xrightarrow{P} V$.

Definition 1.3.2 *Definition of convergence in distribution*

$(U_n)_{n \in \mathbb{N}}$ converges in distribution towards V if

$$\lim_{n \rightarrow +\infty} P(U_n \leq x) = P(V \leq x)$$

for every x at which the function $x \mapsto P(V \leq x)$ is continuous. This is denoted $U_n \xrightarrow{d} V$.

To understand the difference between those two notions, one needs to understand the difference between equality and equality in distribution. Let U and V denote the results from the draw of two fair dices.

Are U and V equal? Do they have the same distribution?

U and V have the same distribution but they will not necessarily be equal: maybe one dice will yield a 3 while the second one will yield a 5. Convergence in probability is related to equality of random variables: when n gets large, U_n and V get close to each other with a high probability. Convergence in distribution is related to equality of distributions: when n gets large, the distributions of U_n and V get close.

Definition 1.3.3 *Consistency of an estimator*

$\hat{\theta}_n$ is a consistent estimator of θ if it converges in probability towards θ .

Theorem 1.3.4 *The weak law of large numbers.*²

Let $(U_n)_{n \in \mathbb{N}}$ be a sequence of iid random variables or random vectors. Let U be a random variable or vector with the same probability distribution as the U_n s. If $E(|U|) < +\infty$, then the sample mean $\bar{U}_n = \frac{1}{n} \sum_{i=1}^n U_i$ converges in probability towards $E(U)$ when n goes to infinity.

Can someone explain with words what this theorem means?

The sample mean of n iid random variables will get arbitrarily close to their expectation with an arbitrarily large probability when the sample size gets large. The sample mean, which is a random variable, i.e. something unpredictable, with a variance, converges to a constant when the sample gets large.

The expectation of a uniform distribution on $[0, 1]$ is equal to 0.5. Assume you run the code below in Stata. Do you expect `r1` or `r2` to be closer to 0.5? Why?

```
set obs 10
gen u=uniform()
sum u
scalar r1=r(mean)
drop _all
set obs 1000
gen u=uniform()
sum u
scalar r2=r(mean)
di r1, r2
```

²This section introduces several theorems from asymptotic statistics. I do not prove those theorems, and you do not need to know the proofs. If you are interested, a good (but tough) reference is van der Vaart (2000), chapter 2.

r2 should be closer to 0.5 than r1. This follows from the weak law of large numbers: with a large sample (e.g. 1000 observations), the **sample mean of n iid random variables will get arbitrarily close to their expectation**. On the other hand, with a small sample size (e.g. 10 observations), the sample mean might be far from the expectation.

To **prove that an estimator is consistent, we usually use the weak law of large numbers**.

Use the weak law of large numbers to prove that $\hat{E}(Y)$ is a consistent estimator of $E(Y)$.

$$\hat{E}(Y) = \frac{1}{n} \sum_{i=1}^n Y_i \xrightarrow{P} E(Y).$$

The convergence arrow follows from the fact that the **random variables $(Y_n)_{n \in \mathbb{N}}$ are iid**, from the fact that $E(|Y|) < +\infty$,³ and from the weak law of large numbers.

Definition 1.3.5 $\sqrt{n}$ -consistency of an estimator.

$\hat{\theta}_n$ is a $\sqrt{n}$ -consistent estimator of θ if

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} Z$$

for a random variable Z finite and different from 0 with probability 1. If Z follows a $N(0, \sigma^2)$ distribution, $\hat{\theta}_n$ is an asymptotically normal estimator, and σ^2 is its asymptotic variance.

$\sqrt{n}$ -consistency is stronger than consistency,⁴ because on top of telling us that $\hat{\theta}_n$ converges towards θ , it also tells us the speed at which it does so. Loosely speaking, if $\hat{\theta}_n$ is $\sqrt{n}$ -consistent,

$$\begin{aligned} \sqrt{n}(\hat{\theta}_n - \theta) &\approx Z \\ \Leftrightarrow \hat{\theta}_n - \theta &\approx \frac{Z}{\sqrt{n}}. \end{aligned}$$

With probability 1, $\frac{Z}{\sqrt{n}}$ converges to 0 when n goes to infinity, because Z is finite with probability 1, while $\sqrt{n}$ goes to infinity. Therefore, the difference between $\hat{\theta}_n$ and θ vanishes to 0 at the same speed as $\frac{1}{\sqrt{n}}$.

Assume we have three estimators of θ . All of them are consistent. The first converges to θ at the same speed as $\frac{1}{\ln n}$ goes to 0. The second converges to θ at the same speed as $\frac{1}{e^n}$ goes to 0. The third converges to θ at the same speed as $\frac{1}{\sqrt{n}}$. Which one is the best of those estimators? Which one is the worst?

³We have assumed that $E(Y^2) < +\infty$. One can show that this implies that $E(|Y|) < +\infty$.

⁴One can show that $\sqrt{n}$ -consistency implies consistency.

The best estimator is the second one: it converges towards θ at a faster rate than the other 2. The worst is the first one: it converges towards θ at a slower rate than the other 2. The reason we focus on $\sqrt{n}$ -consistent estimators and not, say, on e^n -consistent estimators is that often $\sqrt{n}$ is the fastest rate of convergence we can achieve.⁵

Theorem 1.3.6 The *central limit theorem*.

Let $(U_n)_{n \in \mathbb{N}}$ be a *sequence of iid random variables or random vectors*. Let U be a random variable or vector with the same probability distribution as the U_n s. If $E(|UU'|) < +\infty$, then

$$\sqrt{n}(\bar{U}_n - E(U)) \xrightarrow{d} N(0, V(U)).$$

When $n \rightarrow +\infty$, what is the limit of $\bar{U}_n - E(U)$? What is the limit of $\sqrt{n}$? According to the central limit theorem, which of these two terms dominates?

To simplify the discussion, let's assume that the U_n s and U are random variables. It follows from the weak law of large numbers that $\bar{U}_n - E(U) \rightarrow 0$ (actually, that follows from the strong law of large numbers, but that's no big deal). On the other hand, $\sqrt{n} \rightarrow +\infty$. None of the two terms dominates the other. The product of the two converges towards a finite real number different from 0: with probability 1, a $N(0, V(U))$ variable is finite and different from 0.⁶ This shows that $\bar{U}_n - E(U) \rightarrow 0$ at the same rate as $\frac{1}{\sqrt{n}}$ goes to 0. If it were going to 0 at a faster rate, $\bar{U}_n - E(U)$ would dominate and $\sqrt{n}(\bar{U}_n - E(U))$ would go to 0. If it were going to 0 at a slower rate, $\sqrt{n}$ would dominate and $\sqrt{n}(\bar{U}_n - E(U))$ would go to ∞ .

⁵Sometimes, we can find estimators converging at a faster rate, in a which case we say that they are super consistent, or super efficient.

⁶Provided $V(U) > 0$.

The expectation of a uniform distribution on $[0, 1]$ is equal to 0.5. Assume you run the code below in Stata. What should the final histogram look like? E.g.: like a dog? like a cat?

```

qui{
matrix A=0
forvalue i=1/200 {
drop _all
set obs 1000
gen u=uniform()
sum u
scalar n=sqrt(1000)*(r(mean)-0.5)
matrix A=A \n
}
}
drop _all
svmat A
drop if _n==1
histogram A1

```

The final histogram should look like a bell, not like a dog or a cat.

To prove $\sqrt{n}$ -consistency, we often use the central limit theorem.

Use the central limit theorem to prove that $\hat{E}(Y)$ is a $\sqrt{n}$ -consistent (and even asymptotically normal) estimator of $E(Y)$.

$$\sqrt{n}(\hat{E}(Y) - E(Y)) = \sqrt{n}(\bar{Y}_n - E(Y)) \xrightarrow{d} N(0, V(Y)).$$

The convergence arrow follows from the following facts. $(Y_n)_{n \in \mathbb{N}}$ is a sequence of iid random variables. Because Y is of 1×1 vector (it's just a scalar random variable), $|YY'| = Y^2$, and we have assumed that $E(Y^2) < +\infty$. Therefore, we can apply the central limit theorem. $N(0, V(Y))$ is finite and different from 0 with probability 1 (we have assumed that $V(Y) > 0$). This proves that $\hat{E}(Y)$ is a $\sqrt{n}$ -consistent estimator of $E(Y)$.

1.4 Confidence intervals

Definition 1.4.1 *Definition of confidence intervals with asymptotic coverage $1 - \alpha$*

$\underline{\theta}_n$ and $\bar{\theta}_n$ define a confidence interval with asymptotic coverage $1 - \alpha$ if they are both functions of $(Y_i)_{1 \leq i \leq n}$ and known real numbers, and if

$$\lim_{n \rightarrow +\infty} P(\theta \in [\underline{\theta}_n, \bar{\theta}_n]) = 1 - \alpha.$$

For instance, if α is equal to, say, 0.05, then when n is large, from an ex-ante perspective θ has a 95% probability of belonging to $[\underline{\theta}_n, \bar{\theta}_n]$.

In the survey in Subsection 1.1.1, where does the randomness in $(Y_i)_{1 \leq i \leq n}$ come from?

The randomness in $(Y_i)_{1 \leq i \leq n}$ comes from the fact we randomly draw the n registered voters we will interview. There are $8,448,674^n$ possible samples with replacement of n voters out of 8,448,674, and we will randomly draw one of those samples.

Therefore, if we derive a 95% confidence interval for $E(Y)$, what does the statement “ $E(Y)$ has a 95% probability of belonging to the confidence interval” mean?

$P(E(Y) \in \text{Confidence interval}) = 0.95$ means that $E(Y)$ belongs to the confidence interval for $0.95 \times 8,448,674^n$ of these $8,448,674^n$ possible samples (if n is large enough). Therefore, from an ex-ante perspective, before we draw the 1 sample out of those $8,448,674^n$ we will get to observe, $E(Y)$ has a 95% probability of belonging to the confidence interval.

Once we draw the sample and the $(Y_i)_{1 \leq i \leq n}$ get realized, if $\underline{\theta}_n = 0.2$ and $\bar{\theta}_n = 0.4$, is the statement “ $E(Y)$ has a 95% probability of being included between 0.2 and 0.4 correct”?

No, this statement is incorrect. $E(Y) \in [0.2, 0.4]$ only involves deterministic quantities, so the probability that $E(Y) \in [0.2, 0.4]$ is either 0 or 1. On the other hand, we can be 95%

“confident” that $E(Y) \in [0.2, 0.4]$. Indeed, from an ex-ante perspective there are only 5% of the samples we can draw for which $E(Y) \notin [\underline{\theta}_n, \bar{\theta}_n]$. So maybe $E(Y) \notin [0.2, 0.4]$, but then we have been quite unlucky and have drawn one of the few samples for which $E(Y) \notin [\underline{\theta}_n, \bar{\theta}_n]$.

For many parameters, the three ingredients we need to build confidence intervals are the central limit theorem, the **continuous** mapping theorem, and the Slutsky lemma.

Theorem 1.4.2 *The **continuous mapping theorem**.*

Let g be a continuous function, and let $(U_n)_{n \in \mathbb{N}}$ be a sequence of random variables or vectors.

1. If $(U_n)_{n \in \mathbb{N}}$ converges in probability towards a random variable or vector U , then $(g(U_n))_{n \in \mathbb{N}}$ converges in probability towards $g(U)$.
2. If $(U_n)_{n \in \mathbb{N}}$ converges in distribution towards a random variable or vector U , then $(g(U_n))_{n \in \mathbb{N}}$ converges in distribution towards $g(U)$.

Use the continuous mapping theorem to prove that $\hat{V}(Y) = \frac{1}{n} \sum_{i=1}^n Y_i^2 - \left(\frac{1}{n} \sum_{i=1}^n Y_i\right)^2$ converges in probability towards $V(Y) = E(Y^2) - (E(Y))^2$.

$(Y_n^2, Y_n)_{n \in \mathbb{N}}$ is an iid sequence of random vectors, and we have assumed that $E(|Y|) < +\infty$ and $E(Y^2) < +\infty$. Therefore, it follows from the **weak law of large numbers** that

$$\left(\frac{1}{n} \sum_{i=1}^n Y_i^2, \frac{1}{n} \sum_{i=1}^n Y_i \right) \xrightarrow{P} (E(Y^2), E(Y))$$

Moreover, $g : (x, y) \mapsto x - y^2$ is a continuous function. Therefore, the continuous mapping theorem implies that

$$\hat{V}(Y) = g \left(\frac{1}{n} \sum_{i=1}^n Y_i^2, \frac{1}{n} \sum_{i=1}^n Y_i \right) \xrightarrow{P} g(E(Y^2), E(Y)) = V(Y).$$

Lemma 1.4.1 *The **Slutsky lemma**.*

Let $(U_n)_{n \in \mathbb{N}}$ be a sequence of random variables or vectors converging in **distribution** towards a random variable or vector X , and let $(V_n)_{n \in \mathbb{N}}$ be a sequence of **iid random variables** or vectors converging in **probability** towards a real number l . Then,

$$\begin{aligned} U_n + V_n &\xrightarrow{d} X + l, \\ U_n V_n &\xrightarrow{d} Xl. \end{aligned}$$

Use the Slutsky lemma to determine the asymptotic distribution of $\sqrt{n}(\hat{E}(Y) - E(Y)) + \hat{E}(Y)$.

$\sqrt{n}(\hat{E}(Y) - E(Y))$ converges in distribution towards a $N(0, V(Y))$ distribution, while $\hat{E}(Y)$ converges in probability towards $E(Y)$. Therefore, $\sqrt{n}(\hat{E}(Y) - E(Y)) + \hat{E}(Y)$ converges in distribution towards the sum of a $N(0, V(Y))$ and $E(Y)$, which follows a $N(E(Y), V(Y))$ distribution (the sum of a normal distribution and a constant follows a normal distribution).

Now, let's build up a confidence interval for $E(Y)$. For every x in $(0, 1)$, let q_x denote the x^{th} quantile of the $N(0, 1)$ distribution: $q_x = \Phi^{-1}(x)$, where Φ denotes the cdf of the $N(0, 1)$ distribution. $q_{0.975}$ is such that a $N(0, 1)$ has a 0.975 probability of being below $q_{0.975}$. Because the $N(0, 1)$ distribution is symmetric, one can show that $q_{1-x} = -q_x$. For instance, $q_{0.975} = 1.96$ and $q_{0.025} = -1.96$.

Theorem 1.4.3 *Confidence interval for an expectation*

$$IC(\alpha) = \left[\hat{E}(Y) - q_{1-\frac{\alpha}{2}} \sqrt{\hat{V}(\hat{E}(Y))}, \hat{E}(Y) + q_{1-\frac{\alpha}{2}} \sqrt{\hat{V}(\hat{E}(Y))} \right]$$

is a confidence interval for $E(Y)$ with asymptotic coverage equal to $1 - \alpha$.

Proof

What we need to prove is

$$\lim_{n \rightarrow +\infty} P \left(E(Y) \in \left[\hat{E}(Y) - q_{1-\frac{\alpha}{2}} \sqrt{\hat{V}(\hat{E}(Y))}, \hat{E}(Y) + q_{1-\frac{\alpha}{2}} \sqrt{\hat{V}(\hat{E}(Y))} \right] \right) = 1 - \alpha.$$

First, using the continuous mapping theorem, we have shown above that $\hat{V}(Y) \xrightarrow{P} V(Y)$. Using once again the continuous mapping theorem, this implies that $\sqrt{\hat{V}(Y)} \xrightarrow{P} \sqrt{V(Y)}$, because $x \mapsto \sqrt{x}$ is a continuous function. Finally, using once again the continuous mapping theorem, this implies that $\frac{1}{\sqrt{\hat{V}(Y)}} \xrightarrow{P} \frac{1}{\sqrt{V(Y)}}$, because $x \mapsto \frac{1}{x}$ is continuous at any $x > 0$ and we have assumed that $V(Y) > 0$.

Then,

$$\begin{aligned} & \frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \\ &= \sqrt{n}(\hat{E}(Y) - E(Y)) \frac{1}{\sqrt{\hat{V}(Y)}} \\ &\xrightarrow{d} N(0, V(Y)) \frac{1}{\sqrt{V(Y)}} \\ &\sim N(0, 1), \end{aligned} \tag{1.4.1}$$

where $\sim$ denotes equality in distribution. First, the convergence arrow follows from the following facts. The central limit theorem implies that $\sqrt{n}(\hat{E}(Y) - E(Y))$ converges in distribution towards a random variable X following a $N(0, V(Y))$ random variable. We have just shown above that $\frac{1}{\sqrt{\hat{V}(Y)}}$ converges in probability towards $\frac{1}{\sqrt{V(Y)}}$. Therefore, it follows from the Slutsky lemma that $\sqrt{n}(\hat{E}(Y) - E(Y)) \frac{1}{\sqrt{\hat{V}(Y)}}$ converges in distribution towards the product of $X \times \frac{1}{\sqrt{V(Y)}}$. Second, the equality in distribution follows from the following arguments. The product of a normal random variable and a real number still follows a normal distribution, so $X \times \frac{1}{\sqrt{V(Y)}}$ follows a normal distribution. The mean of that distribution is $E(X \times \frac{1}{\sqrt{V(Y)}}) = \frac{1}{\sqrt{V(Y)}}E(X) = 0$. The variance of that distribution is $V(X \times \frac{1}{\sqrt{V(Y)}}) = \frac{1}{V(Y)}V(X) = \frac{1}{V(Y)}V(Y) = 1$. Therefore, $X \times \frac{1}{\sqrt{V(Y)}}$ follows a $N(0, 1)$ distribution. Overall, this shows that $\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}}$ converges in distribution towards a $N(0, 1)$.

Then,

$$\begin{aligned} \hat{E}(Y) - q_{1-\frac{\alpha}{2}}\sqrt{\hat{V}(\hat{E}(Y))} &\leq E(Y) \leq \hat{E}(Y) + q_{1-\frac{\alpha}{2}}\sqrt{\hat{V}(\hat{E}(Y))} \\ \Leftrightarrow -q_{1-\frac{\alpha}{2}} &\leq \frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \leq q_{1-\frac{\alpha}{2}}. \end{aligned} \quad (1.4.2)$$

Therefore,

$$\begin{aligned} &\lim_{n \rightarrow +\infty} P \left(E(Y) \in \left[\hat{E}(Y) - q_{1-\frac{\alpha}{2}}\sqrt{\frac{\hat{V}(Y)}{n}}, \hat{E}(Y) + q_{1-\frac{\alpha}{2}}\sqrt{\frac{\hat{V}(Y)}{n}} \right] \right) \\ &= \lim_{n \rightarrow +\infty} P \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \in \left[-q_{1-\frac{\alpha}{2}}, q_{1-\frac{\alpha}{2}} \right] \right) \\ &= \lim_{n \rightarrow +\infty} P \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \leq q_{1-\frac{\alpha}{2}} \right) - \lim_{n \rightarrow +\infty} P \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} < -q_{1-\frac{\alpha}{2}} \right) \\ &= P \left(N(0, 1) \leq q_{1-\frac{\alpha}{2}} \right) - P \left(N(0, 1) < -q_{1-\frac{\alpha}{2}} \right) \\ &= \Phi(q_{1-\frac{\alpha}{2}}) - \Phi(-q_{1-\frac{\alpha}{2}}) \\ &= \Phi(q_{1-\frac{\alpha}{2}}) - \Phi(q_{\frac{\alpha}{2}}) \\ &= 1 - \frac{\alpha}{2} - \frac{\alpha}{2} = 1 - \alpha. \end{aligned}$$

The first equality follows from Equation (1.4.2). The third follows from the definition of convergence in distribution. The fifth follows from the fact that $q_{1-x} = -q_x$.

QED.

Notice that the confidence interval for $E(Y)$ is centered around our estimator. So our interval writes as our estimator $\pm$ some margin of error term. The margin of error, $q_{1-\frac{\alpha}{2}}\sqrt{\frac{\hat{V}(Y)}{n}}$,

vanishes to 0 when n goes to infinity, at the same speed as some constant times $\frac{1}{\sqrt{n}}$. This reflects the fact that our estimator is $\sqrt{n}$ -consistent. The margin of error is increasing in the sample variance: random variables which have a large variance are more difficult to predict and therefore have larger confidence intervals. Finally, it is decreasing in α : a lower α implies a greater $q_{1-\frac{\alpha}{2}}$ and therefore a wider confidence interval. Confidence intervals with a greater degree of confidence (say 99%) are wider than confidence intervals with a lower degree of confidence (say 80%).

Using only the CLT, one can show that

$$\lim_{n \rightarrow +\infty} P \left(E(Y) \in \left[\hat{E}(Y) - q_{1-\frac{\alpha}{2}} \sqrt{\frac{V(Y)}{n}}, \hat{E}(Y) + q_{1-\frac{\alpha}{2}} \sqrt{\frac{V(Y)}{n}} \right] \right) = 1 - \alpha.$$

Is $\left[\hat{E}(Y) - q_{1-\frac{\alpha}{2}} \sqrt{\frac{V(Y)}{n}}, \hat{E}(Y) + q_{1-\frac{\alpha}{2}} \sqrt{\frac{V(Y)}{n}} \right]$ a confidence interval?

No, this is not a confidence interval. The upper and lower bound are not only functions of $(Y_i)_{1 \leq i \leq n}$, they also depend on $V(Y)$. Therefore, even when the $(Y_i)_{1 \leq i \leq n}$ get realized, we cannot compute this confidence interval. The trick is to replace $V(Y)$ by a consistent estimator. The Slutsky lemma ensures that this will yield a valid confidence interval.

In order to derive a confidence interval, we will almost always proceed as follows:

1. Show that the estimator is asymptotically normal.
2. Derive a consistent estimator of its asymptotic variance.
3. Derive the confidence interval invoking the Slutsky lemma.

Finally, notice that in the proof of Theorem 1.4.3, we have shown that

$$\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \xrightarrow{d} N(0, 1).$$

For most of the estimators we are going to consider in this course, we will have that (estimator-estimator)/estimator of the variance of the estimator $\xrightarrow{d} N(0, 1)$. This is the key ingredient we use to build confidence intervals and statistical tests.

Now that we have defined confidence intervals, we can introduce the concept of a p-value.

Definition 1.4.4 *The p-value.*

Let $IC(\alpha)$ be a family of confidence intervals for a target parameter θ with asymptotic coverage $1 - \alpha$. The p-value is $\inf_{\alpha \in (0,1)} \{\alpha : 0 \notin IC(\alpha)\}$.

The p-value is the lowest value of α such that 0 does not belong to $IC(\alpha)$.

1.5 The t-test: level, power function, and minimum detectable difference

1.5.1 ... Assuming normality and known variance.

In this section, we are going to assume that $Y \sim N(E(Y), \sigma^2)$: Y and therefore the Y_i are normally distributed. Moreover, we are going to assume that σ^2 is known. Notice that under this known variance assumption, $V\left(\widehat{E}(Y)\right) = \frac{V(Y)}{n} = \frac{\sigma^2}{n}$ is also known.

Normality and known variance are strong assumptions. In the next subsection, we will relax those assumptions, and that will not change the main result of this subsection much.

Also, throughout this section we assume that the estimand we are interested in is $E(Y)$.

Finally, the following notation will be useful.

Definition 1.5.1 *A notation for the possible probability distributions of $(Y_i)_{1 \leq i \leq n}$ and Y .*

For any event A involving $(Y_i)_{1 \leq i \leq n}$ and/or Y , let $P_x(A)$ denote the probability of the event A if $E(Y) = x$.

What is the value of $P_0\left(\frac{Y}{\sigma} \leq 0\right)$?

$$P_0\left(\frac{Y}{\sigma} \leq 0\right) = P_0\left(\frac{Y - E(Y)}{\sigma} \leq 0\right) = 0.5.$$

$P_0\left(\frac{Y}{\sigma} \leq 0\right)$ is the probability that $\frac{Y}{\sigma} \leq 0$ if $E(Y) = 0$. But if $E(Y) = 0$, having $\frac{Y}{\sigma} \leq 0$ is the same thing as having $\frac{Y - E(Y)}{\sigma} \leq 0$, hence the first equality. Then, we have assumed that Y follows a normal distribution with variance σ^2 . Therefore, $\frac{Y - E(Y)}{\sigma}$ follows a $N(0, 1)$ distribution, hence the second equality.

What is the value of $P_{1.96\sigma}\left(\frac{Y}{\sigma} \leq 0\right)$?

$$\begin{aligned}
P_{1.96\sigma} \left(\frac{Y}{\sigma} \leq 0 \right) &= P_{1.96\sigma} \left(\frac{Y}{\sigma} - 1.96 \leq -1.96 \right) \\
&= P_{1.96\sigma} \left(\frac{Y - 1.96\sigma}{\sigma} \leq -1.96 \right) \\
&= P_{1.96\sigma} \left(\frac{Y - E(Y)}{\sigma} \leq -1.96 \right) \\
&= 0.025.
\end{aligned}$$

$P_{1.96\sigma} \left(\frac{Y - 1.96\sigma}{\sigma} \leq -1.96 \right)$ is the probability that $\frac{Y - 1.96\sigma}{\sigma} \leq -1.96$ if $E(Y) = 1.96\sigma$. But if $E(Y) = 1.96\sigma$, having $\frac{Y - 1.96\sigma}{\sigma} \leq -1.96$ is the same thing as having $\frac{Y - E(Y)}{\sigma} \leq -1.96$, hence the third equality. Then, we have assumed that Y follows a normal distribution with variance σ^2 . Therefore, $\frac{Y - E(Y)}{\sigma}$ follows a $N(0, 1)$ distribution, hence the fourth equality.

Definition 1.5.2 *Statistical tests*

A statistical test T_n is the indicator function of an event involving only $(Y_i)_{1 \leq i \leq n}$ and known real numbers.

$T_n = 1 \{Y_1 + Y_n \geq 3\}$ is a statistical test. Once the $(Y_i)_{1 \leq i \leq n}$ get realized, one can check whether $Y_1 + Y_n$ is indeed greater than 3 or not. $[3; +\infty)$ is called the critical region of the test: T_n is equal to 1 whenever the statistic involved in the test $(Y_1 + Y_n)$ belongs to $[3; +\infty)$. $1 \{Y_2 + E(Y) \geq 3\}$ is not a statistical test: $\{Y_2 + E(Y) \geq 3\}$ involves $E(Y)$, an unknown real number.

A statistical test is always attached to two mutually exclusive hypothesis on an estimand of interest, $E(Y)$ here. One hypothesis is called H_0 , the null hypothesis, while the other is called H_1 , the alternative hypothesis. H_0 and H_1 are two mutually exclusive sets of values for that estimand. For instance, we might be interested in testing $E(Y) \in H_0 = \{0\}$ against $E(Y) \in H_1 = \{2\}$. $H_0 = \{0\}$ and $H_1 = \{2\}$ are mutually exclusive set of possible values for $E(Y)$. We might also be interested in testing $E(Y) \in H_0 = \{0\}$ against $E(Y) \in H_1 = \mathbb{R} \setminus \{0\}$. $H_0 = \{0\}$ and $H_1 = \mathbb{R} \setminus \{0\}$ are mutually exclusive set of values for $E(Y)$.

T_n is a decision rule to choose between $E(Y) \in H_0$ and $E(Y) \in H_1$ once the $(Y_i)_{1 \leq i \leq n}$ get realized. If $T_n = 0$ we will choose $E(Y) \in H_0$ (we also say that we accept $E(Y) \in H_0$), while if $T_n = 1$ we will choose $E(Y) \in H_1$ (we also say that we reject or do not accept $E(Y) \in H_0$). Our decision rule T_n can therefore lead us to make two types of mistakes: choose $E(Y) \in H_1$ while $E(Y) \in H_0$ (this is called type 1 error), and choose $E(Y) \in H_0$ while $E(Y) \in H_1$ (this is called type 2 error). To assess whether a test is good or not, we want to know, from an ex-ante perspective, what the chances are that we make either of those mistakes. This leads us to define the level and the power of a test.

Definition 1.5.3 *Level, power, and power function of a test*

Let T_n be a test intended at testing $E(Y) \in H_0$ against $E(Y) \in H_1$.

The level of T_n is the maximum of $x \mapsto P_x(T_n = 1)$ over all $x \in H_0$.⁷

The power of T_n is the minimum of $x \mapsto P_x(T_n = 1)$ over all $x \in H_1$.⁸

The function $x \mapsto P_x(T_n = 1)$ is called the power function of T_n .

Remember, we are interested in testing $E(Y) \in H_0$ against $E(Y) \in H_1$, where H_0 and H_1 are two mutually exclusive sets of possible values for $E(Y)$.

What does the level of T_n represent?

For any $x \in H_0$, $P_x(T_n = 1)$ is the probability that T_n leads us to choose $E(Y) \in H_1$ if $E(Y) = x$. As $x \in H_0$, this decision would be incorrect. Therefore, for any $x \in H_0$, $P_x(T_n = 1)$ is the probability that T_n leads us to make a type 1 error if $E(Y) = x$. The level of T_n is the maximum of $x \mapsto P_x(T_n = 1)$ over all $x \in H_0$. Therefore, the level of T_n is a worst-case measure of the probability that it leads us to make a type 1 error, over all $x \in H_0$. If test A has a lower level than test B, then test A is better because the (worst-case) probability that A leads you to make a type 1 error is lower than the same (worst-case) probability for B.

What does the power of T_n represent?

For any $x \in H_1$, $P_x(T_n = 1)$ is the probability that T_n leads us to choose $E(Y) \in H_1$ if $E(Y) = x$. As $x \in H_1$, this decision would be correct. Therefore, for any $x \in H_1$, $P_x(T_n = 1)$ is the probability that T_n leads us not to make a type 2 error if $E(Y) = x$. The power of T_n is the minimum of $x \mapsto P_x(T_n = 1)$ over all $x \in H_1$. Therefore, the power of T_n is a worst-case measure of the probability that our test leads us not to make a type 2 error, over all $x \in H_1$. If test A has a greater power than test B, then test A is better because the (worst-case) probability that A leads you not to make a type 2 error is higher than the same (worst-case) probability for B.

Presumably, the test you will most commonly use in your dissertation is the following one.

⁷An issue is that this maximum might not exist. For instance, $x \mapsto x^2$ does not have a maximum on $[0, 1)$. So the rigorous definition of the level of T_n is $\sup_{x \in H_0} P_x(T_n = 1)$.

⁸Here as well, this minimum might not exist. So the rigorous definition of the power of T_n is $\inf_{x \in H_1} P_x(T_n = 1)$.

Definition 1.5.4 *T-test with normally distributed variables and a known variance*

For any $\alpha \in (0, 1)$, let

$$T_n(\alpha) = 1 \left\{ \left| \frac{\widehat{E}(Y)}{\sqrt{V(\widehat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}} \right\}.$$

$T_n(\alpha)$ is called the *t-test for the mean*.

For any $\alpha \in (0, 1)$, $T_n(\alpha)$ indeed is a test as it is the indicator function of an event which only depends on the $(Y_i)_{1 \leq i \leq n}$ and known real numbers (remember, under the assumption that σ^2 is known, $V(\widehat{E}(Y))$ is also known). This event involves $\frac{\widehat{E}(Y)}{\sqrt{V(\widehat{E}(Y))}}$ which is called the t-statistic. The role of α will become clearer soon. $T_n(\alpha)$ is used to test $E(Y) = 0$ against some alternative hypothesis. The goal of what follows is to determine for which alternative hypothesis we can achieve both low level and high power. It follows from the definition of level and power that to answer this question, we need to study the power function of $T_n(\alpha)$, $x \mapsto P_x(T_n(\alpha) = 1)$. Before that, we need to state a Lemma.

Lemma 1.5.1 *Distribution of the standardized mean of normal variables*

If $(Y_i)_{1 \leq i \leq n}$ are n iid random variables following a normal distribution with mean $E(Y)$ and variance σ^2 , then $\frac{\widehat{E}(Y) - E(Y)}{\sqrt{V(\widehat{E}(Y))}}$ follows a $N(0, 1)$ distribution.

Proof

I let you check that $E\left(\frac{\widehat{E}(Y) - E(Y)}{\sqrt{V(\widehat{E}(Y))}}\right) = 0$ and that $V\left(\frac{\widehat{E}(Y) - E(Y)}{\sqrt{V(\widehat{E}(Y))}}\right) = 1$, that's easy. That $\frac{\widehat{E}(Y) - E(Y)}{\sqrt{V(\widehat{E}(Y))}}$ follows a normal distribution is a direct consequence of the fact that if U and V are independent random variables following normal distributions and λ is a real number, $\lambda U + V$ also follows a normal distribution. Proving this latter statement can be achieved by computing the characteristic function of $\lambda U + V$.

QED.

Theorem 1.5.5 *Some properties of the power function of the t-test with normally distributed variables and a known variance*

1. For any $x \neq 0$, $P_x(T_n(\alpha) = 1)$ is increasing in n and $\lim_{n \rightarrow +\infty} P_x(T_n(\alpha) = 1) = 1$.
2. For any $(x, n) \in \mathbb{R} \times \mathbb{N}$, $P_x(T_n(\alpha) = 1) = \Phi\left(-q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\widehat{E}(Y))}}\right) + \Phi\left(-q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}}\right) \dots$
3. $P_0(T_n(\alpha) = 1) = \alpha$.
4. For any $(x, n) \in \mathbb{R} \times \mathbb{N}$, $P_x(T_n(\alpha) = 1) = P_{-x}(T_n(\alpha) = 1)$: $x \mapsto P_x(T_n(\alpha) = 1)$ is symmetric around the y-axis.
5. For any $n \in \mathbb{N}$, $x \mapsto P_x(T_n(\alpha) = 1)$ is continuously differentiable.

6. For any $n \in \mathbb{N}$, $x \mapsto P_x(T_n(\alpha) = 1)$ is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, +\infty)$. Therefore, it reaches a minimum at 0.

7. For any $n \in \mathbb{N}$, $\lim_{x \rightarrow +\infty} P_x(T_n(\alpha) = 1) = \lim_{x \rightarrow -\infty} P_x(T_n(\alpha) = 1) = 1$.

Proof

We start by proving the second statement.

$$\begin{aligned}
P_x(T_n(\alpha) = 1) &= P_x \left(\left| \frac{\hat{E}(Y)}{\sqrt{V(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}} \right) \\
&= P_x \left(\left| \frac{\hat{E}(Y) - x}{\sqrt{V(\hat{E}(Y))}} + \frac{x}{\sqrt{V(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}} \right) \\
&= P_x \left(\left| \frac{\hat{E}(Y) - E(Y)}{\sqrt{V(\hat{E}(Y))}} + \frac{x}{\sqrt{V(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}} \right) \\
&= P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{V(\hat{E}(Y))}} + \frac{x}{\sqrt{V(\hat{E}(Y))}} > q_{1-\frac{\alpha}{2}} \right) \\
&\quad + P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{V(\hat{E}(Y))}} + \frac{x}{\sqrt{V(\hat{E}(Y))}} < -q_{1-\frac{\alpha}{2}} \right) \\
&= 1 - P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{V(\hat{E}(Y))}} \leq q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\hat{E}(Y))}} \right) \\
&\quad + P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{V(\hat{E}(Y))}} < -q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\hat{E}(Y))}} \right) \\
&= 1 - \Phi \left(q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\hat{E}(Y))}} \right) + \Phi \left(-q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\hat{E}(Y))}} \right) \\
&= \Phi \left(-q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\hat{E}(Y))}} \right) + \Phi \left(-q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\hat{E}(Y))}} \right). \tag{1.5.1}
\end{aligned}$$

The third equality follows from the following argument. $P_x \left(\left| \frac{\hat{E}(Y) - x}{\sqrt{V(\hat{E}(Y))}} + \frac{x}{\sqrt{V(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}} \right)$

is the probability that $\left| \frac{\hat{E}(Y) - x}{\sqrt{V(\hat{E}(Y))}} + \frac{x}{\sqrt{V(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}}$ if $E(Y) = x$. But if $E(Y) = x$, having $\left| \frac{\hat{E}(Y) - x}{\sqrt{V(\hat{E}(Y))}} + \frac{x}{\sqrt{V(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}}$ is the same thing as having $\left| \frac{\hat{E}(Y) - E(Y)}{\sqrt{V(\hat{E}(Y))}} + \frac{x}{\sqrt{V(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}}$.

The fourth equality follows from the following fact. Let a be a real number, and let b be a positive real number. Then $|a| > b$ if and only if $a > b$ or $a < -b$. The sixth equality follows from Lemma 1.5.1. The last equality follows from the fact that for any real number z , $\Phi(z) = 1 - \Phi(-z)$ (for instance, $\Phi(1.96) = 0.975$ and $1 - \Phi(-1.96) = 0.975$).

Then, let's prove the third statement.

$$P_0(T_n(\alpha) = 1) = \Phi\left(-q_{1-\frac{\alpha}{2}}\right) + \Phi\left(-q_{1-\frac{\alpha}{2}}\right) = 2\Phi\left(q_{\frac{\alpha}{2}}\right) = 2\frac{\alpha}{2} = \alpha.$$

Then, let's prove the fourth statement.

$$\begin{aligned} P_{-x}(T_n(\alpha) = 1) &= \Phi\left(-q_{1-\frac{\alpha}{2}} + \frac{-x}{\sqrt{V(\widehat{E}(Y))}}\right) + \Phi\left(-q_{1-\frac{\alpha}{2}} - \frac{-x}{\sqrt{V(\widehat{E}(Y))}}\right) \\ &= \Phi\left(-q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}}\right) + \Phi\left(-q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\widehat{E}(Y))}}\right) \\ &= P_x(T_n(\alpha) = 1). \end{aligned}$$

The first equality follows from the second statement of the theorem.

The fifth statement directly follows from the second, and from the fact $x \mapsto \Phi(x)$ is continuously differentiable.

To prove the sixth statement, we start by proving that $x \mapsto P_x(T_n(\alpha) = 1)$ is increasing on $[0, +\infty)$. Differentiating both the lhs and the rhs of the first statement wrt to x yields

$$\frac{\partial P_x(T_n(\alpha) = 1)}{\partial x} = \frac{1}{\sqrt{V(\widehat{E}(Y))}} \left(\phi\left(-q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\widehat{E}(Y))}}\right) - \phi\left(-q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}}\right) \right),$$

where $\phi(\cdot)$ denotes the pdf of the $N(0, 1)$ distribution. At $x = 0$, this derivative is equal to 0. Then, $\phi(\cdot)$ is symmetric around 0, decreasing on $[0, +\infty)$ and increasing on $(-\infty, 0]$.

Therefore, for any two real numbers a and b , $\phi(a) - \phi(b) > 0$ if and only if $|a| < |b|$ (i.e. if and only if a is closer to 0 than b). For any $x > 0$, $\left| -q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\widehat{E}(Y))}} \right| < \left| -q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}} \right|$.

Indeed, $\left| -q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}} \right| = q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\widehat{E}(Y))}}$. Then, $\left| -q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\widehat{E}(Y))}} \right| = q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}}$ if $\frac{x}{\sqrt{V(\widehat{E}(Y))}} \leq q_{1-\frac{\alpha}{2}}$, and $\left| -q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\widehat{E}(Y))}} \right| = \frac{x}{\sqrt{V(\widehat{E}(Y))}} - q_{1-\frac{\alpha}{2}}$ otherwise.

In both cases, $\left| -q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\widehat{E}(Y))}} \right| < \left| -q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}} \right|$. Therefore, for any $x > 0$

$\phi\left(q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}}\right) - \phi\left(-q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}}\right) > 0$ and $\frac{\partial P_x(T_n(\alpha)=1)}{\partial x} > 0$. This proves

that $x \mapsto P_x(T_n(\alpha) = 1)$ is strictly increasing on $[0, +\infty)$. Given the fourth statement of the theorem, this implies that $x \mapsto P_x(T_n(\alpha) = 1)$ is strictly decreasing on $(-\infty, 0]$. This, combined with the fact this function is continuous, implies that $x \mapsto P_x(T_n(\alpha) = 1)$ reaches a global minimum at 0.

Then let's prove the seventh statement.

$$\begin{aligned} \lim_{x \rightarrow +\infty} P_x(T_n(\alpha) = 1) &= \lim_{x \rightarrow +\infty} \Phi \left(-q_{1-\frac{\alpha}{2}} + \frac{x}{\sqrt{V(\widehat{E}(Y))}} \right) + \lim_{x \rightarrow +\infty} \Phi \left(-q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{V(\widehat{E}(Y))}} \right) \\ &= 1. \end{aligned}$$

Similarly, one can show that

$$\lim_{x \rightarrow -\infty} P_x(T_n(\alpha) = 1) = 1.$$

Finally, let's prove the first statement. I will not prove that $P_x(T_n(\alpha) = 1)$ is increasing in n , that can be achieved through the same steps as those followed to prove the sixth statement of the theorem. Now let's prove the result concerning $\lim_{n \rightarrow +\infty} P_x(T_n(\alpha) = 1)$. First, let's assume that $x > 0$.

$$\begin{aligned} \lim_{n \rightarrow +\infty} P_x(T_n(\alpha) = 1) &= \lim_{n \rightarrow +\infty} \Phi \left(-q_{1-\frac{\alpha}{2}} + x \sqrt{\frac{n}{\sigma^2}} \right) + \lim_{n \rightarrow +\infty} \Phi \left(-q_{1-\frac{\alpha}{2}} - x \sqrt{\frac{n}{\sigma^2}} \right) \\ &= 1. \end{aligned}$$

Similarly, one can show that if $x < 0$,

$$\lim_{n \rightarrow +\infty} P_x(T_n(\alpha) = 1) = 1.$$

QED.

According to the first statement of Theorem 1.5.5, does $T_n(\alpha)$ look like a good test of $E(Y) = 0$ versus $E(Y) \neq 0$?

The first statement of Theorem 1.5.5 says that for any $x \neq 0$, $P_x(T_n(\alpha) = 1)$, the probability of not making a type 2 error if $E(Y) = x$, converges to 1 when the sample size goes to infinity. This means that with an infinite sample at hand, we will reject $E(Y) = 0$ with probability 1 if $E(Y) = x$, for any $x \neq 0$. Therefore, the t-test would be a very good test of $E(Y) = 0$ versus $E(Y) \neq 0$ if we had an infinite sample: its power would be equal to 1. When the power function of a test converges to 1 when the sample size goes to infinity for any possible value of the parameter of interest in the alternative hypothesis, we say that the test is consistent

against any alternative. Therefore, the first statement of Theorem 1.5.5 shows that the t-test is consistent against any alternative.

However, we never have an infinite sample of data at hand. The second statement of Theorem 1.5.5 can help us assess the power of the t-test with a finite sample, i.e. when n is fixed. This second statement implies that with normally distributed $(Y_i)_{1 \leq i \leq n}$ with a known variance, for any fixed sample size n we can compute $P_x(T_n(\alpha) = 1)$. We can therefore plot that function. This stata code generates a plot of $x \mapsto P_x(T_n(\alpha) = 1)$ for $n = 500$, $\sigma^2 = 1$, and $\alpha = 0.05$

```
set obs 1001
gen x=( _n-501)/1000
gen power=normal(-1.96+x*sqrt(500))+normal(-1.96-x*sqrt(500))
twoway line power x, xlabel(-0.5(0.1)0.5) ylabel(0(0.1)1) xtitle("E(Y)=x", size(huge)) ytitle("Power of the t-test", size(huge))
```

This plot is shown in Figure 1.1 below.

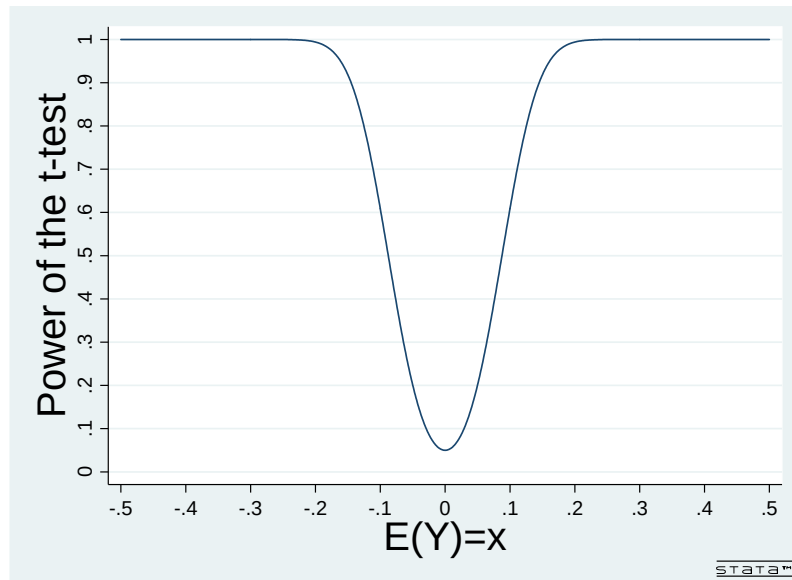


Figure 1.1: An example of a power function

Does the graph confirm the various statements of Theorem 1.5.5?

$x \mapsto P_x(T_n(\alpha) = 1)$ seems symmetric around the y -axis. $x \mapsto P_x(T_n(\alpha) = 1)$ seems continuously differentiable. $x \mapsto P_x(T_n(\alpha) = 1)$ seems decreasing on $(-\infty, 0]$ and increasing on $[0, +\infty)$. $P_0(T_n(\alpha) = 1) = 0.05$. Finally, it seems that $\lim_{x \rightarrow +\infty} P_x(T_n(\alpha) = 1) = \lim_{x \rightarrow -\infty} P_x(T_n(\alpha) = 1) = 1$.

From Figure 1.1, if $E(Y) = 0.3$ what is the probability that the t-test will lead you to reject $E(Y) = 0$ once the $(Y_i)_{1 \leq i \leq n}$ get realized? If $E(Y) = 0.2$? If $E(Y) = 0.1$? If $E(Y) = 0.05$? If $E(Y) = 0.01$? If $E(Y) = 0.001$?

If $E(Y) = 0.3$, this probability is equal to 1 essentially. If $E(Y) = 0.2$, this probability is around 0.99. If $E(Y) = 0.1$, this probability is around 0.65. If $E(Y) = 0.05$, this probability is around 0.15. If $E(Y) = 0.01$, this probability is around 0.06. If $E(Y) = 0.001$, this probability is around 0.05.

If $H_0 = \{0\}$ and $H_1 = \mathbb{R} \setminus \{0\}$, what is the power of $T_n(\alpha)$? What is its level?

Theorem 1.5.6 *Level and power of a t-test of $E(Y) = 0$ against $E(Y) \neq 0$.*

Assume the Y_i follow normal distributions with mean $E(Y)$ and variance σ^2 . If $H_0 = \{0\}$ and $H_1 = \mathbb{R} \setminus \{0\}$, then:

1. *The level of $T_n(\alpha)$ is equal to α .*
2. *The power of $T_n(\alpha)$ is equal to α .*

Proof

The level of $T_n(\alpha)$ is $P_0(T_n(\alpha) = 1)$ (this follows from the definition of the level of a test). This probability is equal to α (see Theorem 1.5.5).

The power of $T_n(\alpha)$ is the minimum value of $x \mapsto P_x(T_n(\alpha) = 1)$ on $\mathbb{R} \setminus \{0\}$. Actually, it follows from Theorem 1.5.6 that this function does not have a minimum on $\mathbb{R} \setminus \{0\}$. Therefore, the power of $T_n(\alpha)$ is the infimum of this function on $\mathbb{R} \setminus \{0\}$. This infimum is the largest number smaller than $P_x(T_n(\alpha) = 1)$ for every $x \neq 0$. This infimum is α . For every $x \neq 0$, $\alpha \leq P_x(T_n(\alpha) = 1)$ (see fourth statement of Theorem 1.5.5). Therefore, α is indeed smaller than $P_x(T_n(\alpha) = 1)$ for every $x \neq 0$. Now assume that there were another number strictly larger than α , $\alpha + \varepsilon$ for some $\varepsilon > 0$, such that for every $x \neq 0$, $P_x(T_n(\alpha) = 1) \geq \alpha + \varepsilon$. Because $P_0(T_n(\alpha) = 1) = \alpha$, this would in turn imply that $x \mapsto P_x(T_n(\alpha) = 1)$ is discontinuous at 0, which would contradict Theorem 1.5.5. Therefore α is the largest number smaller than $P_x(T_n(\alpha) = 1)$ for every $x \neq 0$.

QED.

According to Theorem 1.5.6, is $T_n(\alpha)$ a good test of $E(Y) = 0$ versus $E(Y) \neq 0$?

Theorem 1.5.6 is a negative result. It says that in any finite sample of size n , the t-test cannot discriminate at all between $H_0 : E(Y) = 0$ and $H_1 : E(Y) \neq 0$. As we would like to have small chances of making type 1 error, we would like to pick-up a small α . But if we do so, our test will have a low power too, because its power is equal to α . Therefore, we will have high chances of making a type 2 error. There is a trade-off between level and power, and for every possible value of α , the sum of the probabilities we make type 1 or type 2 error (level + 1-power) is equal to 1.

This is because H_1 contains values for $E(Y)$ which can be arbitrarily close to 0. Theorem 1.5.6 does not say that the test has a low power for every element of H_1 . For instance, with $n = 500$, $\sigma^2 = 1$, and $\alpha = 0.05$, Figure 1.1 shows that the t-test actually has very high power when testing $E(Y) = 0$ against $E(Y) = 0.3$. But the low power of the t-test comes from values of x close to but different from 0, as one can see from Figure 1.1.

A test whose power is equal to its level is called a trivial test. It follows from the Karlin-Rubin theorem that when the Y_i s follow a normal distribution, the t-test is the most powerful test you can build to test $E(Y) = 0$ against $E(Y) \neq 0$.⁹ Therefore, the problem does not lie in the t-test. Rather, achieving both low type 1 and type 2 error is impossible when testing $E(Y) = 0$ against $E(Y) \neq 0$.

We are now going to consider a different alternative hypothesis. Let α and λ be pre-specified values of level and power we want to achieve. Researchers often choose $\alpha = 0.05$ and $\lambda = 0.8$, meaning that they do not want to have more than 5% of chances of making a type 1 error, and 20% of chances of making a type 2 error. Our goal is to find an alternative hypothesis H_1 such that the t-test has a level α and a power λ when testing $E(Y) = 0$ against $E(Y) \in H_1$.

Theorem 1.5.7 *A first characterization of the largest alternative hypothesis that we can test against $E(Y) = 0$ with level α and power λ using the t-test.*

For any $\alpha \in (0, 1)$ and for any $\lambda \in [\alpha, 1)$, let $e(\alpha, \lambda)$ be the unique solution of $P_x(T_n(\alpha) = 1) = \lambda$ on $[0, +\infty)$. $E(Y) \in \mathbb{R} \setminus (-e(\alpha, \lambda), e(\alpha, \lambda))$, or equivalently $|E(Y)| \geq e(\alpha, \lambda)$, is the largest alternative hypothesis that we can test against $E(Y) = 0$ with level α and power λ using the t-test.

⁹I do not prove this but you can check the “Uniformly most powerful test” entry on Wikipedia.

Proof

First, $e(\alpha, \lambda)$ is well-defined. For any $\alpha \in (0, 1)$ and for any $\lambda \in [\alpha, 1)$, $P_x(T_n(\alpha) = 1) = \lambda$ has a unique solution on $[0, +\infty)$. Indeed, $x \mapsto P_x(T_n(\alpha) = 1)$ is strictly increasing and continuous on $[0, +\infty)$. Moreover, $P_0(T_n(\alpha) = 1) = \alpha$ and $\lim_{x \rightarrow +\infty} P_x(T_n(\alpha) = 1) = 1$. Therefore, $P_x(T_n(\alpha) = 1)$ is a bijection from $[0, +\infty)$ onto $[\alpha, 1)$, and for any $\lambda \in [\alpha, 1)$, $P_x(T_n(\alpha) = 1) = \lambda$ has a unique solution on $[0, +\infty)$.

Then, we prove that the t-test indeed has level α and power λ when testing $E(Y) = 0$ against $|E(Y)| \geq e(\alpha, \lambda)$. With this null hypothesis, the level of the t-test is $P_0(T_n(\alpha) = 1) = P_0(T_n(\alpha) = 1) = \alpha$. With this alternative hypothesis, the power of the t-test is the minimum of $x \mapsto P_x(T_n = 1)$ on $\mathbb{R} \setminus (-e(\alpha, \lambda), e(\alpha, \lambda))$. It directly follows from Theorem 1.5.6 that this minimum is attained at $e(\alpha, \lambda)$ (and also at $-e(\alpha, \lambda)$). It follows from the definition of $e(\alpha, \lambda)$ that $P_{e(\alpha, \lambda)}(T_n = 1) = \lambda$.

Finally, we prove that any strictly larger alternative hypothesis than $|E(Y)| \geq e(\alpha, \lambda)$ cannot be tested against $H_0 : E(Y) = 0$ with level α and power λ using the t-test. A strictly larger alternative hypothesis than $|E(Y)| \geq e(\alpha, \lambda)$ must contain at least one value in $(-e(\alpha, \lambda), e(\alpha, \lambda))$. Let x_0 denote one such value. It follows from Theorem 1.5.5 that $P_{x_0}(T_n = 1) < P_{e(\alpha, \lambda)}(T_n = 1) = \lambda$. Therefore, the power of the t-test when testing this strictly larger alternative hypothesis against $H_0 : E(Y) = 0$ is strictly lower than λ .

QED.

Theorem 1.5.7 provides us a graphical method to find the the largest alternative hypothesis that we can test against $E(Y) = 0$ with level α and power λ using the t-test. When $n = 500$ and $\sigma^2 = 1$, use Figure 1.1 to find $e(0.05, 0.8)$.

We just need to find the value of $E(Y)$ such that the power of the test is 0.8. From Figure 1.1, it appears that $e(0.05, 0.8)$ is approximately equal to 0.12. This means that we can test $E(Y) = 0$ against $|E(Y)| \geq 0.12$ with a 5% probability of making a type 1 error and a 20% probability of making a type 2 error. If we want to test $E(Y) = 0$ against $E(Y) = x$ for values of $x \in (-0.12, 0.12)$, we still have a 5% probability of making a type 1 error, but the probability we make a type 2 error is higher than 20%.

Graphical methods are nice, but we would like to have a closed-form expression of $e(\alpha, \lambda)$. That would allow us to compute easily this largest alternative hypothesis. We will not be able to compute $e(\alpha, \lambda)$, but we will be able to derive a very good approximation of it. The

intuition goes as follows. $e(\alpha, \lambda)$ solves

$$\Phi \left(-q_{1-\frac{\alpha}{2}} + \frac{e(\alpha, \lambda)}{\sqrt{V(\widehat{E}(Y))}} \right) + \Phi \left(-q_{1-\frac{\alpha}{2}} - \frac{e(\alpha, \lambda)}{\sqrt{V(\widehat{E}(Y))}} \right) = \lambda. \quad (1.5.2)$$

Deriving a closed form expression of $e(\alpha, \lambda)$ is not easy, because it appears twice in the lhs of the equation. However, maybe we could derive a closed-form expression of an approximation of $e(\alpha, \lambda)$ by solving a modified version of Equation 1.5.2 where we drop one of the two terms with $e(\alpha, \lambda)$. [Which of these two terms should we drop to get the best approximation?](#)

We should drop $\Phi \left(-q_{1-\frac{\alpha}{2}} - \frac{e(\alpha, \lambda)}{\sqrt{V(\widehat{E}(Y))}} \right)$. Given that $e(\alpha, \lambda) \geq 0$, this term is smaller than $\Phi \left(-q_{1-\frac{\alpha}{2}} + \frac{e(\alpha, \lambda)}{\sqrt{V(\widehat{E}(Y))}} \right)$. Therefore, dropping it from the equation will affect the solution less than if we drop $\Phi \left(-q_{1-\frac{\alpha}{2}} + \frac{e(\alpha, \lambda)}{\sqrt{V(\widehat{E}(Y))}} \right)$.

Let $e^*(\alpha, \lambda)$ denote the solution of

$$\Phi \left(-q_{1-\frac{\alpha}{2}} + \frac{e^*(\alpha, \lambda)}{\sqrt{V(\widehat{E}(Y))}} \right) = \lambda.$$

[Derive a closed-form expression of \$e^*\(\alpha, \lambda\)\$.](#)

$$\begin{aligned} & \Phi \left(-q_{1-\frac{\alpha}{2}} + \frac{e^*(\alpha, \lambda)}{\sqrt{V(\widehat{E}(Y))}} \right) = \lambda \\ \Leftrightarrow & -q_{1-\frac{\alpha}{2}} + \frac{e^*(\alpha, \lambda)}{\sqrt{V(\widehat{E}(Y))}} = q_\lambda \\ \Leftrightarrow & e^*(\alpha, \lambda) = \sqrt{V(\widehat{E}(Y))} \left(q_{1-\frac{\alpha}{2}} + q_\lambda \right). \end{aligned}$$

Theorem 1.5.8 *An approximation of $e(\alpha, \lambda)$.*

For any $\alpha \in (0, 1)$ and for any $\lambda \in [\alpha, 1)$, let $e^*(\alpha, \lambda) = \sqrt{V(\hat{E}(Y))} (q_{1-\frac{\alpha}{2}} + q_\lambda)$. $e^*(\alpha, \lambda)$ solves

$$\Phi \left(-q_{1-\frac{\alpha}{2}} + \frac{e^*(\alpha, \lambda)}{\sqrt{V(\hat{E}(Y))}} \right) = \lambda,$$

Proof

Everything was already proven above.

QED.

$e^*(\alpha, \lambda)$ is usually a very good approximation of $e(\alpha, \lambda)$. Indeed, $P_{e^*(\alpha, \lambda)}(T_n(\alpha) = 1)$ is very slightly larger than, but almost equal to, λ . Plugging $e^*(\alpha, \lambda)$ into the lhs of (1.5.2) yields

$$\begin{aligned} & \Phi \left(-q_{1-\frac{\alpha}{2}} + \frac{\sqrt{V(\hat{E}(Y))} (q_{1-\frac{\alpha}{2}} + q_\lambda)}{\sqrt{V(\hat{E}(Y))}} \right) + \Phi \left(-q_{1-\frac{\alpha}{2}} - \frac{\sqrt{V(\hat{E}(Y))} (q_{1-\frac{\alpha}{2}} + q_\lambda)}{\sqrt{V(\hat{E}(Y))}} \right) \\ &= \lambda + \Phi(-2q_{1-\frac{\alpha}{2}} - q_\lambda). \end{aligned}$$

For $\alpha = 0.05$ and $\lambda = 0.8$, $\Phi(-2q_{1-\frac{\alpha}{2}} - q_\lambda) \approx 0.000001$. Therefore, $e^*(\alpha, \lambda)$ “almost” solves (1.5.2) and is therefore very close to $e(\alpha, \lambda)$. Accordingly, $|E(Y)| \geq e^*(\alpha, \lambda)$ is a good approximation of the largest alternative hypothesis that we can test against $E(Y) = 0$ with level α and power λ using the t-test.

When $n = 500$ and $\sigma^2 = 1$, compute $e^*(0.05, 0.8)$. Hint: $q_{0.8} = 0.84$.

With $n = 500$ and $\sigma^2 = 1$, $e^*(0.05, 0.8) = \sqrt{\frac{1}{500}} (1.96 + 0.84) = 0.125$.

Definition 1.5.9 *Minimum detectable difference from 0 (MDD) in a t-test on $E(Y)$ with normally distributed random variables and a known variance*

Let's assume that the $(Y_i)_{1 \leq i \leq n}$ are normally distributed, with a known variance σ^2 . For any $(\alpha, \lambda) \in (0, 1) \times [\alpha, 1)$, we say that $e^*(\alpha, \lambda) = \sqrt{V(\hat{E}(Y))} (q_{1-\frac{\alpha}{2}} + q_\lambda)$ is the minimum detectable difference from 0 (MDD) in a t-test with level α and power λ .

In the survey example in Subsection 1.1.1, there are $8,448,674^n$ possible samples with replacement of n voters out of $8,448,674$, and we will randomly draw one of those samples. The largest alternative hypothesis we can test against $E(Y) = 0$ with level 0.05 and power 0.8

using the t-test is (approximately) $|E(Y)| \geq \sqrt{V(\hat{E}(Y))} (q_{0.975} + q_{0.8})$. This means that if $|E(Y)| \geq \sqrt{V(\hat{E}(Y))} (q_{0.975} + q_{0.8})$, there are at most $0.20 \times 8,448,674^n$ of the $8,448,674^n$ possible samples we can draw for which we will not reject the test. As you will see in the problem set, the MDD is a useful tool to choose the sample size in a survey, depending on the type of assumptions one wants to be able to tell apart at the outset of the survey.

The MDD is a measure of the power of the t-test: if the MDD is large, the t-test only allows us to discriminate between $E(Y) = 0$ and $E(Y) = x$ with reasonable power for values of x very far from 0. The MDD is decreasing in α . Therefore, there is a trade-off taking place between power and level. If you want to achieve low level (α), you will be able to have high power only for values of $E(Y)$ far from 0. The MDD is decreasing with the sample size n (because $V(\hat{E}(Y))$ is decreasing in n : the larger the sample, the more the t-test can discriminate with high power between $E(Y) = 0$ and $E(Y) = x$ for values of x close to 0).

1.5.2 ... Without assuming normality and known variance.

A limitation of the results in the previous section is that they rely on the assumption that Y is normally distributed with a known variance. In many instances the normality assumption is implausible or impossible (think of the survey example in Subsection 1.1.1), and we do not know the variance of the random variables we consider. Without these two assumptions, we can still derive an upper bound for the MDD, but this relies on fairly recent maths papers. The main message is that the MDD formula we derived in the previous subsection under the normality assumption is a good approximation of the true MDD even if the random variables we consider are not normally distributed, for samples of reasonable size.

Let me first redefine our test (we no longer assume that the variance of the Y_i is known, so the previous test is no longer a test as it is no longer the event of an indicator involving only $(Y_i)_{1 \leq i \leq n}$ and known real numbers).

Definition 1.5.10 *T-test*

For any $\alpha \in (0, 1)$, let

$$T_n(\alpha) = 1 \left\{ \left| \frac{\hat{E}(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}} \right\}.$$

$T_n(\alpha)$ is called the *t-test for the mean*.

Theorem 1.5.11 *Berry-Esseen theorem for the t-statistic ((Bentkus & Götze 1996))*.

Let $(U_n)_{n \in \mathbb{N}}$ be a sequence of iid random variables. Let U be a random variable with the same

distribution as the U_n s. If $E(|U - E(U)|^3) < +\infty$, then

$$\sup_{x \in \mathbb{R}} \left| P \left(\sqrt{n} \left(\frac{\bar{U}_n - E(U)}{\hat{V}(U)} \right) \leq x \right) - \Phi(x) \right| \leq \frac{CE(|U - E(U)|^3)}{V(U)^{\frac{3}{2}} \sqrt{n}},$$

for some constant C . Shao (2005) shows that $C < 25$.

The central limit theorem combined with Slutsky implies that for any x , the difference between $P \left(\sqrt{n} \left(\frac{\bar{U}_n - E(U)}{\hat{V}(U)} \right) \leq x \right)$, the cdf of $\frac{\bar{U}_n - E(U)}{\hat{V}(U)}$, and $\Phi(x)$, the cdf of a $N(0, 1)$ variable, converges to 0 when the sample size n goes to infinity. Berry-Esseen refines this by giving you an upper bound for the difference between $P \left(\sqrt{n} \left(\frac{\bar{U}_n - E(U)}{\hat{V}(U)} \right) \leq x \right)$ and $\Phi(x)$ when n is fixed, i.e. when the size of the sample is finite.

Theorem 1.5.12 *Minimum detectable difference from 0 (MDD) in a t-test on $E(Y)$*

Let $BE = \frac{25E(|Y - E(Y)|^3)}{V(Y)^{\frac{3}{2}} \sqrt{n}}$. For any $\alpha \in (0, 1)$ and for any $\lambda \in [\alpha, 1)$, let

$$\tilde{e}(\alpha, \lambda) = \sqrt{\hat{V}(\hat{E}(Y))} (q_{1-\frac{\alpha}{2}} + q_{\lambda+BE}).$$

One can test the null hypothesis $E(Y) = 0$ against $|E(Y)| \geq \tilde{e}(\alpha, \lambda)$ with power at least λ using the t-test.

Proof

To prove the result, we need to show that for every $x \geq \tilde{e}(\alpha, \lambda)$ or $x \leq -\tilde{e}(\alpha, \lambda)$, $P_x(T_n(\alpha) = 1) \geq \lambda$. We only prove the result for $x \geq \tilde{e}(\alpha, \lambda)$, the proof is similar for $x \leq -\tilde{e}(\alpha, \lambda)$.

For every $x \geq \tilde{e}(\alpha, \lambda)$,

$$\begin{aligned}
P_x(T_n(\alpha) = 1) &= P_x \left(\left| \frac{\hat{E}(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}} \right) \\
&= P_x \left(\left| \frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} + \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right| > q_{1-\frac{\alpha}{2}} \right) \\
&= P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} + \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} > q_{1-\frac{\alpha}{2}} \right) \\
&+ P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} + \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} < -q_{1-\frac{\alpha}{2}} \right) \\
&\geq P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} > q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) \\
&= 1 - P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \leq q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) \\
&= 1 - \Phi \left(q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) \\
&+ \Phi \left(q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) - P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \leq q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) \\
&\geq 1 - \Phi \left(q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) - BE \\
&\geq 1 - \Phi \left(q_{1-\frac{\alpha}{2}} - \frac{\tilde{e}(\alpha, \lambda)}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) - BE \\
&= 1 - \Phi \left(q_{1-\frac{\alpha}{2}} - \frac{\sqrt{\hat{V}(\hat{E}(Y))}(q_{1-\frac{\alpha}{2}} + q_{\lambda+BE})}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) - BE \\
&= 1 - \Phi(-q_{\lambda+BE}) - BE \\
&= 1 - \Phi(q_{1-\lambda-BE}) - BE \\
&= 1 - (1 - \lambda - BE) - BE \\
&= \lambda.
\end{aligned} \tag{1.5.3}$$

The second inequality follows from Theorem 1.5.11, which implies that

$$\Phi \left(q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) - P_x \left(\frac{\hat{E}(Y) - E(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} \leq q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) \geq -BE.$$

The third inequality follows from the fact $x \mapsto 1 - \Phi \left(q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) - BE$ is increasing in x . Therefore, for $x \geq \tilde{e}(\alpha, \lambda)$,

$$1 - \Phi \left(q_{1-\frac{\alpha}{2}} - \frac{x}{\sqrt{\hat{V}(\hat{E}(Y))}} \right) - BE \geq 1 - \Phi \left(-\tilde{e}(\alpha, \lambda) \frac{\sqrt{n}}{\sqrt{\hat{V}(Y)}} + q_{1-\frac{\alpha}{2}} \right) - BE.$$

QED.

Unsurprisingly, the MDD without assuming normality is greater than the MDD under normality, because $BE \geq 0$. But this term vanishes to 0 when $n \rightarrow +\infty$, which implies that in large samples the two quantities should be close. Therefore, in a large sample, the MDD formula derived under normality will not be far from the true MDD.¹⁰

To conclude, I would like to emphasize once more the logic underlying statistical tests. Statistical tests are some kind of probabilistic “reductio ad absurdum” argument. In logic, reductio ad absurdum works as follows. You want to prove that statement A, say “rocks have no weight”, is wrong. You know from previous deductions or experience that statement B, say “rocks do not float in the air”, is true. To prove that statement A is wrong, you can then show that if A were true, this would contradict B: “If rocks had no weight, then they would float in the air, which I know is not true, so it cannot be true that rocks have no weight”. Statistical tests are a probabilistic version of this. When a t-test leads you to reject the null hypothesis $E(Y) = 0$, your rejection is based on the following reasoning: “If $E(Y) = 0$ were true, $\frac{\hat{E}(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}}$ would follow a $N(0, 1)$ distribution. In the sample $(Y_i)_{1 \leq i \leq n}$ I get to observe, $\frac{\hat{E}(Y)}{\sqrt{\hat{V}(\hat{E}(Y))}} = 3$. I know that the probability that a $N(0, 1)$ is greater in absolute value than 1.96 is only equal to 5%, so either $E(Y) = 0$ is false or I am observing something which had less than 5% probability of happening. Therefore, I choose to reject $E(Y) = 0$.”

¹⁰However, in practice the sample should actually be fairly large. For instance, whenever $n \leq (25)^2 = 625$, the bound we just derived is equal to $+\infty$. That is because Theorem 1.5.11 is fairly recent, and the upper bound for C is not optimal yet. For another Berry-Esseen theorem, the upper bound for the constant was equal to 7 when the original paper was published in 1941, but it has been refined over the years, and the latest paper on this topic showed that it is less than 0.4748. So we should just wait for mathematicians to improve on Shao’s result. In some years from now we will have a 2 instead of a 25 in our definition of BE , and we will be able to assert that the normal MDD is a safe approximation of the true MDD, even in small samples.

Chapter 2

Randomized controlled trials

This chapter borrows from Imbens & Rubin (2015). Most of the theoretical results in this chapter are from Neyman (1923), Fisher et al. (1935), and Abadie et al. (2014).

2.1 Randomized controlled trials in a finite population

2.1.1 Causality, potential outcomes, and our parameter of interest

Assume we want to learn the effect of a **treatment** on the **outcome** of some units. Let's consider an example. Ability-tracked schools are schools where classrooms are formed according to students' ability: if the school has, say, 2 first grade classes, all above-median students go to class A, while all below-median students go to class B. One might want to know the effect of going to a tracked versus a non-tracked school on students' test scores. For now, let's assume that a fixed number of units n are eligible for the treatment. For instance, n first grade students might be eligible to enter tracked or non-tracked primary schools.

For every $i \in \{1, \dots, n\}$, let $y_i(1)$ denote the outcome that unit i will obtain if they receive the treatment, and let $y_i(0)$ denote the outcome they will obtain if they do not receive the treatment. $y_i(1)$ and $y_i(0)$ are the two potential outcomes of unit i . Following Imbens & Rubin (2015), for now I consider these two quantities as real numbers, not as random variables. For instance, in the tracked-school example, let's assume that irrespective of whether they go to a tracked school or not, the n 1st grade students will all take the same standardized test in the end of first grade. $y_i(1)$ is the score that student i will obtain in that test if they go to a tracked school, and $y_i(0)$ is the score that they will obtain if they do not go to a tracked school. These notations were introduced by Neyman (1923), Fisher et al. (1935), and Rubin (1974).

The causal effect of the treatment on unit i is the difference between their two potential outcomes, $y_i(1) - y_i(0)$. $y_i(1) - y_i(0)$ is a ceteris paribus comparison: we compare the same unit with and without treatment, so it really measures the effect of the treatment.

In the tracking example, do you think it is plausible that all students have the same value of $y_i(1) - y_i(0)$?

No, that is not very plausible. Consider the following hypothetical but not implausible scenarios. Assume that student 1 is a very weak student. If they go to a non-tracked school, they will lag well below the median in their class, will not be able to follow the pace of instruction, and will not dare to ask questions when they do not understand. Therefore, $y_1(0) = 0.2$: if they go to a non-tracked school, they will only be able to answer 20% of the questions in the end of 1st grade test. If they go to a tracked school, they will be sent to the weaker class and will therefore be closer to the median in their class, they will have less difficulties to follow the pace of instruction, and they will dare to ask questions when they do not understand. Therefore, $y_1(1) = 0.5$: if they go to a tracked school, they will be able to answer 50% of the questions in the end of 1st grade test. Therefore, the causal effect of tracking on the end of 1st grade test score of that student is $y_1(1) - y_1(0) = 0.3$: tracking increases by 30 points their percentage of correct answers in the end of 1st year test.

Now, assume that student 2 is just above the median in their school. If they go to a non-tracked school, they are very close to the median in their class and can follow the pace of instruction. Therefore, $y_2(0) = 0.6$: if they go to a non-tracked school, they will be able to answer 60% of the questions in the end of 1st grade test. If they go to a tracked school, they will be sent to the stronger class and will therefore be the weakest student in their class: they will have more difficulties to follow the pace of instruction, and they will not dare to ask questions when they do not understand. Therefore, $y_2(1) = 0.4$: if they go to a tracked school, they will be able to answer 40% of the questions in the end of 1st grade test. The causal effect of tracking on the end of 1st grade test score of that student is $y_2(1) - y_2(0) = -0.2$: tracking decreases by 20 points their percentage of correct answers in the end of 1st year test.

Can you compute $y_i(1) - y_i(0)$ for any unit in the sample?

No, you never get to observe the outcome of the same unit without and with the treatment. Either a unit gets treated, in which case you can observe $y_i(1)$ but not $y_i(0)$, or the unit does not get treated in which case you can observe $y_i(0)$ but not $y_i(1)$. In the tracking example, if

a student goes to a tracked school, their end of first grade test score will be $y_i(1)$, while if they go to a non-tracked school, their end of first grade test score will be $y_i(0)$, you will never get to observe their two test scores at the same time. A priori, each of these potential outcomes is observable: $y_i(0)$ can be observed if unit i remains untreated, $y_i(1)$ can be observed if unit i gets treated. But a posteriori, once the assignment of unit i to treatment or non treatment has been decided, only one potential outcome can be observed. This implies that we cannot compute effect of the treatment for any unit in our sample. This is called the fundamental problem of causal inference.

The poem “The Road not Taken” by Robert Frost is about the fundamental problem of causal inference:

“Two roads diverged in a yellow wood,
And sorry I could not travel both
And be one traveler...”

In the end of the poem Frost claims that taking the less travelled road had a large treatment effect on him:

“Two roads diverged in a wood, and I-
I took the one less traveled by,
And that has made all the difference.”

In an economics seminar, the audience would probably argue Frost has no evidence to make such a claim because he gets to observe his life after taking the less traveled road, but he does not get to observe his life if he had taken the other road, so he cannot be sure that taking the less traveled road really made a big difference.

This problem is a missing data problem: we never get to observe simultaneously the two potential outcomes of the same unit. Let D_i be an indicator equal to 1 if unit i receives the treatment, and to 0 otherwise. For each unit, we only observe Y_i which is equal to

$$Y_i = y_i(0) \times (1 - D_i) + y_i(1) \times D_i = y_i(0) + (y_i(1) - y_i(0)) \times D_i. \quad (2.1.1)$$

However, we might be able to compute or at least estimate the average effect of the treatment in the population. Throughout this section, our parameter of interest is going to be $ATE_n = \frac{1}{n} \sum_{k=1}^n (y_i(1) - y_i(0))$, the average effect of the treatment in our population of n units. In the tracking example, ATE_n is the average test score gain produced by tracking, across the n students in our population.

2.1.2 Random assignment of treatment

The following result from combinatorics will be useful. Assume you are going to draw simultaneously $k \leq n$ numbers from $\{1, \dots, n\}$. The following theorem gives the number of possible outcomes of that draw. For any integer j , let $j! = j \times (j - 1) \times \dots \times 1$.

Theorem 2.1.1 Combinations

When drawing simultaneously $k \leq n$ numbers from $\{1, \dots, n\}$, there are $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ different subsets of k numbers from $\{1, \dots, n\}$ you can draw. $\frac{n!}{k!(n-k)!}$ is denoted $\binom{n}{k}$.

Before we see the proof, let's verify that the formula works in a simple example. If $n = 3$ and $k = 2$, drawing simultaneously 2 numbers from $\{1, 2, 3\}$ can yield 3 outcomes. Either we draw $\{1, 2\}$, or we draw $\{1, 3\}$, or we draw $\{2, 3\}$. It turns out that $\binom{3}{2} = \frac{3!}{2!1!} = \frac{3 \times 2 \times 1}{(2 \times 1)1} = 3$. So the formula seems to work.

Proof

First, let's count the number of ways of choosing k numbers sequentially (one after the other) from $\{1, \dots, n\}$. For the first number we choose, we have n choices. For the second number we choose, we have $n - 1$ choices left, and so on and so forth until we choose the k th number, for which we have $n - (k - 1)$ choices left. Therefore, there are $n \times (n - 1) \times \dots \times (n - (k - 1)) = \frac{n!}{(n-k)!}$ ways of choosing sequentially k numbers from $\{1, \dots, n\}$.

Now, choosing k numbers simultaneously is equivalent to choosing them sequentially, except that the order in which numbers are drawn does not matter. For instance, drawing 1, and then 2, ..., and then k , or drawing k , and then $k - 1$, ..., and then 1 counts as two different draws if the ordering matters, but only counts as one draw if the ordering does not matter. Any subset of k numbers can be drawn in $k!$ different orders. Therefore, the number of subsets of k numbers one can simultaneously draw from $\{1, \dots, n\}$ is equal to the number of sequential draws of k numbers from $\{1, \dots, n\}$ divided by $k!$, that is $\frac{n!}{(n-k)!k!}$.

QED.

If you need a refresher on combinations, see <https://en.wikipedia.org/wiki/Combination>.

Now, assume that the n units in our population are volunteers to receive the treatment. Let $(D_1, D_2, \dots, D_n)$ be a vector stacking the treatment indicators of all the units in the population. Each element of this vector can either be equal to 0 or to 1. Therefore, if we do not impose any constraint on assignment to treatment, $(D_1, D_2, \dots, D_n)$ can take 2^n values, because each unit can either have $D_i = 0$ or $D_i = 1$. However, in most real life situations, treatments have capacity constraints: $0 < n_1$ seats for treatment are available. Let's assume that $n_1 < n$: there are more volunteers for treatment than seats available. Then, we must have $\sum_{i=1}^n D_i = n_1$. n_1 units can receive the treatment, while the remaining $n_0 = n - n_1$ units must remain untreated.

If $(D_1, D_2, \dots, D_n)$ must satisfy $\sum_{i=1}^n D_i = n_1$, what is the number of possible values $(D_1, D_2, \dots, D_n)$ can take?

$(D_1, D_2, \dots, D_n)$ can take $\binom{n}{n_1} = \frac{n!}{n_1!(n-n_1)!}$ values. Indeed, there are as many possible values of $(D_1, D_2, \dots, D_n)$ as there are ways of choosing the n_1 units among n that receive the treatment.

Let's verify that the formula works in a simple example. If $n = 3$ and $n_1 = 2$, (D_1, D_2, D_3) can be equal to $(1, 1, 0)$ (units 1 and 2 receive the treatment, 3 does not), $(1, 0, 1)$ (units 1 and 3 receive the treatment, 2 does not), and $(0, 1, 1)$ (units 2 and 3 receive the treatment, 1 does not). Therefore D can take 3 values. It turns out that $\binom{3}{2} = \frac{3!}{2!1!} = \frac{3 \times 2 \times 1}{(2 \times 1)1} = 3$. So the formula seems to work.

Assumption 1 *Random assignment of treatment*

We say that *treatment is randomly assigned* if $(D_1, D_2, \dots, D_n)$ are random variables, and if for any vector of n numbers $(d_1, d_2, \dots, d_n) \in \{0, 1\} \times \{0, 1\} \dots \times \{0, 1\}$ such that $\sum_{i=1}^n d_i = n_1$,

$$P(D_1 = d_1, D_2 = d_2, \dots, D_n = d_n) = \frac{1}{\binom{n}{n_1}}. \quad (2.1.2)$$

Assumption 1 requires that the n_1 units out of n that get treated are chosen randomly, by a lottery. Moreover, each of the $\binom{n}{n_1}$ possible realizations of $(D_1, D_2, \dots, D_n)$ should have the same probability of being selected by the lottery. For instance, assume that $n = 100$, $n_1 = 50$. The Stata code below randomly chooses one realization of the vector $(D_1, D_2, \dots, D_{100})$ satisfying the condition $D_1 + D_2 + \dots + D_{100} = 50$, out of the $\binom{100}{50}$ possible values of $(D_1, D_2, \dots, D_{100})$ satisfying that condition. In this lottery, each possible realization of $(D_1, D_2, \dots, D_{100})$ satisfying the condition $D_1 + D_2 + \dots + D_{100} = 50$ has the same probability of being selected. Therefore this lottery satisfies Assumption 1. The final line “list i d” shows the randomly chosen realization.

```
set obs 100
gen i=_n
gen u=uniform()
sort u
gen d=(_n<=50)
sort i
drop u
list i d
```

If Assumption 1 is satisfied, what is the probability that $D_i = 1$, meaning that unit i is selected by the lottery to get treated?

This probability is equal to $\frac{n_1}{n}$.

Before proving it, let's start verifying that this statement is true in a simple example. Assume that $n_1 = 2$ and $n = 3$. As explained above, (D_1, D_2, D_3) can take three values: $(1, 1, 0)$ (units 1 and 2 receive the treatment, 3 does not), $(1, 0, 1)$ (units 1 and 3 receive the treatment, 2 does not), and $(0, 1, 1)$ (units 2 and 3 receive the treatment, 1 does not). If Assumption 1 is satisfied, each of these values has the same probability of being selected by the lottery, and this probability is equal to $1/3$. Now, $D_1 = 1$ if $(1, 1, 0)$ or $(1, 0, 1)$ gets selected, which has a probability $\frac{2}{3}$ of happening. Therefore, $P(D_1 = 1) = \frac{2}{3}$. What is the value of $P(D_2 = 1)$? Of $P(D_3 = 1)$?

It is easy to check that $P(D_2 = 1) = P(D_3 = 1) = \frac{2}{3}$. In this example, $\frac{n_1}{n} = \frac{2}{3}$, so the formula works.

Theorem 2.1.2 *Random assignment generates uniform treatment probabilities across units*

If Assumption 1 is satisfied, then for every $i \in \{1, \dots, n\}$, $P(D_i = 1) = \frac{n_1}{n}$.

Proof

To compute $P(D_i = 1)$, we need to count the possible realizations of $(D_1, \dots, D_n)$ such that $D_i = 1$, and divide this number by the total number of possible realizations of $(D_1, \dots, D_n)$.

If $D_i = 1$, we still need to choose $n_1 - 1$ of the remaining $n - 1$ units that will receive the treatment. There are $\binom{n-1}{n_1-1}$ different ways of choosing those $n_1 - 1$ units. Therefore, there are $\binom{n-1}{n_1-1}$ possible realizations of $(D_1, \dots, D_n)$ such that $D_i = 1$. We have already seen that there are $\binom{n}{n_1}$ possible realizations of $(D_1, \dots, D_n)$. Thus,

$$P(D_i = 1) = \frac{\binom{n-1}{n_1-1}}{\binom{n}{n_1}} = \frac{\frac{(n-1)!}{(n_1-1)!(n-n_1)!}}{\frac{n!}{n_1!(n-n_1)!}} = \frac{(n-1)!}{n!} \frac{n_1!}{n_1!} = \frac{n_1}{n}.$$

QED.

2.1.3 Estimating ATE_n

Now, assume that a lottery satisfying Assumption 1 is conducted. Some time after the lottery, the observed outcome Y_i is measured for all the units that participated in that lottery. For instance, in the tracking example, a lottery randomly assigns some students to tracked / non-tracked schools in the beginning of first grade, and in the end of first grade we measure the end of first grade test score of all the students that participated in the lottery. Often, some units that participated in the lottery are lost to follow-up and we cannot measure their outcome, but let's put this (important) issue aside. Propose an estimator of ATE_n .

We can use $\widehat{ATE}_n = \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n-n_1} \sum_{i:D_i=0} Y_i$, the comparison of the mean of the observed outcome among treated and untreated units. We will call these two groups the treatment and the control group. Is $\widehat{ATE}_n$ an estimator?

Yes $\widehat{ATE}_n$ is an estimator. An equivalent way of writing $\widehat{ATE}_n$ is

$$\widehat{ATE}_n = \frac{1}{n_1} \sum_{i=1}^n D_i Y_i - \frac{1}{n-n_1} \sum_{i=1}^n (1-D_i) Y_i. \quad (2.1.3)$$

First, does everybody see that $\sum_{i:D_i=1} Y_i = \sum_{i=1}^n D_i Y_i$ and $\sum_{i:D_i=0} Y_i = \sum_{i=1}^n (1-D_i) Y_i$? At the outset of the experiment, we will get to observe the random variables (Y_i, D_i) for every subject. From Equation (2.1.3), one can see that $\widehat{ATE}_n$ is a function of those random variables and of known real numbers. $\widehat{ATE}_n$ is something we will be able to compute at the outset of the experiment, therefore it is an estimator.

Finally, combining Equations (2.2.8) and (2.1.3), we have

$$\widehat{ATE}_n = \frac{1}{n_1} \sum_{i=1}^n D_i y_i(1) - \frac{1}{n-n_1} \sum_{i=1}^n (1-D_i) y_i(0). \quad (2.1.4)$$

For students with $D_i = 1$, the outcome we measure is their $y_i(1)$, while for students with $D_i = 0$, the outcome we measure is their $y_i(0)$. $\widehat{ATE}_n$ compares the average $y_i(1)$ in the treatment group to the average $y_i(0)$ in the control group. $\widehat{ATE}_n$ is a random variable. Where does the randomness in $\widehat{ATE}_n$ come from?

From Equation (2.1.4), one can see that the only source of randomness in $\widehat{ATE}_n$ is $(D_1, D_2, \dots, D_n)$, the outcome of the lottery. For instance, in our example with $n = 3$ and $n_1 = 2$, if $(D_1, D_2, D_3) = (1, 1, 0)$, $\widehat{ATE}_n = \frac{1}{2}(y_1(1) + y_2(1)) - y_3(0)$. If $(D_1, D_2, D_3) = (1, 0, 1)$, $\widehat{ATE}_n = \frac{1}{2}(y_1(1) + y_3(1)) - y_2(0)$. If $(D_1, D_2, D_3) = (0, 1, 1)$, $\widehat{ATE}_n = \frac{1}{2}(y_2(1) + y_3(1)) - y_1(0)$. Therefore, $\widehat{ATE}_n$ is a random variable, with an expectation and a variance.

Show that $\widehat{ATE}_n$ is an unbiased estimator of ATE_n .

Theorem 2.1.3 $\widehat{ATE}_n$ is an unbiased estimator of ATE_n .

If Assumption 1 is satisfied, then $E(\widehat{ATE}_n) = ATE_n$.

Proof

$$\begin{aligned}
E(\widehat{ATE}_n) &= E\left(\frac{1}{n_1} \sum_{i=1}^n D_i y_i(1) - \frac{1}{n - n_1} \sum_{i=1}^n (1 - D_i) y_i(0)\right) \\
&= \frac{1}{n_1} \sum_{i=1}^n E(D_i) y_i(1) - \frac{1}{n - n_1} \sum_{i=1}^n E(1 - D_i) y_i(0) \\
&= \frac{1}{n_1} \sum_{i=1}^n \frac{n_1}{n} y_i(1) - \frac{1}{n - n_1} \sum_{i=1}^n \left(1 - \frac{n_1}{n}\right) y_i(0) \\
&= \frac{1}{n} \sum_{i=1}^n y_i(1) - \frac{1}{n} \sum_{i=1}^n y_i(0) \\
&= ATE_n.
\end{aligned}$$

The first equality follows from Equation (2.1.4). The second come from the linearity of the expectation operator. The third comes from Theorem 2.1.2 and from the fact the D_i s are binary random variables.

QED.

Theorem 2.1.3 shows that $\widehat{ATE}_n$ does not systematically over- or underestimate ATE_n . Its distribution is “centered” around ATE_n . This is because Assumption 1 ensures that all units have the same probability of receiving the treatment. Assume for a minute that Assumption 1 does not hold, and $P(D_1 = 1) = 1$, $P(D_n = 1) = 0$, and $P(D_n = 1) = \frac{n_1 - 1}{n - 1}$. Then, the treatment group is no longer representative of the population: unit 1 is systematically in the treatment group, while unit n is never in the treatment group. You can check that if $P(D_1 = 1) = 1$, $P(D_n = 1) = 0$, and $P(D_n = 1) = \frac{n_1 - 1}{n - 1}$, $E(\widehat{ATE}_n) \neq ATE_n$.

A consequence of Theorem 2.1.3 is that ATE_n , our parameter of interest is identified: it is equal to an estimand. Indeed, $E(\widehat{ATE}_n)$ is an estimand: it is a function of the probability distribution of the random variables we will get to observe at the outset of the experiment (the Y_i s and the D_i s).

2.1.4 Computing the variance of $\widehat{ATE}_n$

As you will remember from the first chapter, a good way to assess the quality of an unbiased estimator is to compute its variance, because the variance of an unbiased estimator is equal to its quadratic risk. However, deriving the variance of $\widehat{ATE}_n$ is cumbersome. Let’s start doing it in the special case where $n = 2$ and $n_1 = 1$. Then, (D_1, D_2) can only take two values, $(1, 0)$ and $(0, 1)$. If Assumption 1 is satisfied, each of these values has a probability $1/2$ of being selected by the lottery. Accordingly, $\widehat{ATE}_2$ can also take only two values: $y_1(1) - y_2(0)$, or

$y_2(1) - y_1(0)$. Finally, remember that $E(\widehat{ATE}_2) = ATE_2 = \frac{1}{2}(y_1(1) + y_2(1)) - \frac{1}{2}(y_1(0) + y_2(0))$.
Derive the variance of $\widehat{ATE}_2$.

$$\begin{aligned}
V(\widehat{ATE}_2) &= E\left(\left(\widehat{ATE}_2 - ATE_2\right)^2\right) \\
&= \frac{1}{2}\left(y_1(1) - y_2(0) - \left(\frac{1}{2}(y_1(1) + y_2(1)) - \frac{1}{2}(y_1(0) + y_2(0))\right)\right)^2 \\
&\quad + \frac{1}{2}\left(y_2(1) - y_1(0) - \left(\frac{1}{2}(y_1(1) + y_2(1)) - \frac{1}{2}(y_1(0) + y_2(0))\right)\right)^2 \\
&= \frac{1}{2}\left(\frac{1}{2}(y_1(1) + y_1(0)) - \frac{1}{2}(y_2(1) + y_2(0))\right)^2 \\
&\quad + \frac{1}{2}\left(\frac{1}{2}(y_2(1) + y_2(0)) - \frac{1}{2}(y_1(1) + y_1(0))\right)^2 \\
&= \frac{1}{4}((y_1(1) + y_1(0)) - (y_2(1) + y_2(0)))^2.
\end{aligned}$$

Now, let's compute the variance of $\widehat{ATE}_n$. Before that, we need a lemma, and quite a bit of notation.

Lemma 2.1.1 *The cross moment of D_i and D_j*

If Assumption 1 is satisfied, then for every $i \neq j$, $E(D_i D_j) = \frac{n_1}{n} \frac{n_1 - 1}{n - 1}$.

Proof

$E(D_i D_j) = P(D_i = 1, D_j = 1)$. To compute $P(D_i = 1, D_j = 1)$, we need to count the possible realizations of $(D_1, \dots, D_n)$ such that $D_i = 1$ and $D_j = 1$, and divide this number by the total number of possible realizations of $(D_1, \dots, D_n)$. If $D_i = 1$ and $D_j = 1$, we still need to choose $n_1 - 2$ of the remaining $n - 2$ units that will receive the treatment. There are $\binom{n-2}{n_1-2}$ different ways of choosing those $n_1 - 2$ units. Therefore, there are $\binom{n-2}{n_1-2}$ possible realizations of $(D_1, \dots, D_n)$ such that $D_i = 1$ and $D_j = 1$. We have already seen that there are $\binom{n}{n_1}$ possible realizations of $(D_1, \dots, D_n)$. Thus,

$$E(D_i D_j) = \frac{\binom{n-2}{n_1-2}}{\binom{n}{n_1}} = \frac{\frac{(n-2)!}{(n_1-2)!(n-n_1)!}}{\frac{n!}{n_1!(n-n_1)!}} = \frac{(n-2)!}{n!} \frac{n_1!}{(n_1-2)!} = \frac{n_1}{n} \frac{n_1 - 1}{n - 1}.$$

QED.

Use this result to compute $cov(D_i, D_j)$. Are D_i and D_j independent?

$$\begin{aligned}
\text{cov}(D_i, D_j) &= E(D_i D_j) - E(D_i)E(D_j) \\
&= \frac{n_1}{n} \frac{n_1 - 1}{n - 1} - \left(\frac{n_1}{n}\right)^2 \\
&= \frac{n_1}{n} \left(\frac{n_1 - 1}{n - 1} - \frac{n_1}{n} \right) \\
&= \frac{n_1}{n} \frac{(n_1 - 1)n - n_1(n - 1)}{(n - 1)n} \\
&= -\frac{n_1(n - n_1)}{n^2(n - 1)}.
\end{aligned}$$

D_i and D_j are not independent, they are negatively correlated. The intuition is that if $D_i = 1$, D_j is less likely to be equal to 1 than if $D_i = 0$: in the first case, there are only $n_1 - 1$ seats for treatment left for $n - 1$ units, while in the second case there are n_1 seats for treatment left for $n - 1$ units.

Let $\bar{y}(0) = \frac{1}{n} \sum_{i=1}^n y_i(0)$. Let $\bar{y}(1) = \frac{1}{n} \sum_{i=1}^n y_i(1)$. Let $S_{y(0),n}^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i(0) - \bar{y}(0))^2$, let $S_{y(1),n}^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i(1) - \bar{y}(1))^2$, and let $S_{y(1)-y(0),n}^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i(1) - y_i(0) - (\bar{y}(1) - \bar{y}(0)))^2$. $S_{y(0),n}^2$, $S_{y(1),n}^2$, and $S_{y(1)-y(0),n}^2$ are respectively the variances of $y_i(0)$, $y_i(1)$, and $y_i(1) - y_i(0)$ in our population of n units.

Theorem 2.1.4 *The variance of $\widehat{ATE}_n$.*

If Assumption 1 is satisfied, then

$$V\left(\widehat{ATE}_n\right) = \frac{S_{y(0),n}^2}{n - n_1} + \frac{S_{y(1),n}^2}{n_1} - \frac{S_{y(1)-y(0),n}^2}{n}.$$

Proof

Let's start by rewriting our estimator a bit.

$$\begin{aligned}
\widehat{ATE}_n &= \frac{1}{n_1} \sum_{i=1}^n D_i y_i(1) - \frac{1}{n - n_1} \sum_{i=1}^n (1 - D_i) y_i(0) \\
&= \frac{1}{n} \sum_{i=1}^n D_i \frac{n}{n_1} y_i(1) - \frac{1}{n} \sum_{i=1}^n (1 - D_i) \frac{n}{n - n_1} y_i(0) \\
&= \frac{1}{n} \sum_{i=1}^n \left(D_i \frac{n}{n_1} y_i(1) - (1 - D_i) \frac{n}{n - n_1} y_i(0) \right) \\
&= \frac{1}{n} \sum_{i=1}^n D_i \left(\frac{n}{n_1} y_i(1) + \frac{n}{n - n_1} y_i(0) \right) - \frac{1}{n} \sum_{i=1}^n \frac{n}{n - n_1} y_i(0)
\end{aligned}$$

To alleviate the notational burden, let $y_i^+ = \frac{n}{n_1} y_i(1) + \frac{n}{n - n_1} y_i(0)$. Therefore,

$$\widehat{ATE}_n = \frac{1}{n} \sum_{i=1}^n D_i y_i^+ - \frac{1}{n} \sum_{i=1}^n \frac{n}{n - n_1} y_i(0). \quad (2.1.5)$$

Finally, let's derive another useful formula.

$$\begin{aligned}
S_{y(1)-y(0),n}^2 &= \frac{1}{n-1} \sum_{i=1}^n (y_i(1) - y_i(0) - (\bar{y}(1) - \bar{y}(0)))^2 \\
&= \frac{1}{n-1} \sum_{i=1}^n (y_i(1) - \bar{y}(1))^2 + \frac{1}{n-1} \sum_{i=1}^n (y_i(0) - \bar{y}(0))^2 \\
&\quad - \frac{2}{n-1} \sum_{i=1}^n (y_i(1) - \bar{y}(1)) (y_i(0) - \bar{y}(0)) \\
&= S_{y(1),n}^2 + S_{y(0),n}^2 - \frac{2}{n-1} \sum_{i=1}^n (y_i(1) - \bar{y}(1)) (y_i(0) - \bar{y}(0)).
\end{aligned}$$

Therefore,

$$\frac{2}{n-1} \sum_{i=1}^n (y_i(1) - \bar{y}(1)) (y_i(0) - \bar{y}(0)) = S_{y(1),n}^2 + S_{y(0),n}^2 - S_{y(1)-y(0),n}^2 \quad (2.1.6)$$

We can now prove the result.

$$\begin{aligned}
& V\left(\widehat{ATE}_n\right) \\
&= \frac{1}{n^2} V\left(\sum_{i=1}^n D_i y_i^+\right) \\
&= \frac{1}{n^2} \sum_{i=1}^n V(D_i) (y_i^+)^2 + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i}^n \text{cov}(D_i, D_j) y_i^+ y_j^+ \\
&= \frac{1}{n^2} \sum_{i=1}^n \frac{n_1}{n} \frac{n-n_1}{n} (y_i^+)^2 - \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i}^n \frac{n_1(n-n_1)}{n^2(n-1)} y_i^+ y_j^+ \\
&= \frac{n_1(n-n_1)}{n^4} \sum_{i=1}^n (y_i^+)^2 - \frac{n_1(n-n_1)}{n^4(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n y_i^+ y_j^+ \\
&= \frac{n_1(n-n_1)}{n^4} \sum_{i=1}^n (y_i^+)^2 + \frac{n_1(n-n_1)}{n^4(n-1)} \sum_{i=1}^n (y_i^+)^2 - \frac{n_1(n-n_1)}{n^4(n-1)} \sum_{i=1}^n (y_i^+)^2 - \frac{n_1(n-n_1)}{n^4(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n y_i^+ y_j^+ \\
&= \frac{n_1(n-n_1)}{n^3(n-1)} \sum_{i=1}^n (y_i^+)^2 - \frac{n_1(n-n_1)}{n^4(n-1)} \sum_{i=1}^n \sum_{j=1}^n y_i^+ y_j^+ \\
&= \frac{n_1(n-n_1)}{n^3(n-1)} \sum_{i=1}^n (y_i^+)^2 - \frac{n_1(n-n_1)}{n^4(n-1)} \left(\sum_{i=1}^n y_i^+ \right)^2 \\
&= \frac{n_1(n-n_1)}{n^2(n-1)} \left(\frac{1}{n} \sum_{i=1}^n (y_i^+)^2 - \left(\frac{1}{n} \sum_{i=1}^n y_i^+ \right)^2 \right) \\
&= \frac{n_1(n-n_1)}{n^2(n-1)} \left(\frac{1}{n} \sum_{i=1}^n \left(y_i^+ - \frac{1}{n} \sum_{i=1}^n y_i^+ \right)^2 \right) \\
&= \frac{n_1(n-n_1)}{n^3(n-1)} \sum_{i=1}^n (y_i^+ - \bar{y}^+)^2 \\
&= \frac{n_1(n-n_1)}{n^3(n-1)} \sum_{i=1}^n \left(\frac{n}{n_1} y_i(1) + \frac{n}{n-n_1} y_i(0) - \frac{n}{n_1} \bar{y}(1) - \frac{n}{n-n_1} \bar{y}(0) \right)^2 \\
&= \frac{n_1(n-n_1)}{n^3(n-1)} \sum_{i=1}^n \left(\frac{n}{n_1} y_i(1) - \frac{n}{n_1} \bar{y}(1) \right)^2 + \frac{n_1(n-n_1)}{n^3(n-1)} \sum_{i=1}^n \left(\frac{n}{n-n_1} y_i(0) - \frac{n}{n-n_1} \bar{y}(0) \right)^2 \\
&\quad + 2 \frac{n_1(n-n_1)}{n^3(n-1)} \sum_{i=1}^n \left(\frac{n}{n_1} y_i(1) - \frac{n}{n_1} \bar{y}(1) \right) \left(\frac{n}{n-n_1} y_i(0) - \frac{n}{n-n_1} \bar{y}(0) \right) \\
&= \frac{(n-n_1)}{nn_1(n-1)} \sum_{i=1}^n (y_i(1) - \bar{y}(1))^2 + \frac{n_1}{n(n-n_1)(n-1)} \sum_{i=1}^n (y_i(0) - \bar{y}(0))^2 \\
&\quad + \frac{2}{n(n-1)} \sum_{i=1}^n (y_i(1) - \bar{y}(1)) (y_i(0) - \bar{y}(0)) \\
&= \frac{(n-n_1)}{nn_1} S_{y(1),n}^2 + \frac{n_1}{n(n-n_1)} S_{y(0),n}^2 + \frac{1}{n} \left(S_{y(1),n}^2 + S_{y(0),n}^2 - S_{y(1)-y(0),n}^2 \right) \\
&= \frac{S_{y(1),n}^2}{n_1} + \frac{S_{y(0),n}^2}{n-n_1} - \frac{S_{y(1)-y(0),n}^2}{n}.
\end{aligned}$$

The first equality follows from Equation (2.1.5). The third follows from Theorem 2.1.2 and

from the formula we derived above for $cov(D_i, D_j)$. The last but one follows from the definition of $S_{y(1),n}^2$ and $S_{y(0),n}^2$ and from Equation (2.1.6).

QED.

Find a sufficient condition to have that $V(\widehat{ATE}) = 0$.

$V(\widehat{ATE}) = 0$ if there exist two real numbers $y(0)$ and $y(1)$ such that for every i , $y_i(0) = y(0)$ and $y_i(1) = y(1)$, meaning that all units have the same untreated and treated outcomes. In that case, one can show that $S_{y(1),n}^2 = S_{y(0),n}^2 = S_{y(1)-y(0),n}^2 = 0$, so $V(\widehat{ATE}) = 0$.

Intuitively, why do we have that $V(\widehat{ATE}) = 0$ when all units have the same untreated and treated outcomes?

In that case, irrespective of which units are assigned to the treatment group, we always have that the average outcome in the treatment group $\frac{1}{n_1} \sum_{i:D_i=1} Y_i$ is equal to $y(1)$, because the treated outcome of every unit is equal to $y(1)$. Similarly, irrespective of which units are assigned to the control group, we always have that the average outcome in the control group $\frac{1}{n-n_1} \sum_{i:D_i=0} Y_i$ is equal to $y(0)$, because the untreated outcome of every unit is equal to $y(0)$. Then, $\widehat{ATE} = \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n-n_1} \sum_{i:D_i=0} Y_i = y(1) - y(0)$, a deterministic quantity, so its variance is equal to zero.

Holding everything else fixed, is $V(\widehat{ATE})$ increasing or decreasing in $S_{y(1),n}^2$ and $S_{y(0),n}^2$?

$V(\widehat{ATE})$ is increasing in both of those quantities.

Intuitively, why do we have that $V(\widehat{ATE})$ is increasing in $S_{y(1),n}^2$ and $S_{y(0),n}^2$?

If $S_{y(1),n}^2$ and $S_{y(0),n}^2$ are large, that means that the treated and untreated outcomes vary a lot across units. Then, lottery draws assigning high $y_i(1)$ units to the treatment and low $y_i(0)$ units to the control will generate substantially larger values of $\widehat{ATE}$ than lottery draws assigning low $y_i(1)$ units to the treatment and high $y_i(0)$ units to the control. Consequently, $\widehat{ATE}$ can vary a lot across lottery draws. On the other hand, if the treated and untreated outcomes do not vary a lot across units, $\widehat{ATE}$ cannot vary as much across lottery draws, so its variance is smaller. In fact, in the extreme case where the treated and untreated outcomes do not vary at all across units, we have seen above that $V(\widehat{ATE}) = 0$.

The discussion above shows that $V(\widehat{ATE})$ is lower if the RCT is conducted in a homogeneous population (units with relatively similar treated and untreated outcomes) than if it is conducted in a heterogeneous population (units with fairly different treated and untreated outcomes). This is the reason why researchers often conduct stratified RCTs. Before randomly assigning the treatment, researchers divide the population into S strata of n/S units, assuming to simplify that n/S is an integer. Units' grouping into strata is based on their characteristics (e.g. their gender, their socio-economic background): units with similar characteristics are grouped into the same strata. Then, in each stratum n_1/S units are randomly assigned to the treatment, assuming again to simplify that n_1/S is an integer. $\widehat{ATE}$ has a lower variance in a stratified than in an unstratified RCT, and the difference can be large if the variances of the treated and untreated outcomes are substantially lower within each stratum than in the full population. This can happen when units' grouping into strata is based on variables that predict their untreated and treated outcomes well, so that units in the same stratum have homogeneous treated and untreated outcomes. In practice, researchers often collect their main outcome of interest at baseline, before randomly assigning some units to the treatment. Then, they stratify their randomization according to the baseline outcome. The baseline outcome usually predicts the treated and untreated outcomes well, so the resulting strata group together units with very similar treated and untreated outcomes, thus lowering the variance of $\widehat{ATE}$. In the tracking example, one would collect students' test scores at the start of first grade. Then, one would form strata of students with similar scores, and randomly assign tracking within each stratum.

Pushed to the extreme, the stratification logic amounts to dividing the population into $n/2$ pairs of 2 units, and randomly assigning one unit to treatment in each pair. Such stratified RCTs are called paired RCTs. Even though the issue is still debated in the academic literature, there is a growing consensus that paired RCTs dominate other stratified RCTs.

Holding everything else fixed, is $V(\widehat{ATE})$ increasing or decreasing in $S_{y(1)-y(0),n}^2$?

$V(\widehat{ATE})$ is decreasing in $S_{y(1)-y(0),n}^2$. In other words, $V(\widehat{ATE})$ is lower if the RCT is conducted in a population where units' treatment effects are heterogeneous than if it is conducted in a population where units' treatment effects are homogeneous. Here is some intuition for this result. Let us start by assuming that $S_{y(1)-y(0),n}^2 = 0$. This implies that the treatment effect is constant: for every i , $y_i(1) = y_i(0) + \tau$ for some real number τ . Then the two potential outcomes are perfectly positively correlated, so if the lottery assigns the highest $y_i(1)$ units to the treatment group, it is also assigning the lowest $y_i(0)$ units to the control group, thus generating a high value of $\widehat{ATE}$. Conversely, if it assigns low $y_i(1)$ units to the treatment group, it is also assigning high $y_i(0)$ units to the control group, thus generating a low value of $\widehat{ATE}$. Accordingly, the variance of $\widehat{ATE}$ is high. Now let us assume that $S_{y(1)-y(0),n}^2 > 0$, meaning that the treatment effect is not constant. For instance, one could have that $y_i(1) = y_i(0) + \varepsilon_i$, where ε_i are realizations of a standard normal distribution. Then the potential outcomes are no longer perfectly correlated. Therefore, some high $y_i(1)$ units have a low $y_i(0)$, and conversely. Then, assigning high $y_i(1)$ units to the treatment groups no longer automatically means one is assigning low $y_i(0)$ units to the control, thus reducing the variance of $\widehat{ATE}$.

Holding everything else fixed, is $V(\widehat{ATE})$ increasing or decreasing in n_1 ?



If n is fixed, increasing the number of treated units decreases the number of untreated units. Accordingly, $\frac{S_{y(1),n}^2}{n_1}$ decreases but $\frac{S_{y(0),n}^2}{n-n_1}$ increases, and the overall effect is ambiguous.

If $S_{y(1),n}^2 = S_{y(0),n}^2$, meaning that the treated and untreated outcomes have the same variances,

$$\begin{aligned}
 V(\widehat{ATE}) &= S_{y(1),n}^2 \left(\frac{1}{n_1} + \frac{1}{n-n_1} \right) - \frac{S_{y(1)-y(0),n}^2}{n} \\
 &= S_{y(1),n}^2 \frac{n}{n_1(n-n_1)} - \frac{S_{y(1)-y(0),n}^2}{n} \\
 &= S_{y(1),n}^2 \frac{1}{n \frac{n_1}{n} \left(1 - \frac{n_1}{n}\right)} - \frac{S_{y(1)-y(0),n}^2}{n} \\
 &= \frac{1}{n} \left(S_{y(1),n}^2 \frac{1}{\frac{n_1}{n} \left(1 - \frac{n_1}{n}\right)} - S_{y(1)-y(0),n}^2 \right). \tag{2.1.7}
 \end{aligned}$$

Then, holding everything else fixed, which value of $\frac{n_1}{n}$ minimizes $V(\widehat{ATE})$?

Minimizing $V(\widehat{ATE})$ as a function of $\frac{n_1}{n}$ is equivalent to maximizing $\frac{n_1}{n}(1 - \frac{n_1}{n})$. Let $g(x) = x(1 - x)$. The first derivative of $x \mapsto g(x)$ is $1 - 2x$ which cancels out at $x = \frac{1}{2}$. Its second derivative is -2 which is strictly negative, thus showing that $x \mapsto g(x)$ is concave so $x = \frac{1}{2}$ is its global maximizer. Accordingly, $V(\widehat{ATE})$ is minimized when 50% of units are assigned to the treatment. The proportion of units assigned to the treatment is not always something the researcher can choose. Often times, this proportion directly follows from the number of treatments seats and the number of eligible applicants. But when $\frac{n_1}{n}$ is left or partly left to the discretion of the researchers, it is optimal to set $\frac{n_1}{n}$ as close as possible to 50%, to minimize $V(\widehat{ATE})$ and $\widehat{ATE}$'s MSE.

If $S_{y(1),n}^2 = S_{y(0),n}^2$, one can show that $S_{y(1),n}^2 \frac{1}{\frac{n_1}{n}(1 - \frac{n_1}{n})} - S_{y(1)-y(0),n}^2 \geq 0$, so it follows from Equation (2.1.7) that $V(\widehat{ATE})$ decreases with n , holding everything else in Equation (2.1.7), and in particular the proportion of treated units, fixed. This implies that the larger the number of participants in our RCT, the more precise our estimator of ATE . Does $V(\widehat{ATE})$ converge to 0 when n , the size of the population, goes to $+\infty$?

It follows from Theorem 2.1.4 that $V(\widehat{ATE})$ converges to 0 when n goes to $+\infty$, provided that n_1 and $n - n_1$ also go to $+\infty$ when n goes to $+\infty$.¹ Assume for instance that $n_1 = \lceil \frac{n}{2} \rceil$, meaning that n_1 is the integer closest to $\frac{n}{2}$. Then $V(\widehat{ATE}) \rightarrow 0$ when $n \rightarrow +\infty$. The variance of $\widehat{ATE}$ is equal to its MSE, so this implies that when the number of RCT participants goes to infinity, the MSE of $\widehat{ATE}$ goes to 0. This means that the expectation of the squared difference between $\widehat{ATE}$ and ATE goes to 0, which suggests that $\widehat{ATE}$ converges towards ATE . This turns out to be the case: using the fact that $V(\widehat{ATE}) \rightarrow 0$ and Markov's inequality, one can show a result that for any real number $\epsilon > 0$, $\lim_{n \rightarrow +\infty} P(|\widehat{ATE} - ATE| \geq \epsilon) = 0$.

2.1.5 Estimating $V(\widehat{ATE}_n)$.

Once we have computed the variance of an estimator, we usually try to find an estimator of that variance. Propose an estimator of $S_{y(0),n}^2$.

¹It should also be the case that $S_{y(1),n}^2$, $S_{y(1),n}^2$, and $S_{y(1)-y(0),n}^2$ all converge towards real numbers or are bounded when $n \rightarrow +\infty$.

To estimate the variance of $y(0)$ in the entire population, we can use the variance of $y(0)$ in our sample of untreated observations. Let $\bar{y}_{D=0}(0) = \frac{1}{n-n_1} \sum_{i:D_i=0} y_i(0)$ be the average of $y_i(0)$ among units that remain untreated following the lottery. Our estimator of $S_{y(0),n}^2$ is $\hat{S}_{y(0)}^2 = \frac{1}{n-n_1-1} \sum_{i:D_i=0} (y_i(0) - \bar{y}_{D=0}(0))^2$. One can show that $\hat{S}_{y(0)}^2$ is an unbiased estimator of $S_{y(0),n}^2$.

Propose an estimator of $S_{y(1),n}^2$.

To estimate the variance of $y(1)$ in the entire population, we can use the variance of $y(1)$ in our sample of treated observations. Let $\bar{y}_{D=1}(1) = \frac{1}{n_1} \sum_{i:D_i=1} y_i(1)$ be the average of $y_i(1)$ among units that get treated following the lottery. Our estimator of $S_{y(1),n}^2$ is $\hat{S}_{y(1)}^2 = \frac{1}{n_1-1} \sum_{i:D_i=1} (y_i(1) - \bar{y}_{D=1}(1))^2$. One can show that $\hat{S}_{y(1)}^2$ is an unbiased estimator of $S_{y(1),n}^2$.

Propose an estimator of $S_{y(1)-y(0),n}^2$.

Estimating the variance of $y(1) - y(0)$ would require that we observe jointly $y_i(1)$ and $y_i(0)$ for some units in the population. However, we never jointly observe $y_i(1)$ and $y_i(0)$. Therefore, we cannot estimate $S_{y(1)-y(0),n}^2$.

Can you estimate $V(\widehat{ATE}_n)$?

$V(\widehat{ATE}_n) = \frac{S_{y(0),n}^2}{n-n_1} + \frac{S_{y(1),n}^2}{n_1} - \frac{S_{y(1)-y(0),n}^2}{n}$. Given that we cannot estimate $S_{y(1)-y(0),n}^2$, we cannot estimate $V(\widehat{ATE}_n)$.

Can you estimate an upper bound of $V(\widehat{ATE}_n)$?

$S_{y(1)-y(0),n}^2 \geq 0$. Therefore, $V(\widehat{ATE}_n) \leq \frac{S_{y(0),n}^2}{n-n_1} + \frac{S_{y(1),n}^2}{n_1}$. This upper bound is equal to $V(\widehat{ATE}_n)$ if the effect of the treatment is constant across units. Otherwise it is strictly larger than $V(\widehat{ATE}_n)$. We can estimate this upper bound using $\widehat{V}(\widehat{ATE}_n)^+ = \frac{\widehat{S}_{y(0)}^2}{n-n_1} + \frac{\widehat{S}_{y(1)}^2}{n_1}$.

2.1.6 Confidence intervals for ATE_n .

Usually, once we have an estimator and an estimator of the variance of that estimator, we try to show that our estimator is asymptotically normal when the number of units goes to infinity, so as to derive confidence intervals for our target parameter. Here, this is harder to do than in the previous chapter. Indeed, $\widehat{ATE}_n = \frac{1}{n} \sum_{i=1}^n \left(\frac{n}{n_1} D_i y_i(1) - \frac{n}{n-n_1} (1-D_i) y_i(0) \right)$. Can we use the usual central limit theorem to study the asymptotic distribution of

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n \left(\frac{n}{n_1} D_i y_i(1) - \frac{n}{n-n_1} (1-D_i) y_i(0) \right) \right)?$$

No we cannot. This theorem applies to averages of iid random variables. Still using a central-limit theorem for sample averages with sampling without replacement, we can show the following result.

Theorem 2.1.5 *Assume that Assumption 1 is satisfied. Assume also that other technical conditions you do not need to know are satisfied. Then,*

$$\frac{\widehat{ATE}_n - ATE_n}{\sqrt{V(\widehat{ATE}_n)}} \xrightarrow{d} N(0, 1).$$

Theorem 2.1.5 implies that if the population in which the experiment is conducted is large enough, $\widehat{ATE}_n$ approximately follows a normal distribution.

If we had an estimator x of $V(\widehat{ATE}_n)$, we could use this result to form a confidence interval for ATE_n with asymptotic coverage equal to $1 - \alpha$, exactly as we did in the previous chapter. This confidence interval would be $IC(\alpha) = \left[\widehat{ATE}_n - q_{1-\frac{\alpha}{2}} \sqrt{x}, \widehat{ATE}_n + q_{1-\frac{\alpha}{2}} \sqrt{x} \right]$. As we discussed above, we cannot estimate $V(\widehat{ATE}_n)$. However, we can estimate an upper bound of $V(\widehat{ATE}_n)$. Use this to propose a confidence interval for $\widehat{ATE}_n$ with asymptotic coverage at least equal to $1 - \alpha$.

$IC(\alpha)_+ = \left[\widehat{ATE}_n - q_{1-\frac{\alpha}{2}} \sqrt{\widehat{V}(\widehat{ATE}_n)^+}, \widehat{ATE}_n + q_{1-\frac{\alpha}{2}} \sqrt{\widehat{V}(\widehat{ATE}_n)^+} \right]$ is a confidence interval for $\widehat{ATE}_n$ with asymptotic coverage at least equal to $1 - \alpha$. Indeed, $\widehat{V}(\widehat{ATE}_n)^+$ is the estimator of the upper bound of $V(\widehat{ATE}_n)$. Therefore, $\widehat{V}(\widehat{ATE}_n)^+$ is larger than x . Accordingly, $IC(\alpha)_+$ is wider than $IC(\alpha)$, and therefore, $\lim_{n \rightarrow +\infty} P(ATE_n \in IC(\alpha)_+) \geq \lim_{n \rightarrow +\infty} P(ATE_n \in IC(\alpha)) = 1 - \alpha$.

Here, the randomness arises from the random assignment of $(D_1, \dots, D_n)$. $P(ATE_n \in IC(\alpha)_+) \geq 1 - \alpha$ means that out of the $\binom{n}{n_1}$ possible realizations of $(D_1, \dots, D_n)$, for at most $\alpha \times \binom{n}{n_1}$ of them $ATE_n \notin IC(\alpha)_+$. Therefore, ex-post, once the lottery has been conducted and the data has been collected, we can be $1 - \alpha\%$ “confident” that $ATE_n \in IC(\alpha)_+$. Indeed, from an ex-ante perspective there are at most $\alpha\%$ of the random assignments of treatment we can draw for which $ATE_n \notin IC(\alpha)_+$.

2.1.7 Testing the null of no-treatment effect for any unit in the population

In this section, we present another method to infer whether the treatment has an effect or not. This method amounts to testing the null hypothesis $y_i(1) = y_i(0)$ for every $i \in \{1, \dots, n\}$.

Let me present this test through a simple example. Assume $n = 4$ and $n_1 = 2$. Assume you run the randomization, and it assigns subject 1 and 4 to the treatment, while subjects 2 and 3 remain untreated. After, say, one year, you measure the outcome you are interested in for these 4 subjects. For each subject, Table 2.1 below gives the value of the potential outcome you observe, while question marks denote the outcome we do not observe.

Table 2.1: What is known/unknown at the outset of a randomized experiment

Subject	D	y(0)	y(1)
1	1	?	3
2	0	2	?
3	0	0	?
4	1	?	1

Compute $\widehat{ATE}_n$ in this experiment.

$$\widehat{ATE}_n = 1.$$

If you assume that $y_i(1) = y_i(0)$ for every $i \in \{1, \dots, 4\}$, can you fill the missing values in Table 2.1?

Yes!

Table 2.2: What is known/unknown at the outset of a randomized experiment, under the assumption that $y_i(1) = y_i(0)$

Subject	D	y(0)	y(1)
1	1	3	3
2	0	2	2
3	0	0	0
4	1	1	1

In this example, (D_1, D_2, D_3, D_4) can take $\binom{4}{2} = 6$ values. For each of these possible values, compute the value of $\widehat{ATE}_4$, under the hypothesis that $y_i(1) = y_i(0)$.

Table 2.3: The probability distribution of $\widehat{ATE}_4$, under the hypothesis that $y_i(1) = y_i(0)$

(D_1, D_2, D_3, D_4)	$\widehat{ATE}_4$
(1,1,0,0)	2
(1,0,1,0)	0
(1,0,0,1)	1
(0,1,1,0)	-1
(0,1,0,1)	0
(0,0,1,1)	-2

(D_1, D_2, D_3, D_4) can take 6 values, each with probability 1/6. For each of these values, under

the null hypothesis $y_i(1) = y_i(0)$ we can compute which value of $\widehat{ATE}_4$ we will observe if that value of (D_1, D_2, D_3, D_4) gets selected by the lottery. This gives us the probability distribution of $\widehat{ATE}_4$ under the null hypothesis $y_i(1) = y_i(0)$. It is presented in Table 2.3. If $y_i(1) = y_i(0)$, then with a probability $\frac{1}{6}$ $\widehat{ATE}_4 = 2$, with a probability $\frac{1}{3}$ $\widehat{ATE}_4 = 0$, etc.

In this example, if you want to have a probability $\frac{1}{3}$ of rejecting the null hypothesis $y_i(1) = y_i(0)$ while it is true, which decision rule should you use?

You should reject the null if the value of $\widehat{ATE}_4$ you get to observe at the end of the experiment is strictly larger than 1, the $(1 - \frac{1}{6})$ th quantile of the distribution of $\widehat{ATE}_4$ under the null, or strictly lower than -1 , the $\frac{1}{6}$ th quantile of that distribution. If the null hypothesis is true, this only has a probability $\frac{1}{3}$ of happening. In our example, $\widehat{ATE}_4 = 1$, so we do not reject the null hypothesis $y_i(1) = y_i(0)$ at the $\frac{1}{3}$ level.

If you want to have a probability α of rejecting the null hypothesis $y_i(1) = y_i(0)$ while it is true, which decision rule should you use?

You should reject the null when the value of $\widehat{ATE}_4$ you get to observe at the end of the experiment is either larger than the $(1 - \frac{\alpha}{2})$ th quantile of the distribution of $\widehat{ATE}_4$ under the null or lower than the $\frac{\alpha}{2}$ th quantile of that distribution.

In practice, $\binom{n}{n_1}$ quickly gets very large, even when n and n_1 are not very large. Therefore, computing all the possible values of $\widehat{ATE}_n$ under the null hypothesis $y_i(1) = y_i(0)$ can quickly get infeasible, even with today's fast computers. Instead, researchers randomly draw a subset of, say 1000, possible values for $(D_1, \dots, D_n)$, estimate $\widehat{ATE}_n$ for these values, and then use the $(1 - \frac{\alpha}{2})$ th quantile and $\frac{\alpha}{2}$ th quantile of these 1000 values of $\widehat{ATE}_n$ to estimate the true $(1 - \frac{\alpha}{2})$ th quantile and $\frac{\alpha}{2}$ th quantile of $\widehat{ATE}_n$ across its $\binom{n}{n_1}$ possible values. This method works very well. In the problem sets, you will see how to implement this test in a real application.

This test is called a randomization inference test, or randomization test. Let's compare it to the t-test of $ATE_n = 0$ we proposed in the previous subsection.

The randomization test has one advantage relative to the t-test. It is valid even if the number of units in the experiment is very small, while the t-test is valid only if this number is large.

On the other hand, the randomization test has one **disadvantage** relative to the t-test: often, the analyst **wants to know whether $ATE_n = 0$, not whether $y_i(1) = y_i(0)$ for every $i \in \{1, \dots, n\}$.**

First, let's clarify the difference between these two null hypothesis. If $y_i(1) = y_i(0)$ for every $i \in \{1, \dots, n\}$, then $ATE_n = 0$. Conversely, it might be the case that $y_i(1) \neq y_i(0)$ for some $i \in \{1, \dots, n\}$, while $ATE_n = 0$. For instance, some units benefit from the treatment while other units are harmed by it, and those benefits and losses compensate each other. Therefore, the null hypothesis " $y_i(1) = y_i(0)$ for every $i \in \{1, \dots, n\}$ " is stronger than " $ATE_n = 0$ ", and the test of " $y_i(1) = y_i(0)$ for every $i \in \{1, \dots, n\}$ " might get rejected while " $ATE_n = 0$ ".

Now the issue is that from a decision-theoretic perspective, it's difficult to motivate why one might be interested in knowing whether $y_i(1) = y_i(0)$ for every $i \in \{1, \dots, n\}$, while it's very easy to motivate why one might be interested in knowing ATE_n . Let's assume that a social planner wants to decide whether they should give the treatment to their population or not. If their welfare function is the average utility of their population, and if units' utility is additively separable in their potential outcomes, what the planner needs to know to decide whether they should give the treatment to their population is ATE_n .

Overall, the randomization test can be a very useful complement to the t-test of $ATE_n = 0$. In field or in laboratory experiments, samples are often quite small (say less than 200 units). Then, the asymptotic approximation underlying the t-test might not apply. With a small sample, maybe a 5% level t-test of $ATE_n = 0$ has a higher probability of leading you to make a type 1 error than its 5% nominal value, because the sample is not large enough. In a field or lab experiment with a small sample, if the t-test leads you to reject $ATE_n = 0$, it is advisable to also run the randomization test. Indeed, if ATE_n is really $\neq 0$, then it must also be the case that $y_i(1) \neq y_i(0)$ for some $i \in \{1, \dots, n\}$. Therefore, the randomization test should also be rejected. If the randomization test is indeed rejected, this suggests that the rejection of the t-test of $ATE_n = 0$ is robust and does not come from the fact your sample is too small to use the t-test.

In a recent paper, Young (2015) conducts this exercise for 53 experimental papers published in journals of the American Economic Association. He finds that for 35% of all the 5%-level t-tests of $ATE_n = 0$ which are rejected in these papers, the corresponding randomization test of the null hypothesis that $y_i(1) = y_i(0)$ is not rejected. This suggests that 35% of these 5%-level tests are rejected due to type 1 error. 35% is a much higher type 1 error rate than what these 5%-level t-tests are supposed to yield. However, these t-tests yield a 5% type 1 error rate only if they are used with a large enough sample. Young's findings suggest that in many of these papers, the sample is actually not large enough to rely on these t-tests.

2.2 Randomized controlled trials in a sample drawn from an infinite population

2.2.1 A refresher: the conditional expectation function (CEF), and the conditional variance function

For any vectors of random variables U and V , let $f_U(\cdot)$ denote the density of U ,² and let $f_{U|V}(\cdot|v)$ denote the density of U conditional on $V = v$. Let also $\mathcal{S}(U)$ denote the support of U , i.e. the set of real numbers u such that $f_U(u) > 0$.

Let Y be a random variable, and let X be a random vector.

Definition 2.2.1 *Expectation of a conditional distribution*

$E(Y|X = x)$ is the expectation of Y , with respect to the distribution of $Y|X = x$. If Y is a continuous random variable, it is equal to

$$E(Y|X = x) = \int_{\mathcal{S}(Y|X)} y f_{Y|X}(y|x) dy.$$

If Y is a discrete random variable, it is equal to

$$E(Y|X = x) = \sum_{y \in \mathcal{S}(Y|X)} y P(Y = y|X = x).$$

For a discrete Y , $E(Y) = \sum_{y \in \mathcal{S}(Y)} y P(Y = y)$, so the only difference between $E(Y)$ and $E(Y|X = x)$ is that in $E(Y)$, y receives a weight equal to its probability in the unconditional distribution of Y , while in $E(Y|X = x)$ it receives a weight equal to its probability in the conditional distribution of Y . $E(Y|X = x)$ is just a regular expectation, so it satisfies all the properties of expectations (linearity etc.)...

Theorem 2.2.2 *Another formula for the expectation of a conditional distribution*

When X is a discrete variable,

$$E(Y|X = x) = \frac{E(Y1\{X = x\})}{P(X = x)}$$

This formula will sometimes be useful. You will prove it in a problem set.

Definition 2.2.3 *Conditional expectation*³

The CEF of Y conditional on X is denoted $E(Y|X)$. It is a random variable which is a function of X , and which is equal to $E(Y|X = x)$ when $X = x$.

²When U is discrete, $f_U(u) = P(U = u)$.

³This not a very rigorous definition, but it will do the job for our purposes, and it does not require invoking sophisticated concepts from measure theory. If you want to find a more rigorous definition, a good (but tough) reference is Billingsley (2012), Chapter 6.

For instance, if X is a discrete random variable taking J possible values $x_1 < \dots < x_J$, then

$$E(Y|X) = \sum_{j=1}^J 1\{X = x_j\} E(Y|X = x_j). \quad (2.2.1)$$

Consider a binary variable variable D , and assume that $E(Y|D = 0) = 3$ and $E(Y|D = 1) = 5$. Which values can the random variable $E(Y|D)$ take?

$E(Y|D)$ is either equal to 3 when $D = 0$, or to 5 when $D = 1$. Write $E(Y|D)$ as a function of D , 3, and 5.

One has:

$$E(Y|D) = 3 \times (1 - D) + 5 \times D.$$

You can check that the above formula is true for each possible value of D . What is the probability that $E(Y|D)$ is equal to 3?

$E(Y|D) = 3 \Leftrightarrow D = 0$. Therefore, $P(E(Y|D) = 3) = P(D = 0)$.

Below we are going to see many equalities involving functions of X . $f(X)$ and $g(X)$ are equal if for every possible value of X x_j , $f(x_j) = g(x_j)$.

Theorem 2.2.4 *Laws of iterated expectations.*

$$E(Y) = E(E(Y|X)) \quad (2.2.2)$$

$$E(Y|Z) = E(E(Y|X, Z)|Z) \quad (2.2.3)$$

Proof

I prove only (2.2.2) when Y and X are random variables taking a finite number of values

$x_1 < \dots < x_J$ and $y_1 < \dots < y_K$ respectively.

$$\begin{aligned}
E(E(Y|X)) &= \sum_{j=1}^J E(Y|X = x_j)P(X = x_j) \\
&= \sum_{j=1}^J \left(\sum_{k=1}^K y_k P(Y = y_k|X = x_j) \right) P(X = x_j) \\
&= \sum_{j=1}^J \left(\sum_{k=1}^K y_k P(Y = y_k|X = x_j)P(X = x_j) \right) \\
&= \sum_{k=1}^K \left(\sum_{j=1}^J y_k P(Y = y_k|X = x_j)P(X = x_j) \right) \\
&= \sum_{k=1}^K y_k \left(\sum_{j=1}^J P(Y = y_k|X = x_j)P(X = x_j) \right) \\
&= \sum_{k=1}^K y_k \left(\sum_{j=1}^J P(Y = y_k, X = x_j) \right) \\
&= \sum_{k=1}^K y_k P(Y = y_k) \\
&= E(Y)
\end{aligned}$$

The first equality comes from the fact that a) $E(Y|X)$ is a discrete random variable which can take J different values $E(Y|X = x_1), \dots, E(Y|X = x_J)$, b) the probability that $E(Y|X) = E(Y|X = x_j)$ is $P(X = x_j)$, and c) the expectation of a discrete random variable Z taking J values $z_1, \dots, z_J$ with probability $p_1, \dots, p_J$ is $\sum_{j=1}^J z_j p_j$. The second equality comes from the definition of $E(Y|X = x_j)$. The last but one equality comes from the fact the sum of $P(Y = y_k, X = x_j)$ over all possible values of x_j is $P(Y = y_k)$.

QED.

The first law of iterated expectations states that the mean of Y is the same as that of $E(Y|X)$. Let Y be wages in the population of american adults between 20 and 65 years old. Let X be a binary variable equal to 1 for males and to 0 for females.

$$E(E(Y|X)) = E(Y|X = 0)P(X = 0) + E(Y|X = 1)P(X = 1).$$

The equality comes from the fact $E(Y|X)$ is a random variable which can take two values: $E(Y|X = 0)$ when $X = 0$ (this has a probability $P(X = 0)$ of taking place), and $E(Y|X = 1)$ when $X = 1$ (this has a probability $P(X = 1)$ of taking place). The first law of iterated expectations therefore states that

$$E(Y) = E(Y|X = 0)P(X = 0) + E(Y|X = 1)P(X = 1).$$

This merely means that the average wage in the american population is a weighted average of the average wage among american females ($E(Y|X = 0)$) and among american males

$(E(Y|X = 1))$, in which the average wage among american females is weighted by their share in the american population, while the average wage among american males is weighted by their share in the population. If X is age in years, $E(Y) = E(E(Y|X))$ merely means that the average wage among american adults between 20 and 65 is equal to the weighted average of the average wage among 20 years old people, the average wage among 21 years old people,... and the average age among 65 years old people, in which each group is weighted by the percentage it accounts for in the population.

The **second law of iterated expectations generalizes this to two conditioning variables**. It states that the **conditional mean of Y knowing Z is the same as that of $E(Y|X, Z)$** . Let **X be a gender binary** variable equal to 1 for males, and let **Z be a race** binary variable variable equal to 1 for black and to 0 for non-black. Using the same reasoning as above, we have

$$\begin{aligned} E(E(Y|X, Z = 0)|Z = 0) &= E(Y|X = 0, Z = 0)P(X = 0|Z = 0) + E(Y|X = 1, Z = 0)P(X = 1|Z = 0) \\ E(E(Y|X, Z = 1)|Z = 1) &= E(Y|X = 0, Z = 1)P(X = 0|Z = 1) + E(Y|X = 1, Z = 1)P(X = 1|Z = 1). \end{aligned}$$

Therefore, the second law of iterated expectations states that

$$\begin{aligned} E(Y|Z = 0) &= E(Y|X = 0, Z = 0)P(X = 0|Z = 0) + E(Y|X = 1, Z = 0)P(X = 1|Z = 0) \\ E(Y|Z = 1) &= E(Y|X = 0, Z = 1)P(X = 0|Z = 1) + E(Y|X = 1, Z = 1)P(X = 1|Z = 1). \end{aligned}$$

The second identity merely means that the average wage among blacks should be equal to the average wage among black females, multiplied by the percentage of females among blacks, plus the average wage among black males, multiplied by the percentage of males among blacks. So overall, laws of iterated expectations merely state that the average of a random variable Y is also a weighted average of the average of Y among subgroups of the population, in which each subgroup is weighted according to the share of the population it accounts for.

Theorem 2.2.5 *Some rules of algebra for CEF.*

$$E(Y + Z|X) = E(Y|X) + E(Z|X) \quad (2.2.4)$$

$$E(f(X)|X) = f(X) \quad (2.2.5)$$

$$E(f(X)Y|X) = f(X)E(Y|X) \quad (2.2.6)$$

Proof

I will prove these properties for a discrete random variable X taking J possible values $x_1 < \dots < x_J$. It follows from Equation (2.2.1) that

$$\begin{aligned} E(Y + Z|X) &= \sum_{j=1}^J 1\{X = x_j\}E(Y + Z|X = x_j) \\ &= \sum_{j=1}^J 1\{X = x_j\}(E(Y|X = x_j) + E(Z|X = x_j)) \\ &= \sum_{j=1}^J 1\{X = x_j\}E(Y|X = x_j) + \sum_{j=1}^J 1\{X = x_j\}E(Z|X = x_j) \\ &= E(Y|X) + E(Z|X). \end{aligned}$$

The second equality holds because $E(Y + Z|X = x_j)$ is just a regular expectation, so it is linear. This proves (2.2.4).

It follows from Equation (2.2.1) that

$$\begin{aligned} E(f(X)|X) &= \sum_{j=1}^J 1\{X = x_j\} E(f(X)|X = x_j) \\ &= \sum_{j=1}^J 1\{X = x_j\} E(f(x_j)|X = x_j) \\ &= \sum_{j=1}^J 1\{X = x_j\} f(x_j) \\ &= f(X). \end{aligned}$$

The last equality holds because for any possible value of X , $f(X)$ and $\sum_{j=1}^J 1\{X = x_j\} f(x_j)$ are equal. This proves (2.2.5).

It follows from Equation (2.2.1) that

$$\begin{aligned} E(f(X)Y|X) &= \sum_{j=1}^J 1\{X = x_j\} E(f(X)Y|X = x_j) \\ &= \sum_{j=1}^J 1\{X = x_j\} E(f(x_j)Y|X = x_j) \\ &= \sum_{j=1}^J 1\{X = x_j\} f(x_j) E(Y|X = x_j) \\ &= f(X) E(Y|X). \end{aligned}$$

The last equality holds because for any possible value of X , $f(X)E(Y|X)$ and $\sum_{j=1}^J 1\{X = x_j\} f(x_j) E(Y|X = x_j)$ are equal. This proves (2.2.6).

QED.

Definition 2.2.6 *Variance of a conditional distribution*

$V(Y|X = x_j)$ is the variance of the probability distribution of $Y|X = x_j$. For instance, if Y is a discrete random variable taking K possible values $y_1 < \dots < y_K$, then

$$\begin{aligned} V(Y|X = x_j) &= E((Y - E(Y|X = x_j))^2|X = x_j) \\ &= \sum_{k=1}^K (y_k - E(Y|X = x_j))^2 P(Y = y_k|X = x_j). \end{aligned}$$

As $V(Y|X = x_j)$ is a standard variance operator, we have

$$V(Y|X = x_j) = E(Y^2|X = x_j) - (E(Y|X = x_j))^2.$$

Definition 2.2.7 *Conditional variance*

The *variance of Y conditional on X* is a random variable which is a function of X and which is equal to $V(Y|X = x_j)$ when $X = x_j$. It satisfies the following equality:

$$V(Y|X) = E((Y - E(Y|X))^2|X) = E(Y^2|X) - (E(Y|X))^2.$$

For instance, if X is a discrete random variable taking J possible values $x_1 < \dots < x_J$, then

$$\begin{aligned} V(Y|X) &= \sum_{j=1}^J 1\{X = x_j\} V(Y|X = x_j) \\ &= \sum_{j=1}^J 1\{X = x_j\} E((Y - E(Y|X = x_j))^2|X = x_j) \\ &= \sum_{j=1}^J 1\{X = x_j\} (E(Y^2|X = x_j) - (E(Y|X = x_j))^2) \\ &= \sum_{j=1}^J 1\{X = x_j\} E(Y^2|X = x_j) - \sum_{j=1}^J 1\{X = x_j\} (E(Y|X = x_j))^2. \end{aligned}$$

In the derivation above, which equality proves that $V(Y|X) = E((Y - E(Y|X))^2|X)$? Which equality proves that $V(Y|X) = E(Y^2|X) - (E(Y|X))^2$?

The second equality proves that $V(Y|X) = E((Y - E(Y|X))^2|X)$. Indeed, $\sum_{j=1}^J 1\{X = x_j\} E((Y - E(Y|X = x_j))^2|X = x_j) = E((Y - E(Y|X))^2|X)$. You can check that for every possible value of X x_j this equality is true. The fourth equality proves that $V(Y|X) = E(Y^2|X) - (E(Y|X))^2$. Indeed, $\sum_{j=1}^J 1\{X = x_j\} E(Y^2|X = x_j) - \sum_{j=1}^J 1\{X = x_j\} (E(Y|X = x_j))^2 = E(Y^2|X) - (E(Y|X))^2$. You can check that for every possible value of X x_j this equality is true.

2.2.2 Back to randomized controlled trials in a sample drawn from an infinite population

In the previous section, we have assumed that the population of units eligible to participate in the randomized experiment is fixed. Consequently, we treated the $y_i(0)$ s and $y_i(1)$ s as real numbers, not as random variables. This modelling framework was that adopted by Neyman and Fisher, the two statisticians who “invented” RCTs in the beginning of the 20th century.

This contrasts with the presentation of RCTs given in most contemporary econometrics textbooks (except for (Imbens & Rubin 2015)), where it is (implicitly) assumed that in a first

step, n vectors of potential outcomes $(Y_i(0), Y_i(1))$ are drawn with replacement from an infinite “super-population” of units, exactly like in a Monte-Carlo simulation. As these potential outcomes are drawn with replacement, the $(Y_i(0), Y_i(1))$ s are iid. Moreover, our sample of n units is representative of the “super-population”, because it is randomly drawn from it. Let

$$\mathcal{Y} = (Y_1(0), Y_1(1), Y_2(0), Y_2(1), \dots, Y_n(0), Y_n(1))$$

denote a vector stacking together all these potential outcomes. Then, in a second step treatment is randomly assigned to these n_1 units, and the treatment assignment vector $(D_1, D_2, \dots, D_n)$ satisfies the following assumption:

Assumption 2 *Conditional random assignment of treatment*

For any vector of n numbers $(d_1, \dots, d_n) \in \{0, 1\} \times \dots \times \{0, 1\}$ such that $\sum_{i=1}^n d_i = n_1$,

$$P(D_1 = d_1, \dots, D_n = d_n) = \frac{1}{\binom{n}{n_1}}, \text{ and } (D_1, \dots, D_n) \perp\!\!\!\perp \mathcal{Y}. \quad (2.2.7)$$

Assumption 2 requires that treatment be assigned randomly, and independently of units potential outcomes. It implies that for all i , $D_i \perp\!\!\!\perp (Y_i(0), Y_i(1))$. $D_i \perp\!\!\!\perp (Y_i(0), Y_i(1))$ is satisfied as long as the treatment assignment mechanism satisfies Assumption 2, something the researcher can be sure of if they are the one who actually carried out the randomization. Therefore, we say that in RCTs, $D_i \perp\!\!\!\perp (Y_i(0), Y_i(1))$ is satisfied **by design**. In this course, we will study other non-experimental (or observational) methods to infer treatment effects. For all of them, we will have to **assume** that $D_i \perp\!\!\!\perp (Y_i(0), Y_i(1))$, or that some other type of independence assumption is satisfied. This is in sharp contrast with **RCTs where we do not need to assume anything as $D_i \perp\!\!\!\perp (Y_i(0), Y_i(1))$ is satisfied by design.**

In some instances, the modelling framework presented in this section describes more accurately the RCT under consideration, while in other instances the modelling framework in the previous section describes it more accurately. **Assume you are interested in measuring the effect of a training program on the probability that unemployed people have found a job less than one year after the start of their unemployment spell.** There are millions of unemployed people in the US, and in principle all of them could participate in this experiment. However, to measure the final outcome of interest (whether the participants in the experiment have found a job in less than a year), you will need to survey all the participants in your experiment. That is costly to do. Therefore, you cannot include all US unemployed in your experiment. Instead, you can randomly draw a sample of n unemployed people from that population, and then allocate the n_1 seats for training among those n units using a lottery satisfying Assumption 2. If we omit the fact that you would probably draw the n unemployed chosen to participate in your experiment without replacement from the unemployed population, the modelling framework in this section describes very accurately this experiment.

On the other hand, assume you are interested in measuring the effect of a charter school on students’ test scores. n students have applied to this charter school, and n_1 seats are

available. You allocate the n_1 seats among those n applicants by means of a lottery satisfying Assumption 1. Here, the n applicants are not randomly drawn from a larger population of applicants. It's not clear that those n applicants are representative of any larger population than just themselves. Therefore, the modelling framework in the previous section describes more accurately this experiment than the modelling framework in this section.

Let

$$Y_i = Y_i(0) \times (1 - D_i) + Y_i(1) \times D_i = Y_i(0) + (Y_i(1) - Y_i(0)) \times D_i \quad (2.2.8)$$

denote the outcome we will get to observe for each unit at the outset of the experiment. Let $(Y(0), Y(1))$ denote a vector with the same probability distribution as all the $(Y_i(0), Y_i(1))$. You can think of $(Y(0), Y(1))$ as denoting the potential outcomes of a “representative” individual in the super population. Let $ATE = E(Y(1) - Y(0))$ be the average treatment effect in the super population. Let $ATE_n = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))$ be the average treatment effect in the sample we draw.

Is ATE_n an estimator?

No, ATE_n is not an estimator. At the outset of the experiment, the random variables we get to observe for each unit are (Y_i, D_i) . An estimator must be a function of those random variables and known real numbers. ATE_n is a function of the $Y_i(1) - Y_i(0)$ s for each unit, but we will not get to observe $Y_i(1) - Y_i(0)$ for any unit.

ATE_n is equivalent to the parameter ATE_n we considered in the previous section, except that then we considered that the potential outcomes were deterministic because our population of n units was fixed, while now we consider that the potential outcomes are random variables because we randomly draw our sample of n units from the super population. Therefore, ATE_n is now a random variable. The randomness in ATE_n arises from the sampling of the n individuals that will participate in the experiment. It does not arise from the random assignment of n_1 units to the treatment group and of $n - n_1$ units to the control group. ATE_n does not depend on the D_i s, so irrespective of who receives / does not receive the treatment, ATE_n is left unchanged.

In this subsection, our parameter of interest will not be ATE_n . Instead, we are going to be interested in learning ATE , the average effect of the treatment in the super-population. For instance, in the job training RCT example above, ATE_n is the average effect of the job training in the sample of n job seekers we randomly draw from the US population of job seekers. ATE is the average effect of the job training in the entire population of job seekers.

In the proofs below, we will use the two following formulas for the expectation and variance of ATE_n .

$$\begin{aligned}
E(ATE_n) &= E\left(\frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))\right) \\
&= \frac{1}{n} \sum_{i=1}^n E(Y_i(1) - Y_i(0)) \\
&= \frac{1}{n} \sum_{i=1}^n E(Y(1) - Y(0)) \\
&= ATE.
\end{aligned}$$

The second equality follows from our assumption that the $(Y_i(0), Y_i(1))$ s are identically distributed.

$$\begin{aligned}
V(ATE_n) &= V\left(\frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))\right) \\
&= \frac{1}{n^2} \sum_{i=1}^n V(Y_i(1) - Y_i(0)) \\
&= \frac{1}{n^2} \sum_{i=1}^n V(Y(1) - Y(0)) \\
&= \frac{V(Y(1) - Y(0))}{n}.
\end{aligned}$$

The second equality follows from our assumption that the $(Y_i(0), Y_i(1))$ s are independent. The third equality follows from our assumption that the $(Y_i(0), Y_i(1))$ s are identically distributed.

Finally, let

$$\begin{aligned}
\widehat{ATE}_n &= \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n - n_1} \sum_{i:D_i=0} Y_i \\
&= \frac{1}{n_1} \sum_{i:D_i=1} Y_i(1) - \frac{1}{n - n_1} \sum_{i:D_i=0} Y_i(0) \\
&= \frac{1}{n_1} \sum_{i=1}^n D_i Y_i(1) - \frac{1}{n - n_1} \sum_{i=1}^n (1 - D_i) Y_i(0).
\end{aligned}$$

denote our estimator of ATE in this section. Notice that now, there are two sources of randomness in $\widehat{ATE}_n$: $\widehat{ATE}_n$ is a function of the $(Y_i(0), Y_i(1))$ s, which are randomly drawn from the super population, and it is also a function of the D_i s, which are randomly determined by the lottery.

If we condition on the value of the $(Y_i(0), Y_i(1))$ s, the only source of randomness left are the D_i s, like in the previous section. Therefore, a number of results we prove in the previous section now hold conditional on each possible realization of the potential outcomes.

Theorem 2.2.8 *The mean of $\widehat{ATE}_n$ conditional on $\mathcal{Y}$ is ATE_n .*

If Assumption 2 is satisfied, then for any possible value $(y_1(0), y_1(1), \dots, y_n(0), y_n(1))$ of the vector $\mathcal{Y}$,

$$E\left(\widehat{ATE}_n \mid \mathcal{Y} = (y_1(0), y_1(1), \dots, y_n(0), y_n(1))\right) = \frac{1}{n} \sum_{i=1}^n y_i(1) - y_i(0).$$

Therefore,

$$E\left(\widehat{ATE}_n \mid \mathcal{Y}\right) = \frac{1}{n} \sum_{i=1}^n Y_i(1) - Y_i(0) = ATE_n.$$

Theorem 2.2.9 *The variance of $\widehat{ATE}_n$, conditional on $\mathcal{Y}$.*

If Assumption 2 is satisfied, then for any possible value $(y_1(0), y_1(1), \dots, y_n(0), y_n(1))$ of the vector $\mathcal{Y}$,

$$\begin{aligned} V\left(\widehat{ATE}_n \mid \mathcal{Y} = (y_1(0), y_1(1), \dots, y_n(0), y_n(1))\right) &= \frac{\frac{1}{n-1} \sum_{i=1}^n (y_i(0) - \bar{y}(0))^2}{n - n_1} + \frac{\frac{1}{n-1} \sum_{i=1}^n (y_i(1) - \bar{y}(1))^2}{n_1} \\ &\quad - \frac{\frac{1}{n-1} \sum_{i=1}^n (y_i(1) - y_i(0) - (\bar{y}(1) - \bar{y}(0)))^2}{n}. \end{aligned}$$

Therefore,

$$\begin{aligned} V\left(\widehat{ATE}_n \mid \mathcal{Y}\right) &= \frac{\frac{1}{n-1} \sum_{i=1}^n (Y_i(0) - \bar{Y}(0))^2}{n - n_1} + \frac{\frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - \bar{Y}(1))^2}{n_1} \\ &\quad - \frac{\frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - Y_i(0) - (\bar{Y}(1) - \bar{Y}(0)))^2}{n}. \end{aligned}$$

Theorems 2.2.8 and 2.2.9 are straightforward generalizations of Theorems 2.1.3 and 2.1.4. If we condition on $\mathcal{Y} = (y_1(0), y_1(1), \dots, y_n(0), y_n(1))$, $\mathcal{Y}$ is a vector of real numbers, and the only randomness left arises from the lottery to assign the treatment. Therefore, we are back to the same setting as that we considered in the previous section, and all the results we prove there still apply.

Then, one can show the following results.

Theorem 2.2.10 *If treatment is randomly assigned, ATE is identified.*

If Assumption 2 is satisfied, $E(Y_i \mid D_i = 1) - E(Y_i \mid D_i = 0) = ATE$.

Proof

$$\begin{aligned} E(Y_i \mid D_i = 1) - E(Y_i \mid D_i = 0) &= E(Y_i(1) \mid D_i = 1) - E(Y_i(0) \mid D_i = 0) \\ &= \frac{E(Y_i(1)1\{D_i = 1\})}{E(1\{D_i = 1\})} - \frac{E(Y_i(0)1\{D_i = 0\})}{E(1\{D_i = 0\})} \\ &= \frac{E(Y_i(1))E(1\{D_i = 1\})}{E(1\{D_i = 1\})} - \frac{E(Y_i(0))E(1\{D_i = 0\})}{E(1\{D_i = 0\})} \\ &= E(Y_i(1)) - E(Y_i(0)) \\ &= ATE. \end{aligned} \tag{2.2.9}$$

The first equality follows from Equation (2.2.8). The second equality follows from Theorem 2.2.2. The third equality follows from the fact treatment is independent of potential outcomes.

QED.

Theorem 2.2.10 shows that if Assumption 2 is satisfied, ATE is identified by the estimand $E(Y_i|D_i = 1) - E(Y_i|D_i = 0)$. A natural estimator of this estimand is $\widehat{ATE}_n$. This estimator is unbiased, as was already the case in the previous section where we regarded the population as fixed.

Theorem 2.2.11 $\widehat{ATE}_n$ is an unbiased estimator of ATE

$$E(\widehat{ATE}_n) = ATE.$$

Proof

$$\begin{aligned} E(\widehat{ATE}_n) &= E[E(\widehat{ATE}_n|\mathcal{Y})] \\ &= E[ATE_n] \\ &= ATE. \end{aligned}$$

The first equality follows from the first law of iterated expectations. The second follows from Theorem 2.2.8. The third follows from the fact $E[ATE_n] = ATE$, something we have shown above.

QED.

On the other hand, moving from a fixed population to a sample randomly drawn from a super population changes the variance of our estimator.

Theorem 2.2.12 The variance of $\widehat{ATE}_n$.

If Assumption 2 is satisfied, then

$$V(\widehat{ATE}_n) = \frac{V(Y(0))}{n - n_1} + \frac{V(Y(1))}{n_1}.$$

Proof

$$\begin{aligned}
& V(\widehat{ATE}_n) \\
&= E \left[\left(\widehat{ATE}_n - E(\widehat{ATE}_n) \right)^2 \right] \\
&= E \left[\left(\widehat{ATE}_n - ATE \right)^2 \right] \\
&= E \left[\left(\widehat{ATE}_n - ATE_n + ATE_n - ATE \right)^2 \right] \\
&= E \left[\left(\widehat{ATE}_n - ATE_n \right)^2 \right] + E \left[\left(ATE_n - ATE \right)^2 \right] + 2E \left[\left(\widehat{ATE}_n - ATE_n \right) (ATE_n - ATE) \right] \\
&= E \left[E \left[\left(\widehat{ATE}_n - ATE_n \right)^2 \middle| \mathcal{Y} \right] \right] + E \left[\left(ATE_n - E(ATE_n) \right)^2 \right] + 2E \left[E \left[\left(\widehat{ATE}_n - ATE_n \right) (ATE_n - ATE) \middle| \mathcal{Y} \right] \right] \\
&= E \left[E \left[\left(\widehat{ATE}_n - E(\widehat{ATE}_n | \mathcal{Y}) \right)^2 \middle| \mathcal{Y} \right] \right] + V(ATE_n) + 2E \left[(ATE_n - ATE) E \left[\left(\widehat{ATE}_n - ATE_n \right) \middle| \mathcal{Y} \right] \right] \\
&= E \left[V(\widehat{ATE}_n | \mathcal{Y}) \right] + \frac{V(Y(1) - Y(0))}{n} \\
&= E \left[\frac{\frac{1}{n-1} \sum_{i=1}^n (Y_i(0) - \bar{Y}(0))^2}{n - n_1} + \frac{\frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - \bar{Y}(1))^2}{n_1} - \frac{\frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - Y_i(0) - (\bar{Y}(1) - \bar{Y}(0)))^2}{n} \right] \\
&+ \frac{V(Y(1) - Y(0))}{n} \\
&= \frac{V(Y(0))}{n - n_1} + \frac{V(Y(1))}{n_1} - \frac{V(Y(1) - Y(0))}{n} + \frac{V(Y(1) - Y(0))}{n} \\
&= \frac{V(Y(0))}{n - n_1} + \frac{V(Y(1))}{n_1}.
\end{aligned}$$

The second equality follows from the fact $\widehat{ATE}_n$ is an unbiased estimator of ATE , as shown in Theorem 2.2.11. The fifth equality follows from the first law of iterated expectations, and from the fact $E(ATE_n) = ATE$. The sixth follows from the fact $E(\widehat{ATE}_n | \mathcal{Y}) = ATE_n$ as shown in Theorem 2.2.8, and from the fact $ATE_n - ATE$ is a function of $\mathcal{Y}$ combined with the third rule of algebra for CEFs. The seventh follows from the definition of a conditional variance, from the fact $V(ATE_n) = \frac{V(Y(1) - Y(0))}{n}$ as we have shown above, and from the fact $E[(\widehat{ATE}_n - ATE_n) | \mathcal{Y}] = 0$, as shown in Theorem 2.2.8. The eighth equality follows from Theorem 2.2.9. The ninth equality follows from the fact that for any sequence of iid random variables $(X_n)_{n \in \mathbb{N}}$, $E\left(\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right) = V(X)$.

QED.

Is it possible to estimate $V(\widehat{ATE}_n)$?

Yes, one can use $\widehat{V}(\widehat{ATE}_n) = \frac{\frac{1}{n-n_1-1} \sum_{i:D_i=0} (Y_i - \bar{Y}_{D=0})^2}{n - n_1} + \frac{\frac{1}{n_1-1} \sum_{i:D_i=1} (Y_i - \bar{Y}_{D=1})^2}{n_1}$. If we assume that our sample is drawn from an infinite super population instead of being fixed, the

variance of $\widehat{ATE}_n$ no longer depends on the variance of the treatment effect, and therefore we can estimate it. Randomly drawing the sample from the super population adds a term to $\widehat{V}(\widehat{ATE}_n)$, the variance of ATE_n , which cancels out the variance of the treatment effect that appeared in the previous formula. The assumptions we make on how our data is generated have an impact on the properties of the estimators we use.

The formula for $\widehat{V}(\widehat{ATE}_n)$ is exactly the one Stata uses when it computes the variance of the coefficient of a binary treatment variable D in a linear regression of Y on D , provided the “robust” option is specified. Using $\widehat{V}(\widehat{ATE}_n)$, one can derive a confidence interval for ATE , and one can derive a formula for the minimum detectable difference from 0 in a test of $ATE = 0$. The formulas would be exactly the same as those we had derived for the confidence intervals and MDD of ATE_n in the previous section, so I will not repeat those steps here.

The formula in Theorem 2.2.12 is probably one of the most famous formulas in econometrics and statistics. It is rarely proven, while it is very far from being trivial, look at all the steps it took us to get there. Along the way, we have slowly moved from the modelling framework used in the 1930s by Neyman and Fischer of a fixed population, to the modern framework of a random sample drawn from the population. To be able to deal with this framework, we had to introduce mathematical objects like the conditional expectation and the conditional variance. In the following chapters, we will stick to that framework. It is well-suited to observational studies where researchers use surveys, as the sample of the survey is typically drawn from a larger population.

2.3 Randomized experiments or self-selection?

Instead of randomly assigning the treatment, another option to estimate the ATE could be to let units self-select into the treatment, and then compare the mean outcome of units who have chosen to receive the treatment to the mean outcome of a control group of units who have chosen not to receive it. For instance, if we were interested in measuring the effect of health insurance on health, we could compare the health of people who have chosen to get insured to the health of people who have chosen to remain uninsured.

Do you think this comparison can give you a reliable measure of the effect of the treatment?

While this methodology can sometimes work, it will often fail. As economists, we believe that if people can self-select into a treatment, they will do so if the benefit they expect from the

treatment is greater than its cost. Following Roy (1951), one may for instance assume that

$$D_i = 1\{Y_i(1) - Y_i(0) > 0\}, \quad (2.3.1)$$

meaning that unit i will get treated iff their treatment effect is strictly positive. Then, the estimator comparing the mean outcome of untreated and treated units will be equal to

$$\frac{1}{n_1} \sum_{i: Y_i(1) - Y_i(0) > 0} Y_i(1) - \frac{1}{n - n_1} \sum_{i: Y_i(1) - Y_i(0) \leq 0} Y_i(0),$$

where n_1 is the number of treated units. The quantity in the previous display is very likely to differ from

$$\frac{1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{n} \sum_{i=1}^n Y_i(0),$$

the average treatment effect. For instance, $\frac{1}{n_1} \sum_{i: Y_i(1) - Y_i(0) > 0} Y_i(1)$ is likely to be larger than $\frac{1}{n} \sum_{i=1}^n Y_i(1)$. Therefore, comparing the mean outcome of treated and untreated units when units can self-select into the treatment usually gives a poor measure of the true treatment effect.

More intuitively, if units self-select into the treatment, treated and untreated units are likely to differ on many dimensions, besides the fact that one group received the treatment while the other remained untreated. In the health insurance example, insured people are on average better educated and richer than uninsured people. Better educated people tend to smoke less and exercise more. Presumably, insured people would have been in better health than uninsured people even if they had not been insured, for instance because they smoke less and exercise more. Therefore, the comparison of the average health in the two groups will pick up both the effect of health insurance, and the effect of other variables (e.g.: the fact that insured people smoke less and exercise more). On the other hand, randomization ensures that the randomly chosen treatment and control groups are comparable on every dimension, but for the fact the treatment group receives the treatment while the control group remains untreated. Therefore, the comparison of the mean outcome in the treatment and in the control group really picks up the effect of the treatment, not the effect of something else.

While a naive comparison of a self-selected treatment and control group will seldom give a reliable measure of the effect of the treatment, more sophisticated versions of this method can sometimes do a good job, as we will discuss in the next chapters.

2.4 Quantile treatment effects

So far, we have focused on the average treatment effect as our main parameter of interest. Instead of looking at the effect of the treatment on the average of the outcome, one could also be interested in looking at how the treatment affects quantiles of the distribution of the outcome. Let $F_{Y(0)}(y) = P(Y(0) \leq y)$ and $F_{Y(1)}(y) = P(Y(1) \leq y)$ denote the cumulative

distribution functions of the two potential outcomes of the representative individual in the super population. If the potential outcomes are continuously distributed with a strictly positive density everywhere, then these two functions are strictly increasing, and one can define their inverse. The q^{th} quantile of $Y(0)$ is then defined as $F_{Y(0)}^{-1}(q)$. q is the unique solution to $F_{Y(0)}(y) = q$. It is the number such that $q\%$ of the super population has a $Y(0)$ lower than that number. Quantile treatment effects are then defined as $\tau_q = F_{Y(1)}^{-1}(q) - F_{Y(0)}^{-1}(q)$.

When treatment is randomly assigned, quantile treatment effects τ_q are identified. Indeed, we have

$$P(Y \leq y | D = 0) = P(Y(0) \leq y | D = 0) = P(Y(0) \leq y) = F_{Y(0)}(y),$$

where the third equality follows from the fact $D \perp\!\!\!\perp (Y(0), Y(1))$. We can estimate the function $y \mapsto P(Y \leq y | D = 0)$, by looking at the percentage of the control group which has $\{Y_i \leq y\}$, for every possible value of y . Therefore, we can estimate $F_{Y(0)}(y)$ from the data, which implies we can estimate its inverse. Using a similar reasoning, it is easy to see that one can estimate $F_{Y(1)}(y)$ and its inverse. Therefore, we can estimate τ_q for each possible value of q . For instance, to estimate the effect of the treatment on the median of the outcome, one can merely compare the median outcome among treated and untreated subjects.

In most of this course, we will focus on average effects, but we will occasionally discuss quantile treatment effects.

2.5 Limits of randomized controlled trials

Threats to internal validity: attrition, through the example of a boarding school.

Threats to internal validity: general equilibrium effects, through the example of job training programs.

Threats to external validity, through the example of blinded resumes.

It is not possible to conduct an RCT to answer all questions of interest: effect of minimum wage on employment.

For all of those reasons, we need other tools to answer causal questions. We will review some of them during next sessions. However, to assess the value of those tools, we should wonder whether they provide a valid counterfactual for the situation the treated group would have experienced had it remained untreated, as RCTs do.

2.6 A little bit of history

1747: James Lind, surgeon on board of the *Salisbury*, a ship of the British fleet, ran the first medical trial. He divided 12 sailors affected by scurvy into 6 groups of 2, to which he gave 6

different treatments. Group one was given a quart of cider daily, group two twenty-five drops of elixir of vitriol (sulfuric acid), group three six spoonfuls of vinegar, group four half a pint of seawater, group five received two oranges and one lemon, and the last group a spicy paste plus a drink of barley water. Lind did not use a lottery to form the groups so this was not a RCT. The treatment of group five stopped after six days when they ran out of fruit, but by that time one sailor was fit for duty while the other had almost recovered. In spite of this success, citrus treatment for scurvy was generalized only from the beginning of the 20th century.

Late 19th century: RCT start being used in the sciences of education.

1920's-1930's: RCT start being used in agronomy, following works by statisticians Fischer and Neyman.

1948: results from the first RCT in medicine are published in the British Medical Journal.

1962: the American Food and Drug administration requires that pharmaceutical laboratories demonstrate the efficacy of their new drugs through RCT before granting them market approval.

1960's-1980's: several RCTs measuring the effects of various health, educational, and labor policies are led in the US (Job Corps, the Perry preschool, the RAND experiment...)

2000's: a large number of RCTs are conducted in developing countries to measure the efficacy of various health and education programs.

2000's: Private sector companies like Facebook, Amazon, Google, Capital One start using RCTs to maximize (respectively) the time people spend on Facebook, the quantity of products they buy on Amazon, the probability they click on a sponsored link when doing a search on Google, the probability they take up a Capital One credit.

Chapter 3

The OLS regression estimand

In this chapter, we assume we are going to observe a realization of (Y, X) , where Y is a scalar random variable while X is a vector of random variables. Here as well, we place ourselves from an ex-ante perspective, when we have not observed the realization of (Y, X) yet. This modelling framework for instance applies if we are going to randomly draw an individual from the US phonebook and ask their their wage, which will be denoted Y , and their years of education, which will be denoted X . Then, remember that $E(Y)$ and $E(X)$ are just the average wage and years of education in the population (see Theorem 1.1.1). One can also show that $E(Y|X = x)$ is the average wage among people with x years of education in the population. The results that follow apply beyond this simple example, however this simple example might help you understand the meaning of those results.

3.1 More properties of the conditional expectation and of the conditional variance

Theorem 3.1.1 *The decomposition property of the CEF.*

$$Y = E(Y|X) + \varepsilon, \tag{3.1.1}$$

with $E(\varepsilon|X) = 0$ and $\text{cov}(\varepsilon, h(X)) = 0$ for any function of X .

Proof

Let $\varepsilon = Y - E(Y|X)$.

$$\begin{aligned} E(\varepsilon|X) &= E(Y - E(Y|X)|X) \\ &= E(Y|X) - E(E(Y|X)|X) \\ &= E(Y|X) - E(Y|X) \\ &= 0. \end{aligned}$$

The second equality follows from (2.2.4), the third follows from the fact that $E(Y|X)$ is a function of X and from (2.2.5). This proves the first statement.

Then, the first statement combined with the law of iterated expectations implies that $E(\varepsilon) = 0$. Therefore, to prove that $cov(\varepsilon, h(X)) = 0$, it is sufficient to show that $E(\varepsilon h(X)) = 0$.

$$\begin{aligned} E(\varepsilon h(X)) &= E(E(\varepsilon h(X)|X)) \\ &= E(h(X)E(\varepsilon|X)) \\ &= E(h(X) \times 0) \\ &= 0. \end{aligned}$$

The first equality follows from the law of iterated expectations, the second one follows from (2.2.6).

QED.

This property means that Y can be decomposed into two parts, one part which is explained by X , the CEF, and one part which is unrelated/uncorrelated/orthogonal to X .

Theorem 3.1.2 *The CEF prediction property.*

$$E(Y|X) = \underset{m(.)}{\operatorname{argmin}} (E((Y - m(X))^2)). \quad (3.1.2)$$

Proof

$$\begin{aligned} E[(Y - m(X))^2] &= E[(Y - E(Y|X) + E(Y|X) - m(X))^2] \\ &= E[(Y - E(Y|X))^2] \\ &\quad + 2E[(Y - E(Y|X))(E(Y|X) - m(X))] \\ &\quad + E[(E(Y|X) - m(X))^2] \\ &= E[(Y - E(Y|X))^2] \\ &\quad + E[(E(Y|X) - m(X))^2] \end{aligned}$$

The last equality follows from Theorem 3.1.1: $Y - E(Y|X) = \varepsilon$, and $E(Y|X) - m(X)$ is a function of X . We want to minimize the expectation of this last expression with respect to $m(.)$. $m(.)$ does not enter the first term so we can forget about it. The second term is minimized at $m(X) = E(Y|X)$, which proves the result.

QED.

If you want to predict Y based on X , which function of X should you use?

The minimization problem in Theorem 3.1.2 can be interpreted as finding the best predictor for Y once we know X . Assume we are going to draw an individual from the US phonebook

and ask their wage (Y) and education and experience (X), and we would like to make a guess for their wage based on their number of years of education (X). Obviously, our guess will not be the same for someone who completed highschool than for someone with a masters degree. Therefore, our predictor for Y will be a function of X , $m(X)$. Also, a good predictor should be such that in expectation across all the possible draws of that individual, $m(X)$ is not far from Y . So, as a predictor, we want to choose a function $m(X)$ which minimizes the expected value of the distance between Y and $m(X)$, $E(d(Y, m(X)))$. One possible distance function is $d(x, y) = (x - y)^2$. As per this distance, our predictor will be the function $m(\cdot)$ which minimizes $E((Y - m(X))^2)$. Therefore, Theorem 3.1.2 shows that the CEF is the **best prediction** we can make for Y when we know X , according to a mean squared error criterion.¹

Theorem 3.1.3 *The ANOVA formula.*

$$V(Y) = V(E(Y|X)) + E(V(Y|X)) \quad (3.1.3)$$

Proof

$$\begin{aligned} V(Y) &= E[(Y - E(Y))^2] \\ &= E[E((Y - E(Y))^2|X)] \\ &= E[E((Y - E(Y|X) + E(Y|X) - E(Y))^2|X)] \\ &= E[E((Y - E(Y|X))^2 + 2(Y - E(Y|X))(E(Y|X) - E(Y)) + (E(Y|X) - E(Y))^2|X)] \\ &= E[E((Y - E(Y|X))^2|X)] \\ &\quad + 2E[E((Y - E(Y|X))(E(Y|X) - E(Y))|X)] \\ &\quad + E[(E(Y|X) - E(Y))^2] \\ &= E(V(Y|X)) + V(E(Y|X)). \end{aligned}$$

The second equality follows from the law of iterated expectation, and the fifth from the first rule of algebra for conditional expectations. It follows from Theorem 3.1.1 that the second term in the fifth equality is 0: $E(Y|X) - E(Y)$ is a function of X , and $Y - E(Y|X) = \varepsilon$.

QED.

The ANOVA formula says that the variance of Y can be decomposed into the sum of two terms. Give an interpretation to each of those two terms.

This formula says that the variance of Y can be decomposed into two terms: the variance of the mean of Y conditional on X (that's the variance of Y "explained" by X) and the average

¹This theorem shows that the CEF can be defined as the projection of Y on the set of functions of X .

value of the variance of Y for each value of X (that's the variance of Y “unexplained” by X). The ANOVA formula will be useful to calculate the variance of random variables which write as function of others.

Theorem 3.1.4 *Rule of algebra for conditional variances.*

$$V(f(X)Y + g(X)|X) = f(X)^2 V(Y|X). \quad (3.1.4)$$

You will prove this result in the exercises.

3.2 The regression function

Let X be a $K \times 1$ vector of random variables, whose first coordinate is equal to 1.

Definition 3.2.1 *The OLS coefficient, and the regression function*

β , the ordinary least squares (OLS) regression coefficient of Y on X , is a $K \times 1$ vector of real numbers defined as

$$\beta = \underset{b}{\operatorname{argmin}} E((Y - X'b)^2). \quad (3.2.1)$$

The regression function is defined as $X'\beta$.

Following the interpretation we outlined for the CEF, the regression function is the best linear prediction for Y based on X . Y is called the **dependent variable**, while the vector X are the **explanatory variables** or the **regressors** of the regression.

Before deriving an explicit expression of β , we need to introduce an assumption.

Assumption 3 *Rank condition*

$E(XX')$ is invertible.

Theorem 3.2.2 *If the rank condition is verified,*

$$\beta = E(XX')^{-1}E(XY). \quad (3.2.2)$$

Proof

As the objective function is convex, to find β it is necessary and sufficient to differentiate the objective function with respect to b and cancel out the first order conditions. The FOC are

$$E(X(Y - X'b)) = 0. \quad (3.2.3)$$

$E(X(Y - X'b))$ is a $k \times 1$ vector whose j th coordinate is $E(X_j(Y - X'b))$. $E(X_j(Y - X'b)) = 0$ comes from the fact that the partial derivative of the objective function with respect to b_j must be equal to 0. Solving finally yields

$$\beta = E(XX')^{-1}E(XY).$$

QED.

Lemma 3.2.1 *The residuals of a regression are uncorrelated with the explanatory variables.*

Let β denote the solution to (3.2.1), and let $e = Y - X'\beta$ denote the residual of the regression.

$$\text{cov}(e, X_k) = 0 \quad (3.2.4)$$

Proof

First, notice that the foc of the OLS minimization problem with respect to b_1 yields $E(X_1(Y - X'b)) = E((Y - X'b)) = E(e) = 0$. Therefore, to prove the result it suffices to show that $E(Xe) = 0$. This directly follows from (3.2.3).

QED.

Theorem 3.2.3 *The decomposition property of the regression function.*

Let β denote the solution to (3.2.1). Then,

$$Y = X'\beta + e, \quad (3.2.5)$$

with $\text{cov}(X, e) = 0$.

If β is the solution to (3.2.1), $Y = X'\beta + e$ with $\text{cov}(X, e) = 0$ is true by assumption or by construction?

By construction. This directly follows from the definition of β .

The formula for the regression coefficient is not very insightful but I will give much more intuition on what it actually captures later. When $X' = (1, X_2)$ only includes a constant term and one explanatory variable the formula is easier to understand. Indeed, the two regression coefficients are then equal to

$$\begin{aligned} \beta_2 &= \frac{\text{cov}(Y, X_2)}{V(X_2)} \\ \beta_1 &= E(Y) - \beta_2 E(X_2). \end{aligned}$$

3.3 The regression function and the CEF

The regression function is sometimes equal to the CEF.

Theorem 3.3.1 *The linear CEF theorem.*

Suppose that there is a $K \times 1$ function of X $f(X)$ and a $K \times 1$ vector of real numbers $\tilde{\beta}$ such that $E(Y|X) = f(X)'\tilde{\beta}$. Then, $\tilde{\beta}$ is equal to the coefficient of $f(X)$ in a linear regression of Y on $f(X)$:

$$\tilde{\beta} = E(f(X)f(X)')^{-1}E(f(X)Y)$$

Proof

$E(Y|X) = f(X)' \tilde{\beta}$. Then, it follows from Theorem 3.1.1 that $E\left(f(X)(Y - f(X)' \tilde{\beta})\right) = 0$. Solving for $\tilde{\beta}$ yields

$$\tilde{\beta} = E(f(X)f(X)')^{-1}E(f(X)Y).$$

QED.

This theorem is important, because $E(Y|X) = f(X)' \tilde{\beta}$ whenever X is a vector of random variables taking a finite number of values. Indeed, if the support of X is $\{x_1, \dots, x_J\}$,

$$E(Y|X) = \sum_{j=1}^J 1\{X = x_j\} E(Y|X = x_j) = f(X)' \tilde{\beta},$$

where $f(X)' = (1\{X = x_1\}, \dots, 1\{X = x_J\})$ and $\tilde{\beta}' = (E(Y|X = x_1), \dots, E(Y|X = x_J))$. Therefore, it follows from Theorem 3.3.1 that the coefficients of $1\{X = x_1\}, \dots, 1\{X = x_J\}$ in a linear regression of Y on $(1\{X = x_1\}, \dots, 1\{X = x_J\})$ are respectively equal to $E(Y|X = x_1), \dots, E(Y|X = x_J)$.

Now, in economics we often want to compare the mean of Y in some groups with respect to some reference group. Assume that your reference group are units with $\{X = x_1\}$. One has

$$\begin{aligned} E(Y|X) &= \sum_{j=1}^J 1\{X = x_j\} E(Y|X = x_j) \\ &= E(Y|X = x_1) + \sum_{j=2}^J 1\{X = x_j\} (E(Y|X = x_j) - E(Y|X = x_1)) = f(X)' \tilde{\beta}, \end{aligned}$$

where

$$\begin{aligned} f(X)' &= (1, 1\{X = x_2\}, \dots, 1\{X = x_J\}) \\ \tilde{\beta}' &= (E(Y|X = x_1), E(Y|X = x_2) - E(Y|X = x_1), \dots, E(Y|X = x_J) - E(Y|X = x_1)). \end{aligned}$$

Therefore, it follows from Theorem 3.3.1 that the coefficients of 1 (the constant), $1\{X = x_2\}, \dots, 1\{X = x_J\}$ in a linear regression of Y on $(1, 1\{X = x_2\}, \dots, 1\{X = x_J\})$ are respectively equal to $E(Y|X = x_1)$, $E(Y|X = x_2) - E(Y|X = x_1)$, $\dots$, $E(Y|X = x_J) - E(Y|X = x_1)$. The coefficient of the the constant is equal to the expectation of Y in the reference group, while the coefficients of the $1\{X = x_j\}$ indicators are equal to the difference between the expectation of Y in the subgroup with $X = x_j$ and in the reference group.

Assume X is a binary variable, and you regress Y on $(1, X)$. What is the value of the coefficient of 1 in that regression? What is the value of the coefficient of X ?

If X is a binary variable,

$$\begin{aligned} E(Y|X) &= 1\{X = 0\}E(Y|X = 0) + 1\{X = 1\}E(Y|X = 1) \\ &= E(Y|X = 0) + X(E(Y|X = 1) - E(Y|X = 0)) \\ &= f(X)'\tilde{\beta}, \end{aligned}$$

where $f(X)' = (1, X)$ and $\tilde{\beta}' = (E(Y|X = 0), E(Y|X = 1) - E(Y|X = 0))'$. Then, it follows from Theorem 3.3.1 that when you regress Y on a constant and a binary variable, the coefficient of the constant is equal to the expectation of Y conditional on $X = 0$, and the coefficient of X is equal to the difference between the expectation of Y conditional on $X = 1$ and the expectation of Y conditional on $X = 0$.

Assume $X = (X_1, X_2)$ where X_1 and X_2 are two binary variables, and you regress Y on $(1, X_1, X_2, X_1X_2)$. What is the value of the coefficient of 1 in that regression? What is the value of the coefficient of X_1 ? What is the value of the coefficient of X_2 ? What is the value of the coefficient of X_1X_2 ?

$$\begin{aligned} &E(Y|X_1, X_2) \\ &= 1\{X_1 = 0, X_2 = 0\}E(Y|X_1 = 0, X_2 = 0) + 1\{X_1 = 1, X_2 = 0\}E(Y|X_1 = 1, X_2 = 0) \\ &+ 1\{X_1 = 0, X_2 = 1\}E(Y|X_1 = 0, X_2 = 1) + 1\{X_1 = 1, X_2 = 1\}E(Y|X_1 = 1, X_2 = 1) \\ &= E(Y|X_1 = 0, X_2 = 0) \\ &+ X_1(E(Y|X_1 = 1, X_2 = 0) - E(Y|X_1 = 0, X_2 = 0)) + X_2(E(Y|X_1 = 0, X_2 = 1) - E(Y|X_1 = 0, X_2 = 0)) \\ &+ X_1X_2(E(Y|X_1 = 1, X_2 = 1) - E(Y|X_1 = 1, X_2 = 0) - (E(Y|X_1 = 0, X_2 = 1) - E(Y|X_1 = 0, X_2 = 0))) \\ &= f(X)'\tilde{\beta}, \end{aligned}$$

with $f(X)' = (1, X_1, X_2, X_1X_2)$, and

$$\begin{aligned} \tilde{\beta}' &= (E(Y|X_1 = 0, X_2 = 0), E(Y|X_1 = 1, X_2 = 0) - E(Y|X_1 = 0, X_2 = 0), E(Y|X_1 = 0, X_2 = 1) - E(Y|X_1 = 0, X_2 = 0), \\ &E(Y|X_1 = 1, X_2 = 1) - E(Y|X_1 = 1, X_2 = 0) - (E(Y|X_1 = 0, X_2 = 1) - E(Y|X_1 = 0, X_2 = 0))). \end{aligned}$$

For instance, if X_1 is a gender binary variable and X_2 is a race binary variable (black/non black), then the coefficient of the constant is the expected wage for non-black females, the coefficient of the gender binary variable is equal to the difference between the expected wage across non-black males and females, the coefficient of the race binary variable is equal to the difference between the expected wage across black and non-black females, and the coefficient of the product of the two dummies is equal to the expected wage difference across black and non-black males minus the same difference for females. This coefficient is called a difference-in-differences (DID).

Overall, Theorem 3.3.1 is important because it allows you to write the coefficients of **saturated** linear regressions (linear regressions that perfectly model the conditional expectation) as functions of values of the conditional expectation function.

Theorem 3.3.2 *The best linear approximation theorem.*

The regression function is the best linear approximation to the CEF:

$$\beta = \underset{b}{\operatorname{argmin}} E((E(Y|X) - X'b)^2). \quad (3.3.1)$$

Proof

$$\begin{aligned} (Y - X'b)^2 &= (Y - E(Y|X) + E(Y|X) - X'b)^2 \\ &= (Y - E(Y|X))^2 \\ &\quad + 2(Y - E(Y|X))(E(Y|X) - X'b) \\ &\quad + (E(Y|X) - X'b)^2 \end{aligned}$$

b does not enter the first term in the last expression. The expectation of the second term is equal to 0, following Theorem 3.1.1 ($Y - E(Y|X) = \varepsilon$, and $E(Y|X) - X'b$ is a function of X). Therefore, minimizing $E((Y - X'b)^2)$ or $E((E(Y|X) - X'b)^2)$ with respect to b will yield the same solution. This proves the result.

QED.

When X can take an infinite number of values, thus implying that one does not have $E(Y|X) = f(X)'\tilde{\beta}$ for some $K \times 1$ function of X $f(X)$ and for some $K \times 1$ vector of real numbers $\tilde{\beta}$, Theorem 3.3.2 still shows that the regression function is the best linear approximation to the CEF. For instance, a linear regression of wages on individuals height and weight will give you the best linear approximation to the conditional expectation of wages given height and weight. Even when your regressor takes a finite number of values, say years of education, you might prefer running the simple regression of wage on years of education rather than the complicated regression of wages on indicators for each year of education. The simple regression estimates the best linear approximation of the complicated regression. The coefficient of years of education in that simple regression is merely the slope of the line which fits best the relationship between years of education and the average of wages conditional on years of education. It therefore measures by how much the average of wages increases when education increases by one unit. This regression coefficient is a weighted average of $E(Wage|Education = x + 1) - E(Wage|Education = x)$ across all values of x .

3.4 Interpreting a multivariate regression coefficient

Theorem 3.4.1 *The Frisch-Waugh theorem.*

Let β_k denote the coefficient of X_k , the k th coordinate of X , in a regression of Y on X . Let $\tilde{X}_k$ denote the residual of the regression of X_k on all the other variables included in the regression.

$$\beta_k = \frac{\text{cov}(Y, \tilde{X}_k)}{V(\tilde{X}_k)}. \quad (3.4.1)$$

Proof

$$\begin{aligned} \text{cov}(Y, \tilde{X}_k) &= \text{cov}(X'\beta + e, \tilde{X}_k) \\ &= \text{cov}(X'\beta, \tilde{X}_k) \\ &= \text{cov}(X_k\beta_k, \tilde{X}_k) \\ &= \text{cov}(\tilde{X}_k\beta_k, \tilde{X}_k) \\ &= \beta_k V(\tilde{X}_k). \end{aligned}$$

The second equality follows from the fact that

$$\tilde{X}_k = X_k - \rho_1 X_1 - \dots - \rho_{k-1} X_{k-1} - \rho_{k+1} X_{k+1} - \dots - \rho_K X_K$$

where ρ denotes the $K-1 \times 1$ vector of regression coefficients of X_k on all the other explanatory variables. Therefore, $\text{cov}(e, \tilde{X}_k) = 0$ as $\tilde{X}_k$ is merely a linear combination of the explanatory variables in the regression of Y on X , and e is the residual of that regression.

The third equality follows from the fact that by construction $\tilde{X}_k$ is uncorrelated with

$$(X_1, \dots, X_{k-1}, X_{k+1}, \dots, X_K),$$

as shown in Lemma 3.2.1.

Where does the fourth equality come from?

QED.

In a simple univariate regression of Y on X_k , the coefficient of X_k is $\frac{\text{cov}(Y, X_k)}{V(X_k)}$. Frisch-Waugh shows that in a multivariate regression of Y on X , the coefficient of X_k is the same as the coefficient you would get from a univariate regression of Y on $\tilde{X}_k$, the residual from a regression of X_k on all the other regressors. $\tilde{X}_k$ is uncorrelated with all the other regressors. Therefore, the multivariate regression nets out the part of X_k which is correlated with the other regressors, and then measures how the remaining part, the part uncorrelated with the other regressors, is correlated with the outcome. Instead of using the raw correlation between the outcome and X_k , the multivariate regression uses only the part of that correlation which is specifically attributable to X_k , and not to the other regressors.

The Frisch-Waugh theorem motivates the interpretation of multivariate regression coefficients as measures of the correlation between Y and X_k , holding the other variables constant. The following theorem builds upon Frisch-Waugh to derive an even more interpretable formula for a certain class of multivariate regression coefficients.

Theorem 3.4.2 *Regression as a weighted average of conditional mean differences (Angrist 1998)*

Let D be a binary variable. Let X be a vector of random variables taking a finite number of values $\{x_1, \dots, x_J\}$. Assume that $0 < P(D = 1|X = x_j) < 1$ for every $1 \leq j \leq J$.² Let $\delta(x_j) = E(Y|D = 1, X = x_j) - E(Y|D = 0, X = x_j)$, and let $\delta(X) = E(Y|D = 1, X) - E(Y|D = 0, X)$.³ The coefficient β_D of D in a regression of Y on D and $(1, 1\{X = x_2\}, \dots, 1\{X = x_J\})$ is equal to

$$\sum_{j=1}^J \delta(x_j) w_{x_j}, \quad (3.4.2)$$

where

$$w_{x_j} = \frac{P(D = 1|X = x_j)(1 - P(D = 1|X = x_j))P(X = x_j)}{\sum_{j=1}^J P(D = 1|X = x_j)(1 - P(D = 1|X = x_j))P(X = x_j)}$$

are positive weights summing up to 1.

Proof: See PS 3.

In a univariate regression of Y on a constant and D , the coefficient of D would be equal to $E(Y|D = 1) - E(Y|D = 0)$, the comparison of the expected value of Y conditional on $D = 1$ and conditional on $D = 0$. Theorem 3.4.2 shows that in a multivariate regression of Y on D and X , the coefficient of D is equal to a weighted average of the $E(Y|D = 1, X = x_j) - E(Y|D = 0, X = x_j)$ terms, which compare the expected value of Y conditional on $D = 1$ and conditional on $D = 0$, **within units with $X = x_j$, i.e. holding X constant**. Therefore β_D measures by how much the mean of Y increases when D increases by one unit, holding the other variables in the regression fixed. The weights w_{x_j} are not very intuitive, we will discuss them in more details later.

Let's see in a simple example how a multivariate regression coefficient is computed. Assume you regress wages on a constant and two binary variable variables, gender (1 for males, 0 for females) and race (1 for black people, 0 for non-black), respectively denoted by D and X . The following table presents the CEF of wages in the four subgroups defined by those two variables.

Table 3.1: From the CEF to multivariate regression coefficients

	Females	Males
Non-Black	$E(Y D = 0, X = 0) = 2000$	$E(Y D = 1, X = 0) = 2500$
Black	$E(Y X = 0, X = 1) = 1400$	$E(Y D = 1, X = 1) = 1600$

²This is called the support condition. It ensures that the conditional expectations in $\delta(x_j)$ are well defined.

³It follows from those definitions that $\delta(X)$ is a random variable equal to $\delta(x_j)$ when $X = x_j$.

Use Theorem 3.4.2 and the Table above to give a formula for β_D , the coefficient of D in a regression of Y on a constant, D , and X as a function of some weights w_0 and w_1 .

It follows from Theorem 3.4.2 that

$$\beta_D = w_0 \times (2500 - 2000) + w_1 \times (1600 - 1400).$$

2500 – 2000 and 1600 – 1400 respectively correspond to $\delta(0)$ and $\delta(1)$ in Theorem 3.4.2. We cannot compute the weights w_0 and w_1 from the table. We will discuss the interpretation of these weights later.

Finally, the following result shows how the coefficient of a variable changes from a univariate regression to a multivariate regression where more variables are added to the regression.

Theorem 3.4.3 *From short to long regressions*

Let β_s denote the coefficient of a variable D in a regression of Y on a constant and D . Let β_ℓ denote the coefficient of D in a regression of Y on a constant, D , and another variable X . Let β_X denote the coefficient of X in that second regression, and let γ be the coefficient of D in a regression of X on a constant and D .

$$\beta_s = \beta_\ell + \gamma\beta_X.$$

Proof

$$\begin{aligned}\beta_s &= \frac{\text{cov}(Y, D)}{V(D)} \\ &= \frac{\text{cov}(\alpha + \beta_\ell D + \beta_X X + u, D)}{V(D)} \\ &= \beta_\ell + \beta_X \frac{\text{cov}(X, D)}{V(D)} \\ &= \beta_\ell + \beta_X \gamma.\end{aligned}$$

Second equality: regression decomposition of Y wrt to D and X . Third equality: u uncorrelated with X by construction, and algebra.

QED.

When a regression coefficient decreases when more variables are added to the regression, that means $\beta_X \gamma > 0$, so β_X and γ must be of the same sign.

Chapter 4

The OLS regression estimator

In this chapter, we assume we are going to observe n mutually independent random vectors (Y_i, X_i) , with the same probability distribution as (Y, X) . This modelling framework for instance applies if we are going to randomly draw with replacement n individuals from the US phonebook and ask them their wages, their years of education, etc. If sampling is without replacement, this modelling framework approximately applies, provided n is small relative to the size of the population. The goal of this chapter is to estimate, and draw inference on,

$$\beta = E(XX')^{-1}E(XY).$$

Before we do so, just note that $e_i = Y_i - X_i'\beta$ are iid and have the same distribution as $e = Y - X'\beta$. This follows from the fact that (Y_i, X_i) are iid with the same distribution as (Y, X) .

4.1 Estimating β .

Use the method of moments to propose an estimator of β .

To estimate β , we are going to use the following estimator

$$\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \frac{1}{n} \sum_{i=1}^n X_i Y_i. \quad (4.1.1)$$

$\hat{\beta}$ is a $K \times 1$ vector. Its coordinates are denoted $\hat{\beta}_k$. It is an estimator from the method of moments, because it amounts to replacing the unknown expectations in β by sample means. $\hat{Y}_i = X_i' \hat{\beta}$ is the predicted value for Y_i as per our regression, and $\hat{e}_i = Y_i - X_i' \hat{\beta}$ is the estimated residual.

Now, call X the $N \times K$ matrix whose i^{th} line is equal to X'_i , and call Y the $N \times 1$ vector whose i^{th} element is Y_i . One can check that $\sum_{i=1}^n X_i X'_i = X'X$ and $\sum_{i=1}^n X_i Y_i = X'Y$. Using this, and the fact that we can forget about the $\frac{1}{n}$ in (4.1.1) as they appear both in the numerator and in the denominator, we get a second formula for $\hat{\beta}$:

$$\hat{\beta} = (X'X)^{-1} X'Y. \quad (4.1.2)$$

This is probably the most famous formula in econometrics, and you will find it in most textbooks. With respect to (4.1.1), its main advantage is that it is very compact because we are stacking our X_i vectors into a matrix. Its main drawback is that in (4.1.2) it is no longer immediately visible that $\hat{\beta}$ is a simple function of sample moments whose asymptotic behavior is easy to study. I personally find (4.1.1) easier to work with than (4.1.2).

When we regress Y_i on a constant and a variable X_{i2} , the formula for $\hat{\beta}$ is much simpler:

$$\begin{aligned} \hat{\beta}_2 &= \frac{cov_e(Y_i, X_{i2})}{V_e(X_{i2})} \\ \hat{\beta}_1 &= \bar{Y}_i - \hat{\beta}_2 \bar{X}_{i2}, \end{aligned}$$

where $cov_e(Y_i, X_{i2}) = \frac{1}{n} \sum_{i=1}^n Y_i X_{i2} - (\frac{1}{n} \sum_{i=1}^n Y_i) (\frac{1}{n} \sum_{i=1}^n X_{i2})$ is the sample covariance of Y_i and X_{i2} , and $V_e(X_{i2}) = \frac{1}{n} \sum_{i=1}^n X_{i2}^2 - (\frac{1}{n} \sum_{i=1}^n X_{i2})^2$ is the sample variance of X_{i2} .

An alternative estimator of β could be

$$\hat{\beta}^* = \underset{b}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n ((Y_i - X'_i b)^2), \quad (4.1.3)$$

the argmin of the sample version of the population-minimization problem defining β . The FOC of this minimization problem are:

$$\sum_{i=1}^n X_i (Y_i - X'_i b) = 0, \quad (4.1.4)$$

which finally yields

$$\hat{\beta}^* = \left(\frac{1}{n} \sum_{i=1}^n X_i X'_i \right)^{-1} \frac{1}{n} \sum_{i=1}^n X_i Y_i = \hat{\beta}.$$

This shows that $\hat{\beta}^*$ and $\hat{\beta}$ are the same: substituting sample mean to expectations in the minimization problem defining β is the same as operating this substitution directly in the closed form expression for β .

FOC displayed in (3.2.3) are the only thing we used to derive the Frisch-Waugh theorem. $\hat{\beta}$ verifies (4.1.4), the same FOC but for the sample minimization problem. One can therefore prove the following lemma and theorem, which are the sample counterparts of Lemma 3.2.1 and Theorem 3.4.1.

Lemma 4.1.1 *The estimated residuals of a regression are uncorrelated in the sample with the explanatory variables.*

$$cov_e(\hat{e}_i, X_{ik}) = 0$$

Theorem 4.1.1 *The sample Frisch-Waugh theorem.*

Let $\hat{X}_{ik}$ denote the **estimated** residuals from a regression of X_{ik} on all the other variables included in the regression. Then,

$$\hat{\beta}_k = \frac{\text{cov}_e(Y_i, \hat{X}_{ik})}{V_e(\hat{X}_{ik})}.$$

Theorem 4.1.2 *The sample “regression as a weighted average of conditional mean differences” theorem.*

Let D be a binary variable. Let X be a vector of random variables taking a finite number of values $\{x_1, \dots, x_J\}$. Assume that $0 < \hat{P}(D = 1|X = x_j) < 1$ for every $1 \leq j \leq J$, and let $\hat{\delta}(x_j) = \hat{E}(Y|D = 1, X = x_j) - \hat{E}(Y|D = 0, X = x_j)$. The coefficient $\hat{\beta}_D$ of D_i in a regression of Y_i on D_i and $(1, 1\{X_i = x_2\}, \dots, 1\{X_i = x_J\})$ is equal to

$$\sum_{j=1}^J \hat{\delta}(x_j) \hat{w}_{x_j}, \quad (4.1.5)$$

where

$$\hat{w}_{x_j} = \frac{\hat{P}(D = 1|X = x_j)(1 - \hat{P}(D = 1|X = x_j))\hat{P}(X = x_j)}{\sum_{j=1}^J \hat{P}(D = 1|X = x_j)(1 - \hat{P}(D = 1|X = x_j))\hat{P}(X = x_j)}$$

are positive weights summing up to 1.

4.2 Consistency and $\sqrt{n}$ -consistency of $\hat{\beta}$

Theorem 4.2.1 *Consistency of $\hat{\beta}$.*

$\hat{\beta}$ is a consistent sequence of estimators of β .

Proof

Let g be a function taking as arguments a $K \times K$ invertible matrix A and a $K \times 1$ vector B and such that $g(A, B) = A^{-1}B$. This function is continuous, as matrix inversion and the multiplication of a matrix and a vector are continuous applications. Then, note that

$$\hat{\beta} = g\left(\frac{1}{n} \sum_{i=1}^n X_i X_i', \frac{1}{n} \sum_{i=1}^n X_i Y_i\right)$$

and

$$\beta = g(E(XX'), E(XY))$$

The law of large numbers implies that

$$\left(\frac{1}{n} \sum_{i=1}^n X_i X_i', \frac{1}{n} \sum_{i=1}^n X_i Y_i\right) \xrightarrow{P} (E(XX'), E(XY))$$

Therefore, the continuous mapping theorem implies that

$$\widehat{\beta} = g\left(\frac{1}{n} \sum_{i=1}^n X_i X_i', \frac{1}{n} \sum_{i=1}^n X_i Y_i\right) \xrightarrow{P} g(E(X X'), E(X Y)) = \beta.$$

This proves the result.

QED.

Before proving asymptotic normality of $\widehat{\beta}$, I have to state a result about normal vectors.

Theorem 4.2.2 *The product of a deterministic matrix and a normally distributed vector is a normally distributed vector*

If $U \sim N(0, V(U))$ is a $k \times 1$ normal vector and B is a $k \times k$ deterministic matrix, then $BU \sim N(0, BV(U)B')$.

Proof

That BU is normally distributed follows from the definition of normal vectors. Indeed, a vector is normally distributed if and only if every linear combination of its coordinates follows a normal distribution. The coordinates of BU are linear combinations of that of U . Therefore, linear combinations of the coordinates of BU are also linear combinations of that of U which are normally distributed because U is a normal vector.

Then, it follows from the linearity of the expectation operator that $E(BU) = BE(U) = 0$.

Finally,

$$\begin{aligned} V(BU) &= E((BU - E(BU))(BU - E(BU))') \\ &= E(BU(BU)') \\ &= E(BUU' B') \\ &= BE(UU')B' \\ &= BV(U)B'. \end{aligned}$$

QED.

Theorem 4.2.3 *Asymptotic normality of $\widehat{\beta}$.*

$\widehat{\beta}$ is an asymptotically normal sequence of estimators of β :

$$\sqrt{n}(\widehat{\beta} - \beta) \xrightarrow{d} N(0, E(X X')^{-1} E(X X' e^2) E(X X')^{-1}).$$

Proof

Write $Y_i = X_i' \beta + e_i$. Substitute this into (4.1.1). This yields

$$\begin{aligned} \widehat{\beta} &= \left(\frac{1}{n} \sum_{i=1}^n X_i X_i'\right)^{-1} \frac{1}{n} \sum_{i=1}^n X_i (X_i' \beta + e_i). \\ &= \beta + \left(\frac{1}{n} \sum_{i=1}^n X_i X_i'\right)^{-1} \frac{1}{n} \sum_{i=1}^n X_i e_i. \end{aligned}$$

Therefore,

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i e_i \right). \quad (4.2.1)$$

As $\left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \xrightarrow{P} E(XX')^{-1}$ (why?), the Slutsky lemma implies that

$$\left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i e_i \right) \xrightarrow{d} E(XX')^{-1} \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i e_i \right). \quad (4.2.2)$$

Since $E(Xe) = 0$ (why?), the central limit theorem implies that

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i e_i \right) \xrightarrow{d} N(0, V(Xe)). \quad (4.2.3)$$

Because $E(Xe) = 0$,

$$V(Xe) = E((Xe - E(Xe))(Xe - E(Xe))') = E(XX'e^2). \quad (4.2.4)$$

Combining Equations (4.2.1), (4.2.2), (4.2.3), (4.2.4) and Theorem 4.2.2 yields the result (once noted that $(E(XX')^{-1})' = E(XX')^{-1}$).

QED.

4.3 Consistent estimation of the asymptotic variance of $\hat{\beta}$...

4.3.1 ...under no assumption

To be able to construct confidence intervals and hypothesis tests for the regression coefficients, we will need a consistent estimator of $E(XX')^{-1}E(XX'e^2)E(XX')^{-1}$.

Propose an estimator of $E(XX')^{-1}E(XX'e^2)E(XX')^{-1}$.

Can you use $\left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' e_i^2 \right) \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1}$?

One can use $\left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' e_i^2 \right) \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1}$. This estimator proves to be consistent, as shown in the next Theorem.

Theorem 4.3.1 *Consistent estimation of $E(XX')^{-1}E(XX'e^2)E(XX')^{-1}$.*

Let $\hat{e}_i$ denote the estimated residual from the regression: $Y_i - X_i'\hat{\beta}$. Then,

$$\left(\frac{1}{n}\sum_{i=1}^n X_i X_i'\right)^{-1} \left(\frac{1}{n}\sum_{i=1}^n X_i X_i' \hat{e}_i^2\right) \left(\frac{1}{n}\sum_{i=1}^n X_i X_i'\right)^{-1} \xrightarrow{P} E(XX')^{-1}E(XX'e^2)E(XX')^{-1}.$$

Proof

As $\frac{1}{n}\sum_{i=1}^n X_i X_i' \xrightarrow{P} E(XX')$, it follows from the continuous mapping theorem that proving the result amounts to proving that $\frac{1}{n}\sum_{i=1}^n X_i X_i' \hat{e}_i^2 \xrightarrow{P} E(XX'e^2)$. Proving this is slightly more involved than just invoking the law of large numbers, because $(X_i X_i' \hat{e}_i^2)$ is not iid (why?).

$$\frac{1}{n}\sum_{i=1}^n X_i X_i' \hat{e}_i^2 = \frac{1}{n}\sum_{i=1}^n X_i X_i' Y_i^2 - 2\frac{1}{n}\sum_{i=1}^n X_i X_i' X_i' \hat{\beta} Y_i + \frac{1}{n}\sum_{i=1}^n X_i X_i' X_i' \hat{\beta} \hat{\beta}' X_i. \quad (4.3.1)$$

Similarly,

$$E(XX'e^2) = E(XX'Y^2) - 2E(XX'X'\beta Y) + E(XX'X'\beta\beta'X) \quad (4.3.2)$$

To show that (4.3.1) converges in probability towards (4.3.2), we will show that each sample mean in (4.3.1) converges to the corresponding expectation in (4.3.2).

As $X_i X_i' Y_i^2$ are iid matrixes of random variables with an expectation, it follows from the law of large numbers that the first summation in (4.3.1) converges to the first expectation in (4.3.2).

Let's show that the third summation in (4.3.1) converges to the third expectation in (4.3.2). Consider the element on the j^{th} line and k^{th} column of $\frac{1}{n}\sum_{i=1}^n X_i X_i' X_i' \hat{\beta} \hat{\beta}' X_i$. It is equal to

$$\begin{aligned} \frac{1}{n}\sum_{i=1}^n X_{ik} X_{ij} X_i' \hat{\beta} \hat{\beta}' X_i &= \frac{1}{n}\sum_{i=1}^n \hat{\beta}' X_i X_{ik} X_{ij} X_i' \hat{\beta} \\ &= \hat{\beta}' \left(\frac{1}{n}\sum_{i=1}^n X_i X_{ik} X_{ij} X_i' \right) \hat{\beta}. \end{aligned} \quad (4.3.3)$$

The first equality follows from the fact that $\hat{\beta}' X_i$ is a real number, and a real number is such that $A\lambda = \lambda A$ for any matrix A . The second equality follows from the fact that matrix product is distributive over addition: $ABD + ACD = A(B+C)D$. As $X_i X_{ik} X_{ij} X_i'$ are iid random matrixes with an expectation, it follows from the law of large numbers that $\frac{1}{n}\sum_{i=1}^n X_i X_{ik} X_{ij} X_i' \xrightarrow{P} E(XX_k X_j X')$. Then, as $\hat{\beta} \xrightarrow{P} \beta$, it follows from the continuous mapping theorem that

$$\hat{\beta}' \left(\frac{1}{n}\sum_{i=1}^n X_i X_{ik} X_{ij} X_i' \right) \hat{\beta} \xrightarrow{P} \beta' E(XX_k X_j X') \beta.$$

Finally,

$$\beta' E(XX_k X_j X') \beta = E(\beta' X X_k X_j X' \beta) = E(X_k X_j X' \beta \beta' X).$$

The first equality follows from the linearity of the expectation operator (you can check that $\lambda'E(W) = E(\lambda'W)$ for a 2×1 real vector λ and a 2×2 random vector W). The second one

follows from the fact $\beta'X$ is a real number. The right hand side of the previous display is the element on the j^{th} line and k^{th} column of $E(XX'X'\beta\beta'X)$. This proves that the third summation in (4.3.1) converges to the third expectation in (4.3.2).

Using arguments very similar to those used in the previous paragraph, one can show that the second summation in (4.3.1) converges to the second expectation in (4.3.2). This is left as an exercise.

Combining the three pieces together we finally get the result.

QED.

Theorem 4.3.1 shows that

$$\left(\frac{1}{n} \sum_{i=1}^n X_i X_i'\right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \hat{e}_i^2\right) \left(\frac{1}{n} \sum_{i=1}^n X_i X_i'\right)^{-1}$$

is a consistent estimator of the asymptotic variance of $\hat{\beta}$. The square root of the diagonal terms of this matrix can be used as estimates of the standard errors of the regression coefficients $\hat{\beta}_k$. Those standard errors are known as White (1980) standard errors, or as Eicker-Huber-White (EHW) standard errors in recognition of the fact that Eicker (1967) and Huber (1967) have derived similar results to White's.

4.3.2 ...under an homoscedasticity assumption.

It is often assumed that the residual from the regression is homoscedastic:

$$E(e^2|X) = E(e^2). \quad (4.3.4)$$

This assumption means that the variance of the residual from the regression does not vary with X_i . To realize what this means, try to draw a (X_i, Y_i) scatter plot in which homoscedasticity is verified, and one in which it is violated.

Under this assumption, the formula of the asymptotic variance of $\hat{\beta}$ gets much simpler. Indeed, we then have that

$$\begin{aligned} E(XX')^{-1}E(XX'e^2)E(XX')^{-1} &= E(XX')^{-1}E(XX'E(e^2|X))E(XX')^{-1} \\ &= E(e^2)E(XX')^{-1}E(XX')E(XX')^{-1} \\ &= E(e^2)E(XX')^{-1}. \end{aligned}$$

The following theorem gives a consistent estimator of $E(e^2)E(XX')^{-1}$. Remember that this is equal to the asymptotic variance of $\hat{\beta}$ only under the homoscedasticity assumption.

Theorem 4.3.2 *Consistent estimation of $E(e^2)E(XX')^{-1}$.*

$$\frac{1}{n} \sum_{i=1}^n \hat{e}_i^2 \left(\frac{1}{n} \sum_{i=1}^n X_i X_i'\right)^{-1} \xrightarrow{P} E(e^2)E(XX')^{-1}.$$

4.3.3 Homoscedasticity or heteroscedasticity?

$\frac{1}{n} \sum_{i=1}^n \hat{e}_i^2 \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1}$ is a valid estimator of the asymptotic variance of $\hat{\beta}$ only under homoscedasticity. On the contrary, EHW standard errors do not rely on that assumption, which is the reason why they are also called robust standard errors: they are robust to deviations from homoscedasticity, which are called heteroscedasticity.

In many applications, homoscedasticity is not a very credible assumption. It is likely to be violated whenever the conditional expectation function is not linear. Indeed, in such instances,

$$\begin{aligned} E(e^2|X) &= E((Y - X'\beta)^2|X) \\ &= E((Y - E(Y|X) + E(Y|X) - X'\beta)^2|X) \\ &= V(Y|X) + (E(Y|X) - X'\beta)^2. \end{aligned} \tag{4.3.5}$$

Why is the third equality correct? Even if $V(Y|X)$ is constant, there will be variation in $(E(Y|X) - X'\beta)^2$ whenever the CEF is not linear.

Even if the CEF is linear, it might still be the case that homoscedasticity is violated. An important example is when the outcome variable is binary. A linear regression with a binary dependent variable is called a linear probability model (LPM), because in such instances $E(Y|X) = P(Y = 1|X)$. In such instances, equation (4.3.5) rewrites

$$E(e^2|X) = V(Y|X) = P(Y = 1|X)(1 - P(Y = 1|X)).$$

This cannot be constant across all values of X (unless X induces no variation in Y , making it an uninteresting regressor). Therefore, LPMs are always heteroscedastic.

However, the default standard errors implemented in most standard statistical softwares are those obtained under homoscedasticity. In Stata, to get EHW standard errors, you have to write “robust” in the end of your regression command. Therefore, my recommendation would be to add the “robust” option whenever you run a regression. However, if you forget, this should matter little, as those two standard errors are usually fairly close (on this, see page 47 of Angrist & Pischke (2008)).

The only circumstance in which you might prefer to use non robust standard errors is when you have a small sample (say less than 200 observations). Indeed, the estimator of robust standard errors we derived in 4.3.1 is biased in small samples, so if the homoscedasticity assumption is not too far from being satisfied, you might be better off using the unbiased estimator of slightly incorrect standard errors than using a biased estimator of the correct standard errors.

Even though the homoscedasticity assumption is often not perfectly realistic, it is helpful to simplify formulae of asymptotic variances in order to perform comparative statics exercises.

4.4 Confidence intervals for β

Let $\hat{\sigma}_k$ denote the square-root of the k^{th} diagonal term of a consistent estimator of the asymptotic variance of $\hat{\beta}$.¹

Theorem 4.4.1 *Confidence intervals for the regression coefficients*

$$IC(\alpha) = \left[\hat{\beta}_k - q_{1-\frac{\alpha}{2}} \frac{\hat{\sigma}_k}{\sqrt{n}}, \hat{\beta}_k + q_{1-\frac{\alpha}{2}} \frac{\hat{\sigma}_k}{\sqrt{n}} \right]$$

is a confidence interval for β_k with asymptotic coverage equal to $1 - \alpha$.

Proof

As shown in Theorems 4.3.1 and 4.3.2, $\hat{\sigma}_k$ is a consistent estimator of the square root of the k^{th} diagonal element of the asymptotic variance of $\hat{\beta}$. Therefore, it follows from Theorem 4.2.3 and from the Slutsky lemma that

$$\sqrt{n} \left(\frac{\hat{\beta}_k - \beta_k}{\hat{\sigma}_k} \right) \xrightarrow{d} N(0, 1). \quad (4.4.1)$$

Then,

$$\begin{aligned} \hat{\beta}_k - q_{1-\frac{\alpha}{2}} \frac{\hat{\sigma}_k}{\sqrt{n}} \leq \beta_k \leq \hat{\beta}_k + q_{1-\frac{\alpha}{2}} \frac{\hat{\sigma}_k}{\sqrt{n}} \\ \Leftrightarrow -q_{1-\frac{\alpha}{2}} \leq \sqrt{n} \left(\frac{\hat{\beta}_k - \beta_k}{\hat{\sigma}_k} \right) \leq q_{1-\frac{\alpha}{2}}. \end{aligned}$$

Therefore,

$$P \left(\beta_k \in \left[\hat{\beta}_k - q_{1-\frac{\alpha}{2}} \frac{\hat{\sigma}_k}{\sqrt{n}}, \hat{\beta}_k + q_{1-\frac{\alpha}{2}} \frac{\hat{\sigma}_k}{\sqrt{n}} \right] \right) = P \left(\sqrt{n} \left(\frac{\hat{\beta}_k - \beta_k}{\hat{\sigma}_k} \right) \in \left[-q_{1-\frac{\alpha}{2}}, q_{1-\frac{\alpha}{2}} \right] \right)$$

It follows from (4.4.1) that the limit of the right hand side of the last equation is equal to $P \left(N(0, 1) \in \left[-q_{1-\frac{\alpha}{2}}, q_{1-\frac{\alpha}{2}} \right] \right) = 1 - \frac{\alpha}{2} - \frac{\alpha}{2} = 1 - \alpha$.

QED.

4.5 Hypothesis tests

In this context, a typical set of hypothesis we would like to test is $H_0 : \beta_k = 0$ against $H_1 : \beta_k \neq 0$. On that purpose we consider the following test.

Definition 4.5.1 *T-test for a regression coefficient*

For any $\alpha \in (0, 1)$, let

$$T_{nk}(\alpha) = 1 \left\{ \left| \frac{\hat{\beta}_k}{\frac{\hat{\sigma}_k}{\sqrt{n}}} \right| > q_{1-\frac{\alpha}{2}} \right\}.$$

$T_{nk}(\alpha)$ is called the *t-test for a regression coefficient*.

¹If you do not assume homoscedasticity, this estimator is the one introduced in Theorem 4.3.1. If you work under homoscedasticity, this is the one of Theorem 4.3.2.

In research papers, people usually show $\hat{\beta}_k$, the regression coefficient they are interested in, and $\frac{\hat{\sigma}_k}{\sqrt{n}}$ below it between parentheses. So to assess whether $\hat{\beta}_k$ is significantly different from 0 or not, you should divide it by $\frac{\hat{\sigma}_k}{\sqrt{n}}$. This will give you the t-statistic of that coefficient. If this number is greater than 1.96 or lower than -1.96 , you can assert with 95% confidence that the regression coefficient is different from 0.

Chapter 5

Regression and causality

This chapter borrows from Angrist & Pischke (2008).

The regression function is a powerful tool to make predictions on Y based on X . Banks or insurance companies might be interested in making predictions, e.g. on the probability that a customer will default on his loan, or have a car accident, conditional on the variables contained in his application form. But our goal is to study statistical tools which can help us infer causal relationships between variables. Even though this is not obvious from its definition, regression is an useful tool for our purposes, because under a conditional independence assumption (CIA), regression coefficients can receive a causal interpretation. Moreover, regression can help us measuring treatment effects more precisely in randomized experiments.

5.1 Causal inference under the CIA assumption

5.1.1 Conditional independence

Conditional independence generalizes the notion of independence to conditional probabilities.

Definition 5.1.1 *Independence*

Two random variables X and Y are independent ($X \perp\!\!\!\perp Y$) if and only if for every function such that the following expectations are well defined, $E(f(X)g(Y)) = E(f(X))E(g(Y))$. When the random variables are discrete, X and Y are independent if and only if $P(X = x, Y = y) = P(X = x)P(Y = y)$ for every $(x, y) \in \mathcal{S}(X, Y)$.

You will show in an exercise that $X \perp\!\!\!\perp Y$ implies $E(Y|X) = E(Y)$.

Definition 5.1.2 *Conditional independence*

X and Y are independent conditional on Z ($X \perp\!\!\!\perp Y|Z$) if and only if for every function such that the following conditional expectations are well defined, $E(f(X)g(Y)|Z) = E(f(X)|Z)E(g(Y)|Z)$. When X , Y , and Z are discrete, X and Y are independent condi-

tional on Z if and only if $P(X = x, Y = y|Z = z) = P(X = x|Z = z)P(Y = y|Z = z)$ for every $(x, y, z) \in \mathcal{S}(X, Y, Z)$.

One can show that $X \perp\!\!\!\perp Y|Z$ implies that $E(Y|X, Z) = E(Y|Z)$.

Theorem 5.1.3 *Joint independence implies conditional independence*

If $X \perp\!\!\!\perp (Y, Z)$, then $X \perp\!\!\!\perp Y|Z$.

Proof

I prove the result for discrete random variables only. Assume $X \perp\!\!\!\perp (Y, Z)$.

$$\begin{aligned}
P(X = x, Y = y|Z = z) &= \frac{P(X = x, Y = y, Z = z)}{P(Z = z)} \\
&= \frac{P(X = x)P(Y = y, Z = z)}{P(Z = z)} \\
&= \frac{P(X = x)P(Z = z)P(Y = y, Z = z)}{P(Z = z)P(Z = z)} \\
&= \frac{P(X = x, Z = z)P(Y = y, Z = z)}{P(Z = z)P(Z = z)} \\
&= P(X = x|Z = z)P(Y = y|Z = z).
\end{aligned}$$

QED.

5.1.2 Our running example in this chapter: returns to schooling

Now, let's get back to a causal question, similar to those we introduced in Chapter 2: the effect of years of schooling on wage. Assume we are going to draw randomly one individual from a population. Let $Y(d)$ denote the potential wage that this individual will obtain if they complete d years of schooling. $Y(d) - Y(d - 1)$ is the effect of completing d instead of $d - 1$ years of schooling on the wage of that individual. Because we never get to observe jointly $Y(d)$ and $Y(d - 1)$, the parameters we are going to focus on are $ATE_d = E(Y(d) - Y(d - 1))$ for any d between 1 and 20 (we assume that nobody can get more than 20 years of education). ATE_d is the expectation of $Y(d) - Y(d - 1)$. As the randomness arises from the fact we randomly draw one individual from a population, it follows from Theorem 1.1.1 that ATE_d is the average of changes in earnings that individuals in the population experience if they go from $d - 1$ to d years of schooling.

The actual number of years of schooling of the individual we will draw is denoted D and their actual wage is denoted Y . When we will draw that individual, we will get to observe their schooling and wage (D, Y) , but not their potential wages. However, the following equation relates their observed wage, their schooling, and their potential wages: $Y = \sum_{d=0}^{20} 1\{D = d\}Y(d)$. Finally, let's assume we will also get to observe X , a vector of covariates (IQ at age 3, parental background...), for that individual.

Throughout this section, we will maintain the following assumption.

Assumption 4 *The conditional independence assumption (CIA)*

$$D \perp\!\!\!\perp (Y(0), \dots, Y(20)) | X. \quad (5.1.1)$$

In a randomized experiment, we had that by design D was independent of individuals' potential outcomes. Here, schooling is not going to be independent of individuals' potential outcomes by design, because we cannot randomly select a group of individuals who are obligated not to receive any year of education, another group of individual who are obligated to follow one year of education, ..., and a last group of individuals who are obligated to complete a PhD.

We could **assume** that the schooling of the individual we will draw is independent of their potential wages, but this not a very plausible assumption. For instance, it is fairly plausible to assume that $E(Y(0)|D = 16) > E(Y(0)|D = 12)$, which means that the average wage without any schooling is higher among individuals who complete college than among individuals who only complete high school. High-IQ individuals are probably more prone to going to college, as studying is less costly for them (effort wise), and they are also more likely to have a higher $Y(0)$: even without any education they could do well on the labor market. This would imply that $Y(0)$ and D are not independent.

The CIA does not require independence, but conditional independence only. It requires that individuals with the same X (say IQ at age 3, parental background...) but with different educational attainments have the same distribution of their potential outcomes. Putting it in other words, the reason which leads two individuals with the same X to make different educational choices should not be related with their potential wages. The richer the set of covariates, the more credible the CIA assumption. If the set of covariates X includes a large number of variables, 2 individuals with the same X are similar on a large number of dimensions, which makes it more likely that despite the fact they made different educational decisions they still have the same potential wages.

5.1.3 Identification under CIA and the linear and constant effect model

As we will see later, CIA alone is sufficient for regression to identify a causal parameter. However, I now introduce a supplementary assumption which will help us understand what a regression identifies when CIA is violated.

Assumption 5 *Linear and constant effect model (LCEM)*

$$Y(d) = \alpha + \rho d + U, \quad (5.1.2)$$

where α and ρ are two real numbers, and U is a random variable with an expectation equal to 0.

Assumption 5 can actually be decomposed into two assumptions. The first one is that for every unit in the population, potential outcomes are an affine function of schooling, though

the slope of this affine function could vary between individuals:

$$Y(d) = \alpha + Rd + U.$$

where α is a real number and R is a random variable representing returns to education for the individual we draw, and U is a random variable with an expectation equal to 0.

The second one is that $R = \rho$, a real number. This means that the effect of schooling is constant for all individuals in the population. Therefore irrespective of which individual we draw, that individual will always have $R = \rho$.

Equation (5.1.2) is called the “linear and constant treatment effect model”.

Do you think the linear and constant treatment effect model is plausible in the context of returns to education?

The two assumptions of the linear and constant treatment effect model are likely to be violated in the case of returns to schooling. For instance, graduating from college has probably a greater impact on your wage than going from 2nd to 3rd grade, which would violate linearity. Similarly, the effect of schooling is probably heterogeneous across individuals. One could argue for instance that high IQ individuals have a lower effect, as they would do well on the labor market anyway, even without schooling. We will relax those assumptions later.

Let’s interpret what α represents. Under Assumption 5, we have $Y(0) = \alpha + U$. Given that we have assumed $E(U) = 0$,¹ $\alpha = E(Y(0))$. α is the average wage that individuals in the population would make without any education.

Let’s interpret what U represents. Given that $Y(0) = \alpha + U$ and $\alpha = E(Y(0))$, $U = Y(0) - E(Y(0))$ represents the difference between the wage that the individual we will draw would get without any schooling and the average wage without schooling in the population. U is therefore referred to as a measure of innate ability. If U is large, this means that the individual we drew has a much higher level of innate ability than the average person in the population: if they do not receive any schooling, they make much more money than what the average person would make without any schooling.

Under the LCEM assumption, we have

$$Y = \sum_{d=0}^{20} 1\{D = d\}Y(d) = \alpha + \rho D + U. \quad (5.1.3)$$

¹Assuming that $E(U) = 0$ is without loss of generality. Assuming $E(U) \neq 0$ would not change the results below, but that would change the interpretation of α .

This equality is important. To check that it is true, you just need to check that it holds for every possible value of D . Moreover, under LCEM we also have

$$ATE_d = E(Y(d) - Y(d-1)) = E(\alpha + \rho d + U - \alpha - \rho(d-1) - U) = \rho.$$

Therefore, we no longer have 20 different parameters of interest, the ATE_d for $1 \leq d \leq 20$. We have only one parameter of interest, ρ , the return to one year of schooling, which is assumed to be constant both across years of schooling and across individuals.

Theorem 5.1.4 *Causal interpretation of regression under LCEM and CIA*

Under Assumptions 4 and 5, and if $E(U|X) = X'\lambda$, β_D , the coefficient of D in a regression of Y on D and X is equal to ρ .

Proof

Write the CEF decomposition of U with respect to X : $U = E(U|X) + \varepsilon$, where $E(\varepsilon|X) = 0$. As by assumption $E(U|X) = X'\lambda$, we have

$$U = X'\lambda + \varepsilon. \tag{5.1.4}$$

It follows from Theorem 3.1.1 that by construction

$$\text{cov}(X, \varepsilon) = 0. \tag{5.1.5}$$

Under LCEM, the CIA assumption implies that $D \perp\!\!\!\perp \alpha + \rho d + U|X$. This is equivalent to $D \perp\!\!\!\perp U|X$, because only U is random in $\alpha + \rho d + U$. Using Equation (5.1.4), $D \perp\!\!\!\perp U|X$ is equivalent to $D \perp\!\!\!\perp X'\lambda + \varepsilon|X$. Conditional on X , $X'\lambda$ is a constant, so only ε is random in $X'\lambda + \varepsilon$, so $D \perp\!\!\!\perp X'\lambda + \varepsilon|X$ is equivalent to

$$D \perp\!\!\!\perp \varepsilon|X. \tag{5.1.6}$$

This implies that

$$\text{cov}(D, \varepsilon) = E(D\varepsilon) = E(DE(\varepsilon|D, X)) = E(DE(\varepsilon|X)) = 0, \tag{5.1.7}$$

where the last but one equality follows from Equation (5.1.6), while the last equality follows from Theorem 3.1.1.

Let $X'\theta$ denote the regression function of D on X . Let also $D = X'\theta + \tilde{D}$ denote the regression

decomposition of D with respect to X .

$$\begin{aligned}
\beta_D &= \frac{\text{cov}(Y, \tilde{D})}{V(\tilde{D})} \\
&= \frac{\text{cov}(\alpha + \rho D + U, \tilde{D})}{V(\tilde{D})} \\
&= \frac{\text{cov}(\alpha + \rho D + X'\lambda + \varepsilon, \tilde{D})}{V(\tilde{D})} \\
&= \rho \frac{\text{cov}(D, \tilde{D})}{V(\tilde{D})} + \frac{\text{cov}(X'\lambda, \tilde{D})}{V(\tilde{D})} + \frac{\text{cov}(\varepsilon, \tilde{D})}{V(\tilde{D})} \\
&= \rho \frac{\text{cov}(X'\theta + \tilde{D}, \tilde{D})}{V(\tilde{D})} + \frac{\text{cov}(X'\lambda, \tilde{D})}{V(\tilde{D})} + \frac{\text{cov}(\varepsilon, D - X'\theta)}{V(\tilde{D})} \\
&= \rho \frac{\text{cov}(\tilde{D}, \tilde{D})}{V(\tilde{D})} + 0 + 0 \\
&= \rho.
\end{aligned}$$

The first equality follows from Frisch-Waugh. The second follows from (5.1.3). The third follows from Equation (5.1.4). The fifth follows from the regression decomposition of D with respect to X . The sixth follows from the two following facts: a) $\tilde{D}$ is by construction uncorrelated with all of the elements of X , which implies that $\text{cov}(X'\theta, \tilde{D}) = \text{cov}(X'\lambda, \tilde{D}) = 0$, b) ε is by construction uncorrelated with all the elements of X (cf. Equation (5.1.5)), and by assumption uncorrelated with D (cf. Equation (5.1.7)), so $\text{cov}(\varepsilon, D - X'\theta) = 0$.

QED.

The parameter ρ in a linear and constant effect model, and the coefficient β_D of D in a regression of Y on D and X are two different mathematical objects. ρ is defined as $Y(d) - Y(d-1)$, so ρ is a function of the potential outcomes $Y(d)$ and $Y(d-1)$. β_D is the solution to $\underset{b_D, b_X}{\text{argmin}} E((Y - Db_D - X'b_X)^2)$, so it is a function of the joint probability distribution of (Y, D, X) . A priori, there is no connection between these two objects. The previous theorem shows that under CIA and the assumption that $E(U|X)$ is linear, these two objects are actually equal.

A number of econometrics textbooks prove the same result as the one I showed above, but instead of imposing CIA, they only impose that ε be uncorrelated with D , which is weaker. Here, I prefer to derive this result under the CIA assumption, because zero correlation has no clear economic meaning, while independence has. Because many people have been taught econometrics this way, you will very often see equations such as $Y = \alpha + \rho D + X'\lambda + \varepsilon$ in the seminar room, and you will hear people arguing on whether ε , which in our example represents the part of innate ability which cannot be explained by X , is independent of D or not. That's because if that is the case, then the regression can receive a causal interpretation, as the previous theorem shows.

LCEM is not necessary for regression coefficients to receive a causal interpretation (we will see in next section that CIA alone is sufficient). But LCEM proves very useful to understand

what a regression measures when CIA is violated. This is very likely to arise in practice, as we can often not include in our regression some conditioning variables which would be necessary for CIA to hold. In the wage and education example, assume that for the CIA to be true we would need to condition on IQ at age 6. But unfortunately our data does not include this information, so we can only regress wages on educational attainment. Then, our regression might not identify a causal parameter, as shown in the next theorem.

Theorem 5.1.5 *The two omitted variable formulae*

Let X be a scalar random variable. Under Assumptions 4 and 5, and if $E(U|X) = X\lambda$,

$$\frac{\text{cov}(Y, D)}{V(D)} = \rho + \frac{\text{cov}(U, D)}{V(D)} \quad (5.1.8)$$

$$= \rho + \lambda\gamma, \quad (5.1.9)$$

where γ is the coefficient you would get from a regression of X on D and a constant.

Proof

$$\begin{aligned} \frac{\text{cov}(Y, D)}{V(D)} &= \frac{\text{cov}(\alpha + \rho D + U, D)}{V(D)} \\ &= \rho + \frac{\text{cov}(U, D)}{V(D)}. \end{aligned}$$

The first equality follows from (5.1.3). This proves the first result.

The second result follows from Theorem 3.4.3. Indeed, under the Assumptions of the Theorem, it follows from Theorem 5.1.4 that β_ℓ in Theorem 3.4.3 is equal to ρ . Moreover, under the assumptions of the theorem, one has $E(Y|D, X) = \alpha + \rho D + \lambda X$ (as shown in the proof of Theorem 5.1.4), so β_X in Theorem 3.4.3 is equal to λ .

QED.

The first omitted variable formula is useful to assess the direction of the bias when you suspect your regression does not capture a causal parameter. It indeed shows that if schooling is correlated with innate ability, the coefficient of a univariate regression of Y on D will not capture the target parameter ρ . If D and U are positively correlated, it will be greater than ρ , while if they are negatively correlated it will be lower than ρ .

In the returns to schooling example, do you think the regression would under or overestimate ρ ?

Schooling decisions are probably positively correlated with innate ability (undertaking graduate studies is less costly for high IQ people, and less risky as they have higher chances of

graduating from college). If that is the case, then our regression coefficient will overestimate ρ .

The second omitted variable formula can also help us assess if a regression coefficient over- or under-estimates a causal effect, though the first one may be more useful for that purpose. An interesting consequence of this second formula is that it shows that even if we omit from the regression a variable correlated to potential outcomes (i.e. such that $\lambda \neq 0$), it can still identify a causal effect if that variable is not correlated to the treatment variable ($\gamma = 0$). Omitting a variable from the regression will yield biased estimates of causal effects, if this variable is a) correlated with the outcome variable and b) correlated with the treatment variable.

5.1.4 Identification under CIA alone

In this subsection, for the sake of simplicity we assume that treatment D is binary (e.g.: completing primary school or not). CIA then rewrites as

Assumption 6 *CIA for binary treatment*

$$D \perp\!\!\!\perp (Y(0), Y(1)) | X. \quad (5.1.10)$$

Let's also assume that the vector of covariate X takes a finite number of values $\{x_1, \dots, x_J\}$. Let $ATE(x_j) = E(Y(1) - Y(0) | X = x_j)$ denote the average effect of the treatment among individuals with $X = x_j$ in the population. The $ATE(x_j)$ s parameters are called conditional average treatment effects.

Theorem 5.1.6 *Regression as a weighted average of conditional average treatment effects.*

Let D be a binary variable. Let X be a vector of random variables taking a finite number of values $\{x_1, \dots, x_J\}$. Assume that $0 < P(D = 1 | X = x_j) < 1$ for every $1 \leq j \leq J$. If Assumption 6 holds, the coefficient β_D of D in a regression of Y on D and $(1, 1\{X = x_2\}, \dots, 1\{X = x_J\})$ is equal to

$$\sum_{j=1}^J ATE(x_j) w_{x_j}, \quad (5.1.11)$$

where the weights w_{x_j} are those defined in Theorem 3.4.2.

Proof

It follows from Theorem 3.4.2 that

$$\beta_D = \sum_{j=1}^J \delta(x_j) w_{x_j}.$$

Then,

$$\begin{aligned}
\delta(x_j) &= E(Y|D = 1, X = x_j) - E(Y|D = 0, X = x_j) \\
&= E(Y(1)|D = 1, X = x_j) - E(Y(0)|D = 0, X = x_j) \\
&= E(Y(1)|X = x_j) - E(Y(0)|X = x_j) \\
&= ATE(x_j).
\end{aligned}$$

QED.

This proves that under CIA alone, the regression in Theorem 5.1.6 identifies a weighted average of the conditional average treatment effects parameters. Here we derived the result for a binary treatment, but this result can be extended to multivariate and ordered treatment variables such as schooling (see pages 77 to 80 of Angrist & Pischke (2008)).

Note that β_D is not equal to $E(Y(1) - Y(0))$. Indeed, it follows from the law of iterated expectations that

$$\begin{aligned}
E(Y(1) - Y(0)) &= E(E(Y(1) - Y(0)|X)) \\
&= \sum_{j=1}^J E(Y(1) - Y(0)|X = x_j)P(X = x_j).
\end{aligned}$$

$E(Y(1) - Y(0))$ is also a weighted average of the conditional average treatment effect parameters. However, what is the difference between $E(Y(1) - Y(0))$ and the weighted average of conditional average treatment effect parameters identified by a regression?

$E(Y(1) - Y(0))$ gives to each of the $E(Y(1) - Y(0)|X = x_j)$ parameters a weight equal to the percentage of the population with $X = x_j$. On the other hand, β_D gives a stronger weight to the $E(Y(1) - Y(0)|X = x_j)$ s for which $P(D = 1|X = x_j)(1 - P(D = 1|X = x_j))$ is higher than its average value in the population, i.e. to cells where treated and non treated units are the most balanced. This parameter is often called a variance-weighted average of conditional average treatment effects, because $P(D = 1|X = x_j)(1 - P(D = 1|X = x_j))$ is the variance of D conditional on $X = x_j$.

Finally, note that Theorem 5.1.6 only applies to regressions saturated in X . If continuous covariates are included in a regression, the regression can no longer be saturated in X . When the regression is not saturated in X , β_D no longer identifies a weighted average of conditional average treatment effect parameters. Instead, in some cases one can show that β_D identifies a weighted sum of those parameters. Assume for instance that X is a variable taking J ordered value $\{x_1, \dots, x_J\}$, such that $E(Y|D = 0, X) = \alpha_0 + \alpha_1 X$. Let $D = \gamma_0 + \gamma_1 X + \tilde{D}$ denote the

regression decomposition of D with respect to X . Then, one can show that β_D , the coefficient of D in a regression of Y on a constant, X , and D , is equal to

$$\sum_{j=1}^J ATE(x_j)w'_{x_j}, \quad (5.1.12)$$

where

$$w'_{x_j} = \frac{P(D=1|X=x_j)(1-(\gamma_0+\gamma_1x_j))P(X=x_j)}{\sum_{j=1}^J P(D=1|X=x_j)(1-(\gamma_0+\gamma_1x_j))P(X=x_j)}.$$

Are all the weights w'_{x_j} necessarily positive?

Some of the weights may be negative, if there are values of x_j such that $1 < \alpha_0 + \alpha_1 x_j$. When that happens β_D does not identify an interpretable causal parameter. Instead of identifying a weighted average of conditional average treatment effects, it identifies a weighted difference of conditional average treatment effects. It could be the case that all the conditional average treatment effects are positive, but β_D is negative. E.g.: assume that X can be equal to 0, 1, or 2, that $ATE(0) = ATE(1) = 1$, and $ATE(2) = 3$, and that $w'_0 = w'_1 = 1$ while $w'_2 = -1$, then $\beta_D = -1$, while all the conditional average treatment effects are positive.

In the context of a randomized experiment, are all the weights w'_{x_j} positive?

Yes. Indeed, in an experiment $\gamma_1 = 0$: there is no correlation between the treatment and X . And $\gamma_0 = E(D)$.

Bottom line: Under CIA, a linear regression of Y on D and X only identifies a (relatively) interpretable causal parameter if the regression is saturated in X . When it is not possible to saturate your regression in X , it is probably better to use a matching estimator.

5.1.5 Identification of the average treatment effect under CIA: matching

Theorem 5.1.7 *Full matching with discrete covariates*

Assume CIA holds for a set of discrete covariates X taking J values $x_1, \dots, x_J$, and assume

also that $0 < P(D = 1|X = x_j) < 1$ for every $1 \leq j \leq J$. Then,

$$E(Y(1) - Y(0)) = \sum_{j=1}^J \delta(x_j) P(X = x_j) \quad (5.1.13)$$

Proof

As in the proof of Theorem 5.1.6, we have

$$\delta(x_j) = E(Y(1) - Y(0)|X = x_j).$$

Therefore,

$$\begin{aligned} \sum_{j=1}^J \delta(x_j) P(X = x_j) &= \sum_{j=1}^J E(Y(1) - Y(0)|X = x_j) P(X = x_j) \\ &= E(E(Y(1) - Y(0)|X)) \\ &= E(Y(1) - Y(0)). \end{aligned}$$

The second step follows from the fact that $E(Y(1) - Y(0)|X)$ is a random variable which can take J different values, $E(Y(1) - Y(0)|X = x_1), \dots$, and $E(Y(1) - Y(0)|X = x_J)$, with probability $P(X = x_1), \dots$, and $P(X = x_J)$, respectively.

QED.

The intuition underlying this theorem is as follows. Under CIA, observations with $D = 1, X = x_j$ and $D = 0, X = x_j$ are representative of each other: they share the same X , so they must have the same distribution of $Y(0)$ and $Y(1)$ despite the fact they made different treatment decisions. That's the assumption under which matching strategies rely: within subgroups of units with the same X , we can do as if it had been a lottery which had assigned people to take or not the treatment. Because of this, $\delta(x_j)$ is equal to $E(Y(1) - Y(0)|X = x_j)$. Then, to get the ATE one just needs to compute a weighted average of those $E(Y(1) - Y(0)|X = x_j)$, using as weights the percentage of the population with $X = x_j$.

Theorem 5.1.7 implies that if CIA holds for a discrete set of covariates X , and if one observes an iid sample (Y_i, D_i, X_i) with the same distribution as (Y, D, X) , one can estimate the ATE as follows:

1. Divide the sample into J subgroups of individuals: those with $\{X_i = x_1\}$, those with $\{X_i = x_2\}, \dots$, those with $\{X_i = x_J\}$.
2. In each subgroup, compute the difference between the mean of Y among treated and untreated individuals: $\hat{\delta}(x_j) = \hat{E}(Y|D = 1, X = x_j) - \hat{E}(Y|D = 0, X = x_j)$.
3. For every $j \in \{1, \dots, J\}$, estimate $P(X = x_j)$ using the percentage of individuals with $\{X_i = x_j\}$ in the sample: $\hat{P}(X = x_j) = \frac{1}{n} \sum_{i=1}^n 1\{X_i = x_j\}$.
4. Finally, use $\widehat{ATE} = \sum_{j=1}^J \hat{\delta}(x_j) \hat{P}(X = x_j)$ to estimate the average treatment effect.

A draw back of this procedure to estimate the average treatment effect is that it has many steps. To estimate the ATE in less steps, one can use the following result.

Theorem 5.1.8 *Propensity score matching with discrete covariates*

Assume CIA holds for a set of discrete covariates X taking J values $x_1, \dots, x_J$, and assume also that $0 < P(D = 1|X = x_j) < 1$ for every $1 \leq j \leq J$. Then,

$$E(Y(1) - Y(0)) = E\left(Y \frac{P(D=1)}{P(D=1|X)} \middle| D=1\right) - E\left(Y \frac{1-P(D=1)}{1-P(D=1|X)} \middle| D=0\right)$$

Proof

First,

$$\begin{aligned} E\left(Y \frac{P(D=1)}{P(D=1|X)} \middle| D=1\right) &= E\left(E\left[Y \frac{P(D=1)}{P(D=1|X)} \middle| X, D=1\right] \middle| D=1\right) \\ &= E\left(\frac{P(D=1)}{P(D=1|X)} E[Y|X, D=1] \middle| D=1\right) \\ &= \sum_{j=1}^J \frac{P(D=1)}{P(D=1|X=x_j)} E[Y|X=x_j, D=1] P(X=x_j|D=1) \\ &= \sum_{j=1}^J \frac{P(D=1)}{P(D=1|X=x_j)} E[Y(1)|X=x_j, D=1] P(X=x_j|D=1) \\ &= \sum_{j=1}^J E[Y(1)|X=x_j] \frac{P(D=1)}{\frac{P(D=1, X=x_j)}{P(X=x_j)}} \frac{P(D=1, X=x_j)}{P(D=1)} \\ &= \sum_{j=1}^J E[Y(1)|X=x_j] P(X=x_j) \\ &= E(E[Y(1)|X]) \\ &= E(Y(1)). \end{aligned} \tag{5.1.14}$$

Second, one can follow the same steps to show that

$$E\left(Y \frac{1-P(D=1)}{1-P(D=1|X)} \middle| D=0\right) = E(Y(0)). \tag{5.1.15}$$

The result follows combining Equations (5.1.14) and (5.1.15).

QED.

The weights $\frac{P(D=1)}{P(D=1|X)}$ and $\frac{1-P(D=1)}{1-P(D=1|X)}$ are called propensity score weights. The key intuition underlying Theorem 5.1.8 is that $\frac{P(D=1)}{P(D=1|X=x_j)} P(X=x_j|D=1) = P(X=x_j)$. Therefore, reweighting treated units by $\frac{P(D=1)}{P(D=1|X=x_j)}$ makes the distribution of $X|D=1$ equal to the distribution of X in the entire population.

Find a variable such that if you regress that variable on $(1, D)$, then the (population) coefficient of D is equal to $E\left(Y \frac{P(D=1)}{P(D=1|X)} \middle| D=1\right) - E\left(Y \frac{1-P(D=1)}{1-P(D=1|X)} \middle| D=0\right)$.

It follows from the discussion of saturated regressions in Chapter 3 that the coefficient of D in a regression of $Y^* = DY \frac{P(D=1)}{P(D=1|X)} + (1-D)Y \frac{1-P(D=1)}{1-P(D=1|X)}$ on $(1, D)$ is equal to

$$E(Y^*|D=1) - E(Y^*|D=0) = E\left(Y \frac{P(D=1)}{P(D=1|X)} \middle| D=1\right) - E\left(Y \frac{1-P(D=1)}{1-P(D=1|X)} \middle| D=0\right).$$

In practice can you regress Y^* on D ?

No, because you do not observe $P(D=1)$ and $P(D=1|X=x_j)$ for every $j \in \{1, \dots, J\}$. However, if you observe an iid sample $(Y_i, D_i, X_i)_{1 \leq i \leq n}$ with the same distribution as (Y, D, X) , you can use the share of treated individuals in your sample to estimate $P(D=1)$, and the share of treated individuals in the subgroup with $X_i = x_j$ to estimate $P(D=1|X=x_j)$. Then you can generate $\hat{Y}_i^* = D_i Y_i \frac{\hat{P}(D=1)}{\hat{P}(D=1|X=X_i)} + (1-D_i) Y_i \frac{1-\hat{P}(D=1)}{1-\hat{P}(D=1|X=X_i)}$ for every individual in your sample, and finally you can regress $\hat{Y}_i^*$ on $(1, D_i)$. One can show that the coefficient of D_i in this regression is equal to $\widehat{ATE}$. But the method I just described to compute $\widehat{ATE}$ has less steps than that described above: you just need to estimate $P(D=1)$ and the $P(D=1|X=x_j)$ s, generate a new variable, and then estimate a simple regression.

Is the standard error of the coefficient of D_i in that regression a valid estimator of the standard error of $\widehat{ATE}$?

No, because this standard error does not account for the fact $\hat{P}(D=1)$ and $\hat{P}(D=1|X=X_i)$ are the result of a first step estimation with its own estimation error which should be accounted for. A method to get a valid estimate of the standard error of $\widehat{ATE}$ goes as follows. It amounts to performing, say 200 hundred times, the following computations: a) draw with replacement from the sample a new sample with the same number of units, b) estimate $P(D=1)$ and $P(D=1|X=x_j)$ in this new sample, c) create a new variable $\hat{Y}_i^{*}$ using these new estimators of $P(D=1)$ and $P(D=1|X=x_j)$, d) regress this new $\hat{Y}_i^{*}$ on $(1, D_i)$ and store the coefficient of D_i somewhere, e) once the 200 replications of this procedure are terminated, compute the standard error of the coefficient of D_i across those 200 replications and use it as your estimator of the standard error of $\widehat{ATE}$.

A rigorous study of the bootstrap is beyond the scope of this course, but let me give you a little bit of intuition on why this procedure yields a valid estimator of the standard error of $\widehat{ATE}$. In

each bootstrap replication, you draw with replacement a new sample from your original sample. Because you draw observations with replacement, each time you draw a new bootstrap sample you will most likely draw twice or thrice some observations, and zero times others, implying that all of your 200 bootstrap samples will be different from each other. Within each of these different samples, you will run the whole procedure to estimate $\widehat{ATE}$. These 200 estimators will not be the same, as they are all based on different samples of observations. The variability across these 200 bootstrap estimates of $\widehat{ATE}$ is a good estimate of the true variance of $\widehat{ATE}$, because the bootstrap sampling with replacement from the sample mimicks the sampling with replacement from the total population which has generated your data. Also, notice that in the procedure described above, $P(D = 1)$ and $P(D = 1|X = x_j)$ are re-estimated in each bootstrap sample. Therefore, the variability of $\widehat{ATE}$ across bootstrap samples accounts for the estimation error coming from the estimation of $P(D = 1)$ and $P(D = 1|X = x_j)$.

With covariates taking a finite number of values, matching with covariates and propensity score matching are the same thing. One advantage of propensity score matching is that Theorem 5.1.8 remains true if X can take an infinity of values, thus implying that propensity score matching can also be used in this case. However, in such instances a practical issue arises: one can no longer use the share of treated individuals in the subgroup with $X_i = x_j$ to estimate $P(D = 1|X = x_j)$. Instead, people usually estimate $P(D = 1|X)$ with a logit regression, and include some polynomials in X to account for potential non-linearities in the relationship between X and D . But if $P(D = 1|X)$ does not follow this logit model, this logit regression will not estimate $P(D = 1|X)$, but its best approximation by a flexible logit model. So our estimator of $P(D = 1|X)$ will converge towards the best approximation of $P(D = 1|X)$ by a flexible logit model, not towards $P(D = 1|X)$, so it will not be consistent. As a result, our propensity score estimator of the ATE will also be inconsistent. An alternative solution is to use non-parametric regression techniques to estimate $P(D = 1|X)$. Studying non-parametric propensity score matching estimators is technically challenging and goes beyond the scope of this course. There are Stata packages to estimate propensity score matching estimators.

Another issue with propensity score matching is that there may be values of X within which there are very few untreated or treated observations. Accordingly, the comparison of treated and untreated observations in those cells may be estimated very imprecisely, and including those cells in your estimation may actually increase the variance of your estimator. Despite the fact that dropping those cells decreases your sample size, your treatment/control comparison in those cells is so poorly estimated that doing so may actually reduce the variance of your estimator. Crump et al. (2009) study this trade-off formally, and show that to minimize the variance of a matching estimator, one should only use cells where $\alpha < P(D = 1|X = x) < 1 - \alpha$, for some α whose formula can be found in their paper. In practice, α is often close to 0.1. So the bottom line is that if you use a matching estimator, you may want a drop from your analysis observations that have an estimated propensity score $\widehat{P}(D = 1|X = X_i)$ lower than 0.1 or higher than 0.9, a procedure called trimming.

5.1.6 Matching or regression?

Under CIA, we have two possible methods to estimate treatment effects: a regression saturated in X , or matching.

Regression has one drawback: it identifies a tough to interpret variance weighted average of ATE_{x_j} , instead of the ATE . As we discussed in Chapter 2, one can rationalize the study of treatment effects by assuming we are trying to help a social planner decide whether they should give the treatment to their population or not. For simple welfare functions, knowing the ATE is sufficient to answer this question. On the other hand, it is harder to rationalize why a social planner would be interested in knowing the variance weighted average of ATE_{x_j} identified by a regression. This parameter gives a stronger weight to units with values of X for which $P(D = 1|X = x_j)(1 - P(D = 1|X = x_j))$, but a social planner has no obvious reason to value more the welfare of those individuals than the welfare of individuals whose X is such that $P(D = 1|X = x_j)(1 - P(D = 1|X = x_j))$ is low.

But if the social planner takes into account the fact they will only get an estimate of the parameter they choose, then they might prefer regression over matching. Indeed, in such instances, they might choose the estimator $\hat{\theta}$ with the lowest quadratic risk to estimate the ATE : $L(\hat{\theta}) = E\left(\left(\hat{\theta} - ATE\right)^2\right)$. Let θ denote the probability limit of $\hat{\theta}$. Under the assumption that $E(\hat{\theta}) = \theta$, one has $L(\hat{\theta}) = V(\hat{\theta}) + (\theta - ATE)^2$. If the social planner chooses to do matching, they will use $\widehat{ATE}$, which converges towards ATE . Then, their quadratic risk is $L(\widehat{ATE}) = V(\widehat{ATE}) + (ATE - ATE)^2 = V(\widehat{ATE})$. If they choose to run a regression, they will use $\hat{\beta}_D$, which converges towards β_D , the regression coefficient in Theorem 5.1.6. Then, their quadratic risk is $L(\hat{\beta}_D) = V(\hat{\beta}_D) + (\beta_D - ATE)^2$. From (5.1.13) and (9.3.2), it appears that β_D gives more weight than ATE to subgroups $\{X = x_j\}$ where the proportion of treated and untreated units is more balanced, i.e. to subgroups in which the estimator of $\delta(x_j)$ has a lower variance. Because of this, one often has $V(\hat{\beta}_D) < V(\widehat{ATE})$. So even though β_D is not the ATE , the social planner might prefer to use a regression than matching to estimate the ATE , because the gain on variance sometimes compensates the fact that $\hat{\beta}_D$ is an inconsistent estimator of the ATE . This is called a bias-variance trade-off.

When the sample size goes to infinity, both $V(\widehat{ATE})$ and $V(\hat{\beta}_D)$ go to 0 because both are consistent estimators of ATE and β_D . As a result, the limit of $L(\widehat{ATE})$ is lower than the limit of $L(\hat{\beta}_D)$. But in finite samples, the regression estimator might have a lower quadratic risk than the matching estimator.

To decide which estimator to use, one should compare their two quadratic risks. This is not feasible: for instance $L(\hat{\beta}_D)$ is not observed as it depends on β_D and ATE , which are population quantities. However, one could use $\widehat{V}(\hat{\beta}_D) + \left(\hat{\beta}_D - \widehat{ATE}\right)^2$ to estimate it. Actually, it turns out that $\widehat{V}(\hat{\beta}_D) + \left(\hat{\beta}_D - \widehat{ATE}\right)^2$ is an upward biased estimator of $L(\hat{\beta}_D)$. On the

other hand, in De Chaisemartin & D'Haultfoeuille (2018) we show that $\widehat{L}(\widehat{\beta}_D) = \widehat{V}(\widehat{\beta}_D) + \left(\widehat{\beta}_D - \widehat{ATE}\right)^2 - \widehat{V}(\widehat{\beta}_D - \widehat{ATE})$ is an unbiased estimator of $L(\widehat{\beta}_D)$. [How can you estimate \$L\(\widehat{ATE}\)\$?](#)

Using the bootstrap procedure described above. Then, a reasonable decision rule would be to use regression when $\widehat{L}(\widehat{\beta}_D) \leq \widehat{L}(\widehat{ATE})$, and to use covariates or propensity score matching otherwise. In practice, people rarely bother making this comparison, and often use regressions as they are simpler to implement.

5.1.7 When to use / not to use regression and matching

LaLonde (1986) studied the effect of a job training program on the earnings of participants. This program had been assigned by means of a randomized experiment, so a first way to estimate its effect was just to compare the average earnings of the experimental treatment and control groups. Then, he used matching to construct a non-experimental control group: he chose individuals that did not participate in the experiment, but that looked very similar to those who participated on a large number of demographic and labor market history variables. He then used a regression controlling for all those variables to compare the earnings of the treatment and of the non-experimental control group. This regression yielded an estimated effect of the training very different from that obtained comparing the experimental treatment and control groups. The validity of this non-experimental regression relied on the CIA assumption, while the experimental comparison relied on no assumption. Therefore, his paper showed that CIA was violated in this context, because the non-experimental estimate fell far from the true effect unveiled by the experiment. Following his paper, the popularity of methods relying on the CIA assumption diminished.

Still, there are some situations where matching and regression estimators can be useful, because CIA holds by construction. In randomized experiments, lotteries often take place within X cells. When admitting students to a school through a lottery, you must still make sure that your lottery does not only admit say boys, or first graders. There will therefore most often be one lottery within each grade $\times$ sex cell. Then, while

$$D \perp\!\!\!\perp (Y(0), Y(1))$$

might not hold,

$$D \perp\!\!\!\perp (Y(0), Y(1)) | X \tag{5.1.16}$$

will be true by construction (if X represents the covariates used to determine the lottery a student will participate in, i.e. gender and grade in the example above). Then, one will be able to estimate the average effect of that school through regression or matching controlling for those covariates.

Moreover, in instances where the CIA assumption does not hold by construction, one can use placebo tests to assess its plausibility. Imbens et al. (2001) study the effect of non-labor income on labor income. On that purpose, they worked together with the IRS. The IRS sampled people playing regularly the lottery. Their sampling scheme ensured that they interviewed 50% of people who had won large prizes, and 50% of people who never won. When they compared the characteristics of the two groups, Imbens and Sacerdote found that the winners and losers differed on several characteristics: the winners played the lottery more often than the losers, they had completed fewer years of schooling, their earnings prior to winning were lower... Then, would comparing the labor income of the two groups after the winners have won yield a good estimate of the effect of winning the lottery on labor income?

No, because the winners and the losers are not comparable, so the average labor income of the losers is not a good counterfactual of the average income the winners would have obtained if they had not won the lottery. Instead, Imbens and Sacerdote implement a matching estimator. To assess the plausibility of the CIA assumption, they take advantage of the fact that the IRS data contains earnings for every individual in their sample the year after the winners won the lottery, but also during the 6 years before. To estimate the propensity score, they only use 3 covariates: the number of lottery tickets bought by individuals, their years of schooling, and their earnings six years before the year when winners won. Then, the CIA assumption is the following:

$$D \perp\!\!\!\perp (Y(0), Y(1)) | \text{Lottery tickets bought, years of schooling, earnings 6 years before.}$$

Lottery winners and losers that purchased the same number of tickets, that have the same numbers of years of schooling, and that had the same earnings 6 years ago should also have the same potential labor incomes if they win / do not win the lottery.

To test the plausibility of this assumption, Imbens and Sacerdote use placebo tests. The CIA assumption implies that $D \perp\!\!\!\perp Y(0) | X$ and $D \perp\!\!\!\perp Y(1) | X$, something which is not testable, because we only observe the $Y(0)$ of untreated units, and the $Y(1)$ of treated units, so we cannot assess whether treated and untreated units with the same X have the same $Y(0)$ and $Y(1)$. However, we can test whether a condition pretty close from $D \perp\!\!\!\perp Y(0) | X$ holds. For every $k \in \{1, \dots, 5\}$, let $Y_{i,-k}$ denote the labor income of unit i in the sample k years

before the winners won the lottery, and let Y_{-k} denote the labor income of a representative individual. k years before the winners won, nobody had won, so $Y_{-k} = Y_{-k}(0)$. The idea of the placebo tests in Imbens and Sacerdote is to assess whether $Y_{-k}(0) \perp\!\!\!\perp D|X$: in the years before the lottery winners won, was it the case that losers and winners with the same X had the same labor income? To test this condition, one can proceed as follows. Let $\hat{Y}_{i,-k}^* = D_i Y_{i,-k} \frac{\hat{P}(D=1)}{\hat{P}(D=1|X=X_i)} + (1 - D_i) Y_{i,-k} \frac{1 - \hat{P}(D=1)}{1 - \hat{P}(D=1|X=X_i)}$. Let $\hat{\beta}_P$ be the coefficient of D_i in a regression of $\hat{Y}_{i,-k}^*$ on a constant and D_i . If $Y_{-k}(0) \perp\!\!\!\perp D|X$, should $\hat{\beta}_P$ be close to or far from zero? Why?

$\hat{\beta}_P$ should be close to 0. Indeed, one can show that the probability limit of $\hat{\beta}_P$ is $E(E(Y_{-k}(0)|D = 1, X) - E(Y_{-k}(0)|D = 0, X))$, which is equal to 0 if $Y_{-k}(0) \perp\!\!\!\perp D|X$. $\hat{\beta}_P$ is called a placebo estimator: we use the same estimation technique as that we are gonna use to estimate the effect of the treatment (e.g. propensity score matching estimator) to estimate the effect of the treatment on a variable that cannot be affected by it (labor income before the treatment group got treated). If the assumptions underlying our estimator are valid, the placebo estimator should be close to 0.

Imbens and Sacerdote compute placebo estimators using individuals' earnings 1, 2, 3, 4, and 5 years before the treatment. All of those estimators are small and insignificant, meaning that treated and control individuals with the same X have similar earnings before the treatment. This shows that $Y_{-k}(0) \perp\!\!\!\perp D|X$, which itself lends support to the CIA assumption. Then, they estimate their propensity score estimator with earnings during the year after the lottery winners won as the outcome, and their estimator is negative and significant. Overall, they provide plausible evidence that winning the lottery reduces labor income.

Could Imbens and Sacerdote have used earnings 6 years before the treatment in their placebo analysis?

No! That variable is one of those they use to match treated and control units. Therefore, we have that by construction $E(E(Y_{-6}(0)|D = 1, X) - E(Y_{-6}(0)|D = 0, X)) = 0$, irrespective of whether the CIA assumption is plausible in this application.

Imbens and Sacerdote are in an ideal situation to test the CIA assumption: they have 6

measures of their outcome of interest made prior to the treatment. They can use one to match treated and control units, and the remaining 5 to run placebo tests of that assumption. In other applications, lagged values of the outcome of interest may not be available. In such instances, one can still run placebo tests by computing “placebo” propensity score matching estimators, where one replaces the outcome variable by a pre-determined covariate not included in the set of covariates on which treated and control units are matched, and that is likely to be correlated with the potential outcomes. One should find that the placebo estimator is small, and not significantly different from 0. In the Imbens and Sacerdote example, if lagged measures of earnings had not been available, one could have used “years of schooling” to conduct the placebo, because that’s a strong determinant of wages so if treated and control units have the same years of schooling conditional on X , that’s good evidence that they may also have the same potential wages conditional on X . But then years of schooling should be excluded from the set of matching covariates.

5.2 Regression and randomized experiments

In randomized experiments, one can use a regression to estimate the ATE. Indeed, it follows from our discussion of saturated regressions that the coefficient of D in a regression of Y on D and a constant is equal to $E(Y|D = 1) - E(Y|D = 0) = E(Y(1) - Y(0)) = ATE$. So we can use a regression to estimate the ATE in one step instead of having to estimate $E(Y|D = 1)$, then $E(Y|D = 0)$, and finally compute their difference (3 steps).

Moreover, if treatment is assigned independently of some individuals’ covariates X , the second omitted variable formula tells us that regressing Y on D or Y on D and X will leave the coefficient of D unaffected, because $X \perp\!\!\!\perp D$. But as I will show below after a somewhat lengthy derivation, including controls might result in important gains in statistical precision.

Definition 5.2.1 *The population R^2 .*

Let β denote the regression coefficient of Y on X .

$$R_{Y,X}^2 = \frac{V(X'\beta)}{V(Y)}.$$

$R_{Y,X}^2$ is included between 0 and 1.

It is straightforward that $R_{Y,X}^2 \geq 0$. To see that it is lower than 1, write the regression decomposition of Y with respect to X : $Y = X'\beta + e$. $V(Y) = V(X'\beta) + V(e) \geq V(X'\beta)$ because X and e are uncorrelated by construction. The R^2 is a measure of the share of the variance of Y captured by the regression. The higher it is, the more explanatory power our regression has, and the more precise our predictions for Y based on the regression will be.

We will now investigate how the asymptotic variance of a regression coefficient changes when more variables are added to the regression. Let $Y = \alpha + \beta D + X'\gamma + e$ be the regression

decomposition of Y wrt D and X . Let $Y = \eta + \mu D + u$ be the regression decomposition of Y wrt D . Note that the second omitted variable formula shows that β and μ are not necessarily equal. Let $\hat{\beta}$ and $\hat{\mu}$ denote their OLS estimators computed from an iid sample $(Y_i, D_i, X_i)_{1 \leq i \leq n}$ with the same probability distribution as (Y, D, X) . Let $\tilde{D}$ be the residual of D from a regression decomposition of D wrt X .

Theorem 5.2.2 *Comparative statics for asymptotic variances*

If $E(Y|X, D) = \alpha + \beta D + X'\gamma$, $E(Y|D) = \eta + \mu D$, $V(e|D, X) = V(e)$, and $V(u|D) = V(u)$, then

$$\begin{aligned}\sqrt{n}(\hat{\beta} - \beta) &\xrightarrow{d} N\left(0, \frac{E(e^2)}{V(D)(1 - R_{D,X}^2)}\right) \\ \sqrt{n}(\hat{\mu} - \mu) &\xrightarrow{d} N\left(0, \frac{E(u^2)}{V(D)}\right).\end{aligned}$$

Therefore, the ratio of the first and second asymptotic variances is

$$\frac{1 - R_{Y,(D,X)}^2}{1 - R_{Y,D}^2} \times \frac{1}{1 - R_{D,X}^2}.$$

Proof

$$\begin{aligned}\hat{\beta} &= \frac{\text{cov}_e(Y_i, \hat{\tilde{D}}_i)}{V_e(\hat{\tilde{D}}_i)} \\ &= \frac{\text{cov}_e(\alpha + \beta D_i + X_i'\gamma + e_i, \hat{\tilde{D}}_i)}{V_e(\hat{\tilde{D}}_i)} \\ &= \beta + \frac{\text{cov}_e(e_i, \hat{\tilde{D}}_i)}{V_e(\hat{\tilde{D}}_i)} \\ &= \beta + \frac{\frac{1}{n} \sum_{i=1}^n e_i \hat{\tilde{D}}_i}{\frac{1}{n} \sum_{i=1}^n \hat{\tilde{D}}_i^2}.\end{aligned}$$

The first equality follows from the sample Frisch-Waugh theorem. The second one follows from the regression decomposition of Y with respect to D and X . The third one follows from the following facts. $D = X'\hat{\rho} + \hat{\tilde{D}}_i$, where $\hat{\rho}$ is the estimator of ρ , the regression coefficient of D on X . Therefore, $\text{cov}_e(D, \hat{\tilde{D}}_i) = V_e(\hat{\tilde{D}}_i)$, because residuals from a regression must be uncorrelated in the sample with the regressors (see Lemma 4.1.1). For the same reason, $\text{cov}_e(X'\gamma, \hat{\tilde{D}}_i) = 0$. The last one follows from the fact that the estimated residual from a regression has a sample mean equal to 0 (see Lemma 4.1.1).

As a result,

$$\sqrt{n}(\hat{\beta} - \beta) = \frac{\sqrt{n} \frac{1}{n} \sum_{i=1}^n e_i \hat{\tilde{D}}_i}{\frac{1}{n} \sum_{i=1}^n \hat{\tilde{D}}_i^2}.$$

Using the fact that $\hat{D}_i = D_i - X_i' \hat{\rho}$, $\tilde{D}_i = D_i - X_i' \rho$, and $\hat{\rho}$ is a consistent estimator of ρ , one can show after a few algebraic manipulations, uses of the Slutsky lemma and of the continuous mapping theorem, that $\sqrt{n}(\hat{\beta} - \beta)$ has the same asymptotic distribution as

$$\frac{\sqrt{n} \frac{1}{n} \sum_{i=1}^n e_i \tilde{D}_i}{E(\tilde{D}^2)}.$$

Since $E(e\tilde{D}) = 0$ ($\tilde{D}$ is a linear function of D and X , it is therefore uncorrelated with e , following Lemma 3.2.1), the central limit theorem implies that the quantity in the above display converges to a

$$N\left(0, \frac{V(e\tilde{D})}{E(\tilde{D}^2)^2}\right)$$

distribution.

Finally,

$$\begin{aligned} V(e\tilde{D}) &= E(V(e\tilde{D}|X, D)) + V(E(e\tilde{D}|X, D)) \\ &= E(\tilde{D}^2 V(e|X, D)) + V(E((Y - E(Y|X, D))|X, D)\tilde{D}) \\ &= E(\tilde{D}^2) V(e) \\ &= E(\tilde{D}^2) E(e^2). \end{aligned}$$

The first step follows from the ANOVA formula. For the $E(V(.))$ term, the second step follows from Theorem 3.1.4 as $\tilde{D}$ is a function of X, D , the third follows from the homoscedasticity assumption: $V(e|D, X) = V(e)$, the fourth from the fact that $E(e) = 0$. For the $V(E(.))$ term, the second equality follows from the fact that since by assumption $E(Y|X, D)$ is linear, $E(Y|X, D)$ is the regression function, so $\tilde{D} = Y - E(Y|X, D)$. The third equality follows from rules of algebra for conditional expectations.

This completes the proof of the first result, once noted that $E(\tilde{D}^2) = V(\tilde{D}) = V(D)(1 - R_{D,X}^2)$. One can follow similar steps to prove the second result.

QED.

This theorem is derived under somewhat restrictive assumptions (linear CEFs and homoscedasticity) but it is still probably a good first order approximation of what happens even when those assumptions do not hold.

It unveils an interesting trade-off taking place in asymptotic variances when going from a short to a long regression. The first term of the ratio of the asymptotic variances is lower than 1, while the second one is greater than 1.² $\frac{1 - R_{Y,(D,X)}^2}{1 - R_{Y,D}^2}$ will be much lower than 1 if adding X to

²To see that the first term is lower than 1, it suffices to notice that it also equal to $\frac{E(e^2)}{E(u^2)}$. $E(e^2) = \min_{a,b,g} E((Y - a - bD - gX)^2)$, while $E(u^2) = \min_{a,b} E((Y - a - bD)^2)$. Therefore, $E(e^2) \leq E(u^2)$: if a^* and b^* are the solutions of the second minimization problem, then we can do at least as well in the second one by setting $(a, b, g) = (a^*, b^*, 0)$. This merely amounts to saying that the R^2 mechanically increases when we add more variables to a regression.

the regression of Y on D alone increases a lot the R^2 . Conversely, $\frac{1}{1-R_{D,X}^2}$ will be much higher than 1 if the X are strongly correlated with D . Therefore, if you add to your regression some X s correlated with Y but not correlated with D , this will diminish the asymptotic variance of the coefficient of D , which will in turn diminish the width of its confidence interval. Conversely, if the explanatory variables you add to the regression are unrelated to Y but correlated with D , this will reduce statistical precision.

In a randomized experiment, the regression decomposition of D wrt to $(1, X)$ is $E(D)$: all the X have a coefficient equal to 0 in the regression because $D \perp\!\!\!\perp X$. Therefore, $R_{D,X}^2 = 0$, so that the ratio of the asymptotic variances is equal to $\frac{1-R_{Y,(D,X)}^2}{1-R_{Y,D}^2}$. This shows that in a randomized experiment, adding control variables to the regression will always increase statistical power, all the more so when the covariates are strong predictors of the outcome variable.

This has important practical implications. When you work on a randomized (field or lab) experiment, you should make sure to collect baseline covariates which are good predictors of the outcome you are interested in, so as to increase power. Assume you are working on economics of education and you are interested in the effect of some program on students' test scores. Which baseline covariates should you collect to increase the statistical power of your design?

Students' baseline test scores, before the randomization. That should be a very strong predictor of their test score in the end of the experiment.

Chapter 6

M-estimation and the generalized method of moments

This chapter borrows from van der Vaart (2000) and Hansen (1982).

6.1 A few more tools from real analysis and probability theory

For any $k \times 1$ vector x , let $\|x\|$ denote its Euclidean norm: $\|x\|^2 = x'x = x_1^2 + \dots + x_k^2$. For any $k \times 1$ vectors x_1 and x_2 , let $d(x_1, x_2) = \|x_1 - x_2\|$ denote their Euclidean distance. For any function $f : \mathbb{R} \mapsto \mathbb{R}$ twice differentiable, let $\dot{f}$ and $\ddot{f}$ denote its first and second order derivatives. For any function $f : \mathbb{R}^k \mapsto \mathbb{R}$ differentiable at x , let $\dot{f}(x)$ denote its gradient evaluated at x , i.e. the $k \times 1$ vector containing its k first order derivatives evaluated at x :

$$\dot{f}(x) = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_k}(x) \right)'.$$

For any function $f : \mathbb{R}^k \mapsto \mathbb{R}$ twice differentiable at x , let $\ddot{f}(x)$ denote its Hessian evaluated at x , i.e. the $k \times k$ matrix containing its k second order and $k(k-1)$ cross derivatives evaluated at x :

$$\begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2}(x) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(x) & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_k}(x) \\ \frac{\partial^2 f}{\partial x_1 \partial x_2}(x) & \frac{\partial^2 f}{\partial x_2^2}(x) & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_k}(x) \\ \dots & \dots & \dots & \dots \\ \frac{\partial^2 f}{\partial x_1 \partial x_k}(x) & \frac{\partial^2 f}{\partial x_2 \partial x_k}(x) & \dots & \frac{\partial^2 f}{\partial x_k^2}(x) \end{pmatrix}.$$

Definition 6.1.1 *Deterministic $o()$*

Let $g : \mathbb{R}^k \mapsto \mathbb{R}$ and $h : \mathbb{R}^k \mapsto \mathbb{R}$ be two functions. $g(x) = o(h(x))$ if $\lim_{\|x\| \rightarrow 0} \frac{g(x)}{h(x)} = 0$.

For instance, you can show that $x^2 = o(x)$.

Theorem 6.1.2 *Order 1 Taylor expansions*

- Let I be a real interval and let $f : I \mapsto \mathbb{R}$ be differentiable at $a \in I$. Then, for any x in I $f(x) = f(a) + \dot{f}(a)(x - a) + o(x - a)$.
- Let I be a closed real interval and let $f : I \mapsto \mathbb{R}$ be continuous and differentiable on I . Then, for any x in I there exists a number ε included between x and a such that $f(x) = f(a) + \dot{f}(\varepsilon)(x - a)$.
- Let I be a subset of $\mathbb{R}^k$ and let $f : I \mapsto \mathbb{R}$ be differentiable at $a \in I$. Then, for any x in I $f(x) = f(a) + \dot{f}(a)'(x - a) + o(\|x - a\|)$.
- Let I be a closed ball of $\mathbb{R}^k$ and let $f : I \mapsto \mathbb{R}$ be continuous and differentiable on I . Then, for any x in I there exists a vector ε “included” between x and a (for vectors, this means that each coordinate of ε should be included between that of x and a) such that $f(x) = f(a) + \dot{f}(\varepsilon)'(x - a)$.

The first statement of the theorem merely says that when a function is differentiable at some point a , the difference between this function and its tangent at a , $f(a) + \dot{f}(a)(x - a)$, gets very small for values of x close to a . $f(x)$ can be well approximated by $f(a) + \dot{f}(a)(x - a)$ close to a . The second statement asserts that for any (x, a) , there is an ε included between the two such that the slope of the line connecting $(a, f(a))$ to $(x, f(x))$ is the slope of the tangent of f at ε , $\dot{f}(\varepsilon)$. The third statement generalizes the first to functions of several variables. The only difference is that now, $(x - a)$ is a $k \times 1$ vector, $\dot{f}(a)'$ is $1 \times k$, so $\dot{f}(a)'(x - a)$ is an inner product (do you all know what an inner product is?). For instance, if $k = 2$, we have $f(x_1, x_2) = f(a_1, a_2) + \left(\frac{\partial f}{\partial x_1}(a_1, a_2), \frac{\partial f}{\partial x_2}(a_1, a_2) \right) (x_1 - a_1, x_2 - a_2)' + o(\|x - a\|)$. The fourth statement generalizes the second to functions of several variables.

Theorem 6.1.3 Dominated convergence theorems

Let W be a random vector in $\mathbb{R}^l$.

- (a) Let $\Theta \subseteq \mathbb{R}^k$, and let $f : \mathbb{R}^l \times \Theta \mapsto \mathbb{R}$ be such that for every $w \in \mathbb{R}^l$, $f(w, \theta)$ is continuous in θ , and for every $\theta \in \Theta$, $|f(w, \theta)| \leq g(w)$ for some function $g(w)$ such that $E(g(W)) < +\infty$. Then $h(\theta) = E(f(W, \theta))$ is continuous in θ .
- (b) Let $\Theta \subseteq \mathbb{R}^k$, and let $f : \mathbb{R}^l \times \Theta \mapsto \mathbb{R}$ be such that for every $w \in \mathbb{R}^l$, $f(w, \theta)$ is differentiable in θ , and for every $\theta \in \Theta$, $|\dot{f}(w, \theta)| \leq g(w)$ for some function $g(w)$ such that $E(g(W)) < +\infty$.¹² Then $h(\theta) = E(f(W, \theta))$ is differentiable in θ , with $\dot{h}(\theta) = E(\dot{f}(W, \theta))$.

Dominated convergence theorems are tools to invert the lim and expectation operators. Take the first statement. Showing that $h(\theta)$ is continuous is equivalent to showing that for any sequence (θ_n) converging towards θ , $\lim_{n \rightarrow +\infty} h(\theta_n) = h(\theta)$. If we could revert the lim and the

¹ $\dot{f}(w, \theta)$ is the vector of the k first order derivatives of f with respect to θ , not to w .

² $\dot{f}(w, \theta)$ and $g(w)$ are vectors. The inequality means that the absolute value of each coordinate of $\dot{f}(w, \theta)$ should be lower than the corresponding coordinate of $g(w)$.

expectation operators, the proof would directly follow from the fact that f is continuous. Indeed, we would have

$$\lim_{n \rightarrow +\infty} h(\theta_n) = \lim_{n \rightarrow +\infty} E(f(W, \theta_n)) = E(\lim_{n \rightarrow +\infty} f(W, \theta_n)) = E(f(W, \theta)) = h(\theta),$$

because for every w , $f(w, \theta)$ is continuous, so $f(w, \theta_n)$ converges towards $f(w, \theta)$. The dominated convergence theorem gives us a condition under which one can indeed revert these two operators: if $|f(W, \theta)|$ is bounded by a function of W which does not depend on θ and has a finite expectation, one can revert these two operators. The second statement gives you a condition under which you can revert the expectation and the derivative operators. Its logic is the same as for the first statement, as derivatives are limits. One possible way of showing that f satisfies the second condition listed in point a) is to show that $E\left(\sup_{\theta \in \Theta} |f(W, \theta)|\right) < +\infty$. Then, $g(w) = \sup_{\theta \in \Theta} |f(w, \theta)|$ is such that $|f(w, \theta)| \leq g(w)$, and $E(g(W)) < +\infty$. Similarly, f will satisfy the second condition listed in point b) if $E\left(\sup_{\theta \in \Theta} |\dot{f}(W, \theta)|\right) < +\infty$.

I introduce a number of notations, which we will use as shortcuts in our derivations of the proofs of this chapter.

Definition 6.1.4 *Probabilistic $o_P()$ and $O_P()$*

Let (X_n) and (Y_n) be two sequences of iid random variables or vectors.

- $X_n = o_P(1)$ if X_n converges in probability towards 0.
- $X_n = o_P(Y_n)$ if there exists a sequence of random variables or vectors (R_n) which is an $o_P(1)$ and such that $X_n = Y_n R_n$ (if X_n , Y_n , and R_n are vectors, the equality must hold coordinate by coordinate).
- $X_n = O_P(1)$ if X_n converges in distribution towards a random variable or vector X finite with probability 1.
- $X_n = O_P(Y_n)$ if there exists a sequence of random variables or vectors (R_n) which is an $O_P(1)$ and such that $X_n = Y_n R_n$ (if X_n , Y_n , and R_n are vectors, the equality must hold coordinate by coordinate).

An $o_P(1)$ is merely a sequence of random variables converging towards 0 in probability. For instance, $\widehat{E(Y)} - E(Y)$ is an $o_P(1)$. $X_n = o_P(Y_n)$ if $\frac{X_n}{Y_n}$ is an $o_P(1)$. For instance, $\widehat{(E(Y) - E(Y))^2} = o_P(\widehat{E(Y)} - E(Y))$. An $O_P(1)$ sequence is merely a sequence converging in distribution towards a finite random variable or vector X . Finally, $X_n = O_P(Y_n)$ if $\frac{X_n}{Y_n}$ is an $O_P(1)$.

Theorem 6.1.5 *Rules of algebra for $o_P(1)$ and $O_P(1)$*

- $o_P(1) + o_P(1) = o_P(1)$

- $o_P(1) \times o_P(1) = o_P(1)$
- $O_P(1) + O_P(1) = O_P(1)$
- $O_P(1) \times O_P(1) = O_P(1)$
- $o_P(1) + O_P(1) = O_P(1)$
- $o_P(1) \times O_P(1) = o_P(1)$
- $o_P(Y_n) = Y_n o_P(1)$

The first four statements follow from the continuous mapping theorem, for convergence in probability (1 and 2) and in distribution (3 and 4). Which result should you use to prove 5 and 6? How should you prove the 7th?

Lemma 6.1.1 If $X_n \xrightarrow{P} 0$, $o(\|X_n\|) = o_P(\|X_n\|)$.

Proof

Let X_n be a sequence of vectors in $\mathbb{R}^k$ converging towards 0 in probability. Let $f : \mathbb{R}^k \mapsto \mathbb{R} : f(x) = o(\|x\|)$. Let $g(x) = \frac{f(x)}{\|x\|}$ if $x \neq 0$ and $g(0) = 0$. Notice that g is continuous at 0 (why?). Then, it follows from the continuous mapping theorem that $g(X_n)$ converges in probability towards 0: $g(X_n) = o_P(1)$. Since $f(X_n) = g(X_n)\|X_n\|$, $f(X_n) = o_P(1)\|X_n\|$. Therefore, $f(X_n) = o_P(\|X_n\|)$.

QED.

6.2 The delta method

The delta method is a very convenient tool to infer the limiting distribution of $\sqrt{n}(\Phi(\hat{\theta}) - \Phi(\theta))$ from that of $\sqrt{n}(\hat{\theta} - \theta)$, provided Φ is continuously differentiable. This method relies on a Taylor expansion of Φ around θ , hence the need to introduce Taylor expansions, $o()$, $o_P()$, and the link between them in the previous section. Before stating and proving formally the delta method theorem in the general case where θ can be a k -dimensional parameter, I start giving you a heuristic presentation of the argument on which this method relies for a real parameter. Assume $\hat{\theta}$ is a $\sqrt{n}$ -consistent and asymptotically normal estimator of θ , with asymptotic variance σ^2 , and Φ is a continuously differentiable function. Because $\sqrt{n}$ -consistency implies consistency, $\hat{\theta}$ converges in probability towards θ . As a result, when n gets large the two get

closer and closer to each other, which in turn implies that

$$\begin{aligned}
& \frac{\Phi(\hat{\theta}) - \Phi(\theta)}{\hat{\theta} - \theta} \approx \dot{\Phi}(\theta) \\
\Leftrightarrow & \frac{\sqrt{n}(\Phi(\hat{\theta}) - \Phi(\theta))}{\sqrt{n}(\hat{\theta} - \theta)} \approx \dot{\Phi}(\theta) \\
\Leftrightarrow & \sqrt{n}(\Phi(\hat{\theta}) - \Phi(\theta)) \approx \sqrt{n}(\hat{\theta} - \theta)\dot{\Phi}(\theta) \\
\Leftrightarrow & \sqrt{n}(\Phi(\hat{\theta}) - \Phi(\theta)) \approx N(0, \sigma^2)\dot{\Phi}(\theta) \\
\Leftrightarrow & \sqrt{n}(\Phi(\hat{\theta}) - \Phi(\theta)) \approx N(0, \sigma^2 \times \dot{\Phi}(\theta)^2).
\end{aligned}$$

This “proves” that $\sqrt{n}(\Phi(\hat{\theta}) - \Phi(\theta))$ has the same limiting distribution as $\sqrt{n}(\hat{\theta} - \theta)$, except that one has to multiply its asymptotic variance by $\dot{\Phi}(\theta)^2$. This implies that

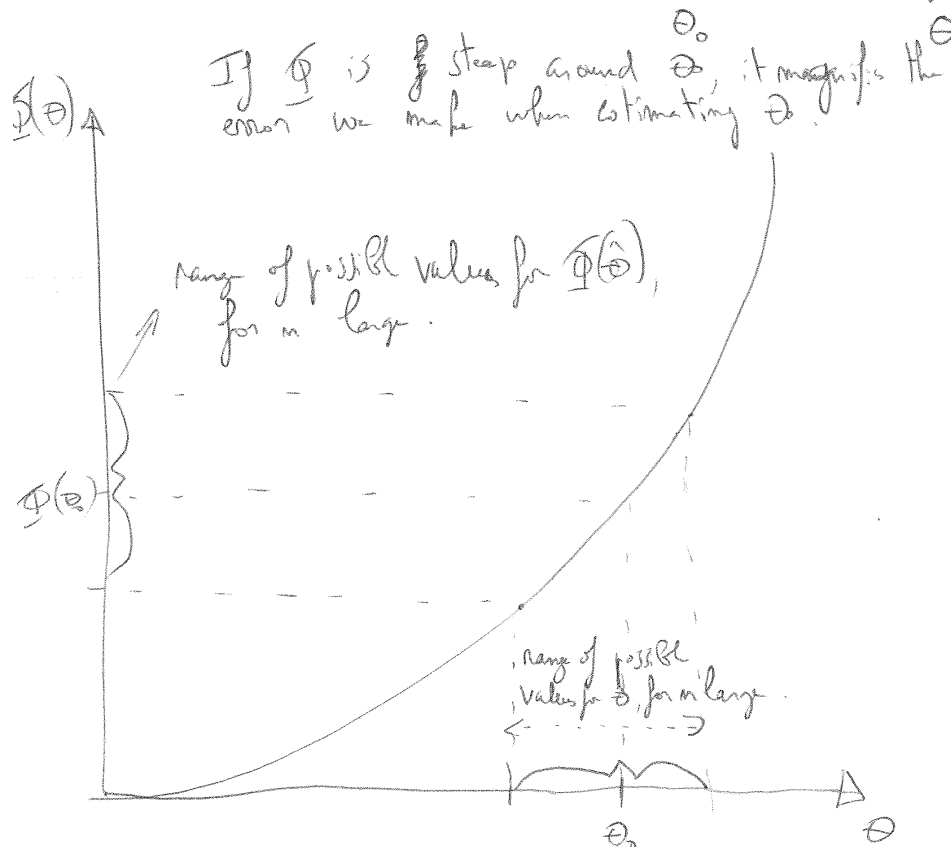
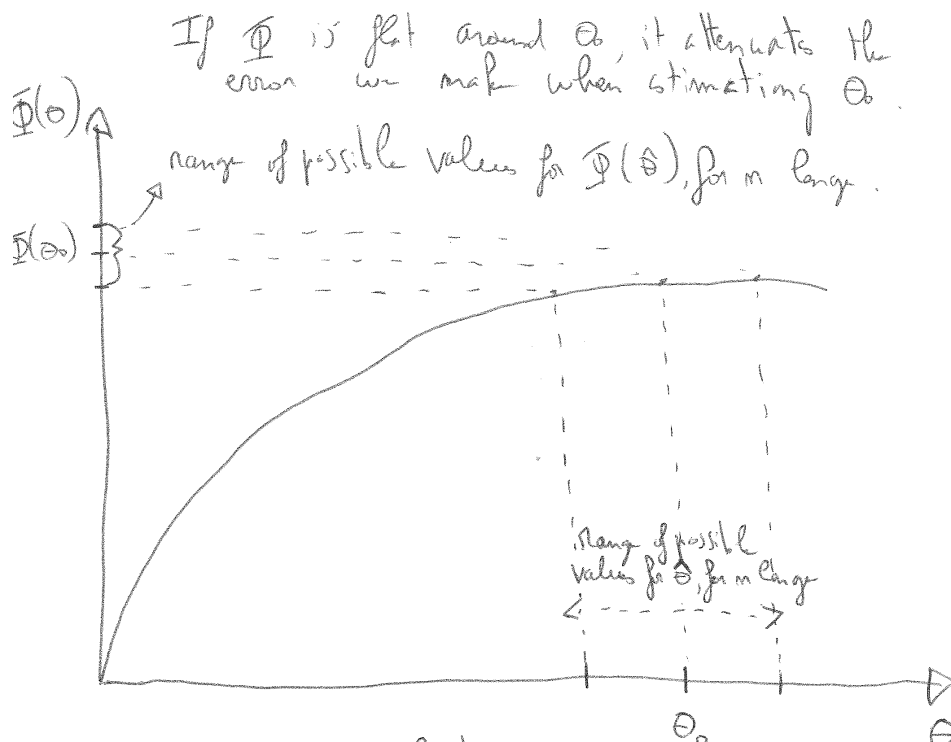
$$\Phi(\hat{\theta}) \approx \Phi(\theta) + \frac{N(0, 1)}{\sqrt{n}} \sigma \dot{\Phi}(\theta),$$

while

$$\hat{\theta} \approx \theta + \frac{N(0, 1)}{\sqrt{n}} \sigma.$$

If $\dot{\Phi}(\theta) < 1$, the convergence from $\Phi(\hat{\theta})$ to $\Phi(\theta)$ is faster than that from $\hat{\theta}$ to θ . [What is the intuition for this result?](#)

If Φ is very flat around θ , the difference between $\Phi(\hat{\theta})$ and $\Phi(\theta)$ will be smaller than the difference between $\hat{\theta}$ and θ : Φ attenuates the error we make when estimating θ . On the contrary, if $\dot{\Phi}(\theta) > 1$, Φ is very steep around θ , so it magnifies the error we make when estimating θ .



Theorem 6.2.1 *The delta method.*

Let $\hat{\theta}$ be an estimator of some parameter θ belonging to Θ , a compact subset of $\mathbb{R}^k$ and such that $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V)$. Let $\Phi: \Theta \mapsto \mathbb{R}$ be continuously differentiable. Then,

$$\sqrt{n}(\Phi(\hat{\theta}) - \Phi(\theta)) \xrightarrow{d} N(0, \dot{\Phi}(\theta)' V \dot{\Phi}(\theta)).$$

Proof

It follows from the third Taylor formula in 6.1.2 that

$$\Phi(\hat{\theta}) - \Phi(\theta) = \dot{\Phi}(\theta)' (\hat{\theta} - \theta) + o(\|\hat{\theta} - \theta\|).$$

Since $\hat{\theta} - \theta$ converge towards 0 in probability, it follows from Lemma 6.1.1 that $o(\|\hat{\theta} - \theta\|)$ is an $o_P(\|\hat{\theta} - \theta\|)$, which is also equal to $\|\hat{\theta} - \theta\|_{o_P(1)}$, according to the 7th point of Theorem 6.1.5. Therefore,

$$\sqrt{n} \left(\Phi(\hat{\theta}) - \Phi(\theta) \right) = \dot{\Phi}(\theta)' \sqrt{n} (\hat{\theta} - \theta) + \sqrt{n} \|\hat{\theta} - \theta\|_{o_P(1)}. \quad (6.2.1)$$

Let's study the behavior of the last term in the previous equation.

$$\begin{aligned} \sqrt{n} \|\hat{\theta} - \theta\|_{o_P(1)} &= \|\sqrt{n}(\hat{\theta} - \theta)\|_{o_P(1)} \\ &= O_P(1) o_P(1) \\ &= o_P(1). \end{aligned}$$

That $\|\sqrt{n}(\hat{\theta} - \theta)\|$ is an $O_P(1)$ follows from the fact that $\sqrt{n}(\hat{\theta} - \theta)$ converges in distribution towards a normal vector X , so by the continuous mapping theorem $\|\sqrt{n}(\hat{\theta} - \theta)\| \xrightarrow{d} \|X\|$. The last equality follows from Theorem 6.1.5, point 6.

Plugging the last equality into Equation (6.2.1) yields

$$\sqrt{n} \left(\Phi(\hat{\theta}) - \Phi(\theta) \right) = \dot{\Phi}(\theta)' \sqrt{n} (\hat{\theta} - \theta) + o_P(1).$$

Therefore, it follows from the Slutsky lemma that $\sqrt{n} \left(\Phi(\hat{\theta}) - \Phi(\theta) \right)$ has the same asymptotic distribution as $\dot{\Phi}(\theta)' \sqrt{n} (\hat{\theta} - \theta)$. Since $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V)$, this distribution is the same as the one of $\dot{\Phi}(\theta)' N(0, V)$. This yields the result, following Theorem 4.2.2.

QED.

The Delta method is an interesting tool per se, because many estimators of interest write as $\Phi(\hat{\theta})$, where $\hat{\theta}$ is a simple estimator whose limiting distribution is easily obtained using for instance the central limit theorem, while $\Phi(\cdot)$ is a continuously differentiable function. But the Delta method is also a good pedagogical introduction to M and GMM estimators, whose asymptotic behavior is also obtained through Taylor expansions.

6.3 Definition and examples of M-estimators

6.3.1 Definition of M-estimands and M-estimators

Many parameters of interest in statistics and econometrics write as solutions of a population minimization problem. Natural candidates to estimate this type of parameters are the vectors

which minimize the sample counterpart of this population minimization problem. This type of estimators are called M-estimators. Let $(W_i)_{1 \leq i \leq n}$ be n iid random variables or vectors with the same distribution as some random variable or vector W .

Definition 6.3.1 *M-estimands and m-estimators*

1. An estimand $\theta_0 \in \mathbb{R}^k$ is called an m-estimand if it satisfies the following condition.³

$$\theta_0 = \underset{\theta \in \Theta}{\operatorname{argmin}} E(m(W, \theta)), \quad (6.3.1)$$

for a vector of random variables $W \in \mathbb{R}^l$, a set $\Theta \subseteq \mathbb{R}^k$ the estimand is known to belong to, and a known function $m : \mathbb{R}^l \times \Theta \mapsto \mathbb{R}$.

2. The m-estimator $\hat{\theta}$ of θ_0 is defined as⁴

$$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n m(W_i, \theta). \quad (6.3.2)$$

M-estimators are natural candidates to estimate m-estimands. By the law of large numbers, $\frac{1}{n} \sum_{i=1}^n m(W_i, \theta)$ should not be too far from $E(m(W, \theta))$. Therefore, minimizing $\frac{1}{n} \sum_{i=1}^n m(W_i, \theta)$ should not be too different from minimizing $E(m(W, \theta))$, which finally implies that $\hat{\theta}$ should not be too different from θ_0 . This heuristic argument suggests that $\hat{\theta}$ should be a good estimator of θ_0 . In the next section, we will show that this heuristic argument happens to be true under some assumptions. Before this, I review examples of very well-known estimands and estimators which can be viewed as m-estimands and m-estimators.

6.3.2 Population and sample mean

Let Y be a random variable, and let $(Y_i)_{1 \leq i \leq n}$ be an iid sample with the same distribution as Y .

Example 1 *Population and sample mean*

Let Y be a random variable admitting a second moment.

1. $E(Y)$ is an m-estimand, with $\Theta = \mathbb{R}$, $W = Y$, and $m(y, \theta) = (y - \theta)^2$.
2. The corresponding m-estimator is the sample mean $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$.

Proof

³For now, I am somewhat loose in my notations, as I do not consider the possibility that the minimizer might not be unique. I will deal rigorously with this issue in the next section.

⁴Here as well, I am somewhat loose and I do not consider the possibility that there could be multiple minima.

I only prove the first statement, the proof of the second statement is left as an exercise. For any real number θ ,

$$\begin{aligned}
E(m(Y, \theta)) &= E((Y - \theta)^2) \\
&= E((Y - E(Y))^2) + 2E[(Y - E(Y))(E(Y) - \theta)] + E((E(Y) - \theta)^2) \\
&= E((Y - E(Y))^2) + (E(Y) - \theta)^2 \\
&\geq E((Y - E(Y))^2) = E(m(Y, E(Y))).
\end{aligned}$$

QED.

6.3.3 Population and sample median

Let Y be a random variable with a continuous and strictly increasing cdf, and let $(Y_i)_{1 \leq i \leq n}$ be an iid sample with the same distribution as Y . The population median is denoted $me(Y)$, and it is the unique solution of $P(Y \leq y) = \frac{1}{2}$. It follows from this definition that $P(Y < me(Y)) = P(Y > me(Y))$. The sample median is the smallest observation in the sample $(Y_i)_{1 \leq i \leq n}$ such that at least half of the sample is smaller than it. It is denoted $\widehat{me}(Y)$. If the distribution of the $(Y_i)_{1 \leq i \leq n}$ is continuous with no mass point, the probability of having two equal observations is equal to 0: we can assume that all the observations in the sample are different from each other. Let $Y_{(i)}$ denote the i^{th} largest observation in the sample. $\widehat{me}(Y) = Y_{(\frac{n+1}{2})}$ if the number of observations in the sample is odd, and $\widehat{me}(Y) = Y_{(\frac{n}{2})}$ if this number is even. You can consider examples of samples with 3 and 4 observations to check this point. $E(Y) = \underset{\theta \in \Theta}{\operatorname{argmin}} E((Y - \theta)^2)$. $me(Y)$ is also the solution to some population minimization problem. Which one?

Example 2 *Population and sample median.*

Let Y be a continuous random variable admitting a first moment.

1. $me(Y)$ is an m -estimand, with $\Theta = \mathbb{R}$, $W = Y$, and $m(y, \theta) = |y - \theta|$.
2. If the sample size is odd, $\widehat{me}(Y)$ is the corresponding m -estimator. If the sample size is even, any number included between $\widehat{me}(Y) = Y_{(\frac{n}{2})}$ and $Y_{(\frac{n}{2}+1)}$ solves the sample minimization problem.

Proof

For any real number θ strictly greater than $me(Y)$,

$$\begin{aligned}
& E(m(Y, \theta)) - E(m(Y, me(Y))) \\
&= E(|Y - \theta|) - E(|Y - me(Y)|) \\
&= E(|Y - \theta| | Y \geq \theta) P(Y \geq \theta) + E(|Y - \theta| | me(Y) \leq Y < \theta) P(me(Y) \leq Y < \theta) \\
&+ E(|Y - \theta| | Y < me(Y)) P(Y < me(Y)) \\
&- E(|Y - me(Y)| | Y \geq \theta) P(Y \geq \theta) - E(|Y - me(Y)| | me(Y) \leq Y < \theta) P(me(Y) \leq Y < \theta) \\
&- E(|Y - me(Y)| | Y < me(Y)) P(Y < me(Y)) \\
&= E(Y - \theta | Y \geq \theta) P(Y \geq \theta) + E(\theta - Y | me(Y) \leq Y < \theta) P(me(Y) \leq Y < \theta) \\
&+ E(\theta - Y | Y < me(Y)) P(Y < me(Y)) \\
&- E(Y - me(Y) | Y \geq \theta) P(Y \geq \theta) - E(Y - me(Y) | me(Y) \leq Y < \theta) P(me(Y) \leq Y < \theta) \\
&- E(me(Y) - Y | Y < me(Y)) P(Y < me(Y)) \\
&= (me(Y) - \theta) P(Y \geq \theta) + E(\theta + me(Y) - 2Y | me(Y) \leq Y < \theta) P(me(Y) \leq Y < \theta) + (\theta - me(Y)) P(Y < me(Y)) \\
&= (\theta - me(Y)) (P(Y < me(Y)) - P(Y \geq \theta)) + E(\theta + me(Y) - 2Y | me(Y) \leq Y < \theta) P(me(Y) \leq Y < \theta) \\
&= (\theta - me(Y)) (P(Y > me(Y)) - P(Y \geq \theta)) + E(\theta + me(Y) - 2Y | me(Y) \leq Y < \theta) P(me(Y) \leq Y < \theta) \\
&= (\theta - me(Y)) P(me(Y) < Y < \theta) + E(\theta + me(Y) - 2Y | me(Y) \leq Y < \theta) P(me(Y) \leq Y < \theta) \\
&= P(me(Y) \leq Y < \theta) (\theta - me(Y) + E(\theta + me(Y) - 2Y | me(Y) \leq Y < \theta)) \\
&= P(me(Y) \leq Y < \theta) 2E(\theta - Y | me(Y) \leq Y < \theta) \\
&> 0.
\end{aligned}$$

The second equality follows from the law of iterated expectations. The last inequality holds because if the distribution of Y is continuous and $\theta > me(Y)$, so $P(me(Y) \leq Y < \theta) > 0$. Moreover, when $me(Y) \leq Y < \theta$, $\theta - Y$ is strictly positive. Therefore, $E(\theta - Y | me(Y) \leq Y \leq \theta)$ is the expected value of a random variable which is always strictly positive, therefore it is strictly positive.

Using a similar reasoning, one can show that for any $\theta < me(Y)$, $E(m(Y, \theta)) - E(m(Y, me(Y))) < 0$ as well. I leave it to you as an exercise. This proves the first statement.

To prove the second statement for an odd sample size, one first needs to show that for any $\theta > Y_{\frac{n+1}{2}}$,

$$\frac{1}{n} \sum_{i=1}^n m(Y_i, \theta) - \frac{1}{n} \sum_{i=1}^n m(Y_i, Y_{\frac{n+1}{2}}) = \frac{1}{n} \sum_{i=1}^n m(Y_{(i)}, \theta) - \frac{1}{n} \sum_{i=1}^n m(Y_{(i)}, Y_{\frac{n+1}{2}}) \geq 0.$$

This can be achieved splitting the two sums into three parts, distinguishing the i such that $Y_{(i)} \geq \theta$, those such that $Y_{\frac{n+1}{2}} \leq Y_{(i)} < \theta$, and those such that $Y_{(i)} < Y_{\frac{n+1}{2}}$. Then one can use a similar argument to show the same thing for any $\theta < Y_{\frac{n+1}{2}}$. This proof is left as an exercise.

QED.

To compute the sample median, is it useful to know that it solves $\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n |Y_i - \theta|$?

No. Knowing that the sample mean and the sample median are m-estimators is not practically useful. To compute, say, the sample median, you will never solve the minimization problem $\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n |Y_i - \theta|$. It's much faster to order the observations according to their value of Y_i to find the median. However, given that the sample mean and sample median are very useful estimators, and they both happen to be m-estimators, maybe we can propose other useful estimators by considering the solutions to other potentially interesting minimization problems. Therefore, a first merit of m-estimation theory is to guide us in the construction of new, useful estimators. We are going to illustrate this in the remainder of this section. Moreover, maybe we can study the properties of all estimators that satisfy (6.3.2), irrespective of the specific form of the function $m(\cdot)$ involved in the minimization problem. This would allow us to derive simultaneously the asymptotic properties of the sample mean, the sample median, and all the other type of m-estimators we are going to introduce now, without having to study each estimator separately. We will pursue this goal in the next section.

6.3.4 OLS estimand and estimator

Example 3 *Population and sample OLS regression coefficient.*

Let Y be a random variable, let X be a $K \times 1$ vector of random variables, and assume that both Y and X have second moments.

1. β , the OLS regression coefficient of Y on X , is an m-estimand, with $\Theta = \mathbb{R}$, $W = (Y, X)$, and $m(y, x, \theta) = (y - x'\theta)^2$.
2. The corresponding m-estimator is $\hat{\beta}$, the sample OLS regression coefficient of Y on X .

Proof

The first result follows from Equation (3.2.1). The second result follows from Equation (4.1.3).

QED.

6.3.5 Median regression estimand and estimator

The OLS regression function $X'\beta$ is the best linear approximation of the conditional mean function $E(Y|X)$. Instead of modelling how the mean of Y varies with X , one might be interested in modelling how the median of Y varies with X , for instance because the median is less sensitive to outliers. This requires defining the conditional median function. Let Y be a random variable such that $P(Y \leq y|X = x)$ is continuous and strictly increasing in y for every value of x . Let $me(Y|X = x)$ be the median of Y in the distribution of $Y|X = x$. You can think of $me(Y|X = x)$ as the median value of Y among people with $X = x$. More formally, $me(Y|X = x)$ is the unique solution of $P(Y \leq y|X = x) = \frac{1}{2}$. Let $me(Y|X)$ be a random variable which is a function of X and which is equal to $me(Y|X = x)$ when $X = x$.

Following the exact same steps as those we used to prove that $me(Y) = \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} E(|Y - \theta|)$, one can show that $me(Y|X = x) = \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} E(|Y - \theta||X = x)$. ($me(Y|X = x)$ is just the median of a conditional distribution.) This in turn implies that $me(Y|X) = \underset{m(\cdot)}{\operatorname{argmin}} E(|Y - m(X)|)$: $me(Y|X)$ is the function of X which minimizes the expected absolute value of $|Y - m(X)|$. Let's prove it in the case where X can take a finite number of values $\{x_1, \dots, x_J\}$. For any function $m(\cdot)$,

$$\begin{aligned} & E(|Y - m(X)|) - E(|Y - me(Y|X)|) \\ &= \sum_{j=1}^J P(X = x_j) (E(|Y - m(X)||X = x_j) - E(|Y - me(Y|X)||X = x_j)) \\ &= \sum_{j=1}^J P(X = x_j) (E(|Y - m(x_j)||X = x_j) - E(|Y - me(Y|X = x_j)||X = x_j)) \\ &> 0. \end{aligned}$$

Where does the last inequality come from?

From the fact $me(Y|X = x_j) = \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} E(|Y - \theta||X = x_j)$.

So we have $me(Y|X) = \underset{m(\cdot)}{\operatorname{argmin}} E(|Y - m(X)|)$. In Chapter 3, as we had $E(Y|X) = \underset{b \in \mathbb{R}^k}{\operatorname{argmin}} E((Y - m(X))^2)$, we chose to approximate $E(Y|X)$ by $X'\beta$, with $\beta = \underset{b \in \mathbb{R}^k}{\operatorname{argmin}} E((Y - X'b)^2)$. $X'\beta$ prove to be a good approximation of $E(Y|X)$. First, when $E(Y|X)$ is linear, $X'\beta = E(Y|X)$. Second, even when $E(Y|X)$ is not linear, $X'\beta$ is its best linear approximation from a mean squared-error perspective.

Which linear function of X should provide a good approximation of the conditional median function?

Following the same logic as for OLS regression, we will then define the median regression function as $X'\lambda$, with $\lambda = \underset{l \in \mathbb{R}^k}{\operatorname{argmin}} E(|Y - X'l|)$. λ is called the median regression coefficient of Y on X .

Here as well, $X'\lambda$ proves to be a good approximation of $me(Y|X)$. First, if $me(Y|X)$ is a linear function of X , then $me(Y|X) = X'\lambda$. This follows from the fact that $me(Y|X) =$

$\operatorname{argmin}_{m(\cdot)} E(|Y - m(X)|)$. If $me(Y|X)$ is linear, $\operatorname{argmin}_{m(\cdot)} E(|Y - m(X)|) = \operatorname{argmin}_{l \in \mathbb{R}^k} E(|Y - X'l|)$: the two minimization problems have the same solution because we have assumed that the solution to the left hand side one, which screens a large set of functions, actually lies within the set of functions screened in the right hand side minimization problem. An interesting consequence of this result is that in a randomized experiment, the median regression of Y on a constant and D will identify $me(Y(1)) - me(Y(0))$, the effect of the treatment on the median of the outcome. Indeed, $me(Y|D) = me(Y|D = 0) + D(me(Y|D = 1) - me(Y|D = 0))$ is a linear function of a constant and of the binary variable for treatment. Therefore, the coefficient of the constant in a median regression of Y on $(1, D)$ will be equal to $me(Y|D = 0)$, while the coefficient of D will be equal to $me(Y|D = 1) - me(Y|D = 0)$. Then,

$$\begin{aligned}
me(Y|D = 1) - me(Y|D = 0) &= me(Y(1)|D = 1) - me(Y(0)|D = 0) \\
&= \operatorname{argmin}_{\theta \in \mathbb{R}} E(|Y(1) - \theta||D = 1) - \operatorname{argmin}_{\theta \in \mathbb{R}} E(|Y(0) - \theta||D = 0) \\
&= \operatorname{argmin}_{\theta \in \mathbb{R}} E(|Y(1) - \theta|) - \operatorname{argmin}_{\theta \in \mathbb{R}} E(|Y(0) - \theta|) \\
&= me(Y(1)) - me(Y(0)).
\end{aligned}$$

Where does the third equality come from?

In a randomized experiment, $Y(1) \perp\!\!\!\perp D$. Therefore, $|Y(1) - \theta| \perp\!\!\!\perp D$ for any real number θ . This in turn implies $E(|Y(1) - \theta||D = 1) = E(|Y(1) - \theta|)$.

Second, even if $me(Y|X)$ is not linear, Angrist et al. (2006) show that $X'\lambda$ is the best linear approximation of the conditional median from a weighted mean squared-error perspective: $\lambda = \operatorname{argmin}_{l \in \mathbb{R}^k} E(w(l, X)(me(Y|X) - X'l)^2)$, where $w(l, X)$ are some weights. If we omit the weights, for a uni-dimensional X this implies that λ is the slope of the regression line approximating the best the scatter plot of $me(Y|X)$ and X .

A natural estimator of λ is $\hat{\lambda} = \operatorname{argmin}_l \frac{1}{n} \sum_{i=1}^n |Y_i - X_i'l|$. $\hat{\lambda}$ is called the sample median regression coefficient of Y on X . Finding the minimizer of the previous function is not as straightforward as finding the minimizer of the OLS sample problem, because the objective function is not differentiable everywhere. However, Koenker & Bassett Jr (1978) have shown that this problem is actually equivalent to a linear programming problem, something a computer can easily solve.

Example 4 *Population and sample median regression coefficient*

Let Y be a random variable, let X be a $K \times 1$ vector of random variables, and assume that both Y and X have first moments.

1. λ , the median regression coefficient of Y on X is an m -estimand, with $\Theta = \mathbb{R}$, $W = (Y, X)$, and $m(y, x, \theta) = |y - x'\theta|$.
2. The corresponding m -estimator is $\hat{\lambda}$, the sample median regression coefficient of Y on X .

Proof

The two results directly follow from the definition of λ and $\hat{\lambda}$.

QED.

6.3.6 Maximum likelihood estimand and estimator

Parametric models

In many applications of interest, especially in structural econometrics, it is convenient to posit parametric models in which the distribution of the data is known up to a parameter of finite dimension. Positing a parametric model on a random vector W amounts to assuming that the probability density function f_W of W belongs to a family of densities indexed by a finite dimensional parameter θ : $f_W \in \{d_\theta : \theta \in \Theta \subseteq \mathbb{R}^k\}$. Therefore, $f_W = d_{\theta_0}$ for some $\theta_0 \in \Theta$. We call θ_0 the estimand in a parametric model.

When one posits a parametric model, one usually uses the maximum likelihood estimator to estimate θ_0 . Ex-post, once the random vector $(W_i)_{1 \leq i \leq n}$ gets realized and is equal to $(w_i)_{1 \leq i \leq n}$, the maximum likelihood estimator is

$$\hat{\theta} = \operatorname{argmax}_{\theta \in \Theta} \prod_{i=1}^n d_\theta(w_i).$$

When W can take a countable number of values, the likelihood function $\prod_{i=1}^n d_\theta(w_i)$ has a simple interpretation. In this case, for every w in the support of W , $d_\theta(w)$ is equal to $P_\theta(W = w)$, the probability that $W = w$ if the unknown parameter θ_0 is equal to θ . Therefore, the likelihood is just $\prod_{i=1}^n P_\theta(W = w_i)$. What does $\prod_{i=1}^n P_\theta(W = w_i)$ represent?

$\prod_{i=1}^n P_\theta(W = w_i)$ is the probability that n iid random variables with the same distribution as W are equal to $(w_i)_{1 \leq i \leq n}$. Therefore, $\prod_{i=1}^n P_\theta(W = w_i)$ is the probability to observe the realization of $(W_i)_{1 \leq i \leq n}$ we observe in our data if $\theta_0 = \theta$. Therefore, the maximum likelihood estimator picks the value of θ under which we had the highest ex-ante probability of observing the realization of $(W_i)_{1 \leq i \leq n}$ we indeed observe.

Ex-ante, the maximum likelihood estimator is

$$\hat{\theta} = \operatorname{argmax}_{\theta \in \Theta} \prod_{i=1}^n d_{\theta}(W_i).$$

Equivalently, this estimator maximizes the logarithm of the likelihood multiplied by $\frac{1}{n}$:

$$\hat{\theta} = \operatorname{argmax}_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \ln(d_{\theta}(W_i)).$$

As usual, we will study the properties of that estimator from an ex-ante perspective.

Example: Probit model

To fix ideas, let's frame our discussion of the Probit model in a simple example. Assume we are interested in measuring the effect of a treatment whereby senior high school students are shown estimates of returns to going to college. We run a randomized experiment during which some students are exposed to that information while some other students are not. If we also get to observe Y , their decision to enrol into college, we can measure the effect of this treatment of the proportion of students that choose to enroll by running a linear regression of Y on a constant and D . The population coefficient of D in that regression is equal to $E(Y(1) - Y(0)) = P(Y(1) = 1) - P(Y(0) = 1)$. However, we might also be interested in measuring the effect of this treatment on the parameters students use when they make their cost-benefit analysis of whether they should go to college or not. Doing so is interesting, because that can tell us by how much the treatment changes students' valuation of a college degree (in principle, this treatment should not affect their perception of the cost of a college degree).

Assume that

$$\begin{aligned} Y(0) &= 1\{\tilde{Y}(0) \geq 0\} \\ Y(1) &= 1\{\tilde{Y}(1) \geq 0\}. \end{aligned}$$

$\tilde{Y}(0)$ represents the difference between a student's expected costs and benefits of going to college, if that student does not receive the information on returns to going to college. $\tilde{Y}(1)$ represents the difference between a student's expected costs and benefits of going to college, if that student receives the information on returns to going to college. $\tilde{Y}(1) - \tilde{Y}(0)$ is the effect of that information on a student's cost benefit analysis. We will make the following Assumption:

Assumption 7 *Probit model assumptions*

$$\begin{aligned} \tilde{Y}(1) - \tilde{Y}(0) &= \rho_0 \\ \tilde{Y}(0) &\perp\!\!\!\perp D \\ \tilde{Y}(0) &\sim N(\alpha_0, 1). \end{aligned}$$

Let $\varepsilon = \tilde{Y}(0) - \alpha_0$. Under Assumption 7, one has

$$\begin{aligned}
Y &= 1\{D\tilde{Y}(1) + (1-D)\tilde{Y}(0) \geq 0\} \\
&= 1\{D(\tilde{Y}(0) + \rho_0) + (1-D)\tilde{Y}(0) \geq 0\} \\
&= 1\{\tilde{Y}(0) + \rho_0 D \geq 0\} \\
&= 1\{\alpha_0 + \rho_0 D + \varepsilon \geq 0\}.
\end{aligned}$$

Then,

$$\begin{aligned}
P(Y = 1|D = 1) &= P(\alpha_0 + \rho_0 D + \varepsilon \geq 0|D = 1) \\
&= P(\alpha_0 + \rho_0 + \varepsilon \geq 0|D = 1) \\
&= P(\alpha_0 + \rho_0 + \varepsilon \geq 0) \\
&= P(-\varepsilon \leq \alpha_0 + \rho_0) \\
&= \Phi(\alpha_0 + \rho_0).
\end{aligned}$$

Why can we go from line 2 to 3?

Because $\tilde{Y}(0) \perp\!\!\!\perp D$.

Similarly,

$$P(Y = 1|D = 0) = \Phi(\alpha_0).$$

Let $p_0 = P(D = 1)$. We have

$$\begin{aligned}
P(Y = 0, D = 0) &= P(Y = 0|D = 0)P(D = 0) = (1 - \Phi(\alpha_0))(1 - p_0) \\
P(Y = 1, D = 0) &= P(Y = 1|D = 0)P(D = 0) = \Phi(\alpha_0)(1 - p_0) \\
P(Y = 0, D = 1) &= P(Y = 0|D = 1)P(D = 1) = (1 - \Phi(\alpha_0 + \rho_0))p_0 \\
P(Y = 1, D = 1) &= P(Y = 1|D = 1)P(D = 1) = \Phi(\alpha_0 + \rho_0)p_0.
\end{aligned}$$

Now, for any $(\alpha, \rho, p) \in \mathbb{R}^2 \times [0, 1]$, let $d_{\alpha, \rho, p}$ be a function from $\{0, 1\}^2$ onto $\mathbb{R}$ and such that

$$\begin{aligned}
d_{\alpha, \rho, p}(0, 0) &= (1 - \Phi(\alpha))(1 - p) \\
d_{\alpha, \rho, p}(0, 1) &= \Phi(\alpha)(1 - p) \\
d_{\alpha, \rho, p}(1, 0) &= (1 - \Phi(\alpha + \rho))p \\
d_{\alpha, \rho, p}(1, 1) &= \Phi(\alpha + \rho)p.
\end{aligned}$$

The true density function of (Y, D) , $P(Y = y, D = d)$ is equal to $d_{\alpha_0, \rho_0, p_0}$. But because (α_0, ρ_0, p_0) is unknown, it can be any function $d_{\alpha, \rho, p}$, for $(\alpha, \rho, p) \in \mathbb{R}^2$. So under the Probit

model assumptions, the distribution of (Y, D) is known up to a parameter of finite dimension, and (Y, D) follows a parametric model.

Assume we will get to observe a sample $(Y_i, D_i)_{1 \leq i \leq n}$ of n iid observations with the same probability distribution as (Y, D) . Then, if $(\alpha_0, \rho_0, p_0) = (\alpha, \rho, p)$, the likelihood of each observation (Y_i, D_i) is

$$\begin{aligned} d_{\alpha, \rho, p}(Y_i, D_i) &= ((1 - \Phi(\alpha))(1 - p))^{(1-Y_i)(1-D_i)} \\ &\times (\Phi(\alpha)(1 - p))^{Y_i(1-D_i)} \\ &\times ((1 - \Phi(\alpha + \rho))p)^{(1-Y_i)D_i} \\ &\times (\Phi(\alpha + \rho)p)^{Y_i D_i}. \end{aligned}$$

The likelihood of the sample is

$$\begin{aligned} \Pi_{i=1}^n d_{\alpha, \rho, p}(Y_i, D_i) &= \Pi_{i=1}^n \left[((1 - \Phi(\alpha))(1 - p))^{(1-Y_i)(1-D_i)} \right. \\ &\times (\Phi(\alpha)(1 - p))^{Y_i(1-D_i)} \\ &\times ((1 - \Phi(\alpha + \rho))p)^{(1-Y_i)D_i} \\ &\times \left. (\Phi(\alpha + \rho)p)^{Y_i D_i} \right]. \end{aligned}$$

Finally, the log-likelihood of the sample is

$$\begin{aligned} L((Y_i, D_i); \alpha, \rho, p) &= \sum_{i=1}^n [(1 - Y_i)(1 - D_i) (\ln(1 - \Phi(\alpha)) + \ln(1 - p)) \\ &+ Y_i(1 - D_i) (\ln(\Phi(\alpha)) + \ln(1 - p)) \\ &+ (1 - Y_i)D_i (\ln(1 - \Phi(\alpha + \rho)) + \ln(p)) \\ &+ Y_i D_i (\ln(\Phi(\alpha + \rho)) + \ln(p))]. \end{aligned}$$

The maximum likelihood estimator of (α_0, ρ_0, p_0) is $(\hat{\alpha}, \hat{\rho}, \hat{p}) = \underset{(\alpha, \rho, p) \in \mathbb{R}^2 \times [0, 1]}{\operatorname{argmax}} L((Y_i, D_i); \alpha, \rho, p)$.

Notice that maximizing the log-likelihood with respect to (α, ρ) alone will yield the same estimators $(\hat{\alpha}, \hat{\rho})$ as maximizing the log-likelihood with respect to (α, ρ, p) : the problem is additively separable in (α, ρ) and p , so the FOC determining (α, ρ) are the same in the first and in the second problem. When one estimates a probit regression of Y on a constant and D with Stata, the coefficients of the constant and D are $(\hat{\alpha}, \hat{\rho}) = \underset{(\alpha, \rho) \in \mathbb{R}^2}{\operatorname{argmax}} L((Y_i, D_i); \alpha, \rho, p)$.

In the next subsection, we will show that $(\hat{\alpha}, \hat{\rho})$ is a consistent and asymptotically normal estimator of (α_0, ρ_0) (under some conditions).

This shows that under the probit model assumptions, one can consistently estimate ρ_0 , the effect of letting students know the returns to a college degree on their valuation of that degree. The probit model has three assumptions. The first is a constant treatment effect assumption: letting students known the returns to a college degree should have the same effect on the valuation of a college degree of every student. This is a strong assumption. Without it, it is no longer possible to use a probit model to estimate the effect of the treatment. One does

not have that without constant treatment effect, the probit model estimates the average effect of the treatment. The second assumption will automatically be satisfied if D_i is randomly assigned, otherwise it should be carefully discussed. The third assumption can be relaxed. Manski (1975) indeed shows that one can estimate (α_0, ρ_0) without imposing that $\tilde{Y}(0)$ follow a normal distribution, but only imposing $\tilde{Y}(1) - \tilde{Y}(0) = \rho_0$ and $\tilde{Y}(0) \perp\!\!\!\perp D$. I refer you to pages 3270-3272 in the Handbook of Econometrics Chapter by Arellano & Honoré (2001) for a clear and insightful presentation of Manski's idea.

The probit model is the simplest possible example of what economists typically call a discrete choice model. Discrete choice models use choice decisions to infer the preference parameters that led to the observed choice. Provided one imposes a distributional assumption on the random component of the choice, one can write down the likelihood function of the observed choices, and use maximum likelihood to estimate the parameters of the model. As discussed in the Probit example, the distributional assumption is not key for identification. On the other hand, identification in this type of models usually relies on the assumption that the effect of the regressor of interest is constant across units. Discrete choice models are very commonly used in structural econometrics.

The estimand in a parametric model is an M-estimand.

Example 5 *Let W be a random vector whose probability density function d_{θ_0} belongs to a parametric family of densities $\{d_\theta : \theta \in \Theta \subseteq \mathbb{R}^k\}$ such that for any θ , $d_\theta \neq d_{\theta_0}$ on at least a non empty open subset of the support of W .*

1. θ_0 is an m -estimand, with $m(w, \theta) = -\ln(d_\theta(w))$.
2. The corresponding m -estimator is the maximum likelihood estimator.

Proof

To prove the first statement, I assume that W is a continuous random variable with real support (I take this assumption only to make the proof less abstract, the result holds without

it).

$$\begin{aligned}
& E(\ln(d_\theta(W))) - E(\ln(d_{\theta_0}(W))) \\
&= E\left(\ln\left(\frac{d_\theta(W)}{d_{\theta_0}(W)}\right)\right) \\
&\leq E\left(2\sqrt{\frac{d_\theta(W)}{d_{\theta_0}(W)}} - 2\right) \\
&= 2E\left(\sqrt{\frac{d_\theta(W)}{d_{\theta_0}(W)}}\right) - 2 \\
&= 2\int_{-\infty}^{+\infty} \sqrt{\frac{d_\theta(x)}{d_{\theta_0}(x)}} d_{\theta_0}(x) dx - 2 \\
&= 2\int_{-\infty}^{+\infty} \sqrt{d_\theta(x)d_{\theta_0}(x)} dx - 2 \\
&= \int_{-\infty}^{+\infty} 2\sqrt{d_\theta(x)d_{\theta_0}(x)} - d_\theta(x) - d_{\theta_0}(x) dx \\
&= -\int_{-\infty}^{+\infty} \left(\sqrt{d_\theta(x)} - \sqrt{d_{\theta_0}(x)}\right)^2 dx \\
&\leq 0
\end{aligned}$$

The first inequality follows from the fact that for any $x > 0$, $\ln(x) \leq 2\sqrt{x} - 2$ (this is because $x \mapsto 2\sqrt{x} - 2 - \ln(x)$ reaches a minimum at $x = 1$, where it is equal to 0). The next equality follows from the linearity of the expectation operator. The following one follows from the fact that for any function h , $E(h(W)) = \int_{-\infty}^{+\infty} h(w)d_{\theta_0}(w)dw$, because $d_{\theta_0}(w)$ is the probability density of W . The last but one equality holds because $d_\theta(x)$ and $d_{\theta_0}(x)$ are densities, therefore their integrals are equal to 1. This implies that for any θ ,

$$E(\ln(d_\theta(W))) \leq E(\ln(d_{\theta_0}(W))).$$

Notice that for any $\theta \neq \theta_0$, the inequality is strict: by assumption, d_{θ_0} and d_θ differ at least on a non-empty interval, therefore $\int_{-\infty}^{+\infty} \left(\sqrt{d_\theta(x)} - \sqrt{d_{\theta_0}(x)}\right)^2 dx > 0$. This completes the proof of the first statement.

The proof of the second statement directly follows from the definition of the maximum likelihood estimator.

QED.

6.4 Consistency and asymptotic normality of M-estimators

For every θ in Θ , let $M_n(\theta) = \frac{1}{n} \sum_{i=1}^n m(W_i, \theta)$, and let $M(\theta) = E(m(W, \theta))$. With these notations, the m-estimand introduced in Definition 6.3.1 is $\theta_0 = \underset{\theta \in \Theta}{\operatorname{argmin}} M(\theta)$, and its m-estimator is $\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} M_n(\theta)$. In the preceding section I outlined an informal argument

which suggests that $\hat{\theta}$ might converge towards θ : as for every θ , $M_n(\theta) \xrightarrow{P} M(\theta)$, when n gets large minimizing $M_n(\theta)$ is the same thing as minimizing $M(\theta)$, so $\hat{\theta}$ will converge towards θ . This happens to be true under some assumptions.

First, for $\hat{\theta}$ to converge towards θ_0 , θ_0 should be the unique solution of the minimization problem (6.3.1).

Assumption 8 *Uniqueness of the minimizer.*

For any $\theta \neq \theta_0$, $M(\theta_0) < M(\theta)$.

If assumption 8 is not satisfied, there is at least one another value of θ , $\tilde{\theta}_0$, such that $M(\tilde{\theta}_0) = M(\theta_0)$. Assume there is exactly one such point. When n will grow, $M_n(\theta)$ will become close to $M(\theta)$, but this does not imply that $\hat{\theta}$ will become close to θ_0 . Indeed, it follows from the central limit theorem that $M_n(\theta_0) \approx M(\theta_0) + \frac{X_1}{\sqrt{n}}\sigma_{\theta_0}^2$ and $M_n(\tilde{\theta}_0) \approx M(\tilde{\theta}_0) + \frac{X_2}{\sqrt{n}}\sigma_{\tilde{\theta}_0}^2$ where $\sigma_{\theta_0}^2$ and $\sigma_{\tilde{\theta}_0}^2$ are the asymptotic variances of $M_n(\theta_0)$ and $M_n(\tilde{\theta}_0)$, and X_1 and X_2 are two $N(0, 1)$ variables. To simplify, let's assume that $\sigma_{\theta_0}^2 = \sigma_{\tilde{\theta}_0}^2$. Even if $M(\tilde{\theta}_0) = M(\theta_0)$ and even if n is extremely large, as soon as X_1 and X_2 are not perfectly correlated, there is a strictly positive probability that $X_2 < X_1$, in which case we will have $M_n(\tilde{\theta}_0) < M_n(\theta_0)$, so our estimator will be equal to $\tilde{\theta}_0$, not to θ_0 . This explains why Assumption 8 is necessary for $\hat{\theta}$ to converge towards θ_0 .

We also need the parameter space to be compact.

Assumption 9 *Compactness of the parameter space, and interior solution to the minimization problem*

(a) Θ is a compact subset of $\mathbb{R}^k$.

(b) θ_0 is in the interior of Θ .

A subset of $\mathbb{R}^k$ is compact if and only if it is closed and bounded. For instance, $[0, 2]$ is a compact subset of $\mathbb{R}$. On the contrary, $(0, 2]$ and $[2, +\infty)$ are not compact: the first set is not closed, while the second one is not bounded. Similarly, $[0, 2] \times [0, 4]$ is a compact subset of $\mathbb{R}^2$, but $(0, 2] \times [0, 4]$ is not because it is not closed.

Finally, we need to impose regularity conditions on the function $m(\cdot)$.

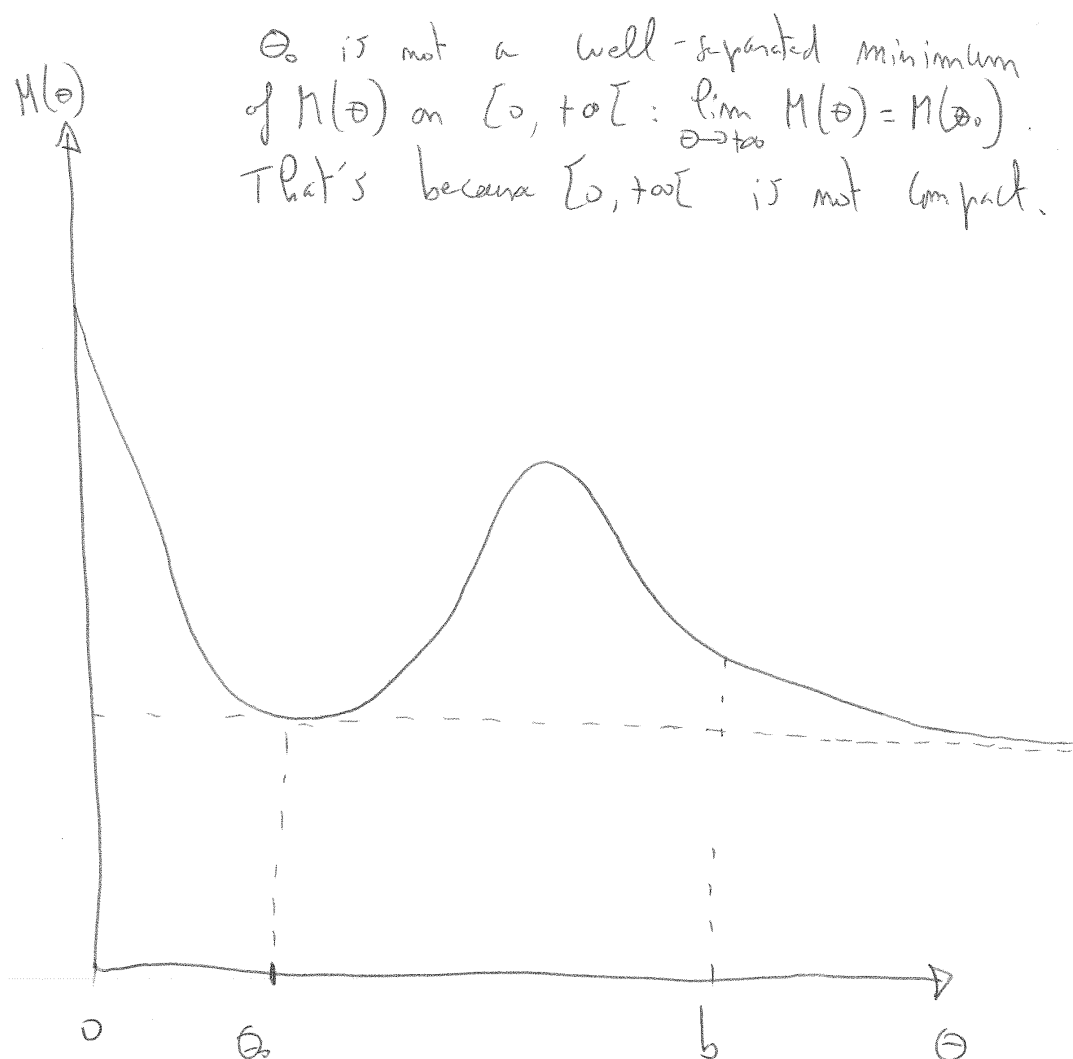
Assumption 10 *Regularity and boundedness conditions on m .*

(a) For every $w \in \mathbb{R}^l$, $\theta \mapsto m(w, \theta)$ is continuous on Θ .

(b) $E \left(\sup_{\theta \in \Theta} |m(W, \theta)| \right) < +\infty$.

Compactness of Θ ensures that θ_0 , the unique minimizer of $M(\theta)$, is a well-separated minimum, i.e. that there does not exist points in Θ far away from θ_0 such that $M(\theta)$ can be made arbitrarily close to $M(\theta_0)$. If that were to be the case, we would “almost” be in the situation

where $M(\theta)$ is minimized at two different points, so we would be back to the problem I described when I explained why Assumption 8 is necessary for consistency.



You can check that on any compact set included in $[0, +\infty[$, θ_0 is a well-separated minimum. For instance, it is well-separated on $[0, b]$.

The next lemma shows that if Θ is compact, θ_0 is well-separated.

Lemma 6.4.1 *The unique minimum of a continuous function on a compact is well separated. If Assumptions 8, 9, and 10, are satisfied, θ_0 is a well separated minimum of $M(\cdot)$: for every $\varepsilon > 0$,*

$$\inf_{\theta: d(\theta, \theta_0) \geq \varepsilon, \theta \in \Theta} M(\theta) > M(\theta_0). \quad (6.4.1)$$

Proof

We will use a reductio ad absurdo reasoning. Assume there is a strictly positive ε such that $\inf_{\theta: d(\theta, \theta_0) \geq \varepsilon, \theta \in \Theta} M(\theta) = M(\theta_0)$. $\{\theta : d(\theta, \theta_0) \geq \varepsilon, \theta \in \Theta\}$ is a closed subset of Θ . Because Θ is compact, $\{\theta : d(\theta, \theta_0) \geq \varepsilon, \theta \in \Theta\}$ is closed and bounded. This set is therefore compact. It follows from Assumption 10 and the dominated convergence theorem that $M(\cdot)$ is continuous. A continuous function on a compact reaches a minimum. Therefore, there is a $\theta^* \in \Theta$ such that $d(\theta^*, \theta_0) \geq \varepsilon$, and $\inf_{\theta: d(\theta, \theta_0) \geq \varepsilon, \theta \in \Theta} M(\theta) = M(\theta^*)$. This implies that $M(\theta^*) = M(\theta_0)$. This contradicts Assumption 8, hence the result.

QED.

Finally, in my informal argument, I asserted that as for every θ , $M_n(\theta) \xrightarrow{P} M(\theta)$, when n gets large minimizing $M_n(\theta)$ is the same thing as minimizing $M(\theta)$. This is not strictly speaking correct. For the minimization of $M_n(\theta)$ to become the same thing as the minimization of $M(\theta)$ when n gets large, we need that $\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{P} 0$, which is stronger than having that $M_n(\theta) \xrightarrow{P} M(\theta)$ for every θ . To have that $\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{P} 0$, we need the uniform weak law of large numbers to apply, which is the reason why we need Assumption 10.

Theorem 6.4.1 *The uniform weak law of large numbers*

Let Θ be a subset of $\mathbb{R}^k$ satisfying part a) of Assumption 9, and let m be a function satisfying Assumption 10. Then,

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n m(W_i, \theta) - E(m(W, \theta)) \right| \xrightarrow{P} 0.$$

Theorem 6.4.2 *Consistency of m -estimators*

If Assumptions 8, 9, and 10 are satisfied, $\hat{\theta} \xrightarrow{P} \theta_0$.

Proof

$$\begin{aligned} |M(\hat{\theta}) - M(\theta_0)| &= M(\hat{\theta}) - M(\theta_0) \\ &= M(\hat{\theta}) - M_n(\theta_0) + M_n(\theta_0) - M(\theta_0) \\ &= M(\hat{\theta}) - M_n(\theta_0) + o_P(1) \\ &\leq M(\hat{\theta}) - M_n(\hat{\theta}) + o_P(1) \\ &\leq \sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| + o_P(1) \\ &= o_P(1) + o_P(1) \\ &= o_P(1). \end{aligned}$$

The first equality follows from the definition of θ_0 . The third follows from the weak law of large numbers, which implies that $M_n(\theta_0) - M(\theta_0) = o_P(1)$. The first inequality follows from the definition of $\hat{\theta}$, which implies that $M_n(\hat{\theta}) \leq M_n(\theta_0)$. The last but one equality follows from the uniform weak law of large numbers, which applies because all of its conditions are satisfied under Assumptions 9 and 10, and which implies that $\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| = o_P(1)$. The last equality follows from Theorem 6.1.5. The previous inequality implies that for any $\eta > 0$, $P(|M(\hat{\theta}) - M(\theta_0)| \geq \eta) \leq P(o_P(1) \geq \eta)$ which converges to 0 by definition of an $o_P(1)$. This shows that $M(\hat{\theta}) \xrightarrow{P} M(\theta_0)$.

Then, for any $\varepsilon > 0$, let $\eta(\varepsilon) = \inf_{\theta: d(\theta, \theta_0) \geq \varepsilon, \theta \in \Theta} M(\theta) - M(\theta_0)$. It follows from Lemma 6.4.1 that $\eta(\varepsilon) > 0$. If $d(\hat{\theta}, \theta_0) \geq \varepsilon$, then $M(\hat{\theta}) - M(\theta_0) \geq \eta(\varepsilon)$. Therefore,

$$0 \leq P(d(\hat{\theta}, \theta_0) \geq \varepsilon) \leq P(M(\hat{\theta}) - M(\theta_0) \geq \eta(\varepsilon)),$$

which converges to 0 as we have shown above. This proves that $P(d(\hat{\theta}, \theta_0) \geq \varepsilon) \rightarrow 0$. The result follows from the definition of convergence in probability.

QED.

Now that we have consistency, we would like to show the asymptotic normality of $\hat{\theta}$. On that purpose, one can impose stronger regularity conditions on $\theta \mapsto m(w, \theta)$, and do a Taylor expansion of $M_n(\hat{\theta})$ around θ_0 , in the spirit of what we did when we considered the delta method. Informally, if $\theta \mapsto m(w, \theta)$ is twice differentiable and satisfies boundedness conditions ensuring that dominated convergence theorems apply, $M_n(\theta)$ is twice differentiable. From the definition of $\hat{\theta}$, we must have $\dot{M}_n(\hat{\theta}) = 0$. Expanding the last equation between $\hat{\theta}$ and θ_0 , yields $\dot{M}_n(\theta_0) + \ddot{M}_n(\theta_0)(\hat{\theta} - \theta_0) \approx 0$. Multiplying by $\sqrt{n}$ and rearranging yields $\sqrt{n}(\hat{\theta} - \theta_0) = -\ddot{M}_n(\theta_0)^{-1} \sqrt{n} \dot{M}_n(\theta_0)$. $\sqrt{n} \dot{M}_n(\theta_0)$ is a standardized sum of iid variables with mean 0, so it converges towards a normal distribution.

I will now state precisely the conditions under which the previous reasoning is indeed correct, and I will then state and prove the result.

Assumption 11 *Further regularity conditions on m .*

(a) *For every $w \in \mathbb{R}^l$, $\theta \mapsto m(w, \theta)$ is twice continuously differentiable on Θ . Let $\theta \mapsto \dot{m}(w, \theta)$ and $\theta \mapsto \ddot{m}(w, \theta)$ denote its first and second derivatives wrt θ .*

(b) $E(\sup_{\theta \in \Theta} |\dot{m}(W, \theta)|) < +\infty$.

(c) $E(\sup_{\theta \in \Theta} |\ddot{m}(W, \theta)|) < +\infty$.

(d) $E(\sup_{\theta \in \Theta} |\dot{m}(W, \theta) \dot{m}(W, \theta)'|) < +\infty$.

(e) $E(\ddot{m}(W, \theta_0))$ is invertible.

Theorem 6.4.3 *Asymptotic normality of m-estimators under strong regularity conditions*
If Assumptions 8, 9, 10, and 11 are satisfied,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, E(\ddot{m}(W, \theta_0))^{-1} E(\dot{m}(W, \theta_0) \dot{m}(W, \theta_0)') E(\ddot{m}(W, \theta_0))^{-1}).$$

Moreover,

$$\begin{aligned} & \left(\frac{1}{n} \sum_{i=1}^n \ddot{m}(W_i, \hat{\theta}) \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \dot{m}(W_i, \hat{\theta}) \dot{m}(W_i, \hat{\theta})' \right) \left(\frac{1}{n} \sum_{i=1}^n \ddot{m}(W_i, \hat{\theta}) \right)^{-1} \\ & \xrightarrow{P} E(\ddot{m}(W, \theta_0))^{-1} E(\dot{m}(W, \theta_0) \dot{m}(W, \theta_0)') E(\ddot{m}(W, \theta_0))^{-1}. \end{aligned}$$

Proof

In this proof, we assume that the parameter space Θ is a closed ball in $\mathbb{R}^k$.⁵ Assumption 11 a) implies that $M_n(\theta)$ is twice continuously differentiable. Then, for n large enough,

$$\begin{aligned} 0 &= \dot{M}_n(\hat{\theta}) \\ &= \dot{M}_n(\theta_0) + \ddot{M}_n(\tilde{\theta})(\hat{\theta} - \theta_0), \end{aligned}$$

for some (random) $\tilde{\theta}$ included between θ_0 and $\hat{\theta}$. Combined with the fact that $M_n(\theta)$ is twice continuously differentiable, the definition of $\hat{\theta}$ implies that $\hat{\theta}$ must satisfy $\dot{M}_n(\hat{\theta}) = 0$, which yields the first equality.⁶ The second equality is obtained as follows. $\dot{M}_n(\hat{\theta})$ is a $k \times 1$ vector whose j^{th} coordinate is $\frac{\partial \dot{M}_n}{\partial x_j}(\hat{\theta})$. $\frac{\partial \dot{M}_n}{\partial x_j}(\theta)$ is continuous and differentiable with respect to θ . Therefore, it follows from the fourth statement of Theorem 6.1.2 that $\frac{\partial \dot{M}_n}{\partial x_j}(\hat{\theta}) = \frac{\partial \dot{M}_n}{\partial x_j}(\theta_0) + \frac{\partial \dot{M}_n}{\partial x_j}(\tilde{\theta})'(\hat{\theta} - \theta_0)$, for some random $\tilde{\theta}$ included between θ_0 and $\hat{\theta}$. This proves the second equality, once noted that the $k \times k$ matrix whose j^{th} line is $\frac{\partial \dot{M}_n}{\partial x_j}(\tilde{\theta})'$, the vector of derivatives of the j^{th} first order derivative, is actually equal to $\ddot{M}_n(\tilde{\theta})$, the matrix of second order derivatives.⁷

Rearranging the last equation and multiplying by $\sqrt{n}$ yields

$$\begin{aligned} \ddot{M}_n(\tilde{\theta})\sqrt{n}(\hat{\theta} - \theta_0) &= -\sqrt{n}\dot{M}_n(\theta_0) \\ &= -\frac{1}{\sqrt{n}} \sum_{i=1}^n \dot{m}(W_i, \theta_0) \\ &= -\frac{1}{\sqrt{n}} \sum_{i=1}^n (\dot{m}(W_i, \theta_0) - E(\dot{m}(W, \theta_0))). \end{aligned} \tag{6.4.2}$$

⁵This is without loss of generality: if Θ is not a closed ball in $\mathbb{R}^k$, we can extend our parameter space to be a closed ball in $\mathbb{R}^k$ including Θ . Given that all of the m-estimands we have considered are actually minimizers of $M(\theta)$ over $\mathbb{R}^k$, extending the parameter space does not change the definition of θ_0 .

⁶For this to be true, $\hat{\theta}$ must lie in the interior of Θ . As θ_0 lies in the interior of Θ by Assumption 9 and $\hat{\theta} \xrightarrow{P} \theta_0$, one can show that $\hat{\theta}$ lies in the interior of Θ with a probability approaching 1 when n grows, hence the qualifier “for n large enough” in the beginning of the derivation.

⁷There is a very slight abuse of notation here: $\tilde{\theta}$ might not be the same for each j . This will eventually prove not to matter, as each of the $\tilde{\theta}$ will converge towards θ_0 .

The first equality follows from the definition of $M_n(\theta_0)$. The second holds because $E(\dot{m}(W, \theta_0)) = 0$. Indeed, it follows from part b) of Assumption 11 and the dominated convergence theorem that $M(\theta)$ is continuously differentiable, with $\dot{M}(\theta) = E(\dot{m}(W, \theta))$. Then, it follows from its definition and from part b) of Assumption 9 that θ_0 must satisfy $E(\dot{m}(W, \theta_0)) = \dot{M}(\theta_0) = 0$.

As part d) of Assumption 11 implies that $\dot{m}(W, \theta_0)$ has a second moment, it follows from the central limit theorem that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (\dot{m}(W_i, \theta_0) - E(\dot{m}(W, \theta_0))) \xrightarrow{d} N(0, V(\dot{m}(W, \theta_0))). \quad (6.4.3)$$

Moreover, let $\ddot{M}_{n,ij}(\tilde{\theta})$ and $E(\ddot{m}_{ij}(W, \theta_0))$ respectively denote the element on the i th line and j th column of $\ddot{M}_n(\tilde{\theta})$ and $E(\ddot{m}(W, \theta_0))$.

$$\begin{aligned} \left| \ddot{M}_{n,ij}(\tilde{\theta}) - E(\ddot{m}_{ij}(W, \theta_0)) \right| &= \left| \frac{1}{n} \sum_{i=1}^n \ddot{m}_{ij}(W_i, \tilde{\theta}) - E(\ddot{m}_{ij}(W, \tilde{\theta})) + E(\ddot{m}_{ij}(W, \tilde{\theta})) - E(\ddot{m}_{ij}(W, \theta_0)) \right| \\ &\leq \left| \frac{1}{n} \sum_{i=1}^n \ddot{m}_{ij}(W_i, \tilde{\theta}) - E(\ddot{m}_{ij}(W, \tilde{\theta})) \right| + \left| E(\ddot{m}_{ij}(W, \tilde{\theta})) - E(\ddot{m}_{ij}(W, \theta_0)) \right| \\ &\leq \sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \ddot{m}_{ij}(W_i, \theta) - E(\ddot{m}_{ij}(W, \theta)) \right| + \left| E(\ddot{m}_{ij}(W, \tilde{\theta})) - E(\ddot{m}_{ij}(W, \theta_0)) \right| \\ &= o_P(1) + o_P(1) \\ &= o_P(1). \end{aligned}$$

That $\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \ddot{m}_{ij}(W_i, \theta) - E(\ddot{m}_{ij}(W, \theta)) \right|$ is an $o_P(1)$ follows from the uniform weak law of large numbers, which we can apply to $\ddot{m}_{ij}(W, \theta)$ under part a) and c) of Assumption 11. That $\left| E(\ddot{m}_{ij}(W, \tilde{\theta})) - E(\ddot{m}_{ij}(W, \theta_0)) \right|$ is an $o_P(1)$ follows from the combination of the three following arguments:

- (i) $\tilde{\theta} \xrightarrow{P} \theta_0$ as $\tilde{\theta}$ is included between θ_0 and $\hat{\theta}$ and $\hat{\theta} \xrightarrow{P} \theta_0$.
- (ii) $E(\ddot{m}_{ij}(W, \theta))$ is continuous in θ . This follows from part c) of Assumption 11, and the dominated convergence theorem.
- (iii) The continuous mapping theorem.

The previous display implies that $\ddot{M}_n(\tilde{\theta}) \xrightarrow{P} E(\ddot{m}(W, \theta_0))$. Once combined with Slutsky lemma this implies that

$$\ddot{M}_n(\tilde{\theta}) \sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} E(\ddot{m}(W, \theta_0)) \sqrt{n}(\hat{\theta} - \theta_0). \quad (6.4.4)$$

Combining Equations (6.4.2), (6.4.3), and (6.4.4) yields

$$E(\ddot{m}(W, \theta_0)) \sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V(\dot{m}(W, \theta_0))).$$

Multiplying each side of the previous display by $E(\ddot{m}(W, \theta_0))^{-1}$ yields

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} E(\ddot{m}(W, \theta_0))^{-1} N(0, V(\dot{m}(W, \theta_0))).$$

$V(\dot{m}(W, \theta_0)) = E(\dot{m}(W, \theta_0)\dot{m}(W, \theta_0)'),$ because $E(\dot{m}(W, \theta_0)) = 0$. Moreover,

$$E(\ddot{m}(W, \theta_0))^{-1}N(0, E(\dot{m}(W, \theta_0)\dot{m}(W, \theta_0)')) \sim N(0, E(\ddot{m}(W, \theta_0))^{-1}E(\dot{m}(W, \theta_0)\dot{m}(W, \theta_0)')E(\ddot{m}(W, \theta_0))^{-1}).$$

This follows from Theorem 4.2.2, and from the fact $E(\ddot{m}(W, \theta_0))^{-1}$ is a symmetric matrix.

This completes the proof of the first statement.

Let us turn to the proof of the second statement.⁸

$$\begin{aligned} \left| \frac{1}{n} \sum_{i=1}^n \ddot{m}(W_i, \hat{\theta}) - E(\ddot{m}(W, \theta_0)) \right| &\leq \left| \frac{1}{n} \sum_{i=1}^n \ddot{m}(W_i, \hat{\theta}) - E(\ddot{m}(W, \hat{\theta})) \right| + \left| E(\ddot{m}(W, \hat{\theta})) - E(\ddot{m}(W, \theta_0)) \right| \\ &\leq \sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \ddot{m}(W_i, \theta) - E(\ddot{m}(W, \theta)) \right| + \left| E(\ddot{m}(W, \hat{\theta})) - E(\ddot{m}(W, \theta_0)) \right|. \end{aligned}$$

It follows from part a) and c) of Assumption 11 that $\ddot{m}$ satisfies the necessary conditions for the uniform weak law of large numbers to apply. Therefore,

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \ddot{m}(W_i, \theta) - E(\ddot{m}(W, \theta)) \right| = o_P(1).$$

Moreover, it follows from part a) and c) of Assumption 11 and the dominated convergence theorem that $\theta \mapsto E(\dot{m}(W, \theta))$ is continuous in θ . As $\hat{\theta} \mapsto \theta_0$, it follows from the continuous mapping theorem that

$$E(\dot{m}(W, \hat{\theta})) \xrightarrow{P} E(\dot{m}(W, \theta_0)).$$

Together with the previous inequality, the two preceding displays imply that

$$\frac{1}{n} \sum_{i=1}^n \ddot{m}(W_i, \hat{\theta}) \xrightarrow{P} E(\ddot{m}(W, \theta_0)).$$

One can use similar arguments to show that $\frac{1}{n} \sum_{i=1}^n \dot{m}(W_i, \hat{\theta})\dot{m}(W_i, \hat{\theta})' \xrightarrow{P} E(\dot{m}(W, \theta_0)\dot{m}(W, \theta_0)').$

Then, the result follows from the continuous mapping theorem.

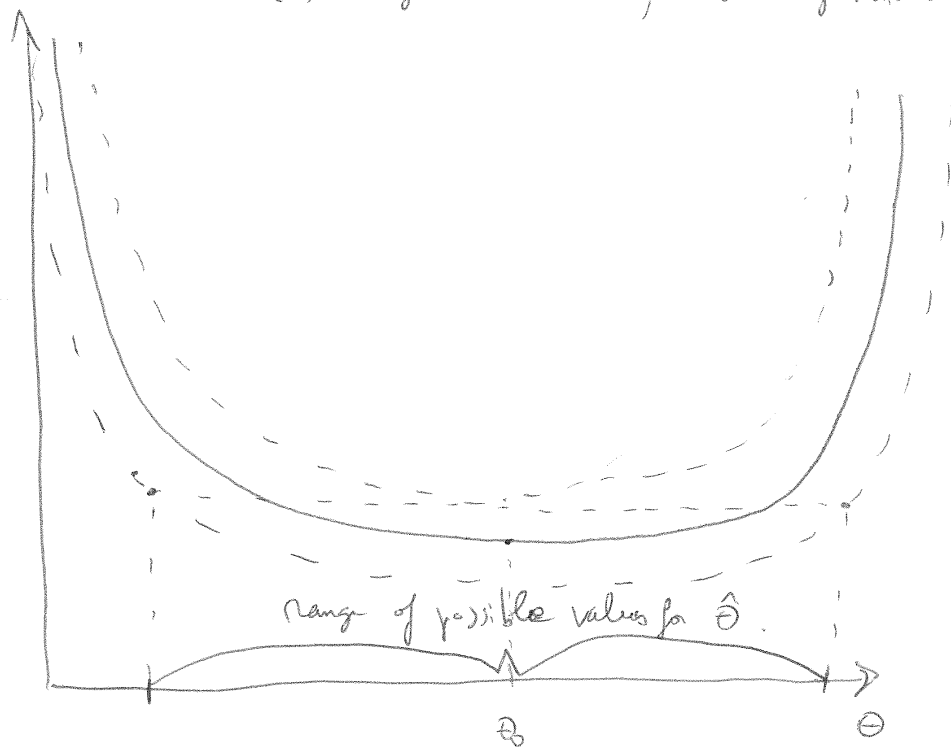
QED.

Once again, the formula for the asymptotic variance of the m-estimator is (relatively) intuitive. To simplify, consider the case where $k = 1$. The asymptotic variance is then equal to $\frac{E(\dot{m}(W, \theta_0)^2)}{(E(\ddot{m}(W, \theta_0)))^2}$. To estimate θ_0 , we need to a) estimate the derivative of $M(\theta)$ and b) find the value of θ which cancels the estimated derivative. a) is the reason why our estimator is not equal to θ_0 : maybe our estimator of the derivative of $M(\theta)$ is far from the true derivative, so cancelling it will be far from cancelling the true derivative. This uncertainty is reflected in the numerator of $\frac{E(\dot{m}(W, \theta_0)^2)}{(E(\ddot{m}(W, \theta_0)))^2}$. Indeed, $E(\dot{m}(W, \theta_0)^2)$ is the asymptotic variance of $\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n \dot{m}(W_i, \theta_0) - E(\dot{m}(W, \theta_0)) \right)$. It is therefore a measure of how fast the estimated derivative of $M(\theta)$, $\frac{1}{n} \sum_{i=1}^n \dot{m}(W_i, \theta_0)$, converges towards the true derivative, $E(\dot{m}(W, \theta_0))$.

⁸With a slight abuse of notation, the following inequalities should be understood as holding for each element of the matrix $\frac{1}{n} \sum_{i=1}^n \ddot{m}(W_i, \hat{\theta}) - E(\ddot{m}(W, \theta_0))$.

Then, b), the search for the value of θ which cancels the estimated derivative, will either magnify or attenuate the error we made in a), depending on the curvature of $M(\theta)$ around θ_0 . If $\ddot{M}(\theta) = E(\ddot{m}(W, \theta_0))$ is large, $M(\theta)$ is fairly steep around θ_0 , so for θ s distant enough from θ_0 , $M(\theta) \gg M(\theta_0)$, so despite the fact $M_n(\theta)$ is an estimate of $M(\theta)$, it is likely that $M_n(\theta) \gg M_n(\theta_0)$ as well, so we will not pick up these distant enough θ s as our estimator of θ_0 . If $\ddot{M}(\theta) = E(\ddot{m}(W, \theta_0))$ is low, $M(\theta)$ is fairly flat around θ_0 , so even for θ s fairly far away from θ_0 , $M(\theta) \approx M(\theta_0)$, so we might pick up these fairly far away θ s as our estimator of θ_0 .

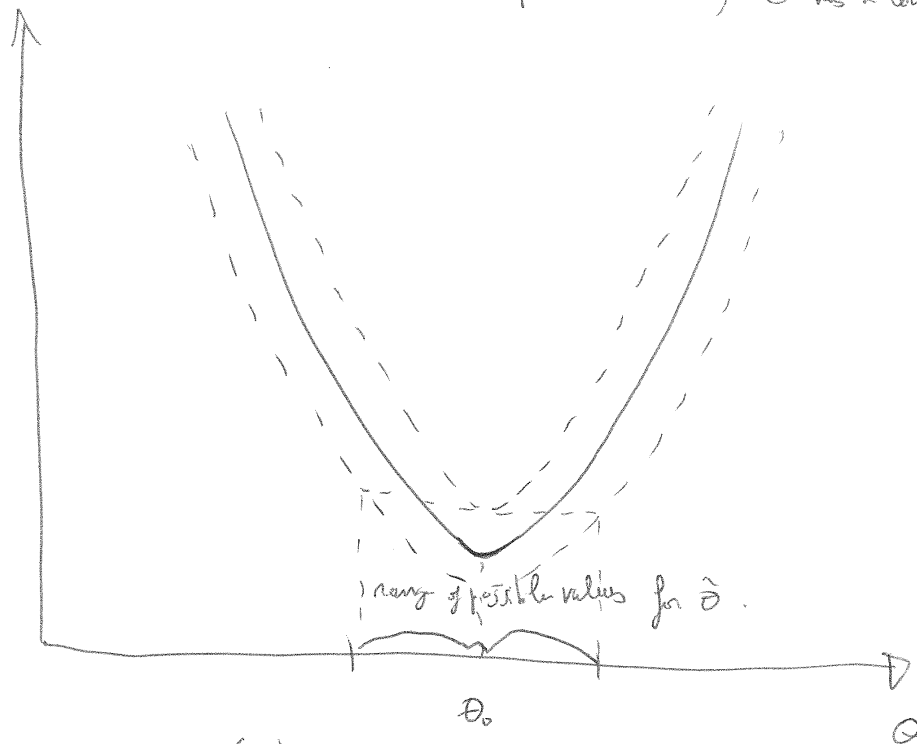
When $\pi(\theta)$ is flat around θ_0 , $\hat{\theta}$ has large variance.



— $\pi(\theta)$

--- range of possible values for $\pi_m(\theta)$, for m large.

When $\eta(\theta)$ is steep around θ_0 , $\hat{\theta}$ has a low variance.



— $\eta(\theta)$

--- Range of possible values for $\eta_n(\theta)$, for n large.

Now that we have shown asymptotic normality of m -estimators and that we have derived a consistent estimator of their variance, we can derive confidence intervals and statistical tests for any type of m -estimand satisfying Assumptions 8 to 11. Let's now assess whether the examples of m -estimands we introduced in the previous section satisfy these conditions.

Assumption 8 is satisfied in Examples 1 to 5. In the proof of Example 1, we have shown that $E((Y - \theta)^2) = E((Y - E(Y))^2) + (\theta - E(Y))^2$. Therefore, for any $\theta \neq E(Y)$, $E((Y - \theta)^2) > E((Y - E(Y))^2)$. This shows that Assumption 8 is satisfied in this example. In the proof of Example 2, we have shown that for any $\theta \neq me(Y)$, $E(|Y - \theta|) > E(|Y - me(Y)|)$, which shows that Assumption 8 is satisfied in this example as well. It is also satisfied in Example 3, provided the standard rank condition is satisfied. Indeed, In Chapter 3, we have shown that if Assumption 3 is satisfied, $\beta = E(XX')^{-1}E(XY)$ is the unique solution to the first order conditions of the OLS minimization problem, implying that β is the unique minimizer of $E((Y - X'\beta)^2)$. One can show that it is also satisfied in Example 4, provided a condition fairly similar to the standard rank condition in linear regression models is satisfied. Finally, we have shown in the proof of example 5 that θ_0 is the unique minimizer of $\theta \mapsto E(-\ln(d_\theta(W)))$.

Strictly speaking, the first part of Assumption 9 does not always apply. For instance, in Example 1, if Y is a bounded variable such as GPA, we know that $E(Y) \in \Theta = [0, 4]$, which is a compact set. But if Y is an unbounded random variable, such as wages, there is no natural upper bound for $E(Y)$. However, imposing ex-ante that $E(Y)$ belongs to a very large compact set, say $[0, 10^{23}]$, does not seem to be a very strong assumption: the average wage in, say, the US is unlikely to be greater than 10^{23} . So imposing part a) of Assumption 9 is only a very minor loss of generality, even for estimands with no natural upper or lower bound. As for the second part of Assumption 9, it only amounts to imposing that all the coordinates of θ_0 are finite. In Example 1, this only rules out $E(Y) = +\infty$ for instance.

The first part of Assumption 10 is satisfied in all the examples we saw together. In Example 1, $\theta \mapsto (y - \theta)^2$ is continuous in θ for every y . In Example 2, $\theta \mapsto |y - \theta|$ is continuous in θ for every y . In Example 3, $\theta \mapsto (y - x'\theta)^2$ is continuous in θ for every (y, x) . In Example 4, $\theta \mapsto |y - x'\theta|$ is continuous in θ for every (y, x) . Finally, in Example 5, $\theta \mapsto \ln(d_\theta(w))$ is continuous in θ for every w in many parametric models. For instance, in the probit model example, $W = (Y, D)$, and

$$\begin{aligned} (\alpha, \rho, p) \mapsto \ln(d_{\alpha, \rho, p}(y, d)) &= (1 - y)(1 - d) \ln((1 - \Phi(\alpha))(1 - p)) \\ &+ y(1 - d) \ln(\Phi(\alpha)(1 - p)) \\ &+ (1 - y)d \ln((1 - \Phi(\alpha + \rho))p) \\ &+ yd \ln(\Phi(\alpha + \rho)p) \end{aligned}$$

is continuous for every $(y, d) \in \{0, 1\}^2$.

The second part of Assumption 10 is usually satisfied in Examples 1 to 5, thanks to Assumption 9. For instance, in Example 1, if $\Theta = [a, b]$, for every $y \geq 0$ we have

$$(y - \theta)^2 = y^2 - 2\theta y + \theta^2 \leq y^2 - 2ya + b^2.$$

For every $y \leq 0$ we have

$$(y - \theta)^2 = y^2 - 2\theta y + \theta^2 \leq y^2 - 2yb + b^2.$$

Therefore,

$$\sup_{\theta \in \Theta} (y - \theta)^2 \leq \max(y^2 - 2ya + b^2, y^2 - 2yb + b^2).$$

This function has a finite expectation under the assumptions we introduced in Example 1. Indeed,

$$\begin{aligned} E(\max(Y^2 - 2Ya + b^2, Y^2 - 2Yb + b^2)) &\leq E(|Y^2 - 2Ya + b^2|) + E(|Y^2 - 2Yb + b^2|) \\ &\leq E(Y^2) + 2aE(|Y|) + b^2 + E(Y^2) + 2bE(|Y|) + b^2 \\ &< +\infty. \end{aligned}$$

One can also show that this assumption is satisfied in the other examples.

Since Assumptions 8, 9, and 10 are satisfied in Examples 1 to 5, Theorem 6.4.2 implies that all the estimators introduced in these examples are consistent.

Asymptotic normality requires a further condition, Assumption 11. This condition is satisfied in Example 1. $m(y, \theta) = (y - \theta)^2$ is twice continuously differentiable with respect to θ for every y . This proves that a) is satisfied. $\dot{m}(y, \theta) = 2(\theta - y)$. If $\Theta = [a, b]$, one can use the same tricks as those used in the last but one paragraph to show that $E\left(\sup_{\theta \in \Theta} |2(\theta - Y)|\right) < +\infty$. This proves that b) is satisfied. $\ddot{m}(y, \theta) = 2$. It is straightforward that $E\left(\sup_{\theta \in \Theta} |2|\right) < +\infty$. This proves that c) is satisfied. $\dot{m}(y, \theta)^2 = 2(\theta - y)^2$. If $\Theta = [a, b]$, one can use again the same tricks as those used in the last but one paragraph to show that $E\left(\sup_{\theta \in \Theta} 2(\theta - Y)^2\right) < +\infty$. This proves that d) is satisfied. Finally, $E(\ddot{m}(Y, \theta)) = 2$ is invertible. This proves that e) is satisfied. Finally,

$$\begin{aligned} &E(\ddot{m}(Y, E(Y))^{-1} E(\dot{m}(Y, E(Y)) \dot{m}(Y, E(Y))') E(\ddot{m}(Y, E(Y))^{-1}) \\ &= \frac{1}{2} E(2(E(Y) - Y) 2(E(Y) - Y)) \frac{1}{2} \\ &= E((Y - E(Y))^2) \\ &= V(Y). \end{aligned}$$

Therefore, it follows from Theorem 6.4.3 that

$$\sqrt{n}(\widehat{E(Y)} - E(Y)) \xrightarrow{d} N(0, V(Y)),$$

something we already knew from the central limit theorem. Assumption 11 is also satisfied in Examples 3 and 5, something you will show in an exercise. This will allow you to derive the asymptotic distribution of the OLS and maximum likelihood estimators.

But Assumption 11 is not satisfied in Examples 2 and 4. $m(y, \theta) = |y - \theta|$ is not differentiable at $\theta = y$. Since it is differentiable everywhere else, one could omit this problem by defining its first order derivative as $\dot{m}(y, \theta) = -sg(y - \theta)$. The proof of Theorem 6.4.3 would go through up to the point where we need to take the inverse of $E(\ddot{m}(y, \theta_0))$. As per our definition of

$\dot{m}(y, \theta)$, $\ddot{m}(y, \theta) = 0$ (except in $\theta = y$, a fact we can once again omit), and 0 does not have an inverse.

To derive asymptotic normality of the sample median and of the median regression coefficient, we must use the following Theorem, which relies on weaker assumptions than Theorem 6.4.3, but which is also more difficult to prove. The proof is much beyond the scope of this course, you will only show as an exercise that Example 2 satisfies all the conditions necessary for the theorem to apply, and use it to derive the asymptotic distribution of the sample median.

Assumption 12 *Weaker further regularity conditions on m .*

(a) $\theta \mapsto m(w, \theta)$ is differentiable at θ_0 with derivative $\dot{m}(w, \cdot)$ for almost every $w \in \mathbb{R}^l$.⁹

(b) For every $(\theta_1, \theta_2) \in \Theta^2$, $|m(w, \theta_1) - m(w, \theta_2)| \leq \|\theta_1 - \theta_2\|$.

(c) $\theta \mapsto M(\theta)$ is twice continuously differentiable.

(d) $\ddot{M}(\theta_0)$ is invertible.

Theorem 6.4.4 *Asymptotic normality of m -estimators under weaker regularity conditions*
If Assumptions 8, 9, 10, and 12 are satisfied,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \ddot{M}(\theta_0)^{-1} E(\dot{m}(W, \theta_0) \dot{m}(W, \theta_0)') \ddot{M}(\theta_0)^{-1}).$$

6.5 The generalized method of moments

A number of parameters of interest in this course cancel out systems of moment conditions.

Definition 6.5.1 *Generalized method of moments estimands*

An estimand $\theta_0 \in \Theta \subseteq \mathbb{R}^k$ is called a GMM estimand if it satisfies the following condition:

$$E(z(W, \theta_0)) = 0, \tag{6.5.1}$$

where W is a vector of random variables in $\mathbb{R}^m$, and z is a function defined on $\mathbb{R}^m \times \mathbb{R}^k$ and taking values in $\mathbb{R}^l$.

For every $1 \leq j \leq l$, let $z_j(w, \theta)$ denote the j th coordinate of $z(w, \theta)$. $E(z(W, \theta_0)) = 0$ means that for every $1 \leq j \leq l$, we should have $E(z_j(W, \theta_0)) = 0$. $E(z(W, \theta_0)) = 0$ therefore defines a system of l equations. This system is called a system of moment conditions.

An important example of a GMM estimand is β , the vector of coefficients of the OLS regression of Y on X .

⁹This means that $\theta \mapsto m(w, \theta)$ might not be differentiable at θ_0 for a countable set of w .

Example 6 *The OLS coefficient is a GMM estimand*

Let Y be a random variable, let X be a $K \times 1$ vector of random variables, and assume that both Y and X have second moments. Let β denote the OLS regression coefficient of Y on X . β satisfies the following system of moment conditions

$$E(X(Y - X'\beta)) = 0. \quad (6.5.2)$$

These moment conditions have the same structure as those introduced in Equation (6.5.1), with $W = (Y, X)$, and $z(y, x, \beta) = x(y - x'\beta)$.

Proof

That β satisfies Equation (6.5.2) directly follows from (3.2.3).

QED.

To estimate GMM estimands, a first solution is to solve the sample counterpart of the system. We could define $\hat{\theta}$ as the solution of the following system of l equations:

$$\frac{1}{n} \sum_{i=1}^n z(W_i, \theta) = 0.$$

When we have as many moment conditions as parameters to be estimated, i.e. when $k = l$, this system will often have exactly one solution, so our estimator will be well-defined. When $l = k$, we say that the system of moment conditions is just identified. In example 6, it follows from Equation (4.1.4) that the solution of the following system of k equations

$$\frac{1}{n} \sum_{i=1}^n X_i(Y_i - X_i'\beta) = 0 \quad (6.5.3)$$

is $\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i'\right)^{-1} \frac{1}{n} \sum_{i=1}^n X_i Y_i$, the sample OLS coefficient.

However, in a number of instances we will focus on GMM estimands whose dimension is lower than the number of moment conditions they satisfy ($k < l$). In such instances, the sample counterpart of the system of moment conditions will not have an exact solution. It will not be possible to find one value of θ such that

$$\frac{1}{n} \sum_{i=1}^n z(W_i, \theta) = 0,$$

because this system has too few unknowns to cancel too many equations. When $l > k$, we say that the system is overidentified: we have more knowledge on the parameters than the amount we would need to estimate them. To be able to cover both the just and the overidentified cases, we will define the GMM estimator of a GMM estimand as the minimizer of the quadratic norm of $\overline{z(W, \theta)} = \frac{1}{n} \sum_{i=1}^n z(W_i, \theta)$ for some positive definite matrix S .¹⁰ Thus, $\hat{\theta}$ will be the value of θ which makes $\overline{z(W, \theta)}$ as close as possible to 0. Before defining the estimator, I have to define positive definite and positive semi-definite matrixes.

¹⁰In this section, I will often use the following notation: for any iid sample of random vectors $(X_i)_{1 \leq i \leq n}$, $\overline{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

Definition 6.5.2 *Positive definite and positive semi-definite matrixes*

Let S be a symmetric $l \times l$ matrix.

1. S is positive definite if and only if for every $l \times 1$ vector x different from 0, $x'Sx > 0$.
2. S is positive definite if and only if there is a positive definite matrix A such that $S = A^2$. A is denoted $S^{\frac{1}{2}}$.
3. S is positive definite if and only if all of its eigen values are strictly positive.
4. S is positive semi-definite if and only if for every $k \times 1$ vector x , $x'Sx \geq 0$.
5. S is positive semi-definite if and only if there is a positive semi-definite matrix A such that $S = A^2$. A is denoted $S^{\frac{1}{2}}$.
6. S is positive semi-definite if and only if all of its eigen values are positive.

For those of you who want to understand why these three definitions are equivalent, this can be seen using the spectral decomposition theorem, which says that any symmetric matrix is diagonalizable in an orthogonal basis: there is a diagonal matrix D and an orthogonal matrix O such that $S = ODO'$.

Definition 6.5.3 *The GMM estimator*

For any $l \times l$ positive definite matrix S , the GMM estimator associated to the moment conditions in Equation (6.5.1) and to S is

$$\hat{\theta}(S) = \underset{\theta \in \Theta}{\operatorname{argmin}} \overline{z(W, \theta)' S z(W, \theta)}. \quad (6.5.4)$$

In the just identified case, there will most often be a $\hat{\theta}^*$ which cancels exactly $\overline{z(W, \theta)}$. Then $\hat{\theta}(S) = \hat{\theta}^*$, irrespective of the choice of S . Therefore, the choice of S does not matter. For instance, it follows from Equation (4.1.4) that $\hat{\beta}$, the sample OLS coefficient, exactly cancels Equation (6.5.3). So for every matrix S ,

$$\underset{b \in \mathbb{R}^k}{\operatorname{argmin}} \overline{X(Y - X'b)' S X(Y - X'b)} = \hat{\beta}.$$

But in the overidentified case, the choice of S will in general matter. To understand this, let's consider two potential choices of S . For $S_1 = I$, the objective function to be minimized in Equation (6.5.4) is $\sum_{j=1}^l \overline{z_j(W, \theta)}^2$. This objective function gives the same weight to each constraint: if $\overline{z_1(W, \theta)}$ is far away from 0, this will penalize the objective function by the same amount as if $\overline{z_2(W, \theta)}$ is far away from 0. Then, let $S_2 = I + 1999 \times A$, where A is an $l \times l$ matrix whose elements are all equal to 0 except the first line and first column one which is equal to 1. For S_2 , the function to be minimized in Equation (6.5.4) is $2000 \overline{z_1(W, \theta)}^2 + \sum_{j=2}^l \overline{z_j(W, \theta)}^2$. From the perspective of this function, cancelling $\overline{z_1(W, \theta)}$ is much more important than cancelling the other coordinates of $\overline{z(W, \theta)}$. Therefore, $\hat{\theta}(S_2)$ will probably be

such that $\overline{z_1(W, \widehat{\theta}(S_2))}$ is closer to 0 than $\overline{z_j(W, \widehat{\theta}(S_2))}$ for $2 \leq j \leq l$, while $\widehat{\theta}(S_1)$ will not give such a priority to cancelling the first coordinate. In the overidentified case, all the sample moments cannot be jointly set to 0. Different S will put different priorities on which sample moment should be set closest to zero. Therefore, S is generally referred to as a weighing matrix, as it determines the weight given to each sample moment in the minimization.

To estimate minimizers of certain type of population functions, we saw in the previous section that one could use m-estimators, which we defined as minimizers of the sample counterpart of the population function. As $E(z(W, \theta_0)) = 0$ and S is positive definite,

$$\theta_0 = \operatorname{argmin}_{\theta \in \Theta} E(z(W, \theta))' S E(z(W, \theta)). \quad (6.5.5)$$

Therefore, GMM estimators are also minimizers of the sample counterpart of some population function minimized at θ_0 . As m-estimators are consistent and asymptotically normal, it sounds reasonable to assume that GMM estimators as well are consistent and asymptotically normal. This proves to be true, but the m-estimators theorems are not readily applicable to GMM estimators, because the minimization problems in (6.3.1) and (6.5.5) do not have the same structure.

$$\operatorname{argmin}_{\theta \in \Theta} E(z(W, \theta)' S z(W, \theta)) \quad (6.5.6)$$

would define an m-estimand, but GMM estimands are not the solution to problem (6.5.6) but to problem (6.5.5), and this problem does not belong to the class defined in (6.3.1). Still, proving consistency and asymptotic normality of GMM estimators relies on very similar arguments to those we used to prove consistency and asymptotic normality of m-estimators (uniform weak law of large numbers, Taylor expansions of the first order conditions of the minimization problem around the true parameter value...). I will therefore only state these theorems, and the assumptions under which they hold, without proving them.

Assumption 13 *Regularity and boundedness conditions on z .*

- (a) For every $w \in \mathbb{R}^m$, $\theta \mapsto z(w, \theta)$ is continuously differentiable on Θ ,¹¹ with first derivative $\theta \mapsto \dot{z}(w, \theta)$. $\dot{z}(w, \theta)$ is an $l \times k$ matrix, whose j^{th} line is $\dot{z}_j(w, \theta)'$, the vector of the k first order derivatives of $\theta \mapsto z_j(w, \theta)$.
- (b) $E(\sup_{\theta \in \Theta} |z(W, \theta)|) < +\infty$.
- (c) $E(\sup_{\theta \in \Theta} |\dot{z}(W, \theta)|) < +\infty$.
- (d) $E(\sup_{\theta \in \Theta} |z(W, \theta) z(W, \theta)'|) < +\infty$.
- (e) Let $G = E(\dot{z}(W, \theta_0))$. $\operatorname{rank}(G) = k$.
- (f) $V = V(z(W, \theta_0))$ is invertible.

¹¹This means that every coordinate of $z(w, \theta)$, $\theta \mapsto z_j(w, \theta)$, should be continuously differentiable.

a) is a regularity condition similar to the one under which we derived the asymptotic normality of m-estimators. b), c) and d) will most often be satisfied under the assumption that Θ is compact. e) and f) essentially mean that there should not be redundant moment conditions.

Assumption 14 *Uniqueness of the solution of the system of moments.*

$$E(z(W, \theta)) = 0 \Leftrightarrow \theta = \theta_0.$$

If Assumption 14 is not satisfied, there is at least another value of θ , $\tilde{\theta}_0$, such that $E(z(W, \tilde{\theta}_0)) = 0$. This implies that θ_0 is not the unique minimizer of (6.5.5). Then, the gmm estimator cannot be a consistent estimator of the gmm estimand, for the exact same reason why a m-estimator cannot be a consistent estimator of an m-estimand if this m-estimand is not the unique minimizer of the objective function.

Theorem 6.5.4 *If Assumptions 9, 13, and 14 are satisfied and S is positive definite,*

$$\hat{\theta}(S) \xrightarrow{P} \theta_0,$$

and

$$\sqrt{n}(\hat{\theta}(S) - \theta_0) \xrightarrow{d} N(0, V_{as}(\hat{\theta}(S))),$$

with $V_{as}(\hat{\theta}(S)) = (G' SG)^{-1} G' S V S G (G' SG)^{-1}$.

Moreover, let $\hat{G} = \overline{z(W, \hat{\theta}(S))}$ and $\hat{V} = \overline{z(W, \hat{\theta}(S)) z(W, \hat{\theta}(S))'}$.

$$(\hat{G}' S \hat{G})^{-1} \hat{G}' S \hat{V} S \hat{G} (\hat{G}' S \hat{G})^{-1} \xrightarrow{P} (G' SG)^{-1} G' S V S G (G' SG)^{-1}.$$

$\hat{\theta}(S) \xrightarrow{P} \theta_0$ irrespective of the choice of the matrix S . So the choice of S does not matter for the convergence of the estimator. However, the choice of S has an effect on the asymptotic variance of the estimator. This raises the question of which matrix S is optimal: for which value of S is the asymptotic variance of $\hat{\theta}(S)$ the smallest? Answering this question requires defining an ordering of variance matrixes. Before this, I will state a simple lemma.

Lemma 6.5.1 *A variance matrix is positive semi-definite.*

Proof

Let X be a $k \times 1$ random vector with second moments and with variance $V(X)$. For any $k \times 1$ vector x ,

$$\begin{aligned} x' V(X) x &= x' E((X - E(X))(X - E(X))') x \\ &= E(x'(X - E(X))(X - E(X))' x) \\ &= E((x' X - E(x' X))(x' X - E(x' X))) \\ &= V(x' X) \geq 0. \end{aligned}$$

QED.

If you have a hard time to get the previous derivation, you can rewrite the steps of the proof with $V(X)$ a 2×2 variance matrix.

Definition 6.5.5 *Ordering of variance matrixes*

Let X and Y be two vectors of random variables with second moments, and let $V(X)$ and $V(Y)$ be their two variance matrixes. We say that $V(X) \geq V(Y)$ if $V(X) - V(Y)$ is positive semi-definite.

This definition follows from the fact that if $V(X) - V(Y)$ is positive semi-definite, any linear combination of the coordinates of X has a greater variance than the same linear combination of the coordinates of Y : $V(\lambda'X) \geq V(\lambda'Y)$ for any vector of real numbers λ of the same size as X and Y . Indeed,

$$V(\lambda'X) - V(\lambda'Y) = \lambda'V(X)\lambda - \lambda'V(Y)\lambda = \lambda'(V(X) - V(Y))\lambda \geq 0.$$

In particular, any element of X will have a larger variance than the corresponding element of Y . For the first element of these two vectors, this can be seen letting $\lambda = (1, 0, \dots, 0)$.

As per this definition, a GMM estimator $\hat{\theta}(S^*)$ is optimal if for any positive definite matrix S , $V_{as}(\hat{\theta}(S)) - V_{as}(\hat{\theta}(S^*))$ is positive semi-definite. Before deriving the optimal GMM estimator, I need to introduce a lemma.

Lemma 6.5.2 *Some properties on positive definite and positive semi-definite matrixes*

1. *An idempotent symmetric matrix is positive semi-definite.*
2. *An invertible positive semi-definite matrix is positive definite.*
3. *The inverse of a positive definite matrix is positive definite.*
4. *If A is positive definite, $(A^{-1})^{\frac{1}{2}} = (A^{\frac{1}{2}})^{-1}$. $(A^{-1})^{\frac{1}{2}}$ is denoted $A^{-\frac{1}{2}}$.*

Proof

Let A be a symmetric $k \times k$ idempotent matrix. A idempotent means that $A^2 = A$. For any $k \times 1$ vector x ,

$$x'Ax = x'A^2x = x'A'Ax = (Ax)'Ax \geq 0.$$

This proves the first statement.

Let A be an invertible positive semi-definite matrix. As A is positive semi-definite, its eigen values are positive. As it is invertible, 0 cannot be one of its eigen values. Therefore, A is positive definite. This proves the second statement.

Let A be positive definite, and let A^{-1} denote its inverse. For any non 0 vector x ,

$$x'A^{-1}x = x'A^{-1}AA^{-1}x = x'(A^{-1})'AA^{-1}x = (A^{-1}x)'A(A^{-1}x) > 0.$$

This proves the third statement.

$(A^{\frac{1}{2}})^{-1} \times (A^{\frac{1}{2}})^{-1} = (A^{\frac{1}{2}}A^{\frac{1}{2}})^{-1} = A^{-1}$. This proves the fourth statement.

QED.

Theorem 6.5.6 *Optimal GMM estimator*

If Assumptions 9, 13, and 14 are satisfied, $\hat{\theta}^* = \hat{\theta}(V^{-1})$ is the optimal GMM estimator.

Proof

V is a variance matrix. It is therefore positive semi-definite (Lemma 6.5.1). As it is invertible by assumption, it is positive definite (Lemma 6.5.2, statement 2). Let $V^{\frac{1}{2}}$ denote its square-root. V^{-1} is also positive definite (Lemma 6.5.2, statement 3), and its square root, $V^{-\frac{1}{2}}$, is the inverse of $V^{\frac{1}{2}}$ (Lemma 6.5.2, statement 4). Let $B = (G'SG)^{-1}G'SV^{\frac{1}{2}}$, and let $C = V^{-\frac{1}{2}}G$. Notice that

$$BC = (G'SG)^{-1}G'SV^{\frac{1}{2}}V^{-\frac{1}{2}}G = I.$$

Then,

$$\begin{aligned} & V_{as}(\hat{\theta}(S)) - V_{as}(\hat{\theta}(V^{-1})) \\ &= (G'SG)^{-1}G'SVSG(G'SG)^{-1} - (G'V^{-1}G)^{-1}G'V^{-1}VV^{-1}G(G'V^{-1}G)^{-1} \\ &= (G'SG)^{-1}G'SVSG(G'SG)^{-1} - (G'V^{-1}G)^{-1} \\ &= (G'SG)^{-1}G'SV^{\frac{1}{2}}V^{\frac{1}{2}}SG(G'SG)^{-1} - (G'V^{-\frac{1}{2}}V^{-\frac{1}{2}}G)^{-1} \\ &= BB' - (C'C)^{-1} \\ &= BB' - BC(C'C)^{-1}C'B' \\ &= B(I - C(C'C)^{-1}C')B'. \end{aligned}$$

$I - C(C'C)^{-1}C'$ is symmetric:

$$\begin{aligned} (I - C(C'C)^{-1}C')' &= I' - (C(C'C)^{-1}C')' \\ &= I - (C')'((C'C)^{-1})'C' \\ &= I - C((C'C)')^{-1}C' \\ &= I - C(C'C)^{-1}C'. \end{aligned}$$

It is also idempotent:

$$\begin{aligned} (I - C(C'C)^{-1}C')(I - C(C'C)^{-1}C') &= I - 2C(C'C)^{-1}C' + C(C'C)^{-1}C'C(C'C)^{-1}C' \\ &= I - 2C(C'C)^{-1}C' + C(C'C)^{-1}C' \\ &= I - C(C'C)^{-1}C'. \end{aligned}$$

Therefore, this matrix is positive semi-definite (Lemma 6.5.2, first statement). Then, for any vector x ,

$$x'B(I - C(C'C)^{-1}C')B'x = (B'x)'(I - C(C'C)^{-1}C')B'x \geq 0.$$

Therefore, $B(I - C(C'C)^{-1}C')B'$ is positive semi-definite. This proves the result.

QED.

The GMM estimator with the lowest asymptotic variance is the solution of the minimization problem in (6.5.4), with $S^* = V^{-1}$. This result is (relatively...) intuitive. To simplify, assume

that V is a diagonal matrix, implying that there is no correlation between $z_j(W, \theta_0)$ and $z_{j'}(W, \theta_0)$. Assume also that $V(z_j(W, \theta)) = \sigma_j^2$, which does not depend on θ . Then V^{-1} is a diagonal matrix, whose j th diagonal element is $\frac{1}{\sigma_j^2}$. Therefore, for $S^* = V^{-1}$, the objective function in the minimization problem (6.5.4) will give a greater weight to the constraints which have a lower variance. It follows from the central limit theorem that $\overline{z_j(W, \theta)} - E(z_j(W, \theta)) \approx \frac{N(0,1)}{\sqrt{n}} \sigma_j$. Therefore, constraints with the lowest variance are also those for which $\overline{z_j(W, \theta)}$ is the best proxy for $E(z_j(W, \theta))$, because the convergence from the former to the latter happens faster. Now assume you want to find the value of θ which makes $\overline{z(W, \theta)}$ as close as possible to 0, because you want to estimate θ_0 , and you know that it is the only value of θ such that $E(z(W, \theta)) = 0$. Then, it makes sense not to give too much weight to the constraints with a large variance, because you know that for these ones $\overline{z_j(W, \theta)}$ might be a poor proxy of $E(z_j(W, \theta))$, so cancelling $\overline{z_j(W, \theta)}$ might be very different from cancelling $E(z_j(W, \theta))$. On the contrary, you should give a large weight to the constraints with a low variance, because for these ones you know that $\overline{z_j(W, \theta)}$ is a good proxy of $E(z_j(W, \theta))$, so cancelling $\overline{z_j(W, \theta)}$ will be close to cancelling $E(z_j(W, \theta))$.

An important issue is that in practice, we cannot compute $\hat{\theta}^*$, because V^{-1} is unknown, as it depends on θ_0 and on population quantities. Solving problem (6.5.4) for $S = V^{-1}$ amounts to finding the minimizer of an unknown function, something which cannot be done. We therefore say that $\hat{\theta}(V^{-1})$, the optimal GMM estimator, is not feasible. The solution to this problem will be to find a feasible estimator (one that we will be able to compute), which will have the same asymptotic properties as the optimal estimator. On that purpose, we will use a two steps procedure. In a first step, we will solve the minimization problem in (6.5.4) for $S = I$. That's a feasible problem because the function to be minimized is a known function of θ . The solution of this minimization problem will give us an estimator $\hat{\theta}(I)$. It follows from Theorem 6.5.4 that this estimator converges in probability towards θ_0 . Therefore, one can show that

$$\overline{z(W, \hat{\theta}(I))z(W, \hat{\theta}(I))'}^{-1} \xrightarrow{P} V^{-1}.$$

$\widehat{V}^{-1} = \overline{z(W, \hat{\theta}(I))z(W, \hat{\theta}(I))'}^{-1}$ is therefore a good proxy for the optimal matrix we would like to use when solving the minimization problem in (6.5.4). Thus, in a second step, we will solve again this minimization problem, with $S = \widehat{V}^{-1}$. Using Slutsky type of arguments, one can show that using a sequence of probabilistic matrixes converging to a fixed matrix S in problem (6.5.4) does not change the asymptotic distribution of the GMM estimator: it will converge to the same distribution as if we had used S . Therefore, $\hat{\theta}_F^* = \hat{\theta}(\widehat{V}^{-1})$ has the same asymptotic distribution as $\hat{\theta}^* = \hat{\theta}(V^{-1})$, which implies it is asymptotically efficient.

Systems of moment conditions with $l > k$ often arise from a model which implies that there must be a $\theta_0 \in \Theta$ satisfying Equation (6.5.1). We will see later in the course examples of such models. The assumptions of this model are testable. If the assumptions of the model are satisfied, we know that if we had an infinity of data at hand we could find a θ_0 such that $E(z(W, \theta_0)) = 0$. So the only reason why we are not able to cancel exactly $\overline{z(W, \hat{\theta}_F^*)}$ is

that we only have a finite sample. Therefore, if our assumptions are correct and if we have a large sample, we should be close to cancelling exactly this vector. If we find that $\overline{z(W, \hat{\theta}_F^*)}$ is actually pretty far from 0 given our sample size, this implies that the assumptions of the model must be violated. The next Theorem formalizes this idea, by presenting a statistical test of Equation (6.5.1). Before stating it, I need to define the chi-squared distribution.

Definition 6.5.7 χ^2 distribution

Z follows a chi-squared distribution with l degrees of freedom if and only if $Z = \sum_{i=1}^l Z_i^2$ where the Z_i are l independent $N(0, 1)$ random variables. This is denoted $Z \sim \chi^2(l)$.

Theorem 6.5.8 Overidentification test

If there is a $\theta_0 \in \Theta$ satisfying Equation (6.5.1), and if Assumptions 9, 13, and 14 are satisfied,

$$\|\sqrt{n}\hat{V}^{-\frac{1}{2}}\overline{z(W, \hat{\theta}_F^*)}\|^2 = \overline{nz(W, \hat{\theta}_F^*)}'\hat{V}^{-1}\overline{nz(W, \hat{\theta}_F^*)} \xrightarrow{d} \chi^2(l - k).$$

Sketch of the Proof*

It follows from a first order Taylor expansion that

$$\overline{z(W, \hat{\theta}_F^*)} \approx \overline{z(W, \theta_0)} + \dot{z}(W, \theta_0)(\hat{\theta}_F^* - \theta_0).$$

Multiplying by $\sqrt{n}$, and replacing by their limits the terms converging in probability towards deterministic values, we get

$$\sqrt{n}\overline{z(W, \hat{\theta}_F^*)} \approx \sqrt{n}\overline{z(W, \theta_0)} + G\sqrt{n}(\hat{\theta}_F^* - \theta_0). \quad (6.5.7)$$

Then, it follows from its definition that $\hat{\theta}_F^*$ must satisfy the first order conditions arising from (6.5.4):

$$0 = \overline{\dot{z}(W, \hat{\theta}_F^*)}'\hat{V}^{-1}\overline{z(W, \hat{\theta}_F^*)}.$$

A first order Taylor expansion of $\overline{z(W, \hat{\theta}_F^*)}$ in the previous equation yields

$$0 \approx \overline{\dot{z}(W, \hat{\theta}_F^*)}'\hat{V}^{-1}\overline{z(W, \theta_0)} + \overline{\dot{z}(W, \hat{\theta}_F^*)}'\hat{V}^{-1}\dot{z}(W, \hat{\theta}_F^*)(\hat{\theta}_F^* - \theta_0).$$

Multiplying by $\sqrt{n}$, rearranging, and replacing by their limits the terms converging in probability towards deterministic values, we get

$$\sqrt{n}(\hat{\theta}_F^* - \theta_0) \approx - (G'V^{-1}G)^{-1}G'V^{-1}\sqrt{n}\overline{z(W, \theta_0)}. \quad (6.5.8)$$

Plugging Equation (6.5.8) into (6.5.7) and multiplying by $V^{-\frac{1}{2}}$ yields

$$\sqrt{n}V^{-\frac{1}{2}}\overline{z(W, \hat{\theta}_F^*)} \approx V^{-\frac{1}{2}} \left(I - G(G'V^{-1}G)^{-1}G'V^{-1} \right) \sqrt{n}\overline{z(W, \theta_0)}.$$

Let

$$P = G(G'V^{-1}G)^{-1}G'V^{-1}.$$

It follows from the central limit theorem that

$$\sqrt{n}V^{-\frac{1}{2}}\overline{z(W, \hat{\theta}_F^*)} \xrightarrow{d} N(0, V^{-\frac{1}{2}}(I - P)V(I - P)'V^{-\frac{1}{2}}).$$

By Lemma 17.1 in van der Vaart (2000) and the continuous mapping theorem,

$$\|\sqrt{n}V^{-\frac{1}{2}}z(W, \widehat{\theta}_F^*)\|^2 \xrightarrow{d} \sum_{i=1}^l \lambda_i Z_i^2,$$

where the λ_i are the eigen values of $V^{-\frac{1}{2}}(I - P)V(I - P)'V^{-\frac{1}{2}}$, and the Z_i are l iid $N(0, 1)$ random variables.

As $VP' = PVP' = G(G'V^{-1}G)^{-1}G'$, we have

$$V^{-\frac{1}{2}}(I - P)V(I - P)'V^{-\frac{1}{2}} = V^{-\frac{1}{2}}(I - P)V^{\frac{1}{2}}.$$

This matrix is a projection matrix:

$$\left(V^{-\frac{1}{2}}(I - P)V^{\frac{1}{2}}\right)^2 = V^{-\frac{1}{2}}(I - P)V^{\frac{1}{2}},$$

whose eigenvalues can either be equal to 0 or 1. Therefore, the asymptotic distribution of $\|\sqrt{n}V^{-\frac{1}{2}}z(W, \widehat{\theta}_F^*)\|^2$ only depends on how many of the λ_i are equal to 1. This quantity is equal to the rank of $V^{-\frac{1}{2}}(I - P)V^{\frac{1}{2}}$. Since $V^{-\frac{1}{2}}$ and $V^{\frac{1}{2}}$ have full rank, the rank of $V^{-\frac{1}{2}}(I - P)V^{\frac{1}{2}}$ is the same as the rank of $I - P$. P is a projection matrix: $P^2 = P$. Therefore the rank of $I - P$ is equal to $\text{rank}(I) - \text{rank}(P) = l - \text{rank}(P)$. As G , $(G'V^{-1}G)^{-1}$, and V^{-1} have full rank, $\text{rank}(P) = \text{rank}(G') = \text{rank}(G) = k$ by assumption. Thus,

$$\|\sqrt{n}V^{-\frac{1}{2}}z(W, \widehat{\theta}_F^*)\|^2 \xrightarrow{d} \chi^2(l - k).$$

Finally, it follows from Slutsky lemma that $\|\sqrt{n}\widehat{V}^{-\frac{1}{2}}z(W, \widehat{\theta}_F^*)\|^2$ has the same asymptotic distribution as $\|\sqrt{n}V^{-\frac{1}{2}}z(W, \widehat{\theta}_F^*)\|^2$. This completes the proof.

QED.

The proof is complicated, so let me summarize its main steps. $\sqrt{n}z(W, \widehat{\theta}_F^*)$ converges towards a normal vector with variance Σ . That's not too surprising, as this quantity is "almost" a standardized sum of iid variables with mean 0 (the "almost" comes from the fact we have $\widehat{\theta}_F^*$ instead of θ_0 , which actually makes an important difference in the asymptotic variance of this sequence). Then it follows from a well-known result in statistics that $\|\sqrt{n}\widehat{V}^{-\frac{1}{2}}z(W, \widehat{\theta}_F^*)\|^2$ converges towards $\sum_{i=1}^l \lambda_i Z_i^2$, where the λ_i are the eigen values of Σ , and the Z_i are iid $N(0, 1)$ random variables. Finally, the result follows from the fact that $l - k$ of the λ_i are equal to 1, while the remaining k are equal to 0.

Let me also try to give you a little bit of intuition on this result. If we knew θ_0 , we could use $\|\sqrt{n}\widehat{V}^{-\frac{1}{2}}z(W, \theta_0)\|^2$ as our test statistic: as $E(z(W, \theta_0)) = 0$, this statistic should be not be too far from zero. It follows from the central limit theorem that $\sqrt{n}z(W, \theta_0)$ converges towards a $N(0, V)$ distribution, so $\sqrt{n}\widehat{V}^{-\frac{1}{2}}z(W, \theta_0)$ converges towards a vector of l independent $N(0, 1)$ variables. Therefore, $\|\sqrt{n}\widehat{V}^{-\frac{1}{2}}z(W, \theta_0)\|^2$ converges towards a chi-squared distribution with l degrees of freedom. Now, we do not know θ_0 , so we use $\|\sqrt{n}\widehat{V}^{-\frac{1}{2}}z(W, \widehat{\theta}_F^*)\|^2$ as our test statistic. $\sqrt{n}z(W, \widehat{\theta}_F^*)$ will converge towards a vector of $N(0, 1)$, k of which write as linear

combinations of the remaining $l - k$. This strong dependency between the elements of the limiting distribution of the vector comes from the fact that $\widehat{\theta}_F^*$ appears in all of the $\overline{z_j(W, \widehat{\theta}_F^*)}$. Therefore, $\|\sqrt{n}\widehat{V}^{-\frac{1}{2}}\overline{z(W, \widehat{\theta}_F^*)}\|^2$ converges towards the sum of the squares of the $l - k$ elements of the limiting distribution of $\sqrt{n}\widehat{V}^{-\frac{1}{2}}\overline{z(W, \widehat{\theta}_F^*)}$ which are not linear combinations of each other, i.e. towards a chi-squared distribution with $l - k$ degrees of freedom.

Following Theorem 6.5.8, the overidentification test amounts to comparing the sample size multiplied by the standardized value of the objective function evaluated at its minimum to the critical value of a chi-squared distribution with $l - k$ degrees of freedom. If this standardized value of the objective function is larger than, say, the 95% percentile of this distribution, we can reject the assumptions of our model, as under these assumptions Equation (6.5.1) should be satisfied for some $\theta \in \Theta$, so our test statistic should follow a chi-squared distribution with $l - k$ degrees of freedom. Intuitively, this test makes sense: if the test statistic is large, this means that we are far from cancelling the objective function, while we should “almost” be able to cancel it if Equation (6.5.1) was satisfied for some $\theta_0 \in \Theta$. Finally, let $c_{l-k}(\alpha)$ denote the $1 - \alpha^{th}$ percentile of a chi-squared distribution with $l - k$ degrees of freedom. We reject a level α overidentification test if $\overline{nz(W, \widehat{\theta}_F^*)' \widehat{V}^{-1} z(W, \widehat{\theta}_F^*)} > c_{l-k}(\alpha)$ i.e. if $\overline{z(W, \widehat{\theta}_F^*)' \widehat{V}^{-1} z(W, \widehat{\theta}_F^*)} > \frac{c_{l-k}(\alpha)}{n}$. For small values of n , we will not reject the test even if our standardized objective function is pretty far from 0. But for a large n , we should “almost” be able to cancel our objective function, so we will reject the test as soon as it is slightly different from 0.

Finally, notice that Equation (6.5.1) and Assumption 14 ensure that the GMM estimand is the unique minimizer of $E(z(W, \theta))'SE(z(W, \theta))$ for any positive definite matrix S . Sometimes, we will encounter parameters which do not necessarily satisfy Equation (6.5.1), but which are the unique minimizers of $E(z(W, \theta))'SE(z(W, \theta))$ for some positive definite matrix S . We could use the minimizer of $\overline{z(W, \theta)'S z(W, \theta)}$, or of $\overline{z(W, \theta)'\widehat{S} z(W, \theta)}$ where $\widehat{S} \xrightarrow{P} S$, to estimate them. This estimator will have the same asymptotic behavior as that of the GMM estimator. Indeed, the derivation of the asymptotic behavior of the GMM estimator only relies on the fact the GMM estimand is the unique minimizer of $E(z(W, \theta))'SE(z(W, \theta))$, not on the fact it satisfies Equation (6.5.1) and Assumption 14.

Theorem 6.5.9 *Let $\theta_0 = \underset{\theta \in \Theta}{\operatorname{argmin}} E(z(W, \theta))'SE(z(W, \theta))$ ¹² for some positive definite matrix S . Let $\widehat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} \overline{z(W, \theta)'\widehat{S} z(W, \theta)}$ for a sequence of random matrixes $\widehat{S}$ converging towards S in probability. If Assumptions 9 and 13 a) to e)¹³ are satisfied,*

$$\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, V_{as}(\widehat{\theta})),$$

with $V_{as}(\widehat{\theta}) = (G'SG)^{-1}G'SVSG(G'SG)^{-1}$. $V_{as}(\widehat{\theta})$ can be consistently estimated, using the same estimator as in Theorem 6.5.4.

¹²Here, I take the = sign at face value: θ_0 should be the unique minimizer of the quadratic form.

¹³f) is only here to ensure the optimal GMM estimator is well-defined. Here, there is no optimal estimator: θ_0 is the minimizer of $E(z(W, \theta))'SE(z(W, \theta))$ for one specific S only, not for all of them. Therefore, we do not need this supplementary condition.

Chapter 7

The two stage least squares estimand, and its estimator

7.1 The two stage least squares estimand

Let Y be a random variable. Let X , D and Z respectively be $H \times 1$, $I \times 1$, and $J \times 1$ vectors of random variables, with $J \geq I$. Let $K = H + I$, and let $L = H + J$. Let $V = (X', D')'$ and $W = (X', Z')'$ respectively be $K \times 1$ and $L \times 1$ vectors of random variables. For reasons which will become clearer later, the elements of D are called endogenous variables, those of Z are called excluded instruments, and those of X are called included instruments. For each coordinate of V V_k , let α_k denote the $L \times 1$ vector of OLS coefficients arising from the regression of V_k on W . The regression function of V_k on W is $W'\alpha_k$, which is also equal to $\alpha_k'W$. The closed form expression of α_k is $E(WW')^{-1}E(WV_k)$. Therefore, $\alpha_k' = E(V_kW')E(WW')^{-1}$. Now, let α be a $K \times L$ matrix whose k th line is α_k' .

$$\alpha = \begin{pmatrix} E(V_1W')E(WW')^{-1} \\ \dots \\ E(V^kW')E(WW')^{-1} \end{pmatrix} = E(VW')E(WW')^{-1}.$$

You can indeed check doing the computations on a sheet of paper that multiplying each line of a 2×2 matrix A by another 2×2 matrix B is the same as multiplying A by B . Moreover, if $V = (V_1, V_2)'$ and W are 2×1 vector of random variables,

$$\begin{pmatrix} E(V_1W') \\ E(V_2W') \end{pmatrix} = E(VW').$$

αW is a $K \times 1$ vector whose k th coordinate is the regression function of V_k on W . Indeed, the k th coordinate of αW is the inner product of the k th line of α and W , i.e. $\alpha_k'W$.

Definition 7.1.1 2SLS coefficients

β_{2SLS} , the vector of coefficients of a two stage least squares (2SLS) regression of Y on V using W as instruments is the $K \times 1$ vector of coefficients arising from the OLS regression of Y

on αW , i.e. of the regression of Y on the $K \times 1$ vector whose k th element is the regression function of V_k on W .

2SLS is a regression technique in which regressors are regression functions. Instead of regressing Y on V , we regress Y on the regression function of V on W . The regressions of the elements of V on W are called the first stage of the 2SLS regression, the regression of Y on the regression function of V on W is called the second stage, and the regression of Y on W is called the reduced form of the 2SLS regression.

I will now give a closed form formula for the 2SLS coefficients.

Theorem 7.1.2 *Closed form formula for the 2SLS estimand.*

Provided all the matrixes inverted in the next display are indeed invertible,

$$\beta_{2SLS} = (E(VW')E(WW')^{-1}E(WV'))^{-1} E(VW')E(WW')^{-1}E(WY). \quad (7.1.1)$$

Proof

β_{2SLS} is just the OLS coefficient from the regression of Y on αW . It follows from Theorem 3.2.2 that

$$\begin{aligned} \beta_{2SLS} &= E(\alpha W(\alpha W)')^{-1}E(\alpha WY) \\ &= E(\alpha WW'\alpha')^{-1}E(\alpha WY) \\ &= (\alpha E(WW')\alpha')^{-1} \alpha E(WY) \\ &= \left(E(VW')E(WW')^{-1}E(WW') (E(VW')E(WW')^{-1})' \right)^{-1} E(VW')E(WW')^{-1}E(WY) \\ &= \left(E(VW') (E(VW')E(WW')^{-1})' \right)^{-1} E(VW')E(WW')^{-1}E(WY) \\ &= (E(VW')E(WW')^{-1}E(WV'))^{-1} E(VW')E(WW')^{-1}E(WY). \end{aligned}$$

The third equality follows from the fact α is a deterministic matrix, and from the linearity of the expectation operator. The fourth equality is obtained substituting the closed form expression of α in the previous equation.

QED.

This formula is very abstract. Actually, 2SLS coefficients have a number of simple characterizations, especially when there is only one endogenous variable and one excluded instrument in the regression. Let me first give a more explicit characterization of the 2SLS coefficients in this case.

Theorem 7.1.3 *2SLS coefficients with one endogenous variable*

Let D and Z be scalar random variables. Let α_X and α_Z be the coefficients arising from the OLS regression of D on $(X', Z)'$. β_{2SLS} , the coefficient of a 2SLS regression of Y on $V = (X', D)'$ using $W = (X', Z)'$ as instruments is the coefficient of an OLS regression of Y on X and $X'\alpha_X + Z\alpha_Z$.

Proof

It follows from Definition 7.1.1 that β_{2SLS} is the coefficient of the regression of Y on the regression function of $V = (X', D)'$ on $W = (X', Z)'$. Let $X'\alpha_X + Z\alpha_Z$ denote the regression function of D on $W = (X', Z)'$. For any element X_h of X , the regression function of X_h on W is merely X_h , because X_h is included in W . Indeed, for any coordinate of X X_h ,

$$E(X_h|W) = E(X_h|X, Z) = X_h.$$

Since this conditional expectation is linear, it is the regression function. Therefore, the regression function of $V = (X', D)'$ on $W = (X', Z)'$ is $(X', X'\alpha_X + Z'\alpha_Z)'$. This proves the result.

QED.

From this Theorem, it is easy to see why we need $J \geq I$. Without Z , β_{2SLS} would not be uniquely defined. Indeed, β_{2SLS} would be the coefficient arising from the regression of Y on X and $X'\alpha_X$. The explanatory variables in that regression do not satisfy the rank condition which is necessary for OLS coefficients to be uniquely defined, as $X'\alpha_X$ is a linear combination of the other variables included in the regression (X).

The previous theorem shows that the coefficients of X in the 2SLS regression, are just regular OLS coefficients for X . The next theorem gives a simple characterization of β_D . It shows that when both D and Z are scalar random variables, β_D is equal to the coefficient of Z in the reduced form regression of Y on X and Z , λ_Z , divided by the coefficient of Z in the first stage regression of D on X and Z , α_Z .

Theorem 7.1.4 *Two stages least squares is the ratio of reduced form and first stage*

1. When Z is a scalar random variable, $\beta_D = \frac{\lambda_Z}{\alpha_Z} = \frac{\text{cov}(Y, \tilde{Z})}{\text{cov}(D, \tilde{Z})}$.
2. When Z is a scalar random variable and there are no X in the regression, $\beta_D = \frac{\lambda_Z}{\alpha_Z} = \frac{\text{cov}(Y, Z)}{\text{cov}(D, Z)}$.

Proof

It follows from Theorem 7.1.3 that $\beta_{2SLS} = (\beta'_X, \beta'_D)'$, with

$$(\beta'_X, \beta'_D)' = \underset{b_X, b_D}{\operatorname{argmin}} E \left((Y - X'b_X - b_D (X'\alpha_X + Z\alpha_Z))^2 \right).$$

We can rearrange a little bit this minimization problem to write

$$(\beta'_X, \beta'_D)' = \underset{b_X, b_D}{\operatorname{argmin}} E \left((Y - X'(b_X + b_D\alpha_X) - \alpha_Z b_D Z)^2 \right).$$

Solving it is equivalent to finding

$$(\tilde{\lambda}'_X, \tilde{\lambda}'_Z)' = \underset{l_X, l_Z}{\operatorname{argmin}} E \left((Y - X'l_X - l_Z Z)^2 \right),$$

and then solving the following system of K equations with K unknowns:

$$\begin{aligned}\tilde{\lambda}_X &= \beta_X + \beta_D \alpha_X \\ \tilde{\lambda}_Z &= \alpha_Z \beta_D.\end{aligned}$$

This yields $\beta_D = \frac{\tilde{\lambda}_Z}{\alpha_Z}$. But the problem that $(\tilde{\lambda}'_X, \tilde{\lambda}'_Z)'$ solves is the reduced form regression of Y on X and Z . Therefore, $\tilde{\lambda}_Z$ is actually equal to λ_Z . This proves that $\beta_D = \frac{\lambda_Z}{\alpha_Z}$. Then, it follows from the Frisch-Waugh theorem that $\lambda_Z = \frac{\text{cov}(Y, \tilde{Z})}{V(\tilde{Z})}$. Similarly, $\alpha_Z = \frac{\text{cov}(D, \tilde{Z})}{V(\tilde{Z})}$. This completes the proof of the first statement. To prove the second statement, it suffices to notice that when there are no covariates, $\lambda_Z = \frac{\text{cov}(Y, Z)}{V(Z)}$ and $\alpha_Z = \frac{\text{cov}(D, Z)}{V(Z)}$. Therefore, $\beta_D = \frac{\text{cov}(Y, Z)}{\text{cov}(D, Z)}$. QED.

Finally, when Z and D are scalar variables and Z is binary, 2SLS coefficients have an even simpler expression.

Lemma 7.1.1 *2SLS and the Wald ratio*

When Z is binary and there are no X in the regression, $\lambda_Z = E(Y|Z = 1) - E(Y|Z = 0)$, $\alpha_Z = E(D|Z = 1) - E(D|Z = 0)$, and $\beta_D = \frac{E(Y|Z=1) - E(Y|Z=0)}{E(D|Z=1) - E(D|Z=0)}$. The rhs of the last equation is called the Wald ratio.

Proof

λ_Z is the coefficient of Z in a regression of Y on Z and a constant. As Z is binary, this regression model is saturated, and it follows from the second example of saturated regression models that $\lambda_Z = E(Y|Z = 1) - E(Y|Z = 0)$. It follows from a similar argument that $\alpha_Z = E(D|Z = 1) - E(D|Z = 0)$. Combining this with Theorem 7.1.4 yields $\beta_D = \frac{E(Y|Z=1) - E(Y|Z=0)}{E(D|Z=1) - E(D|Z=0)}$. This completes the proof.

QED.

7.2 The two stage least squares estimator

For any iid sample of random variables, vectors or matrixes $(Q_i)_{1 \leq i \leq n}$, let $\overline{Q}$ denote their sample mean. Assume that we observe an iid sample of n random variables (Y_i, X_i, D_i, Z_i) following the same distribution as (Y, X, D, Z) . To estimate β_{2SLS} , we are going to follow the closed-form expression for derived in Theorem 7.1.2 and use the following estimator

$$\hat{\beta}_{2SLS} = \left(\overline{VW'} \times \overline{WW'}^{-1} \times \overline{WV'} \right)^{-1} \overline{VW'} \times \overline{WW'}^{-1} \times \overline{WY}. \quad (7.2.1)$$

This estimator can be computed as follows: estimate first an OLS regression of each of the coordinate of V on W , and then regress Y on $\hat{V}$, the predicted value of V as per the first stage regressions. Because the regressors in the second stage regression are estimated, the analysis of the asymptotic behavior of $\hat{\beta}_{2SLS}$ is more complicated than for OLS. Here, we will circumvent

this difficulty by showing that β_{2SLS} satisfies the conditions necessary for Theorem 6.5.9 to apply, provided one is ready to assume ex-ante that β_{2SLS} lies within a (potentially very large) compact subset Θ of $\mathbb{R}^K$, as well as a few other technical assumptions. Let $e = Y - W'\alpha'\beta_{2SLS}$ be the residual from the second stage regression of Y on αW , let $\varepsilon = V - \alpha W$ be the residuals of the first stage regressions of V on W , and let $u = Y - V'\beta_{2SLS} = e - \varepsilon'\beta_{2SLS}$.

Assumption 15 *Assumptions ensuring the asymptotic normality of the 2SLS estimator*

(a) $\beta_{2SLS} \in \Theta$, a compact subset of $\mathbb{R}^K$.

(b) $E(WW')$ is invertible.

(c) $\text{rank}(E(WV')) = K$.

Theorem 7.2.1 *Asymptotic normality of $\hat{\beta}_{2SLS}$.*

If Assumption 15 is satisfied, $\hat{\beta}_{2SLS}$ is an asymptotically normal sequence of estimators of β_{2SLS} :

$$\sqrt{n}(\hat{\beta}_{2SLS} - \beta_{2SLS}) \hookrightarrow N\left(0, V_{as}(\hat{\beta}_{2SLS})\right),$$

with

$$V_{as}(\hat{\beta}_{2SLS}) = (E(VW')E(WW')^{-1}E(WV'))^{-1}E(VW')E(WW')^{-1}V(Wu)E(WW')^{-1}E(WV')(E(VW')E(WW')^{-1}E(WV'))^{-1}.$$

Proof

Let $\theta_0 = \underset{\theta \in \Theta}{\text{argmin}} E(W(Y - V'\theta))'E(WW')^{-1}E(W(Y - V'\theta))$. As Θ is compact, one can apply the dominated convergence theorem to $E(W(Y - V'\theta))$, so the foc conditions associated to this problem are $E(VW')E(WW')^{-1}E(W(Y - V'\theta)) = 0$. Therefore,

$$\theta_0 = (E(VW')E(WW')^{-1}E(WV'))^{-1}E(VW')E(WW')^{-1}E(WY) = \beta_{2SLS}.$$

β_{2SLS} is the unique minimizer of $E(W(Y - V'\theta))'E(WW')^{-1}E(W(Y - V'\theta))$. Similarly, one can show that $\hat{\beta}_{2SLS}$ is (with a probability approaching one) the unique minimizer of $\overline{(W(Y - V'\theta))'} \times \overline{WW'}^{-1} \times \overline{W(Y - V'\theta)}$. $E(WW')$ is definite semi-positive:

$$\lambda'E(WW')\lambda = E(\lambda'WW'\lambda) = E((\lambda'W)^2) \geq 0.$$

As it is invertible by assumption, it is definite positive, and so is $E(WW')^{-1}$. Moreover, $\overline{WW'}^{-1}$ converges in probability towards $E(WW')$. Now, let $z(y, v, w, \theta) = w(y - v'\theta)$. z satisfies points a), b), c), and d) of Assumption 13. $\dot{z}(y, v, w, \theta) = wv'$, so $G = E(WV')$. By Assumption, $\text{rank}(G) = K$, so point e) of Assumption 13 is satisfied. Then, the result follows from Theorem 6.5.9.

QED.

7.3 Two stage least squares in practice

To estimate 2SLS coefficients, a natural idea is to run OLS regressions of V on W , and then of Y on $\widehat{W}$. This procedure will correctly compute $\widehat{\beta}_{2SLS}$, but not the estimator of its asymptotic variance, which implies that confidence intervals and test statistics will not be valid. Indeed, the software will follow Theorem 4.3.1 to estimate the asymptotic variance of the coefficients, and will therefore use $\widehat{V}\widehat{V}'^{-1} \times \widehat{V}\widehat{V}'\widehat{e}^2 \times \widehat{V}\widehat{V}'^{-1}$ instead of $\widehat{V}\widehat{V}'^{-1} \times \widehat{V}\widehat{V}'\widehat{u}^2 \times \widehat{V}\widehat{V}'^{-1}$. That is because the software will not take into account the fact that $\widehat{V}$ is the result of a first estimation, which incorporates some statistical uncertainty. The correct way of estimating 2SLS coefficients and their standard errors it is to use a command specifically designed for 2SLS and which jointly performs the estimation of the first and second stage regression. In Stata, this will be the “2sls” command.

Chapter 8

Two stage least squares and causality in the linear and constant effect model

This chapter and the next borrow from Angrist & Pischke (2008).

Let's get back to the wage and schooling example. Throughout this chapter, we will assume that potential wages follow a linear constant effect model. Then, we have shown that the following equation relates observed wage to observed schooling:

$$Y_i = \alpha + \rho S_i + \eta_i$$

As the first omitted variable formula shows, if schooling and innate ability are correlated, the coefficient of an OLS regression of Y_i on S_i will fail to capture ρ . One strategy could be to add controls to the regression, but then the regression will capture ρ only if CIA holds, and this is often not a credible assumption. An alternative strategy is to find an instrument.

8.1 Identification of ρ with one instrument

Assumption 16 *Instruments in the linear constant effect model*

The scalar variable Z_i is an instrument to measure the effect of S_i on Y_i if either of the two following statements hold:

1. $Z_i \perp\!\!\!\perp \eta_i$ and $\text{cov}(Z_i, S_i) \neq 0$.
2. $Z_i \perp\!\!\!\perp \eta_i | X_i$, $\text{cov}(\tilde{Z}_i, S_i) \neq 0$, and $E(\eta_i | X_i)$ is linear.

The 2 requirements an instrument must satisfy are very different. $\text{cov}(Z_i, S_i) \neq 0$ is testable. How would you test it? $Z_i \perp\!\!\!\perp \eta_i$ is not testable, because η_i is not observed. The credibility of this assumption should therefore be assessed based on intuition, common sense, economic theory... In the wage and schooling example, it means a good instrument should be independent of innate ability.

Theorem 8.1.1 *Identification of ρ using an instrument ((Wright 1928))*

Assume Assumption 5 is satisfied.

1. If Z_i satisfies the first statement of Assumption 16, then ρ is equal to the coefficient of S_i in a 2SLS regression of Y_i on S_i using Z_i as an instrument for S_i .
2. If Z_i satisfies the second statement of Assumption 16, then ρ is equal to the coefficient of S_i in a 2SLS regression of Y_i on S_i and X_i using X_i and Z_i as instruments for S_i .

Proof

Assume Z_i satisfies the first statement of Assumption 16. Then,

$$\text{cov}(Y_i, Z_i) = \text{cov}(\alpha + \rho S_i + \eta_i, Z_i) = \rho \text{cov}(S_i, Z_i).$$

The first equality follows from (5.1.3), the second from the fact that Z_i and η_i are assumed to be independent. Therefore, as $\text{cov}(S_i, Z_i) \neq 0$,

$$\rho = \frac{\text{cov}(Y_i, Z_i)}{\text{cov}(S_i, Z_i)}.$$

It follows from the second statement of Theorem 7.1.4 that the coefficient of a regression of S_i in a 2SLS regression of Y_i on S_i using Z_i as an instrument for S_i is also equal to $\frac{\text{cov}(Y_i, Z_i)}{\text{cov}(S_i, Z_i)}$. This proves the first result.

Assume Z_i satisfies the second statement of Assumption 16. As $E(\eta_i|X_i)$ is assumed to be linear, one can use a CEF decomposition to write $\eta_i = X_i'\lambda + u_i$, with $E(u_i|X_i) = 0$. $\eta_i \perp\!\!\!\perp Z_i|X_i$ then implies that $u_i \perp\!\!\!\perp Z_i|X_i$.

$$\text{cov}(Z_i, u_i) = E(Z_i u_i) = E(E(Z_i u_i|X_i)) = E(E(Z_i|X_i)E(u_i|X_i)) = 0.$$

Similarly,

$$\text{cov}(X_i, u_i) = 0.$$

Then,

$$\text{cov}(Y_i, \tilde{Z}_i) = \text{cov}(\alpha + \rho S_i + X_i'\lambda + u_i, \tilde{Z}_i) = \rho \text{cov}(S_i, \tilde{Z}_i).$$

The first equality follows from (5.1.3) and the CEF decomposition of η_i . The second equality holds because $\text{cov}(X_i, \tilde{Z}_i) = 0$ by construction, while $\text{cov}(u_i, \tilde{Z}_i) = 0$. Why? Therefore, as $\text{cov}(S_i, \tilde{Z}_i) \neq 0$,

$$\rho = \frac{\text{cov}(Y_i, \tilde{Z}_i)}{\text{cov}(S_i, \tilde{Z}_i)}.$$

It follows from the first statement of Theorem 7.1.4 that the coefficient of a regression of S_i in a 2SLS regression of Y_i on S_i using Z_i as an instrument for S_i is also equal to $\frac{\text{cov}(Y_i, \tilde{Z}_i)}{\text{cov}(S_i, \tilde{Z}_i)}$. This proves the second result.

QED.

To answer the difficult question of the effect of education on earnings, Angrist & Krueger (1991) have used quarter of birth as an instrument for schooling. In this setting, an instrument should have an effect on schooling, but it should not be correlated with innate ability. It seems fairly credible to argue that quarter of birth is independent of ability, but how could you test this? Quarter of birth is correlated with education, because of compulsory schooling laws. In the US, children enter kindergarten in September of the civil year during which they have / will turn 5. Because of this, children born in December enter school when they are around 4 years and 9 months old, while those born in January enter at 5 years and 8 months. Schooling being mandatory until 16 years old only, children born in January can drop out during 10th grade, while those born in December must wait until 11th grade. You can check from Table 4.1.2 in Angrist & Pischke (2008) that among men born between 1930 and 1939, those born in the 4th quarter completed 0.15 more years of schooling than those born in the 1st quarter.¹

If the instrument satisfies those two conditions, ρ is equal to the covariance between Y_i and Z_i divided by the covariance between S_i and Z_i . This result was discovered by Philip Wright in 1928, even though it is still somewhat unclear whether it was him or his son, Sewall, who prove the result (apparently Philip was not very good at maths, while Sewall was, so some people say Sewall must have helped his dad, on this see (Stock & Trebbi 2003)).

The intuition of this result is as follows. Z_i is uncorrelated with η_i , which represents all the determinants of Y_i other than S_i . Therefore, if there is some correlation between Z_i and Y_i , it can only come from the fact that S_i is one determinant of Y_i , and S_i and Z_i are correlated. If ρ were equal to 0, Z_i would be uncorrelated with all determinants of Y_i (because $\rho = 0$ means that actually S_i is not a determinant of Y_i), so Y_i and Z_i would be uncorrelated. Therefore, $cov(Y_i, Z_i) \neq 0 \Rightarrow \rho \neq 0$. In the wage / education / quarter of birth example, under the assumption that people born late in the year have the same level of innate ability than people born early in the year, any difference in earnings across those two groups can only come from the fact that those born late in the year completed more schooling.

Then, for a fixed value of ρ , if $cov(S_i, Z_i)$ is low, $cov(Y_i, Z_i)$ will be low too, because the correlation between Y_i and Z_i entirely goes through the correlation between S_i and Z_i . Therefore, to recover ρ one must rescale $cov(Y_i, Z_i)$ by $cov(S_i, Z_i)$. The rescaling argument is probably easier to understand with a binary instrument. In the wage / education / quarter of birth example, the instrument is binary, so Lemma 7.1.1 tells us that ρ is equal to $E(Y_i|Z_i = 1) - E(Y_i|Z_i = 0)$ rescaled by $E(D_i|Z_i = 1) - E(D_i|Z_i = 0)$. People born late in the year earn 1.35% more than those born early. But one must take into account the fact that this only comes from a 0.15 difference in years of schooling completed. This is the reason why we must rescale the difference in earnings if we want to learn the effect of one more year of education. One can then conclude that 1 more year of education increases wages by $\frac{1.35}{0.15} = 8.9\%$.

¹This is a substantial difference which implies that 15% of men of those cohorts dropped out of school when they reached 16.

Finally, Theorem 8.1.1 implies that 2SLS can be used as a device to compute $\hat{\rho}$ and the estimator of its standard error, so as to derive confidence intervals etc..

8.2 Overidentification of ρ with more than one instrument

Assume now that several instruments $(Z_{i1}, \dots, Z_{iJ})$ satisfy the first statement of Assumption 16. Then,

$$\rho = \frac{\text{cov}(Y_i, Z_{i1})}{\text{cov}(S_i, Z_{i1})} = \dots = \frac{\text{cov}(Y_i, Z_{iJ})}{\text{cov}(S_i, Z_{iJ})}. \quad (8.2.1)$$

ρ is **overidentified**: we have several ways of computing it. As a result, there are many possible ways of estimating ρ . One can for instance use any linear combinations of the elements of

$$\left(\frac{\text{cov}_e(Y_i, Z_{i1})}{\text{cov}_e(S_i, Z_{i1})}, \dots, \frac{\text{cov}_e(Y_i, Z_{iJ})}{\text{cov}_e(S_i, Z_{iJ})} \right),$$

with weights summing up to 1. This raises the question of which estimator is optimal. Assumptions 5 and 16 are also testable. As they imply that $\frac{\text{cov}(Y_i, Z_{i1})}{\text{cov}(S_i, Z_{i1})} = \dots = \frac{\text{cov}(Y_i, Z_{iJ})}{\text{cov}(S_i, Z_{iJ})}$, we can reject them when the elements of

$$\left(\frac{\text{cov}_e(Y_i, Z_{i1})}{\text{cov}_e(S_i, Z_{i1})}, \dots, \frac{\text{cov}_e(Y_i, Z_{iJ})}{\text{cov}_e(S_i, Z_{iJ})} \right)$$

are far from each other. This raises the question of how to implement the test.

To answer these two questions, it is easier to see the parameters of the linear and causal effect model as an over-identified GMM estimand. For the sake of generality, let's also incorporate covariates in our analysis. Let X_i be a vector of covariates such that $E(\eta_i|X_i) = X_i'\lambda$. The linear and causal effect model implies that $Y_i = \alpha + \rho S_i + X_i'\lambda + u_i$, with $E(u_i|X_i) = 0$. Let me rewrite this as $Y_i = X_i'\lambda + \rho S_i + u_i$. I am just putting α into λ , and adding a constant term into X_i . Formally, I should call the new λ and X_i with different names, but I keep their old names not to have too heavy notations. Let $Z_i = (Z_{i1}, \dots, Z_{iJ})'$ be a $J \times 1$ vector of excluded instruments. X_i is an $H \times 1$ vector of covariates whose first coordinate is a constant, and let $W_i = (X_i', Z_i')'$ be an $L \times 1$ vector gathering the two.

Theorem 8.2.1 *If Assumption 5 is satisfied, if $Z_{i1}, \dots, Z_{iJ}$ satisfy the second statement of Assumption 16, and if $E(\eta_i|X_i) = X_i'\lambda$ then the parameters $(\lambda', \rho)'$ of the linear and constant effect model are GMM estimands associated to the following system of L moment conditions:*

$$E(W_i(Y_i - X_i'\lambda - \rho S_i)) = 0. \quad (8.2.2)$$

If $J > 1$, this system is overidentified.

Proof

Under the Assumptions of the Theorem, $Y_i - X_i'\lambda - \rho S_i = u_i$. We want to show that for each coordinate of W_i W_{il} , $E(W_{il}u_i) = 0$. The first coordinate of this vector is a constant.

$E(u_i) = 0$ directly follows from the fact u_i is the residual from a CEF decomposition. For any non constant coordinate of X_i X_{ih} ,

$$E(X_{ih}u_i) = E(X_{ih}E(u_i|X_i)) = 0.$$

For any j ,

$$E(Z_{ij}u_i) = E(E(Z_{ij}u_i|X_i)) = E(E(Z_{ij}|X_i)E(u_i|X_i)) = 0,$$

where the second equality follows from Assumption 16.

QED.

This system satisfies Assumptions 9, 13 and 14 provided one is ready to assume (λ', ρ) belongs to some (potentially very large) compact subset of $\mathbb{R}^{H+1}$. Therefore, it follows from Theorem 6.5.4 that for any symmetric positive definite matrix S , the GMM estimators $\hat{\alpha}(S)$ and $\hat{\rho}(S)$ converge towards α and ρ and are asymptotically normal. It follows from the proof of Theorem 7.2.1 that $(\hat{\alpha}(\overline{WW'}^{-1}), \hat{\rho}(\overline{WW'}^{-1}))$ is the estimator of the coefficient of the 2SLS regression of Y on $V = (X', S)'$ using W as instruments. The infeasible optimal GMM estimator associated to this system of moment conditions should use the weighting matrix $S^* = V(W_i u_i)^{-1} = E(W_i W_i' u_i^2)^{-1}$ (the second equality follows from the fact $E(W_i u_i) = 0$, as one can see from Equation 8.2.2). A feasible GMM estimator should therefore use a weighting matrix converging towards S^* . The weighting matrix used by the 2SLS estimator does not converge towards S^* , unless u_i is an homoscedastic residual, in which case $E(W_i W_i' u_i^2) = E(W_i W_i' E(u_i^2|W_i)) = E(W_i W_i') E(u_i^2)$. The 2SLS weighting matrix converges towards $E(W_i W_i')$ instead of $E(W_i W_i') E(u_i^2)$, but that's not an issue, as multiplying the weighting matrix by a constant will not change the solutions of the problems defining the GMM estimand and its estimator. So the take-away from this discussion is that under a linear and constant effect assumption, if one has more instruments than endogenous variables, there are many possible estimators of the treatment effect parameter ρ . 2SLS is one of them, but it will not be optimal, unless the residuals are homoscedastic. So you might be able to reduce the variance of your estimator by estimating the feasible optimal GMM estimator in two steps instead of doing 2SLS. In a first step, you can estimate the 2SLS coefficients, and then use them to derive your second step estimator.

Another consequence of Theorem 8.2.1 is that Assumptions 5 and 16 are refutable (by this, I mean that they can be proven wrong). Indeed, together they imply that there must exist some (λ', ρ) satisfying Equation 8.2.2. As we saw when we discussed Theorem 6.5.8, one can use an overidentification test to test H_0 : there exists (λ', ρ) such that $E(W_i(Y_i - X_i' \lambda - \rho S_i)) = 0$ against H_1 : for every (λ', ρ) , $E(W_i(Y_i - X_i' \lambda - \rho S_i)) \neq 0$. The test amounts to looking at the value of the GMM objective function at its minimum. If H_0 holds, this value should be close to zero. If this test leads us to reject H_0 , this implies that at least one assumption among Assumptions 5 and 16 is violated. If both were satisfied, we have shown earlier that H_0 should hold, so we should not reject it.

Some years ago, overidentification tests were very popular. People thought that when they had more instruments than endogenous variables, this allowed them to test the validity of their instruments. For instance, Angrist & Krueger (1991) use dummies for being born in each quarter of the year, so they have more instruments than endogenous variables, and they use this to implement overidentification tests. In some models, this leads them to reject Assumptions 5 and 16 but not in others. What can we conclude from the results of these tests?

The answer is: not much. Indeed, in the context of 2SLS, the overidentification test has two drawbacks. First, the logical relationship between Assumptions 5 and 16, and H_0 , the condition which is testable from the data, is an implication, not an equivalence. Therefore, it might very well be the case that either Assumption 5 or 16 is violated while H_0 is satisfied. That's why I said that these assumptions are refutable: they can be proven wrong, not right: failing to reject H_0 , even with an infinity of data at hand, does not imply that Assumptions 5 and 16 are satisfied. In practice, this implies that the overidentification test has low "power".² It's only when Assumptions 5 and 16 are very strongly violated that H_0 will be violated too. For mild violations of Assumptions 5 and 16, H_0 will not be rejected, so we will accept our overidentification test. Let's draw a graph to illustrate the situation.

The second drawback of the overidentification test is that it is a joint test of Assumption 5 and 16. Assumption 16 should never be taken for granted, but there are a number of instances in which it might be credible. For instance, it's reasonable to argue that quarter of birth is uncorrelated with innate ability. At least, it's not obviously violated, so that's an assumption worth testing. On the contrary, Assumption 5 is very often not credible. For instance, it is very unlikely that potential wages increase linearly with years of education, and that returns to education are homogenous across people. So even if our overidentification test leads us to reject jointly Assumptions 5 and 16, we will never be able to know whether this is because Assumption 5 or 16 is violated, in which case we will not have learnt much as we knew ex-ante that Assumption 5 was not very credible. To see why the overidentification test heavily relies on the constant treatment effect assumption, assume that instead of being constant across all observations, the effect of schooling is different for men and women. The linear and constant effect model then rewrites

$$Y_i = \alpha + \rho_1 S_i + \rho_2 S_i M_i + \eta_i,$$

where M_i is a binary variable for males. ρ_1 captures females returns to schooling, while ρ_2 captures the difference between males and females returns to schooling. Assume also that we have two instruments, and that we have no controls. Under this modified LCEM, our system

²Power is a slightly misleading word here: the issue I am discussing is not statistical but logical.

of moment conditions (8.2.2) rewrite as follows:

$$\begin{aligned} E(Y_i - \alpha - \rho_1 S_i - \rho_2 S_i M_i) &= 0 \\ E(Z_{i1}(Y_i - \alpha - \rho_1 S_i - \rho_2 S_i M_i)) &= 0 \\ E(Z_{i2}(Y_i - \alpha - \rho_1 S_i - \rho_2 S_i M_i)) &= 0. \end{aligned}$$

This system has three equations and three unknowns. It is therefore possible to solve exactly its sample equivalent, leaving no room for testing. The overidentification test indeed relies on the idea that with more equations than unknowns, the system of sample moment conditions cannot be exactly cancelled but could be made close to 0, and the test amounts to reject Assumption 16 when the solution is too far from 0. But with as many equations as unknowns, the system of sample moment conditions can always be exactly cancelled, irrespective of whether Assumptions 5 and 16 are satisfied.

Overall, it's hard to know what to do with the overidentification tests. When they are not rejected, this could merely be because violations of Assumptions 5 and 16 are not strong enough for H_0 to be violated. When they are rejected, this could very well be because the effect of the treatment is heterogeneous and non-linear, which in many instances is not very surprising.

8.3 Weak instruments: is the cure is worse than the disease?

A seminal paper by Bound et al. (1995) started an extensive literature showing that applied researchers should be cautious when they use weak instruments, i.e. instruments weakly correlated with the endogenous variable. Indeed, weak instruments are very sensitive to small departures from Assumption 16, and even when Assumption 16 holds they will yield biased estimates of ρ .

Postulating a LCEM was very convenient to analyze what OLS capture when CIA is violated. Similarly, here it is convenient to assess how far 2SLS will be from ρ , when Assumption 16 fails to hold.

Theorem 8.3.1 *The bad instrument formula*

If Z_i is not independent of η_i ,

$$\frac{\text{cov}(Y_i, Z_i)}{\text{cov}(S_i, Z_i)} = \rho + \frac{\text{cov}(\eta_i, Z_i)}{\text{cov}(S_i, Z_i)}.$$

In such instances, the coefficient of S_i in a 2SLS regression of Y_i on S_i using Z_i as an instrument for S_i is further away from ρ than the coefficient of S_i in an OLS regression iif

$$\left| \frac{\text{cov}(\eta_i, Z_i)}{\text{cov}(S_i, Z_i)} \right| \geq \left| \frac{\text{cov}(\eta_i, S_i)}{V(S_i)} \right|$$

Proof

$$\text{cov}(Y_i, Z_i) = \text{cov}(\alpha + \rho S_i + \eta_i, Z_i) = \rho \text{cov}(S_i, Z_i) + \text{cov}(\eta_i, Z_i).$$

Dividing by $\text{cov}(S_i, Z_i)$ yields the first result.

Then, the distance between ρ and $\frac{\text{cov}(Y_i, Z_i)}{\text{cov}(S_i, Z_i)}$, the coefficient of S_i in a 2SLS regression of Y_i on S_i using Z_i as an instrument for S_i , is equal to $\left| \frac{\text{cov}(\eta_i, Z_i)}{\text{cov}(S_i, Z_i)} \right|$. As per the first omitted variable formula, the distance between ρ and the coefficient of S_i in an OLS regression of Y_i on S_i is equal to $\left| \frac{\text{cov}(\eta_i, S_i)}{V(S_i)} \right|$.

QED.

The bad instrument formula shows that when Z_i is not independent of η_i , 2SLS will not capture ρ . For instance, it will overestimate it if Z_i and η_i are positively correlated, because 2SLS will attribute all of the correlation between Y_i and Z_i to S_i , while part of it is actually due to η_i . In instances where Assumption 16 is almost true, i.e. when $\text{cov}(\eta_i, Z_i)$ is very small, 2SLS can still fall far from ρ , if the instrument is weak, i.e. if $\text{cov}(S_i, Z_i)$ is small. When $\text{cov}(S_i, Z_i)$ is small, 2SLS might be further away from ρ than OLS, even when the instrument is less correlated with η_i than S_i ($\text{cov}(\eta_i, Z_i) < \text{cov}(\eta_i, S_i)$). Weak instruments might therefore be very misleading when Assumption 16 is violated, even when said violations are small. This is the first reason why people do not like weak instruments.

The second reason is that even when Assumption 16 holds, 2SLS with weak instruments can yield a severely biased estimator. Under the assumption that $E(e_i|X_i)$ is linear, one can show that OLS estimators are unbiased: their expected value is equal to the OLS population vector. On the contrary, the 2SLS estimator of ρ is biased, because one regressor in the second stage equation is estimated. 2SLS will consistently estimate ρ , but in finite samples the expected value of the 2SLS estimator is different from ρ . The derivation of the approximate value of the bias is long and difficult, so I skip it and directly give you the formula. Let Z_i denote a vector of L instruments, Let $S_i = Z_i' \pi + e_i$ be the first stage regression decomposition of S_i , Let $\hat{\rho}$ denote the estimator of ρ arising from a 2SLS regression of Y_i on S_i using Z_i as instruments for S_i , and let $F = \frac{E((Z_i' \pi)^2)}{E(e_i)^2} \frac{n}{L}$ denote the population F statistic of the first stage regression.

Theorem 8.3.2 *An approximation of the 2SLS bias*

$$E(\hat{\rho} - \rho) \approx \frac{\text{cov}(e_i, \eta_i)}{V(e_i)} \frac{1}{F + 1}.$$

2SLS estimators will be severely biased when we use many instruments and those instruments are weak. Without loss of generality, assume that $\text{cov}(e_i, \eta_i) \neq 0$.³ Then, the bias is a

³If e_i and η_i were uncorrelated, CIA would hold: S_i would be uncorrelated with η_i after controlling for Z_i , so instead of using an instrument one could merely regress Y_i on S_i and Z_i and the coefficient of S_i would pick up ρ . We can therefore rule out this scenario, because here we are interested in situations where we cannot use CIA and have to use an instrument.

decreasing function of F . This statistic is equal to the expectation of the square of the regression function divided by the square of the error term, multiplied by n over the number of instruments. With many weak instruments, L is large, $E((Z'_i\pi)^2)$ is low (in the extreme case where $\pi = 0$, this expectation is equal to 0), and $E(e_i)^2$ is large. This implies that F is low, so that the bias is large. When the sample size goes to infinity, the bias vanishes provided $\pi \neq 0$. It is worth mentioning that with one instrument only, 2SLS is approximately unbiased. So the problem arises only with several instruments.

Therefore, it is very important to look at the first stage F-statistic when you run an IV (IV) regression with several instruments (if you have controls in your regression, you should consider the F-statistic on the excluded instruments only). Stock & Yogo (2002) suggest that if this statistic is above 10 the estimator should not be too biased, even though, as Angrist and Pischke put it, “this cannot be a theorem”. If this statistic is below or around 10, you should compare the 2SLS estimator with the limited information maximum likelihood (LIML) estimator which is unbiased. If the two estimators are close, this is reassuring; if they are not, this means that 2SLS is biased, so you should either stick to the unbiased estimator, or try to find another instrument with a better first stage.

A last important thing to know is that the recommendation of using the LIML estimator when the instrument is weak has been recently challenged. Notice that I have embedded the entire discussion on the bias of 2SLS in a LCEM framework. In a very recent paper, Kolesar (2013) shows that in a model with heterogeneous effects, the LIML estimator does not converge towards a well-defined causal effect parameter, so he suggests a new estimator called UJIV which is unbiased and converges towards a well-defined causal parameter.

Chapter 9

Two stage least squares and causality with heterogeneous effects

9.1 The LATE theorems

9.1.1 Set-up, notations, and the LATE assumption

To simplify the discussion, let us start assuming we are using a binary instrument to measure the effect of a binary treatment on some outcome. In Angrist & Evans (1998), the instrument is a binary variable for whether the first two children in a family are of the same sex or not, the treatment is a binary variable for having a third child, and the outcome is mothers' participation to the labor market. As parents have a preference for diversity, those whose first two children are of the same sex are more likely to have a third child, so the instrument positively affects the treatment. But as the sex of the first two children in a family is the outcome of a random draw, we can expect that it is uncorrelated with all the other determinants of mothers' participation to the labor market, which probably includes father's wage, mother's education... Under the LCEM assumption, one can use this instrument to estimate ρ , the (constant) effect of having a third child on mothers' labor market participation. This section investigates whether this result can be extended to a more general model with heterogeneous treatment effects.

Let Y , D , and Z be random variables denoting, respectively, the outcome, the treatment, and the instrument for an individual randomly drawn from, and therefore representative of, the population we are interested in. In our example, we are primarily interested in the effect of having a third child on mothers' labor market participation, and therefore our representative individual is a mother randomly drawn from the population of US mothers.

One can think of the instrument as initiating a causal chain. The instrument has an effect on D which in turn has an effect on Y . Same-sexness of the first two children has an effect on mothers' decision to have a third child or not, which in turn has an effect on their participation

to the labor market. We are not interested in the effect of same-sexness on the decision of having a third child, but we still have to incorporate this causal link in our model because it provides us with a clean source of variation that we will use to measure the effect of having a third child on mothers labor market participation. We must therefore enrich Rubin's model of potential outcomes, to account for the fact we are considering a more complex causal chain.

Let $D(0)$ denote the treatment decision of the individual we draw if they do not receive the instrument. Let $D(1)$ denote their treatment decision if they receive the instrument. In our example, $D(0)$ is the decision to have a third child or not that the mother we draw will make if their first two children are of a different sex. $D(1)$ is the decision to have a third child or not that the mother we draw will make if their first two children are of the same sex. $D(0)$ and $D(1)$ are never jointly observed. We only observe

$$D = Z \times D(1) + (1 - Z) \times D(0). \quad (9.1.1)$$

To describe in full generality this more complex causal chain, we also introduce potential outcomes depending both on Z and D . This yields four potential outcomes, depending on the instrument and the treatment. Those are denoted $Y(z, d)$. In our example, $Y(0, 1)$ is the decision of the mother we draw to work or not if their first two children are of a different sex ($z = 0$) and if they choose to have a third child ($d = 1$). The following equation relates the observed outcome to the potential outcomes, the observed treatment, and the instrument:

$$Y = (1 - Z)(1 - D)Y(0, 0) + Z(1 - D)Y(1, 0) + (1 - Z)DY(0, 1) + ZDY(1, 1) \quad (9.1.2)$$

Notice that as the two potential treatments are binary, $(D(0), D(1))$ can take only four values, which implies we can partition the population into four subgroups. Subjects satisfying $(D(0) = 0, D(1) = 0)$ are called never takers (NT): they will never get treated, irrespective of whether they receive the instrument or not. In our example, those are parents with a low taste for having a third child: even if their first two children are of the same sex, they do not want to have a third child. Subjects satisfying $(D(0) = 1, D(1) = 1)$ are called always takers (AT): they will always get treated, irrespective of whether they receive the instrument or not. In our example, those are parents with a high taste for having a third child. Subjects such that $(D(0) = 0, D(1) = 1)$ are called compliers (C): they will get treated iff they receive the instrument. In our example, those are parents with a preference for diversity: they do not specifically want to have three children, but they want to have at least a boy and a girl. Finally, subjects such that $(D(0) = 1, D(1) = 0)$ are called defiers (F): they will get treated iff they do not receive the instrument. [In the Angrist & Evans \(1998\) example, which type of parents could be defiers?](#)

If there are parents who want to have at least two boys, or parents who want to have at least two girls, they would be defiers. Indeed, those parents will have a third child iff their first two children are of a different sex.

Assumption 17 *Instruments in the heterogeneous effects model*

The binary variable Z is an instrument to measure the effect of D on Y if the four following statements are verified:

1. *First stage:* $P(D = 1|Z = 1) > P(D = 1|Z = 0)$.
2. *Random instrument:* $(Y(0, 0), Y(1, 0), Y(0, 1), Y(1, 1), D(0), D(1)) \perp\!\!\!\perp Z$.
3. *Exclusion restriction:* for every $d \in \{0; 1\}$, $Y(0, d) = Y(1, d) = Y(d)$.
4. *Monotonicity, or no defiers:* $D(1) \geq D(0)$.¹

This assumption is similar to Assumption 16, but for the monotonicity condition.

$P(D = 1|Z = 1) > P(D = 1|Z = 0)$ is similar to the rank condition. With binary instrument and treatment, $cov(D, Z) \neq 0$ is equivalent to having $P(D = 1|Z = 1) \neq P(D = 1|Z = 0)$ (cf. proof of Lemma 7.1.1). Then, one can assume without loss of generality that $P(D = 1|Z = 1) > P(D = 1|Z = 0)$ (if the converse inequality holds, then one can just consider that the treatment variable is $\tilde{D} = 1 - D$). Anyway, this assumption is testable from the data, so we should not worry about it.

Random instrument means that the instrument is as good as randomly assigned, while the exclusion restriction states it cannot have a direct effect on the outcome. This was already there in Assumption 16, because $\eta \perp\!\!\!\perp Z$ is equivalent to exclusion restriction + random instrument. It is clear that $\eta \perp\!\!\!\perp Z$ implies that the instrument must be as good as randomly assigned. But it also implies an exclusion restriction. Indeed, if $\eta = \lambda Z + \varepsilon$ with $\lambda \neq 0$, meaning that Z has a direct effect on the outcome, Z cannot be independent of η . The potential outcomes & treatments formulation of the model just makes it more salient that $\eta \perp\!\!\!\perp Z$ not only means that the instrument should be as good as randomly assigned, but also that it should not have a direct effect on the outcome.

¹ Alternatively, one could assume that $D(1) \leq D(0)$. What is important is that there should not be subjects for which $D(1) > D(0)$ and subjects for which $D(1) < D(0)$.

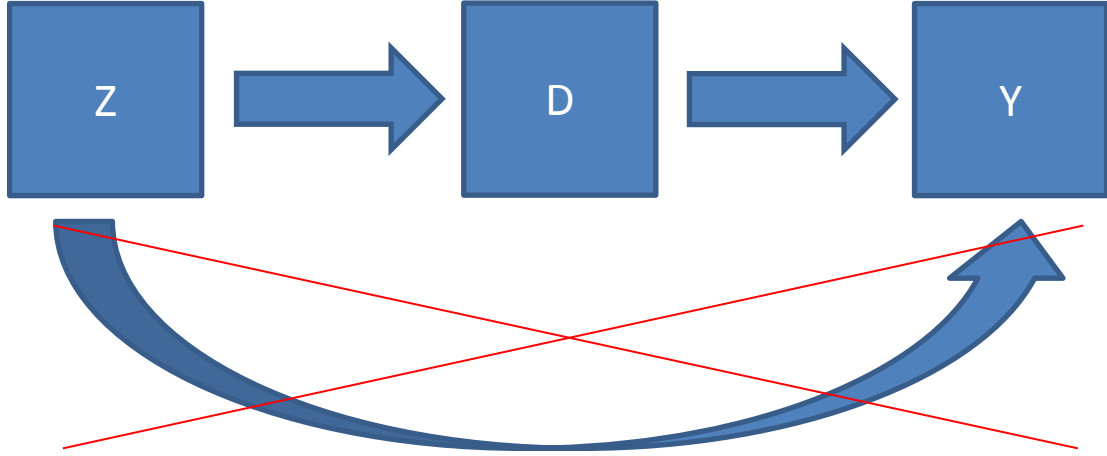


Figure 9.1: Exclusion restriction: Z cannot have a direct effect on Y

Monotonicity is the only new condition in Assumption 17 relative to Assumption 16.

$$\begin{aligned}
 & P(D = 1|Z = 1) - P(D = 1|Z = 0) \\
 = & P(D(1) = 1|Z = 1) - P(D(0) = 1|Z = 0) \\
 = & P(D(1) = 1) - P(D(0) = 1) \\
 = & P(D(0) = 0, D(1) = 1) + P(D(0) = 1, D(1) = 1) \\
 & - P(D(0) = 1, D(1) = 0) - P(D(1) = 1, D(0) = 1) \\
 = & P(C) - P(F).
 \end{aligned} \tag{9.1.3}$$

The first step follows from (9.1.1), the second from the independence assumption, and the third from the law of total probabilities. As D is binary, it follows from Lemma 7.1.1 that $P(D = 1|Z = 1) - P(D = 1|Z = 0)$ is the population first stage coefficient of Z in a 2SLS regression of Y on D using Z as an instrument. If it is positive, this implies that more subjects are compliers than defiers. For instance, in Angrist & Evans (1998), the first stage coefficient is equal to 0.06, meaning that the share of parents having a 3rd child is 6 points higher when their first two children are of the same sex than when they are of a different sex. This means that compliers outnumber defiers by 6%: the share of people who are induced to have a third child because their first two children are of the same sex is 6 percentage points larger than the share of people induced not to have a third child.

But monotonicity is stronger than requiring only that there be more compliers than defiers, as it says there cannot be any defier in the population. In the same sex instrument example, this could be violated, for instance if some parents want at least two boys or two girls.

9.1.2 The LATE theorems

Theorem 9.1.1 *The Local Average Treatment Effect Theorems*

If Z satisfies Assumption 17,

$$\begin{aligned}
P(C) &= P(D = 1|Z = 1) - P(D = 1|Z = 0) \\
E(Y(1) - Y(0)|C) &= \frac{E(Y|Z = 1) - E(Y|Z = 0)}{E(D|Z = 1) - E(D|Z = 0)} \\
P(Y(0) \leq y|C) &= \frac{P(D = 0|Z = 0)P(Y \leq y|D = 0, Z = 0) - P(D = 0|Z = 1)P(Y \leq y|D = 0, Z = 1)}{P(D = 0|Z = 0) - P(D = 0|Z = 1)} \\
P(Y(1) \leq y|C) &= \frac{P(D = 1|Z = 1)P(Y \leq y|D = 1, Z = 1) - P(D = 1|Z = 0)P(Y \leq y|D = 1, Z = 0)}{P(D = 1|Z = 1) - P(D = 1|Z = 0)}.
\end{aligned}$$

Moreover, if X are covariates independent of the instrument and determined prior to the treatment,

$$E(X|C) = \frac{E(XD|Z = 1) - E(XD|Z = 0)}{E(D|Z = 1) - E(D|Z = 0)}.$$

Proof

Combining (9.1.3) and monotonicity yields

$$P(C) = P(D = 1|Z = 1) - P(D = 1|Z = 0). \quad (9.1.4)$$

This proves the first statement.

Then,

$$\begin{aligned}
&E(Y|Z = 1) - E(Y|Z = 0) \\
&= E(Y(1)D + Y(0)(1 - D)|Z = 1) - E(Y(1)D + Y(0)(1 - D)|Z = 0) \\
&= E(Y(1)D(1) + Y(0)(1 - D(1))|Z = 1) - E(Y(1)D(0) + Y(0)(1 - D(0))|Z = 0) \\
&= E(Y(1)D(1) + Y(0)(1 - D(1))) - E(Y(1)D(0) + Y(0)(1 - D(0))) \\
&= E((Y(1) - Y(0))(D(1) - D(0))) \\
&= E[E((Y(1) - Y(0))(D(1) - D(0))|D(1) - D(0))] \\
&= E((Y(1) - Y(0))(D(1) - D(0))|D(1) - D(0) = 1)P(D(1) - D(0) = 1) \\
&+ E((Y(1) - Y(0))(D(1) - D(0))|D(1) - D(0) = 0)P(D(1) - D(0) = 0) \\
&+ E((Y(1) - Y(0))(D(1) - D(0))|D(1) - D(0) = -1)P(D(1) - D(0) = -1) \\
&= E(Y(1) - Y(0)|D(1) - D(0) = 1)P(D(1) - D(0) = 1) \\
&- E(Y(1) - Y(0)|D(1) - D(0) = -1)P(D(1) - D(0) = -1) \\
&= E(Y(1) - Y(0)|C)P(C) - E(Y(1) - Y(0)|F)P(F) \\
&= E(Y(1) - Y(0)|C)P(C).
\end{aligned}$$

The first equality follows from (9.1.2) and the exclusion restriction, the second from (9.1.1), the third from the random instrument condition, the fifth from the law of iterated expectations, and the last from monotonicity. Combining the last equation with (9.1.4) proves the second statement.

Then,

$$\begin{aligned}
& P(D = 0|Z = 0)P(Y \leq y|D = 0, Z = 0) - P(D = 0|Z = 1)P(Y \leq y|D = 0, Z = 1) \\
= & P(D(0) = 0|Z = 0)P(Y(0) \leq y|D(0) = 0, Z = 0) - P(D(1) = 0|Z = 1)P(Y(0) \leq y|D(1) = 0, Z = 1) \\
= & P(D(0) = 0)P(Y(0) \leq y|D(0) = 0) - P(D(1) = 0)P(Y(0) \leq y|D(1) = 0) \\
= & P(Y(0) \leq y, D(0) = 0) - P(Y(0) \leq y, D(1) = 0) \\
= & P(Y(0) \leq y, D(0) = 0, D(1) = 0) + P(Y(0) \leq y, D(0) = 0, D(1) = 1) \\
& - P(Y(0) \leq y, D(0) = 0, D(1) = 0) - P(Y(0) \leq y, D(0) = 1, D(1) = 0) \\
= & P(Y(0) \leq y, D(0) = 0, D(1) = 1) - P(Y(0) \leq y, D(0) = 1, D(1) = 0) \\
= & P(Y(0) \leq y|D(0) = 0, D(1) = 1)P(D(0) = 0, D(1) = 1) \\
& - P(Y(0) \leq y|D(0) = 1, D(1) = 0)P(D(0) = 1, D(1) = 0) \\
= & P(Y(0) \leq y|C)P(C) - P(Y(0) \leq y|F)P(F) \\
= & P(Y(0) \leq y|C)P(C)
\end{aligned}$$

The first equality follows from (9.1.1), (9.1.2) and the exclusion restriction, the second from the random instrument condition and from Lemma 5.1.3, the fourth from the law of total probabilities, and the last from monotonicity. Combining the last equation with the fact that $P(D = 0|Z = 0) - P(D = 0|Z = 1) = P(D = 1|Z = 1) - P(D = 1|Z = 0)$ and (9.1.4) proves the third statement. The proof of the fourth statement is similar, it is left as an exercise.

Finally,

$$\begin{aligned}
E(XD|Z = 1) - E(XD|Z = 0) &= E(XD(1)|Z = 1) - E(XD(0)|Z = 0) \\
&= E(XD(1)) - E(XD(0)) \\
&= E(X(D(1) - D(0))) \\
&= E[E(X(D(1) - D(0))|D(1) - D(0))] \\
&= E(X|D(1) - D(0) = 1)P(D(1) - D(0) = 1) \\
&\quad - E(X|D(1) - D(0) = -1)P(D(1) - D(0) = -1) \\
&= E(X|C)P(C) - E(X|F)P(F) \\
&= E(X|C)P(C).
\end{aligned}$$

The first equality follows from (9.1.1), the second from the random instrument condition, the fourth from the law of iterated expectations, and the last from monotonicity. Combining the last equation with (9.1.4) proves the third statement.

QED.

The theorem above combines results in (Imbens & Angrist 1994), (Angrist et al. 1996), and (Imbens & Rubin 1997).

The intuition underlying the derivation of the second equation goes as follows. The numerator of the rhs of this equation compares $E(Y|Z = 1)$ to $E(Y|Z = 0)$. Irrespective of which value of

the instrument they receive, never takers always choose not to get treated. Moreover, we have assumed that the instrument has no direct effect on the treatment. Combined together, those two facts imply that irrespective of whether $Z = 1$ or $Z = 0$, we always observe never takers' $Y(0)$. Therefore, never takers' Y disappears when we subtract $E(Y|Z = 0)$ to $E(Y|Z = 1)$. For the same reason, always takers also disappear in this comparison. This corresponds to the step between the third and the fourth equality in the proof. When going from $Z = 0$ to $Z = 1$, compliers' Y goes from $Y(0)$ to $Y(1)$, so the difference between $E(Y|Z = 1)$ and $E(Y|Z = 0)$ identifies compliers' treatment effect. Conversely, when going from $Z = 1$ to $Z = 0$, defiers' Y goes from $Y(1)$ to $Y(0)$, so the difference between $E(Y|Z = 1)$ and $E(Y|Z = 0)$ identifies defiers' treatment effect multiplied by minus 1. This corresponds to the steps between the 4th and the 8th equalities. Finally, defiers vanish, because we assume there are no such people.

Observed outcome when $Z=0$	Observed outcome when $Z=1$
Never takers: Y_0	Never takers: Y_0
Always takers: Y_1	Always takers: Y_1
Compliers: Y_0	Compliers: Y_1
Defiers: Y_1	Defiers: Y_0

Figure 9.2: 2 populations vanish in the $Z = 1$ across $Z = 0$ comparison

Combining the second equation of Theorem 9.1.1 with Lemma 7.1.1 shows that if Assumption 17 is satisfied, the coefficient of D in a 2SLS regression of Y on D using Z as an instrument identifies the average effect of the treatment among compliers. Therefore, 2SLS can still identify a treatment effect parameter if treatment effects are heterogeneous across units. With respect to LCEM and Assumption 16, Assumption 17 imposes less assumptions on the effect of the treatment on the outcome as it does not require constant treatment effects, but it imposes more assumptions on the effect of the instrument on the treatment as it requires that there be no defiers.

The third and fourth statements of Theorem 9.1.1 imply that when we have at hand an instrument satisfying Assumption 17, we can not only estimate the average effect of the treatment for compliers, but any quantile treatment effect within this population. Indeed, these statements imply that we can estimate $F_{Y(0)|C}$ and $F_{Y(1)|C}$, which in turn implies we can estimate $\tau_{q|C} = F_{Y(1)|C}^{-1}(q) - F_{Y(0)|C}^{-1}(q)$. This means that we can estimate how the treatment affects the median or the first decile of the outcome among compliers. In practice, estimating

$\tau_{q|C}$ cannot be achieved via 2SLS regressions. One has to use IV-quantile regressions. A thorough study of these regression models goes beyond the scope of this course. For more details, I refer you to Abadie et al. (2002) and Frölich & Melly (2008).

Assume you observe an iid sample (Y_i, D_i, Z_i, X_i) with the same distribution as (Y, D, Z, X) . Can you find which units in this sample are compliers?

No. Compliers are units such that $(D_i(0) = 0, D_i(1) = 1)$. Determining who is a complier requires jointly observing $D_i(0) = 0$ and $D_i(1) = 1$. For units with $Z_i = 0$, we observe $D_i(0)$. Under the no-defiers condition, units with $D_i(0) = 1$ must be always takers. On the other hand, units with $D_i(0) = 0$ can either be never takers or compliers. For units with $Z_i = 1$, we observe $D_i(1)$. Under the no-defiers condition, units with $D_i(1) = 0$ must be never takers. On the other hand, units with $D_i(1) = 1$ can either be always takers or compliers. So the second statement of Theorem 9.1.1 shows that the average treatment effect of compliers is identified, despite the fact there are no unit for which we can be sure that they are a complier. Even more puzzling, the last statement of this theorem says that the mean of any covariate X in that population is identified by the coefficient of D in a 2SLS regression of XD on D using Z as an instrument. So despite the fact we cannot know which units are compliers, we can still describe this population (say in terms of its average age, average income...), and see how it compares to the rest of the population. For instance, Table 4.4.3 in Angrist & Pischke (2008) reports descriptive statistics on the population of compliers in Angrist & Evans (1998).

Let's review one application of Theorem 9.1.1. In Angrist & Evans (1998), the share of women working is 0.8 percentage points lower among mothers whose first two children are of the same sex than among mothers whose first two children are of a different sex: $\hat{E}(Y|Z = 1) - \hat{E}(Y|Z = 0) = -0.008$. Since $\hat{P}(D = 1|Z = 1) - \hat{P}(D = 1|Z = 0) = 0.06$, the estimate of the effect of having a third child on the labor supply of compliant mothers is $\frac{-0.008}{0.06} = -0.13$. 13% of them are induced not to work because they had a third child.

A special case of Theorem 9.1.1 worth mentioning is when $P(D = 1|Z = 0) = 0$, meaning that there are no treated observations among people who did not receive the instrument. $P(D = 1|Z = 0) = 0 \Rightarrow P(D(0) = 1) = 0$, meaning that there are no unit in the population with $D(0) = 1$.

Earlier, we partitioned the population into 4 subgroups. 2 subgroups out of those 4 satisfy $D(0) = 1$. Which are they? Therefore, do we still need to assume that there are no defiers?

Always takers and defiers satisfy $D(0) = 1$. Therefore, if $P(D = 1|Z = 0) = 0$ we can be sure that there are no defiers, we do not need to assume it. Moreover, if $P(D = 1|Z = 0) = 0$ we can also be sure there are no always takers. Therefore, treated units among those with $Z = 1$ can only be compliers. In this special case, we can actually identify the compliers among units that receive the instrument. This result is attributed to Bloom (1984). $P(D = 1|Z = 0) = 0$ is called a one-sided non-compliance situation. Perfect compliance is when $P(D = 1|Z = 0) = 0$ and $P(D = 1|Z = 1) = 1$. Full imperfect compliance is when $P(D = 1|Z = 0) > 0$ and $P(D = 1|Z = 1) < 1$ and one sided non-compliance is when $P(D = 1|Z = 0) = 0$ and $P(D = 1|Z = 1) < 1$ (no always takers) or $P(D = 1|Z = 0) > 0$ and $P(D = 1|Z = 1) = 1$ (no never takers).

9.1.3 Assessing the plausibility of the LATE assumptions

The first-stage assumption is directly testable. The random instrument assumption is also testable. [How could you test that \$Z\$ is randomly assigned?](#)

You can just run balancing checks comparing the mean of some predetermined covariates among units with $Z = 1$ and units with $Z = 0$. If you find many significant differences between these two groups, Z is not randomly assigned. For instance, Buckles & Hungerman (2013) find that mothers giving birth in the last and in the first quarter of the year have significantly different socio-demographic characteristics: winter births are disproportionately realized by teenagers and the unmarried. Therefore, quarter of birth is not a randomly assigned instrument.

Given that first-stage and random assignment are testable, we focus our discussion on exclusion restriction and no-defiers. These assumptions are not fully testable, so it's important to assess their plausibility in any application. In what follows, we review examples of papers which have challenged the validity of exclusion restriction and/or no-defiers for some specific instruments.

Examples of articles which have challenged the exclusion restriction

Some people argue that the exclusion restriction and the no-defiers assumption are not always applicable.

[Do you think that the exclusion restriction is plausible in the context of the same-sex instrument?](#)

Rosenzweig & Wolpin (2000) argue that exclusion restriction might be violated due to sex-specific hand-me-downs. Having a second child of the same sex as the first one reduces the cost of raising him/their, relative to the situation where the second child is of a different sex: in such instances parents do not have to buy, say, new clothes for the second child. This changes the family's budget constraint which could have an effect on the mother's decision to work or not. They show that in India, such sex-specific hand-me-downs account for a substantial share of family budget. [In the context studied in Angrist & Evans \(1998\) do you think these sex-specific hand-me-downs will strongly impact households budget constraints?](#)

In the US, the argument in Rosenzweig & Wolpin (2000) remains valid, but those sex-specific hand-me-downs probably account for a much lower fraction of family income.

Examples of articles which have challenged the no-defiers condition

Many papers have used the differential severity of randomly assigned judges to estimate the effect of, say, incarceration on say, earnings, or recidivism. [Assume that there are only two judges, A and B, and B is on average more severe than A \(B sends a greater proportion of their randomly assigned defendants to jail\). To instrument for whether a defendant is sent to jail, one would use a binary variable for whether that defendant is randomly sent to judge B. In this context what does the no-defiers condition mean? Do you think it is plausible?](#)

No-defiers requires that for every defendant in the population, judge B is more severe than A. From the data, we can see that B is on average more severe than A, but what no-defiers requires is stronger than that: B should be more severe with everyone. Imbens & Angrist (1994) argue that the no-defiers restriction is not plausible in this context. Judge B being on average more severe than judge A does not necessarily imply that for any single defendant, B will also be more severe than A. A might be more liberal than B, and therefore more severe towards defendants from high socio-economics backgrounds, and less severe towards defendants from low-socioeconomic backgrounds. If that's the case, and if the population bears more defendants from low socio-economic backgrounds, B will be on average more severe than A, but A will be more severe than B for some defendants.

The instrument in Angrist & Evans (1998) might also violate the no-defiers assumption, if there are some parents that want at least two boys, or parents that want at least two girls. De Chaisemartin (2017) shows that in the 2012 Peruvian wave of the Demographic and Health Surveys, women were asked their ideal sex sibship composition. Among women whose first two children is a boy and a girl, 1.8% had 3 children or more and retrospectively declare that their ideal sex sibship composition would have been two boys and no girl, or no boy and two girls. These women seem to have been induced to having a third child because their first two children were a boy and a girl.

If the no-defier condition is violated, the 2SLS coefficients in those studies do not estimate causal effects. Indeed, you can see from the proof of Theorem 9.1.1 that without the no-defiers assumption, the 2SLS estimand identifies a weighted difference of the LATEs of compliers and defiers. This weighted difference might lie far both from the LATE of compliers and from the LATE of defiers. It might for instance be positive while the two LATEs are negative. For instance, in Angrist & Evans (1998), let's assume there are 2% of defiers. The first stage coefficient is 0.06. Then, following Equation (9.1.3) this implies that there are 8% of compliers in the population, and following the last-but-one equality in the proof of the second statement of Theorem 9.1.1, we have that the 2SLS coefficient identifies $E(Y(1) - Y(0)|C) \frac{0.08}{0.06} - E(Y(1) - Y(0)|F) \frac{0.02}{0.06}$. Assume for instance that $E(Y(1) - Y(0)|C) = -0.1$ and $E(Y(1) - Y(0)|F) = -0.05$. Then, the 2SLS coefficient will be equal to -0.12 , which is lower than the LATE of compliers and the LATE of defiers.

9.1.4 Testing and weakening the LATE assumptions

The LATE assumptions might sometimes not appear plausible. A first route to address this concern has been to develop a statistical test of Assumption 17. [Under Assumption 17, which properties should](#)

$$y \mapsto \frac{P(D = 0|Z = 0)P(Y \leq y|D = 0, Z = 0) - P(D = 0|Z = 1)P(Y \leq y|D = 0, Z = 1)}{P(D = 0|Z = 0) - P(D = 0|Z = 1)}$$

and

$$y \mapsto \frac{P(D = 1|Z = 1)P(Y \leq y|D = 1, Z = 1) - P(D = 1|Z = 0)P(Y \leq y|D = 1, Z = 0)}{P(D = 1|Z = 1) - P(D = 1|Z = 0)}$$

satisfy?

Under Assumption 17,

$$P(Y(0) \leq y|C) = \frac{P(D=0|Z=0)P(Y \leq y|D=0, Z=0) - P(D=0|Z=1)P(Y \leq y|D=0, Z=1)}{P(D=0|Z=0) - P(D=0|Z=1)}$$

$$P(Y(1) \leq y|C) = \frac{P(D=1|Z=1)P(Y \leq y|D=1, Z=1) - P(D=1|Z=0)P(Y \leq y|D=1, Z=0)}{P(D=1|Z=1) - P(D=1|Z=0)}.$$

Therefore, if Assumption 17 holds,

$$y \mapsto \frac{P(D=0|Z=0)P(Y \leq y|D=0, Z=0) - P(D=0|Z=1)P(Y \leq y|D=0, Z=1)}{P(D=0|Z=0) - P(D=0|Z=1)}$$

and

$$y \mapsto \frac{P(D=1|Z=1)P(Y \leq y|D=1, Z=1) - P(D=1|Z=0)P(Y \leq y|D=1, Z=0)}{P(D=1|Z=1) - P(D=1|Z=0)}$$

should be non-decreasing everywhere. Let H_0 denote this null hypothesis. H_0 is not automatically true, because these functions are differences between two cdfs, so this difference might actually be decreasing for some values of y . This shows that Assumption 17 is testable: it implies a non-automatically true restriction on some estimands.

Using an iid sample (Y_i, D_i, Z_i) with the same distribution as (Y, D, Z) , one can form estimators of these two functions: $y \mapsto \frac{\hat{P}(D=0|Z=0)\hat{P}(Y \leq y|D=0, Z=0) - \hat{P}(D=0|Z=1)\hat{P}(Y \leq y|D=0, Z=1)}{\hat{P}(D=0|Z=0) - \hat{P}(D=0|Z=1)}$ and $y \mapsto \frac{\hat{P}(D=1|Z=1)\hat{P}(Y \leq y|D=1, Z=1) - \hat{P}(D=1|Z=0)\hat{P}(Y \leq y|D=1, Z=0)}{\hat{P}(D=1|Z=1) - \hat{P}(D=1|Z=0)}$, and one can use these estimators to test H_0 . If our test rejects H_0 , then we can also reject Assumption 17, because if it were satisfied then H_0 should be satisfied. [Intuitively, how could you use](#)

$$y \mapsto \frac{\hat{P}(D=0|Z=0)\hat{P}(Y \leq y|D=0, Z=0) - \hat{P}(D=0|Z=1)\hat{P}(Y \leq y|D=0, Z=1)}{\hat{P}(D=0|Z=0) - \hat{P}(D=0|Z=1)}$$

and

$$y \mapsto \frac{\hat{P}(D=1|Z=1)\hat{P}(Y \leq y|D=1, Z=1) - \hat{P}(D=1|Z=0)\hat{P}(Y \leq y|D=1, Z=0)}{\hat{P}(D=1|Z=1) - \hat{P}(D=1|Z=0)}$$

to test H_0 ?

One could compute the largest decrease of the functions

$$\frac{\hat{P}(D=0|Z=0)\hat{P}(Y \leq y|D=0, Z=0) - \hat{P}(D=0|Z=1)\hat{P}(Y \leq y|D=0, Z=1)}{\hat{P}(D=0|Z=0) - \hat{P}(D=0|Z=1)}$$

and

$$\frac{\hat{P}(D=1|Z=1)\hat{P}(Y \leq y|D=1, Z=1) - \hat{P}(D=1|Z=0)\hat{P}(Y \leq y|D=1, Z=0)}{\hat{P}(D=1|Z=1) - \hat{P}(D=1|Z=0)}$$

between two values y_1 and y_2 such that $y_1 \leq y_2$. Under H_0 , the probability limits of these two functions are non-decreasing, so when the sample size goes to infinity this “largest decrease” should converge to 0. With a finite sample, maybe these two functions can decrease a bit, but this decrease can only come from sampling error, and therefore it cannot be “too large”. We can reject H_0 when this largest decrease is too large. Kitagawa (2013) and Mourifie & Wan (2014) have developed statistical tests of H_0 .

Unfortunately, the relationship between Assumption 17 and H_0 is only an implication, not an if and only if. H_0 might very well hold while Assumption 17 is violated. Failing to reject H_0 does not imply that you can safely regard Assumption 17 as valid. Kitagawa (2013) also shows that H_0 is the strongest testable implication of Assumption 17: if H_0 is satisfied, you can exhibit random variables $(\tilde{Y}(1), \tilde{Y}(0), \tilde{D}(1), \tilde{D}(0))$ which might not be the true $(Y(1), Y(0), D(1), D(0))$ generating (Y, D, Z) , the random variables you observe, but which are compatible with what you observe, and which satisfy Assumption 17. In other words, when H_0 is satisfied, you cannot reject Assumption 17. So the tests of Kitagawa (2013) and Mourifie & Wan (2014) are the best possible tests of Assumption 17, but they might fail to detect violations of this assumption. Overall, we say that Assumption 17 is refutable but not fully testable: it can be proven wrong, but not right.

A second route has been to try to weaken the LATE assumptions. For instance, in De Chaisemartin (2017) I show that it is possible to substantially weaken the no-defiers assumption.

Assumption 18 *The compliers-defiers (CD) assumption*

There is a subpopulation of compliers CF which satisfies:

$$P(CF) = P(F) \tag{9.1.5}$$

$$E(Y(1) - Y(0)|CF) = E(Y(1) - Y(0)|F). \tag{9.1.6}$$

Theorem 9.1.2 *Suppose Z satisfies the random instrument and exclusion restriction Assumptions. If a subpopulation of compliers CF satisfies (9.1.5) and (9.1.6), then $CV = C \setminus CF$ satisfies*

$$P(CV) = P(D = 1|Z = 1) - P(D = 1|Z = 0) \tag{9.1.7}$$

$$E(Y(1) - Y(0)|CV) = \frac{E(Y|Z = 1) - E(Y|Z = 0)}{E(D|Z = 1) - E(D|Z = 0)}. \tag{9.1.8}$$

Conversely, if a subpopulation of compliers CV satisfies (9.1.7) and (9.1.8), then $CF = C \setminus CV$ satisfies (9.1.5) and (9.1.6).

Proof

Start from the last but one equation in the derivation of the second statement of the LATE Theorem. If there is a subgroup of compliers CF with same size as defiers and same average

treatment effect, and if you denote CV the remaining part of compliers, then

$$\begin{aligned}
E(Y|Z = 1) - E(Y|Z = 0) &= E(Y(1) - Y(0)|C)P(C) - E(Y(1) - Y(0)|F)P(F) \\
&= E(Y(1) - Y(0)|CV)P(CV) + E(Y(1) - Y(0)|CF)P(CF) \\
&\quad - E(Y(1) - Y(0)|F)P(F) \\
&= E(Y(1) - Y(0)|CV)P(CV).
\end{aligned}$$

Moreover, it follows from (9.1.3) that

$$\begin{aligned}
P(D = 1|Z = 1) - P(D = 1|Z = 0) &= P(C) - P(F) \\
&= P(CV) + P(CF) - P(F) \\
&= P(CV).
\end{aligned}$$

Combining the two last equations yields

$$E(Y(1) - Y(0)|CV) = \frac{E(Y|Z = 1) - E(Y|Z = 0)}{P(D = 1|Z = 1) - P(D = 1|Z = 0)}.$$

This completes the proof.

QED.

This theorem implies that 2SLS still identifies the LATE of a subgroup of units even if there are defiers in the population, provided you can extract from compliers a subgroup similar to defiers. It also implies that the CD condition is the weakest condition under which the 2SLS estimand identifies a LATE: if it is violated, the 2SLS estimand cannot be interpreted as a causal effect, but only as the difference between two causal effects.

Then I give conditions which are sufficient for the CD to hold. If the outcome is binary, CD will automatically be satisfied if $E(Y(1) - Y(0)|F)$, i.e. the LATE of defiers, has the same sign as the 2SLS coefficient of D . This condition is likely to be satisfied in a number of applications. Maestas et al. (2013) study the effect of receiving disability insurance (DI) on labor market participation. They use average allowance rates of randomly assigned examiners as an instrument for receipt of DI. No-defiers might be violated in this context, if examiners on average more severe are not more severe than more lenient examiners for every case. The 2SLS coefficient in this paper is negative, so CD will hold if $E(Y(1) - Y(0)|F) \leq 0$. A sufficient condition for this to be true is the following condition: $Y(1) - Y(0) \leq 0$. Theory suggests this is a credible assumption. By increasing non-labor income, DI should unambiguously diminish labor supply. Aizer & Doyle (2013) study the effect of juvenile incarceration on high school completion. They use average sentencing rates of randomly assigned judges as an instrument for incarceration. Their 2SLS coefficient is negative, so Assumption 18 will hold if $E(Y(1) - Y(0)|F) \leq 0$. This is a credible condition. Being incarcerated disrupts schooling and increases the chances a youth form relationships with non-academically oriented peers. This should increase the chances of drop-out. Prison education programs might have a positive

effect on taste for schooling, but it sounds implausible they can offset schooling disruption and negative peer effects. Moreover, it suffices that incarceration have on average a negative effect among defiers for CD to hold.

One can show that this condition is equivalent to assuming that compliers' and defiers' LATE are of the same sign, and $\frac{E(Y(1)-Y(0)|F)}{E(Y(1)-Y(0)|C)} \leq 1 + \frac{\alpha_Z}{P(F)}$. Given that $1 + \frac{\alpha_Z}{P(F)} \geq 1$, this condition is automatically satisfied if one is ready to assume defiers' LATE is of the same sign as compliers' and is also closer to 0, meaning that defiers are less affected by the treatment.

If the outcome is binary, another condition which ensures that CD is satisfied is:

$$|E(Y(1) - Y(0)|C) - E(Y(1) - Y(0)|F)| \leq \Delta(P(F)) = |\beta_D| \frac{\alpha_Z}{\alpha_Z + P(F)}.$$

This condition places a restriction on the difference between the LATEs of defiers and compliers. The upper bound on LATEs difference, $\Delta(P(F))$, is decreasing in the share of defiers, and increasing in $|\beta_D|$, the absolute value of the 2SLS coefficient of D and α_Z , the first stage coefficient of Z . When $P(F)$ is low, compliers and defiers can have fairly different LATEs under this condition. When $P(F)$ is large, compliers and defiers LATEs should be fairly similar. This condition is also more credible when the instrument has large first and second stages. It is likely to be satisfied in a number of applications. The left axis of Figure 9.3 shows the sample counterpart of $\Delta(P(F))$ for all values of $P(F)$ included between 0 and 37.2%. The right axis shows the same quantity normalized by the standard deviation of the outcome. Assumption 18 is satisfied for values of $P(F)$ and $|E(Y_1 - Y_0|C) - E(Y_1 - Y_0|F)|$ below the green line. For instance, Assumption 18 holds if there are less than 5% of defiers and compliers and defiers LATEs differ by less than 7.2 percentage points.

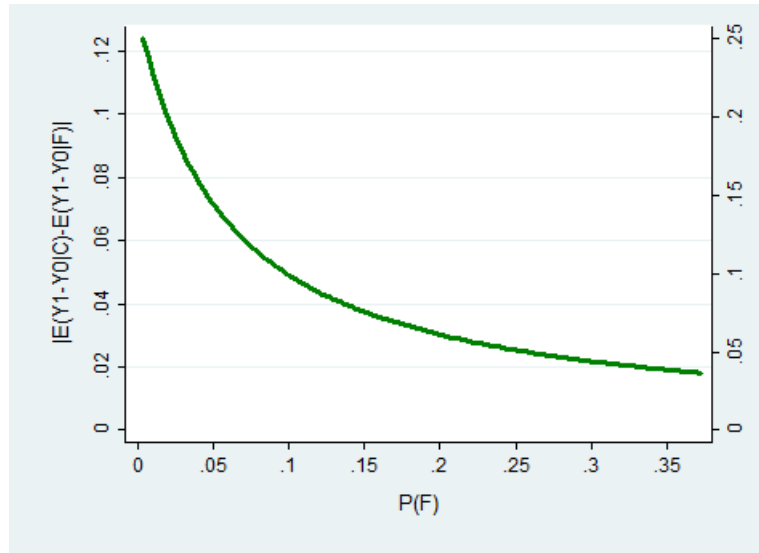


Figure 9.3: For all values of $P(F)$ and $|E(Y(1) - Y(0)|C) - E(Y(1) - Y(0)|F)|$ below the green line, CD is satisfied in Angrist & Evans (1998).

The very limited evidence available suggests that 5% is a reasonably conservative upper bound for the share of defiers in this application. As I said earlier, in the 2012 Peruvian wave of the Demographic and Health Surveys, women were asked their ideal sex sibship composition. Among women whose first two children is a boy and a girl, 1.8% had 3 children or more and retrospectively declare that their ideal sex sibship composition would have been two boys and no girl, or no boy and two girls. These women seem to have been induced to having a third child because their first two children were a boy and a girl. Similar questions have never been asked in an American survey. There are many reasons why 1.8% could under or overestimate the share of defiers in the American population. But this figure is, as of now, the best piece of evidence available to make a guess on the percentage of defiers in Angrist & Evans (1998). 5% therefore sounds like a reasonably conservative upper bound. 7.2 percentage points also sounds like a reasonably conservative upper bound for $|E(Y(1) - Y(0)|C) - E(Y(1) - Y(0)|F)|$ in this application. Compliers are couples with a preference for diversity, while defiers are sex-biased couples. Preference for diversity and sex bias are probably correlated with some of the variables entering into mothers' labor market participation equation in standards models of labor supply (mother's potential wage, preference for leisure...), but they are generally not directly included into this equation. Selection into being a complier or a defier is not directly based on gains from treatment, so compliers' and defiers' LATEs are unlikely to be extremely different.

Finally, irrespective of whether the outcome is binary or not, another condition which ensures that CD holds is: for every δ in the support of $Y(1) - Y(0)$,

$$\frac{f_{Y(1)-Y(0)|F}(\delta)}{f_{Y(1)-Y(0)|C}(\delta)} \leq 1 + \frac{\alpha_Z}{P(F)}.$$

The right hand side of the previous display is decreasing in $P(F)$. If there are many defiers, CD is satisfied if the distributions of $Y(1) - Y(0)$ among compliers and defiers are not too different. Conversely, if these two distributions are very different, CD can be satisfied if there are few defiers. The right hand side of the previous display is also increasing in α_Z : CD is more likely to hold when the instrument has a strong rather than a weak first stage.

Overall, 2SLS identifies:

1. The constant treatment effect parameter in a model that fully restricts treatment effect heterogeneity while leaving the effect of the instrument on the treatment unrestricted.
2. The average treatment effect of compliers in a model that allows for fully heterogeneous treatment effects while restricting the effect of the instrument on the treatment to be monotonic.
3. The average treatment effect of a subset of compliers in a model that imposes some restrictions on the degree of heterogeneity between the LATEs of compliers and defiers while leaving the effect of the instrument on the treatment unrestricted.

From a practical standpoint, my study shows that even when there are defiers, there are three types of studies in which the 2SLS estimand still identifies a LATE. First, CD is plausible in studies where there is little uncertainty on the sign of the treatment effect, as is the case in Maestas et al. (2013) and Aizer & Doyle (2013), for example. My condition is also plausible in studies where selection into being a complier or a defier is not directly based on gains from treatment, as is the case in Angrist & Evans (1998), for example. Finally, CD is plausible in studies with large first-stage coefficients. On the other hand, there are still instances in which even CD could fail, because none of the aforementioned criteria is satisfied. Estimates arising from these studies should be interpreted with caution, as they might not estimate anything else than the difference between two LATEs, and this difference might be far from these two LATEs.

9.2 RCTs with imperfect compliance: you can't always get what you want.

9.2.1 In a RCT with imperfect compliance, one can estimate a LATE...

An important consequence of Imbens & Angrist (1994) is that RCT can identify a well-defined treatment effect parameter under weak assumptions, even when compliance to the random assignment is not perfect, as is most often the case in the social sciences. In medicine, RCTs are double blinded. Patients do not know whether the blue pill they are receiving is the treatment or the placebo. Doctors also do not know who receives the real pill. As a result, patients cannot escape the experimental design, for instance by asking doctors to give them the real pill. In medicine, the treatment actually received is equal to the random assignment. In the social sciences, it is not possible to run double blinded experiments. For instance, in the Star experiment, students and teachers were randomly assigned to small and normal-sized classrooms. Parents of children assigned to normal-sized classrooms could see that their child was not in a small-sized classroom. Therefore, some parents of children initially sent to normal-sized classrooms bargained with principals and managed to have them finally sent to small sized class room. In the social sciences, random assignment is most often not equal to the treatment actually received.

One solution people sometimes use to circumvent this problem is to discard treated subjects in the control group and untreated subjects in the treatment group, and then compare untreated subjects in the control group to treated subjects in the treatment group, to ensure they compare a fully treated group to a fully untreated group. [Is this a good idea?](#)

Nope, this is a really bad idea. This amounts to comparing the always takers and the compliers in the treatment group, to the compliers and the never takers in the control group. If always takers and never takers do not have the same potential outcomes, which is likely to be the case, these two groups are not comparable. In the Star experiment, there was no never takers. There, this solution would amount to comparing always takers and compliers in the treatment group to compliers in the control group. [Explain why these two groups are not comparable.](#)

Always takers get discarded from the control group, while they are kept in the treatment group (because in this group one cannot distinguish always takers from compliers). As a result, the modified control group bears only compliers, while the modified treatment group bears both always takers and compliers. Always takers are likely to be different from compliers. For instance, their parents are likely to be better connected with school principals. Therefore, these two groups are not comparable.

Another solution is to stick to the random assignment and compare the mean of Y among subjects initially sent to the treatment group (even though some of them might finally have opted out of treatment) to the mean of Y among subjects initially sent to the control group (even though some of them might finally have managed to receive the treatment). This is a much better solution, because this amounts to comparing two truly comparable groups. Actually, this amounts to doing the same thing as in an RCT with perfect compliance, except that here the difference between the two groups is not that one received the treatment while the other did not, but that one group was assigned to treatment by the lottery while the other was not. This comparison therefore identifies the effect of intending to treat people, of assigning them to treatment, so we call this parameter the intention to treat parameter (ITT).

A third solution is to use random assignment Z as an instrument for treatment. Z indeed satisfies the random instrument condition by construction. It also seems reasonable to assume it satisfies the exclusion restriction (the result of a lottery should not have per se an effect on outcomes) and the no-defiers condition (defiers would be subjects who manage to get treated if they are assigned to the control group, and who manage not to get treated if they are assigned to the treatment group: such people might exist, but that sounds unlikely). Since it is plausible that Z satisfies Assumption 17, we can use Theorem 9.1.1 to assert that the coefficient of D in a 2SLS regression of Y on D using Z as an instrument will capture the effect of the treatment among subjects who comply with the experimental design. To estimate the ITT, one can merely run the reduced form regression of Y on Z .

Another consequence of Imbens & Angrist (1994) is that when running an experiment raises

ethical issues because it seems too unfair to deprive some people from treatment, we can use a softer experimental design called the encouragement design. Assume you want to measure the effect of a training program for unemployed people on the probability they find a job in less than 6 months. In this context, the encouragement design amounts to selecting randomly a subgroup which will receive an encouragement to get treated (e.g.: some money if people show up), while the other group can show up to the training program but does not receive any incentive to do so. Provided the incentive is successful in raising the share of people following the training program, random assignment to the encouragement group can be used as an instrument to measure the effect of the training program among the compliant subpopulation, i.e. among subjects not motivated enough to participate in the training program if they do not receive an incentive/information, but motivated enough to do so if they receive it.

Notwithstanding, such designs might violate the exclusion restriction and the no-defiers assumption. Paying unemployed so that they participate to a job training program can have a direct effect on their labor supply. If the amount is small, this effect is probably negligible, as in the same-sex instrument example, but with large amounts this could become a concern. Moreover, the financial incentive given to the encouragement group could actually discourage part of it from getting treated, which would violate the no-defiers assumption. Indeed, psychologists have shown in a number of contexts that paying people to do something can decrease their intrinsic motivation and lead them to actually not do it. Benabou & Tirole (2003) give an interesting micro-foundation to this phenomenon. They adopt a principal-agent framework to model the situation. The principal wants the agent to perform a task, whose quality/difficulty is unknown to the agent. The agent therefore seeks to extract information from the principal. If they see that the principal is offering their money to complete the task, they might infer that the task must be boring / difficult, otherwise the principal would not need to pay them so they complete it. In an encouragement design, paying subjects to get treated might lead them not to undertake the treatment, as this might lead them to infer the treatment is not worth undertaking.

9.2.2 ... though we will have low statistical power...

Imperfect compliance to the random assignment does not impair the internal validity of randomized experiments, but it has serious consequences on their statistical power. Assume we randomly assign n subjects randomly drawn from a large population to a control and to a treatment group. Let n_1 and n_0 respectively denote the number of subjects randomly assigned to the control group, and let $p = \frac{n_1}{n_0+n_1}$. Also, let us assume that $V(Y(1)) = V(Y(0)) = \sigma^2$. Then, one can show that in a randomized experiment with imperfect compliance, the Minimum detectable effect for the LATE is approximately equal to

$$\sqrt{\frac{\sigma^2}{np(1-p)}} \frac{1}{(P(D=1|Z=1) - P(D=1|Z=0))} (q_{1-\frac{\alpha}{2}} + q_\lambda).$$

The MDD of the LATE is proportional to 1 over the compliance rate. To perform power computations before starting an experiment, researchers must therefore form expectations on the rate of compliance with the experimental design. That is not always easy: there is some guessing involved in ex-ante power computations.

Non compliance substantially reduces the statistical power of a randomized experiment. When $P(D = 1|Z = 1) - P(D = 1|Z = 0) = 0.25$, meaning that 75% of subjects do not comply with their random assignment, the MDE is multiplied by four relative to a situation with perfect compliance. The intuition for this result goes as follows. In a randomized experiment with perfect compliance, the parameter of interest is the ITT. When compliance is imperfect, the parameter of interest becomes the LATE. The LATE is equal to the ITT divided by $P(D = 1|Z = 1) - P(D|Z = 0)$, so its MDE is the MDE of the ITT divided by $P(D = 1|Z = 1) - P(D|Z = 0)$. If a design can detect a 0.10 effect with perfect compliance, this implies that with imperfect compliance it still can detect a 0.10 ITT effect, which is equivalent to having a 0.40 effect among compliers if compliers account for 25% of the population.

As a side comment, this also shows that encouragement designs often have very low power, unless the encouragement is strong enough to ensure that $P(D = 1|Z = 1) - P(D = 1|Z = 0)$ is large.

9.2.3 ... and studying the LATE is hard to motivate from a decision-theoretic perspective.

Besides power issues, a more substantial concern with randomized experiments with imperfect compliance is that they often have low external validity, because they capture the effect of the treatment for a subgroup only.² While it is very easy to motivate the study of the ATE by considering a utilitarian social planner who tries to decide whether they should give the treatment to their population (cf. Chapter 2), it is less straightforward to motivate why we should care about the LATE. Knowing the LATE will not help the social planner decide whether they should give the treatment to their population. The LATE only applies to compliers, so it might differ from the average effect of the treatment if never takers and always takers have a different average treatment effect than compliers. Using the LATE to determine whether they should give the treatment to the full population might lead the planner to make the wrong decision.

Instead of estimating the LATE, another approach which has been advocated in the literature is to focus on the ATE and try to recover as much information as possible on this parameter under Assumption 17. This is the approach taken in Manski (1990), Balke & Pearl (1997), and Kitagawa (2009). For instance, Manski (1990) shows that if the outcome is bounded, one can

²I discuss the issue of external validity of 2SLS estimates only in the context of randomized experiments, but actually the same concern arises with any 2SLS estimate.

derive bounds for the ATE. Here I consider the result derived in Manski (1990), even though Kitagawa (2009) has shown that one can derive narrower bounds in some circumstances.

Theorem 9.2.1 *Bounds for the ATE with binary outcome*

If Y is binary, and Z satisfies random instrument and exclusion restriction, then

$$B_- \leq ATE \leq B_+,$$

where

$$\begin{aligned} B_- &= P(D = 1|Z = 1)E(Y|D = 1, Z = 1) \\ &\quad - (P(D = 0|Z = 0)E(Y|D = 0, Z = 0) + P(D = 1|Z = 0)) \end{aligned}$$

and

$$\begin{aligned} B_+ &= P(D = 1|Z = 1)E(Y|D = 1, Z = 1) + P(D = 0|Z = 1) \\ &\quad - P(D = 0|Z = 0)E(Y|D = 0, Z = 0). \end{aligned}$$

Proof

Under random instrument and exclusion restriction,

$$\begin{aligned} &P(D = 1|Z = 1)E(Y|D = 1, Z = 1) + P(D = 0|Z = 1)E(Y(1)|D(1) = 0) \\ &= P(D(1) = 1|Z = 1)E(Y(1)|D(1) = 1, Z = 1) + P(D(1) = 0|Z = 1)E(Y(1)|D(1) = 0) \\ &= P(D(1) = 1)E(Y(1)|D(1) = 1) + P(D(1) = 0)E(Y(1)|D(1) = 0) \\ &= E(Y(1)). \end{aligned}$$

As we assumed that the outcome is binary,

$$0 \leq E(Y(1)|D(1) = 0) \leq 1,$$

which in turn implies

$$P(D = 1|Z = 1)E(Y|D = 1, Z = 1) \leq E(Y(1)) \leq P(D = 1|Z = 1)E(Y|D = 1, Z = 1) + P(D = 0|Z = 1).$$

Similarly, one can show that

$$P(D = 0|Z = 0)E(Y|D = 0, Z = 0) \leq E(Y(0)) \leq P(D = 0|Z = 0)E(Y|D = 0, Z = 0) + P(D = 1|Z = 0).$$

Combining the last two equations yields the result.

QED.

In some instances (but not in all instances, as shown in Kitagawa (2009)), the lhs and rhs of the previous display are sharp bounds for the ATE under the random instrument and exclusion restriction assumptions. Sharp means that those bounds exhaust all the information you can derive on the ATE from the distribution of (Y, D, Z) , the variables you observe in the data,

under random instrument and exclusion restriction.³ Indeed, you can exhibit random variables $(\tilde{Y}(1), \tilde{Y}(0), \tilde{D}(1), \tilde{D}(0))$ which might not be the true $(Y(1), Y(0), D(1), D(0))$ generating (Y, D, Z) , the random variables you observe, but which are compatible with the distribution of (Y, D, Z) you observe in the data, which satisfy exclusion restriction and random instrument, and which are such that if $(\tilde{Y}(1), \tilde{Y}(0), \tilde{D}(1), \tilde{D}(0))$ were the true $(Y(1), Y(0), D(1), D(0))$, the ATE would be equal to B_- (you can show the same thing for B_+). Putting it in other words, under exclusion restriction and random instrument, the ATE might be equal to B_- or to B_+ , so in order to derive narrower bounds for the ATE one would have to make more assumptions.

Deriving bounds for parameters of interest is called partial identification or set identification. This approach has been initiated by Manski in several seminal papers in the 90s. For this particular problem of bounding the ATE, this method has not been used so often by applied researchers, because the bounds are usually not very informative.⁴ The width of $[B_-, B_+]$ is equal to $P(D = 0|Z = 1) + P(D = 1|Z = 0)$. In (Angrist & Evans 1998), this is equal 94%. This means that in this paper, if we use bounds we will only be able to say something like “having a third child has an effect on mothers’ labor market participation included between -40% and $+54\%$ ”. Because in the end of a long and tiring research project, it is more satisfactory to reach firm and definitive conclusions than just being able to make such a vague statement, people usually prefer computing the LATE than bounds for the ATE. But this might hide the fact that however unsatisfactory this might be, even in the end of a long and tiring research project there is still some huge uncertainty on the parameter we were interested in the first place.

While the LATE is often not “policy relevant” in randomized experiments with imperfect compliance, in many instances the ITT is. In many cases, the social planner cannot compel people to receive the treatment, so the relevant policy decision to consider is whether they should offer the treatment or not to their population. The ITT is the average effect of offering the treatment, so that’s the “policy relevant” parameter to make that decision.

Overall, in a RCT with imperfect compliance it seems reasonable to report the estimates of the ITT, of the bounds of the ATE, and of the LATE.

9.3 Generalizing the LATE

So far we have shown that in a world with heterogeneous effects, 2SLS with binary instrument and treatment and no control variables identifies a LATE. This section investigates whether this translates to more general settings.

³Notice that we did not invoke the no-defiers assumption when deriving those bounds.

⁴There are other instances in which bounds can be very informative, such as when dealing with survey non response, or differential attrition in randomized experiments.

9.3.1 Multivariate instruments

Some instruments take more than two values. A good example is the lottery number for Vietnam draft, used by Angrist (1990). It was a lottery which determined who had to serve in Vietnam. A number between 1 and 365 was randomly associated to each day of the year, and men whose date of birth was above some threshold were eligible to the draft. For instance, for those born in 1950 the threshold was 195. So we could use a binary variable for being above or below the 195 threshold as an instrument for serving in Vietnam. Using this strategy, Angrist estimates the impact of serving in Vietnam on yearly earnings 30 years later. Men above the 195 cut-off earned 435 dollars less than men below it, and were 15.9 percentage points more likely to serve. Using Theorem 9.1.1, we can therefore say that among compliers, the average effect of serving on later earnings is equal to $\frac{-435}{0.159} = -2771$ dollar, a 15% decrease relative to mean earnings. However, this number is marginally significantly different from 0, as its standard deviation is 1324, implying that the t -statistic is close to 2.

The link between the probability to serve and the lottery number is richer than the binary description outlined above. The cutoff was not known in advance, and it was known that volunteering was a way to serve under better conditions. Because of this, some men below the 195 cutoff volunteered, because they feared they would be drafted and preferred to serve under better conditions. Obviously, the closer to 1 one's lottery number was, the fewer incentives one had to volunteer. Then, one could use the lottery number and not the binary variable described above as an instrument.

Let Z be a random variable such that $\mathcal{S}(Z) = \{0, \dots, J\}$. To describe the causal chain initiated by that random variable, one must introduce $J + 1$ potential treatments $(D(j))_{0 \leq j \leq J}$ depending on which value of the instrument a subject receives, and $2(J + 1)$ potential outcomes $(Y(j, d))_{0 \leq j \leq J, d \in \{0, 1\}}$ depending on instrument and treatment. One can then rewrite Assumption 17 as follows.

Assumption 19 *Multivariate instrument*

A variable Z with finite support $\mathcal{S}(Z) = \{0, \dots, J\}$ is an instrument to measure the effect of D on Y if it satisfies the four following conditions:

1. *Multivariate first stage: for every $j \geq 1$, $P(D = 1|Z = j) > P(D = 1|Z = j - 1)$.*
2. *Random instrument: $((Y(j, d))_{0 \leq j \leq J, d \in \{0, 1\}}, (D(j))_{0 \leq j \leq J}) \perp\!\!\!\perp Z$.*
3. *Exclusion restriction: for every $0 \leq j \neq j' \leq J$ and $d \in \{0, 1\}$, $Y(j, d) = Y(j', d) = Y(d)$.*
4. *Multivariate monotonicity: for every $0 \leq j < j' \leq J$, $D(j') \geq D(j)$.*

The multivariate first stage assumption is testable from the data. It states that $j \mapsto P(D = 1|Z = j)$ should be strictly increasing. The random instrument and exclusion restriction

assumptions are very similar to the corresponding conditions in Assumption 17. The multivariate monotonicity assumption states for each subject in the population, going from a lower to a greater value of the instrument can never have a negative effect on treatment decision.

In this setting, compliers are units who satisfy $D(0) = 0$, but whose treatment decision switches at some point when the value of the instrument they receive increases. For instance, under the multivariate monotonicity assumption, units satisfying $D(j-1) = 0, D(j) = 1$ must also satisfy $D(0) = 0, \dots, D(j-1) = 0, D(j) = 1, \dots, D(J) = 1$. Those are units whose treatment decision switches when the value of the instrument they receive goes from $j-1$ to j . Thereafter, we refer to these units as the $(j-1, j)$ compliers. The union of all the subpopulations of $(j-1, j)$ compliers is $\{D(0) = 0, D(J) = 1\}$, which gathers all the units whose treatment status switches at some point when the value of the instrument they receive moves from 0 to J . For instance, in Angrist (1990), those are all the men who will not go to Vietnam if their lottery number is 1, but who will go to Vietnam if their lottery number is 365, irrespective of the specific lottery number at which this switch occurs.

For every $k < j$, let

$$wald_{j,k} = \frac{E(Y|Z=j) - E(Y|Z=k)}{P(D=1|Z=j) - P(D=1|Z=k)}$$

One can show that $E(Y(1) - Y(0)|D(0) = 0, D(J) = 1) = wald_{J,0}$. Indeed, $wald_{J,0}$ is just the population coefficient of D in a 2SLS regression of Y on D using $1\{Z = J\}$ as an instrument, among units with $Z = J$ or $Z = 0$. It follows from Assumption 19 that $1\{Z = J\}$ satisfies Assumption 17. Therefore, we can estimate $E(Y(1) - Y(0)|D(0) = 0, D(J) = 1)$ by running this 2SLS regression, among the sample of units with $Z = J$ or $Z = 0$. However, this will yield a very imprecise estimate, because this regression discards all observations that do not satisfy $Z = J$ or $Z = 0$. For instance, in Angrist (1990) this regression only makes use of observations with a lottery number equal to 1 or to 365, which presumably account for less than 1% of the sample. Instead, we are going to study two other 2SLS regressions that make use of the entire sample.

To interpret the parameters estimated by these regressions, it will be useful to note that one has

$$E(Y(1) - Y(0)|D(0) = 0, D(J) = 1) = \sum_{j=1}^J wald_{j,j-1} \mu_j,$$

where

$$\mu_j = \frac{P(D=1|Z=j) - P(D=1|Z=j-1)}{\sum_{j=1}^J P(D=1|Z=j) - P(D=1|Z=j-1)}.$$

One can show that under Assumption 19, $E(Y(1) - Y(0)|D(j-1) = 0, D(j) = 1) = wald_{j,j-1}$: $wald_{j,j-1}$ identifies the LATE of the $(j-1, j)$ compliers. Moreover, $\mu_j = \frac{P(D=1|Z=j) - P(D=1|Z=j-1)}{\sum_{j=1}^J P(D=1|Z=j) - P(D=1|Z=j-1)}$ is the share the $(j-1, j)$ compliers account for in the total population of compliers $\{D(0) =$

$0, D(J) = 1\}$. Therefore, the formula above just says that $E(Y(1) - Y(0) | D(0) = 0, D(J) = 1)$, the LATE of the total population of compliers, is a weighted average of the LATE of each subpopulation of $(j - 1, j)$ compliers.

Theorem 9.3.1 *2SLS with multivariate instrument*

Let β_{NS} denote the coefficient of D in a 2SLS regression of Y on D using Z as an instrument.

$$\beta_{NS} = \sum_{j=1}^J \text{wald}_{j,j-1} \lambda_j,$$

where

$$\lambda_j = \frac{(P(D = 1 | Z = j) - P(D = 1 | Z = j - 1)) \sum_{k=j}^J P(Z = k)(k - E(Z))}{\sum_{j=1}^J (P(D = 1 | Z = j) - P(D = 1 | Z = j - 1)) \sum_{k=j}^J P(Z = k)(k - E(Z))}$$

are positive weights summing up to 1.

Proof

Notice first that

$$\begin{aligned} & E(Y | Z = k) - E(Y | Z = 0) \\ &= \sum_{j=1}^k E(Y | Z = j) - E(Y | Z = j - 1) \\ &= \sum_{j=1}^k \text{wald}_{j,j-1} (P(D = 1 | Z = j) - P(D = 1 | Z = j - 1)). \end{aligned} \quad (9.3.1)$$

$\beta_{NS} = \frac{\text{cov}(Y, Z)}{\text{cov}(D, Z)}$. Let us start considering its numerator.

$$\begin{aligned} \text{cov}(Y, Z) &= E(Y(Z - E(Z))) \\ &= E((Z - E(Z))E(Y | Z)) \\ &= \sum_{k=0}^J P(Z = k)(k - E(Z))E(Y | Z = k) \\ &= \sum_{k=0}^J P(Z = k)(k - E(Z))(E(Y | Z = k) - E(Y | Z = 0) + E(Y | Z = 0)) \\ &= \sum_{k=0}^J P(Z = k)(k - E(Z))E(Y | Z = 0) \\ &+ \sum_{k=1}^J P(Z = k)(k - E(Z))(E(Y | Z = k) - E(Y | Z = 0)) \\ &= \sum_{k=1}^J P(Z = k)(k - E(Z)) \left(\sum_{j=1}^k \text{wald}_{j,j-1} (P(D = 1 | Z = j) - P(D = 1 | Z = j - 1)) \right) \\ &= \sum_{j=1}^J \text{wald}_{j,j-1} \left((P(D = 1 | Z = j) - P(D = 1 | Z = j - 1)) \sum_{k=j}^J P(Z = k)(k - E(Z)) \right). \end{aligned}$$

Where do the second and third equalities come from? The sixth equality follows from (9.3.1). Similarly, one can show that

$$\text{cov}(D, Z) = \sum_{j=1}^J \left((P(D=1|Z=j) - P(D=1|Z=j-1)) \sum_{k=j}^J P(Z=k)(k - E(Z)) \right).$$

Then notice that

$$\sum_{k=j}^J P(Z=k)(k - E(Z)) = P(Z \geq j) (E(Z|Z \geq j) - E(Z)) \geq 0.$$

This proves that the λ_j are positive. One can easily see that they sum up to 1. This completes the proof.

QED.

This theorem shows that with a multivariate instrument, 2SLS identifies a weighted average of wald ratios. Under Assumption 19, those can be interpreted as LATEs, which implies that 2SLS identifies a meaningful treatment effect parameter. However, this weighted average is not equal to the average effect of the treatment among the entire population of compliers, $E(Y(1) - Y(0)|D(0)=0, D(J)=1)$, because it does not give to each $\text{wald}_{j,j-1}$ the weight of the $(j-1, j)$ compliers in the total population of compliers. If the effect of the treatment is the same across those subgroups of compliers, implying that the $\text{wald}_{j,j-1}$ parameters are all equal to each other, then $E(Y(1) - Y(0)|D(0)=0, D(J)=1) = \beta_{NS}$. But if the effect of the treatment is heterogeneous across those subgroups of compliers, then β_{NS} might differ from $E(Y(1) - Y(0)|D(0)=0, D(J)=1)$.

An alternative 2SLS regression would amount to saturating the first stage in Z .

Theorem 9.3.2 *2SLS with saturated first stage in the multivariate instrument*

Let β_S denote the coefficient of D in a 2SLS regression of Y on D using $\{1\{Z=1\}, \dots, 1\{Z=J\}\}$ as instruments.

$$\beta_S = \sum_{j=1}^J \text{wald}_{j,j-1} \eta_j,$$

where

$$\eta_j = \frac{(P(D=1|Z=j) - P(D=1|Z=j-1)) \sum_{k=j}^J P(Z=k)(P(D=1|Z=k) - P(D=1))}{\sum_{j=1}^J (P(D=1|Z=j) - P(D=1|Z=j-1)) \sum_{k=j}^J P(Z=k)(P(D=1|Z=k) - P(D=1))}$$

are positive weights summing up to 1.

Proof

This theorem is (almost) a direct consequence of Theorem 9.3.1. The 2nd stage of this 2SLS regression amounts to regressing Y on $P(D=1|Z)$: because the first stage is saturated, the first stage regression function is $P(D=1|Z)$. Moreover, $D - P(D=1|Z)$ is uncorrelated with $P(D=1|Z)$. Therefore, $\beta_S = \frac{\text{cov}(Y, P(D=1|Z))}{\text{cov}(D, P(D=1|Z))}$. This is the same formula as the one for β_{NS}

in Theorem 9.3.1, except that $P(D = 1|Z)$ has been replaced by Z . Using the same steps as in 9.3.1 will then yield the result. You can indeed check that μ_j and η_j are the same, except that k is replaced by $P(D = 1|Z = k)$ and $E(Z)$ by $P(D = 1)$.

QED.

2SLS with a fully saturated first stage estimates a similar weighted average of LATEs as the non-saturated regression does, but the weighing is different and easier to understand. Indeed, the part of η_j which comes on top of the natural weight of the $(j - 1, j)$ subgroup of compliers is equal to $P(Z \geq j) (P(D = 1|Z \geq j) - P(D = 1))$. The $P(Z \geq j)$ term gives more weight to smaller values of j . On the other hand, the $(P(D = 1|Z \geq j) - P(D = 1))$ term gives more weight to higher values of j . Finally, notice that β_S can be interpreted as the slope of the line connecting Y and $P(D|Z)$: under Assumption 19, the only reason why the mean of Y can change across values of Z is that the share of treated observations changes across values of Z .

It follows from Newey (1990) that $\hat{\beta}_S$ has a lower asymptotic variance than $\hat{\beta}_{NS}$. So, with a multivariate instrument at hand, should we estimate β_{NS} or β_S ?

$\hat{\beta}_S$ is asymptotically more efficient, but in finite samples it might be more biased than $\hat{\beta}_{NS}$, because the 2SLS regression giving rise to $\hat{\beta}_S$ has as many instruments as possible values of the instrument. For instance, in Angrist (1990) this regression has 364 instruments. Therefore, in finite samples $\hat{\beta}_S$ could suffer from a many-weak instruments problem and could therefore be biased.

Angrist (1990) chooses to estimate β_S . He finds $\hat{\beta}_S = -2384$, with a standard error of 774. The estimate is not very different from the one derived using a binary instrument, but its standard error is lower.

9.3.2 Several instruments

In some instances, we have several instruments instead of one. Angrist & Evans (1998) use both same-sexness of the first two children in a family and twins second birth as an instrument for the decision of having a third child. As discussed above, the first stage of the same-sex instrument is 0.06 and the Wald ratio associated to it is -0.13 , meaning that among those 6% of women who are induced to have a third child because their first two children are of the same sex, 13% cannot work because of this third child, while they would have worked otherwise. The first stage of the twins instrument is 0.62: all mothers with a twins second birth have a third child, but 38% of them would have ended up having a third child anyway,

so the share of them induced to having a third child because of twins second birth is only 62%. The Wald ratio associated with that instrument is -0.08 , meaning that among mothers induced to having a third child because of a twins second birth, 8% cannot work because they have to take care of this supplementary child, while they would have worked otherwise. [Do you think twins second birth is a valid instrument? How could you check that?](#)

To me, the assumption that sounds the most problematic here is the random instrument assumption. You could assess its plausibility by checking whether mothers with a twin second birth have similar demographic characteristics to other mothers.

Instead of running separately the 2 2SLS regressions described above, a natural idea would be to run one 2SLS regression with two instruments. The next theorem describes what 2SLS identifies in such instances. Let β denote the coefficient of D in a 2SLS regression of Y on D using Z_1 and Z_2 as instruments for D . Let β_1 (resp. β_2) denote the same coefficient when Z_1 only (resp. Z_2 only) is used as an instrument. Let $\alpha_0 + \alpha_1 Z_1 + \alpha_2 Z_2$ denote the first stage regression function of D on Z_1 and Z_2 .

Theorem 9.3.3 *2SLS with two instruments*

$$\beta = w\beta_1 + (1 - w)\beta_2,$$

with

$$w = \frac{\alpha_1 \text{cov}(D, Z_1)}{\alpha_1 \text{cov}(D, Z_1) + \alpha_2 \text{cov}(D, Z_2)}.$$

Proof

β is the coefficient of a univariate regression of Y on $\alpha_0 + \alpha_1 Z_1 + \alpha_2 Z_2$. Moreover, $D = \alpha_0 + \alpha_1 Z_1 + \alpha_2 Z_2 + e$ where e is uncorrelated with $\alpha_1 Z_1 + \alpha_2 Z_2$. This implies that $V(\alpha_0 + \alpha_1 Z_1 + \alpha_2 Z_2) = \text{cov}(D, \alpha_1 Z_1 + \alpha_2 Z_2)$. Therefore,

$$\begin{aligned} \beta &= \frac{\text{cov}(Y, \alpha_1 Z_1 + \alpha_2 Z_2)}{\text{cov}(D, \alpha_1 Z_1 + \alpha_2 Z_2)} \\ &= \frac{\alpha_1 \text{cov}(D, Z_1)}{\alpha_1 \text{cov}(D, Z_1) + \alpha_2 \text{cov}(D, Z_2)} \frac{\text{cov}(Y, Z_1)}{\text{cov}(D, Z_1)} + \frac{\alpha_2 \text{cov}(D, Z_2)}{\alpha_1 \text{cov}(D, Z_1) + \alpha_2 \text{cov}(D, Z_2)} \frac{\text{cov}(Y, Z_2)}{\text{cov}(D, Z_2)} \\ &= w\beta_1 + (1 - w)\beta_2. \end{aligned}$$

QED.

The 2SLS coefficient with 2 instruments is a weighted average of the 2 2SLS coefficients with one instrument only. The weight depends on the relative strengths of the 2 first stages. If the two instruments are binary, β_1 identifies the LATE for “instrument 1 compliers”, while

β_2 identifies the LATE for “instrument 2 compliers”, so β is a weighted average of those two LATES. If the two instruments are multivariates, β is a weighted average of weighted averages of LATES. In the same-sex / twins birth example, the 2SLS coefficient of a regression using the two instruments is equal to -0.098 . It is much closer from the twins birth coefficient (-0.083), because this instrument has a stronger first stage.

9.3.3 Covariates

In Chapter 5, covariates were useful to identify the effect of a treatment as good as randomly assigned conditional on covariates (CIA), and to increase statistical power. The same applies to instruments. Some instruments are as good as randomly assigned only after conditioning for covariates. Moreover, even when Z is randomly assigned, controlling for covariates can substantially increase statistical power.

Before investigating what 2SLS capture when covariates are added to the regression, one first has to define “conditional instruments”.

Assumption 20 *Conditional on X instrument*

Let X be a vector of covariates. A binary variable Z is a conditional on X instrument if the four following statements are verified:

1. *Conditional first stage: $\forall x_j \in \mathcal{S}(X), P(D = 1|Z = 1, X = x_j) > P(D = 1|Z = 0, X = x_j)$.*
2. *Conditional random instrument: $(Y(0,0), Y(1,0), Y(0,1), Y(1,1), D(0), D(1)) \perp\!\!\!\perp Z|X$.*
3. *Exclusion restriction: for every $d \in \{0; 1\}$, $Y(0, d) = Y(1, d) = Y(d)$.*
4. *Monotonicity, or no defiers: $D(1) \geq D(0)$.*

With a conditional on X binary instrument, one could estimate 2SLS regressions of Y on D using Z as an instrument in each X cell, and then take a weighted average of the resulting “within X cells” wald ratios to estimate the LATE. The weights should reflect the size of the X -compliers relative to the total population of compliers, and should therefore be estimates of

$$\frac{(P(D = 1|Z = 1, X = x_j) - P(D = 1|Z = 0, X = x_j))P(X = x_j)}{P(D = 1|Z = 1) - P(D = 1|Z = 0)}.$$

Frölich (2007) shows that instead of doing manually the weighted average, one can estimate it directly through propensity score reweighting. This also generalizes the result to continuous covariates. Let $\widehat{E}(Y(1) - Y(0)|C)$ denote this estimator of the LATE with covariates and propensity score reweighting. Frölich has developed a Stata package computing this estimator.

But even the fairly simple propensity score estimation in Frölich (2007) is more complicated to implement than adding covariates to the standard 2SLS regression, which is often what

practitioners do. Let's investigate what this regression identifies. Let β_{NS} and β_X respectively denote the coefficients of D and X in a regression of Y on X and D using Z as an instrument, with X -saturated first and second stages. For this regression to be feasible, X must take a finite number of values $x_1, \dots, x_J$. Using the κ result in Abadie (2003), one can show that β_{NS} and β_X satisfy

$$(\beta_{NS}, \beta_X) = \underset{(b_{NS}, b_X)}{\operatorname{argmin}} E \left((E(Y|D, X, C) - b_D D - X'_i b_X)^2 | C \right),$$

meaning that this 2SLS regression identifies the regression function of Y on D and X , **inside the population of compliers**, even though that population is not observed. Then using the “regression as matching” theorem (Theorem 3.4.2), it follows that

$$\beta_{NS} = \sum_{j=1}^J (E(Y|D=1, C, X=x_j) - E(Y|D=0, C, X=x_j)) p_{x_j}, \quad (9.3.2)$$

where

$$p_{x_j} = \frac{P(D=1|X=x_j, C)(1 - P(D=1|X=x_j, C))P(X=x_j|C)}{\sum_{j=1}^J P(D=1|X=x_j, C)(1 - P(D=1|X=x_j, C))P(X=x_j|C)}.$$

Finally, notice that since compliers satisfy $Z = D$,

$$\begin{aligned} & E(Y|D=1, C, X=x_j) - E(Y|D=0, C, X=x_j) \\ &= E(Y(1)|D=1, C, X=x_j) - E(Y(0)|D=0, C, X=x_j) \\ &= E(Y(1)|Z=1, C, X=x_j) - E(Y(0)|Z=0, C, X=x_j) \\ &= E(Y(1) - Y(0)|C, X=x_j). \end{aligned}$$

This shows that β_{NS} is a weighted average of X -LATES with weights summing up to 1. As with simple OLS regressions, the weights are not the “good” ones, so β_{NS} is not equal to $E(Y(1) - Y(0)|C)$. But $\hat{\beta}_{NS}$ will often have a lower variance than $\hat{E}(Y(1) - Y(0)|C)$, as β_{NS} gives more weight to cells in which $P(D=1|X=x_j, C)(1 - P(D=1|X=x_j, C))$ is high, implying that $E(Y|D=1, C, X=x_j) - E(Y|D=0, C, X=x_j)$ is estimated more precisely. Therefore, a bias-variance trade-off might justify using $\hat{\beta}_{NS}$.

Saturating in the covariates is not always possible, for instance when you want to include continuous covariates. It is not clear what 2SLS capture when the model is not saturated in covariates. So when you cannot saturate your regression in X , you should use Abadie (2003) kappa's procedure, or Frölich (2007)'s estimator. They always yield easily interpretable parameters, irrespective of whether the model is saturated in X or not.

9.3.4 Multivariate and continuous treatment

Let S be a multivariate treatment with support $\{0, 1, \dots, \bar{s}\}$, such as years of schooling. Let Z be a binary instrument such as quarter of birth, let $S(0)$ and $S(1)$ denote the two potential

treatment choices of the subject we draw, depending on which value of the instrument they receive, and let $Y(z, s)$ denote their potential outcomes.

Assumption 21 *Instrument with multivalued treatment*

In this context, a binary variable Z is an instrument if the four following conditions are verified:

1. *First stage:* $E(S|Z = 1) > E(S|Z = 0)$.
2. *Random instrument:* $((Y(z, s))_{z \in \{0,1\}, s \in \{0, \dots, \bar{s}\}}, S(0), S(1)) \perp\!\!\!\perp Z$.
3. *Exclusion restriction:* for every $s \in \{0, \dots, \bar{s}\}$, $Y(0, s) = Y(1, s) = Y(s)$.
4. *Monotonicity, or no defiers:* $S(1) \geq S(0)$.

Theorem 9.3.4 *Late with multivariate treatment: the average causal response theorem*

If Z satisfies Assumption 21,

$$\begin{aligned} & \frac{E(Y|Z = 1) - E(Y|Z = 0)}{E(S|Z = 1) - E(S|Z = 0)} \\ = & \frac{1}{E(S|Z = 1) - E(S|Z = 0)} \sum_{1 \leq s' < s \leq \bar{s}} E(Y(s) - Y(s')|S(0) = s', S(1) = s) P(S(0) = s', S(1) = s) \\ = & \sum_{s=1}^{\bar{s}} E(Y(s) - Y(s-1)|S(0) < s \leq S(1)) v_s \end{aligned}$$

where

$$v_s = \frac{P(S(0) < s \leq S(1))}{\sum_{s=1}^{\bar{s}} P(S(0) < s \leq S(1))}.$$

Proof

I will only prove that the first term in the Equation is equal to the third one. Proving that the first term is equal to the second one is left as an exercise. We have

$$Y = Y(0)1\{S \geq 0\} + (Y(1) - Y(0))1\{S \geq 1\} + \dots + (Y(\bar{s}) - Y(\bar{s} - 1))1\{S \geq \bar{s}\}.$$

Why is this equality true? Therefore,

$$\begin{aligned} & E(Y|Z = 1) - E(Y|Z = 0) \\ = & E(Y(0)1\{S(1) \geq 0\} + (Y(1) - Y(0))1\{S(1) \geq 1\} + \dots + (Y(\bar{s}) - Y(\bar{s} - 1))1\{S(1) \geq \bar{s}\}|Z = 1) \\ - & E(Y(0)1\{S(0) \geq 0\} + (Y(1) - Y(0))1\{S(0) \geq 1\} + \dots + (Y(\bar{s}) - Y(\bar{s} - 1))1\{S(0) \geq \bar{s}\}|Z = 0) \\ = & \sum_{s=1}^{\bar{s}} E((Y(s) - Y(s-1))1\{S(0) < s \leq S(1)\}). \end{aligned} \tag{9.3.3}$$

The last equality follows from the fact that under monotonicity, $1\{S(1) \geq s\} - 1\{S(0) \geq s\} = 1\{S(0) < s \leq S(1)\}$. Why is that true?

Then,

$$\begin{aligned}
E(S|Z = 1) - E(S|Z = 0) &= E(S(1)) - E(S(0)) \\
&= \sum_{s=1}^{\bar{s}} sP(S(1) = s) - \sum_{s=1}^{\bar{s}} sP(S(0) = s) \\
&= \sum_{s=1}^{\bar{s}} sP(S(1) \geq s) - \sum_{s=1}^{\bar{s}-1} sP(S(1) \geq s+1) \\
&\quad - \left(\sum_{s=1}^{\bar{s}} sP(S(0) \geq s) - \sum_{s=1}^{\bar{s}-1} sP(S(0) \geq s+1) \right) \\
&= \sum_{s=1}^{\bar{s}} sP(S(1) \geq s) - \sum_{s=2}^{\bar{s}} (s-1)P(S(1) \geq s) \\
&\quad - \left(\sum_{s=1}^{\bar{s}} sP(S(0) \geq s) - \sum_{s=2}^{\bar{s}} (s-1)P(S(0) \geq s) \right) \\
&= \sum_{s=1}^{\bar{s}} P(S(1) \geq s) - \sum_{s=1}^{\bar{s}} P(S(0) \geq s) \\
&= \sum_{s=1}^{\bar{s}} P(S(0) < s \leq S(1)). \tag{9.3.4}
\end{aligned}$$

The last step follows from the monotonicity condition. Indeed, under monotonicity, $\{S(0) \geq s\} \subseteq \{S(1) \geq s\}$. Therefore,

$$P(S(0) < s \leq S(1)) = P(S(1) \geq s \setminus S(0) \geq s) = P(S(1) \geq s) - P(S(0) \geq s).$$

Combining (9.3.3) and (9.3.4) yields the result.

QED.

The first equality of the Theorem shows that when treatment is multivariate, the Wald ratio identifies some kind of weighted average of $\frac{\bar{s} \times (\bar{s}+1)}{2}$ LATEs. As in the standard LATE theorem, units whose treatment decision is not affected by the instrument, i.e. those such that $S(0) = S(1)$, vanish in the comparison of $E(Y|Z = 1)$ and $E(Y|Z = 0)$. Under the assumption that the instrument cannot negatively affect treatment, only compliers, i.e. units whose treatment decision is positively affected by the instrument, are not netted out in this comparison. In the multivariate treatment setting, there are $\frac{\bar{s} \times (\bar{s}+1)}{2}$ subgroups of compliers: units who go from 0 to 1 unit of treatment received when they receive the instrument, units who go from 0 to 2, ..., from 0 to $\bar{s}$, from 1 to 2, ..., from 1 to $\bar{s}$, ..., and from $\bar{s} - 1$ to $\bar{s}$. Hence the second equality. The third equality is just a smart rewriting of the double sum in the second equality into a single sum. It shows that the Wald ratio is a weighted average of the effect of going from $s - 1$ to s units of treatment, for compliers whose treatment value is switched from strictly below to above s by the instrument. The weights are fairly natural because v_s is merely the size of those compliers divided by the sum of the size of all such populations across values of s . This sum can still not be rewritten as the average treatment effect for the entire population of compliers

(i.e. those such that $S(0) < S(1)$), because the $S(0) < s \leq S(1)$ populations are not disjoint. For instance, a complier going from $S(0) = 2$ to $S(1) = 5$ will be in the $S(0) < 3 \leq S(1)$, $S(0) < 4 \leq S(1)$, and $S(0) < 5 \leq S(1)$ subpopulations: they will be counted three times. To some extent, this makes sense to count their three times, because the instrument moves their treatment decision up by three units. We can therefore observe 3 1-unit treatment effects for their, as $Y(6) - Y(3) = Y(6) - Y(5) + Y(5) - Y(4) + Y(4) - Y(3)$. As we are interested in the average effect of increasing treatment by one unit, it is natural they appear three times in our formula.

A strong limit of 2SLS with multivariate treatment is that it does not identify the causal parameters $E(Y(s) - Y(s')|S(0) = s', S(1) = s)$ or $E(Y(s) - Y(s-1)|S(0) < s \leq S(1))$ but a weighted average of them. The intuition is that we have only one clean source of variation (the comparison of $E(Y|Z = 1)$ and $E(Y|Z = 0)$) to identify the effect of the treatment among many groups of compliers.

This limits the interpretability of 2SLS coefficients with multivariate treatment. For instance, in the wage and schooling example, it would be interesting to separate the effect of completing high school from the effect of graduating from college. We cannot learn this from the Wald ratio. But at least, we can learn which of the $E(Y(s) - Y(s-1)|S(0) < s \leq S(1))$ receive the greatest weights. Indeed, v_s is identified, because

$$\begin{aligned} P(S(0) < s \leq S(1)) &= P(S(1) \geq s) - P(S(0) \geq s) \\ &= P(S \geq s|Z = 1) - P(S \geq s|Z = 0). \end{aligned}$$

Therefore, the s which receive the most weight are those for which the difference between the survival function of S conditional on $Z = 0$ and $Z = 1$ is the largest.

Acemoglu & Angrist (2001) use child labor laws passed in US states in the beginning of the 20th century as an instrument to estimate returns to schooling. Intuitively, do you think their 2SLS estimate can teach us something about returns to completing a college degree?

The figure below (from Angrist & Pischke (2008)) shows the weights v_s in that application. It appears that child labor laws have a large effect on the share of people completing at least 8, 9, 10, 11, or 12 years of education, but has no effect on the probability of completing strictly more than 12 years of education. Therefore, the Wald ratio of wages on schooling using those laws as an instrument will capture a weighted average of the effect of completing 8 against 7, 9 against 8, 10 against 9, 11 against 10 and 12 against 11 years of schooling, among some subgroups of compliers, while the other $E(Y(s) - Y(s-1)|S(0) < s \leq S(1))$

parameters will receive almost no weight. So even though we cannot separately identify each of the $E(Y(s) - Y(s-1)|S(0) < s \leq S(1))$ parameters, we can at least say which receive the largest weight. As a result, the 0.08 – 0.10 estimates of the returns to completing one more year of schooling in Acemoglu & Angrist (2001) are essentially weighted averages of returns to completing 8 against 7 years of schooling, 9 against 8 years, ..., and 12 against 11 years. This paper tells us something about the returns to completing middle school and high school, but it cannot tell us anything on the returns to graduating from college, because the instrument it uses switches some subjects from non completing to completing middle and highschool, but it does not switch anyone from not graduating to graduating from college. Child labor laws keep in schools children from poor backgrounds for a couple more years while they who would have worked otherwise, but they do not have an influence on college enrolment decisions.

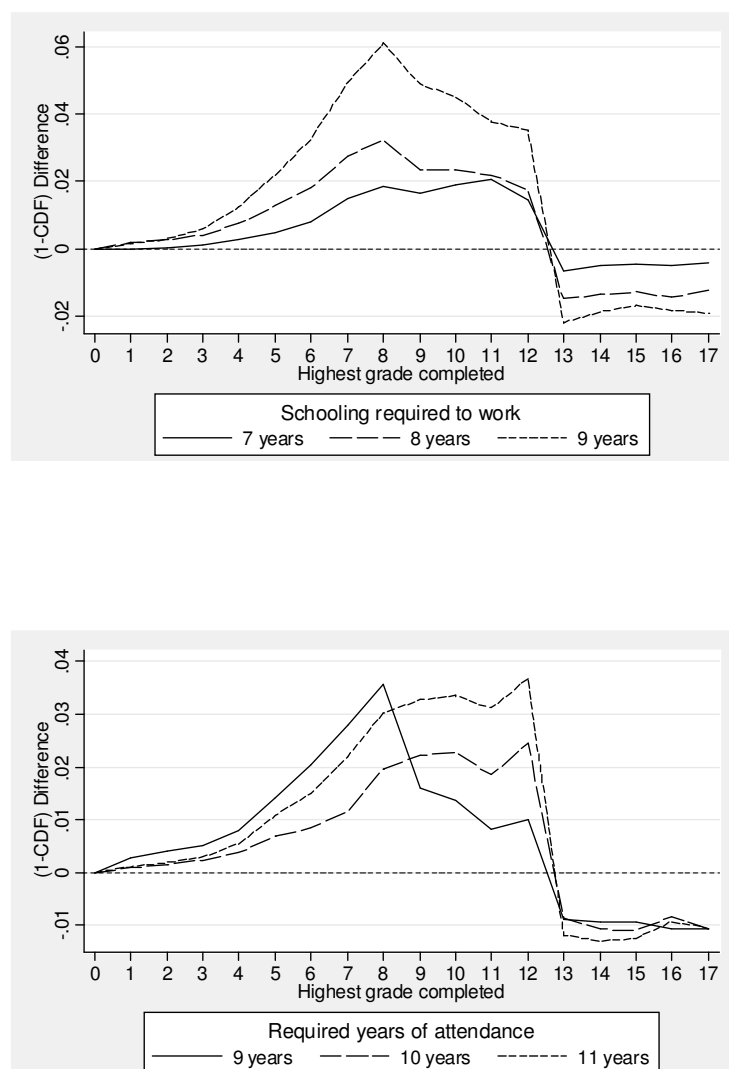


Figure 9.4: The effect of compulsory laws on schooling

Let me give you another example in which the 2SLS estimate with a multivariate treatment is hard to interpret. In Behaghel et al. (2017), we conducted a randomized experiment to

measure the effect of sending high-achievers from deprived neighborhoods in France to a boarding school. We followed the treatment group (children randomly offered a seat in the school) and the control group during two years after the lottery, and had them taking language and maths tests every year. One year after the lottery, 80% of the treatment group children were in the boarding school, and 2 years after the lottery 60% were still there. In the control group, almost no child managed to enrol. In the end of the first year, β_1 , our 2SLS coefficient arising from the regression of students maths tests scores on the number of years spent in the boarding school instrumented by our lottery offer was 0. In the end of the second year, our coefficient β_2 was 0.2, meaning that the average effect of the boarding school across the different groups of compliers was to increase test scores by 20% of a standard deviation per year spent in the school. The one versus two years pattern is interesting, as most papers studying returns to educational policies find linear or concave returns, while here we obtain convex estimates: our estimates of the effect of the school increase over time. But these 2SLS estimates are not sufficient for us to conclude that the school indeed has increasing returns. Let C_1 denote the compliers induced to spend one year in the boarding school by our lottery offer, and let C_2 denote those induced to spend two years. Let γ_{yt}^c denote the average effect of having spent y years in the boarding school t years ago among the population of compliers c . In the end of the first year, treatment is binary (students have spent at most one year in the boarding school). It therefore follows from the standard LATE theorem that

$$\beta_1 = E(Y(1) - Y(0)|C) = \frac{P(C_1)}{P(C_1) + P(C_2)}\gamma_{10}^1 + \frac{P(C_2)}{P(C_1) + P(C_2)}\gamma_{10}^2.$$

In the end of the second year, our treatment variable can take three values, 0, 1, and 2. Therefore, it follows from the second formula in the previous theorem that

$$\beta_2 = \frac{P(C_1)}{P(C_1) + 2P(C_2)}\gamma_{11}^1 + \frac{P(C_2)}{P(C_1) + 2P(C_2)}\gamma_{20}^2.$$

What we observe ($\beta_1 = 0$, $\beta_2 = 0.2$) is perfectly compatible with linear returns to the boarding school among students staying two years ($\gamma_{20}^2 = 2 \times \gamma_{10}^2$), and negative returns among students staying 1 year. The zero effect after one year would be the weighted average of a negative effect among students staying one year, and a positive effect among students staying two years. The positive effect after two years would be the weighted average of a zero effect among students who stayed only one year (one year after they left, they have recovered from their bad experience in the boarding school so their stay in the school no longer has any effect on them), and of a positive effect among students staying two years which is the mere continuation of the pattern of improvement they initiated in their first year. An alternative story is that nobody benefits from their first year ($\gamma_{10}^1 = \gamma_{10}^2 = 0$) because students need to adapt to their environment, which would explain the zero effect after one year, and then in their second year students who stay have overcome this adaptation process and start benefiting from the school. 2SLS cannot tell us which of these two stories (heterogeneity versus convexity of the returns to the boarding school) is the true one.

On a related note, another limitation of instruments with multivariate treatment is that you can no longer get quantile treatment effects. With a binary treatment, we have

$$P(Y(0) \leq y|C) = \frac{P(D=0|Z=0)P(Y \leq y|D=0, Z=0) - P(D=0|Z=1)P(Y \leq y|D=0, Z=1)}{P(D=0|Z=0) - P(D=0|Z=1)}$$

$$P(Y(1) \leq y|C) = \frac{P(D=1|Z=1)P(Y \leq y|D=1, Z=1) - P(D=1|Z=0)P(Y \leq y|D=1, Z=0)}{P(D=1|Z=1) - P(D=1|Z=0)},$$

the cdfs of $Y(0)$ and $Y(1)$ among compliers are identified. With a multivariate treatment, you can check that $\frac{P(D=s|Z=1)P(Y \leq y|D=s, Z=1) - P(D=s|Z=0)P(Y \leq y|D=s, Z=0)}{P(D=s|Z=1) - P(D=s|Z=0)}$ is not equal to the cdf of $Y(s)$ for a well-defined population of units.

A more positive note for 2SLS with multivariate treatment is that Assumption 19 has more testable implications than in the binary case. We showed above that under Assumption 19, we must have

$$P(S(0) < s \leq S(1)) = P(S \geq s|Z=1) - P(S \geq s|Z=0).$$

What does this imply for the rhs of the previous display?

The rhs of the previous display must be positive for every s . This might not necessarily be the case as this is the difference between two probabilities. Rewriting this rhs, we get that we must have

$$P(S \leq s|Z=1) \leq P(S \leq s|Z=0) :$$

the cdf of $S|Z=0$ must stochastically dominate that of $S|Z=1$. Barua & Lang (2010) use this to show that the quarter of birth instrument does not satisfy Assumption 19. Indeed, the cdf of school entry age among children born in the last quarter of the year does not stochastically dominate that of children born in the first quarter, as shown below. This is because parents can strategically manipulate the entry age of their child if they assess they are not mature enough to enter school by the time they would be supposed to, so-called redshirting. For instance, all children born in the 4th quarter should have entered at 4.75 years old. But 40% are actually held back by their parents, who think they are not mature enough to enter, and who eventually send them to school when they are 5.75. This causes the 2 cdfs to cross at 5.5, because almost all children born in the first quarter enter at 5.5 years old: only a very tiny fraction of them are held back by their parents until 6.5. Because the stochastic dominance assumption is violated, some of the weights v_s will be negative in the average causal response function. 2SLS will estimate a weighted average of causal effects with negative weights, a parameter which does not have a clear causal interpretation.

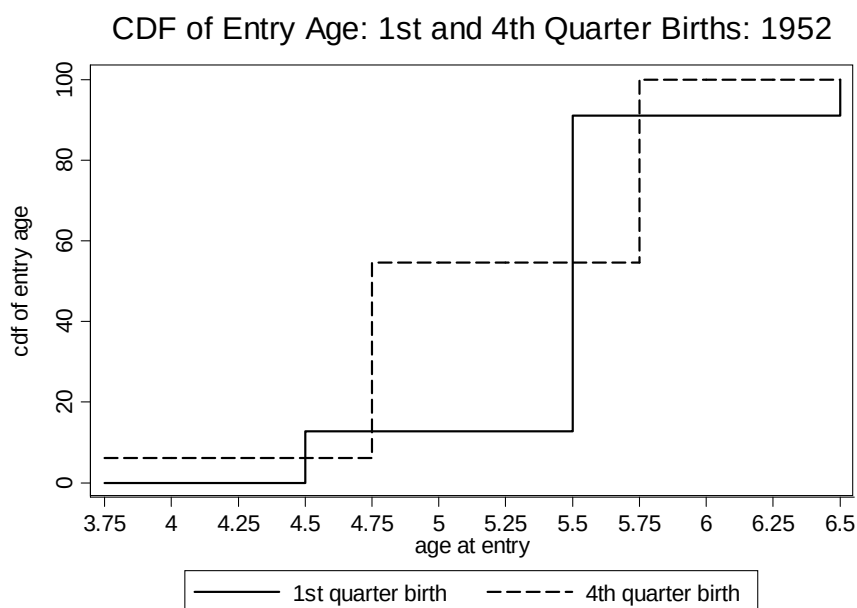


Figure 9.5: The quarter of birth instrument does not satisfy Assumption 19.

Finally, it is worth noting that the average causal response theorem also extends to continuous treatments (see Angrist et al. (2000)).

Chapter 10

Regression discontinuity

10.1 Identification

10.1.1 Sharp designs

Treatment allocation rule: $D = 1\{S \geq s_0\}$. S : running variable. Only people whose running variable above cut-off s_0 get treated.

In RD, identifying assumption is

Assumption 22 *Continuity of $E(Y(0)|S = s)$ and $E(Y(1)|S = s)$*

S is continuous, and $s \mapsto E(Y(0)|S = s)$ and $s \mapsto E(Y(1)|S = s)$ are continuous at s_0 .

Under Assumption 22, $E(Y(1) - Y(0)|S = s_0)$ is identified.

Theorem 10.1.1 *If Assumption 22 is satisfied,*

$$\lim_{s \rightarrow s_0, s > s_0} E(Y|S = s) - \lim_{s \rightarrow s_0, s < s_0} E(Y|S = s) = E(Y(1) - Y(0)|S = s_0).$$

Proof

$$\begin{aligned} & \lim_{s \rightarrow s_0, s > s_0} E(Y|S = s) - \lim_{s \rightarrow s_0, s < s_0} E(Y|S = s) \\ &= \lim_{s \rightarrow s_0, s > s_0} E(Y(1)|S = s) - \lim_{s \rightarrow s_0, s < s_0} E(Y(0)|S = s) \\ &= E(Y(1)|S = s_0) - E(Y(0)|S = s_0) \\ &= E(Y(1) - Y(0)|S = s_0). \end{aligned}$$

The second equality follows from Assumption 22.

QED.

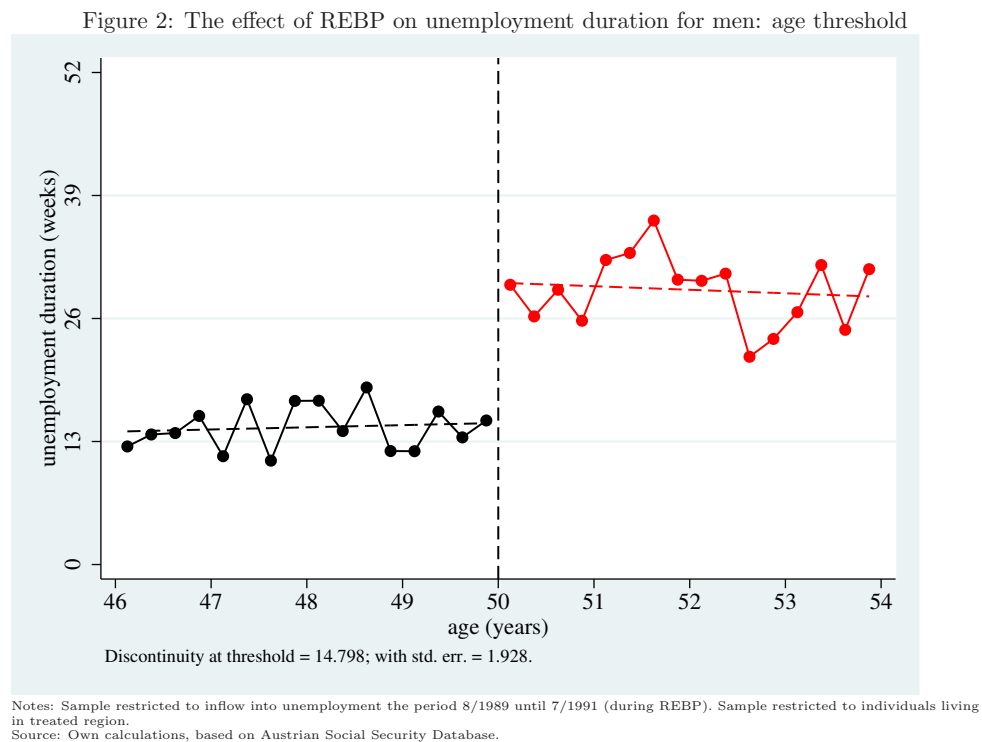
$E(Y(1) - Y(0)|S = s_0)$: very local treatment effect parameter. $S = s_0$ = subpopulation of mass 0! RD: strong internal validity, very low (0!) external validity.

Concerns with identifying assumption: individuals may manipulate their running variable to fall right after cut-off and benefit from treatment. Tests: is density of running variable

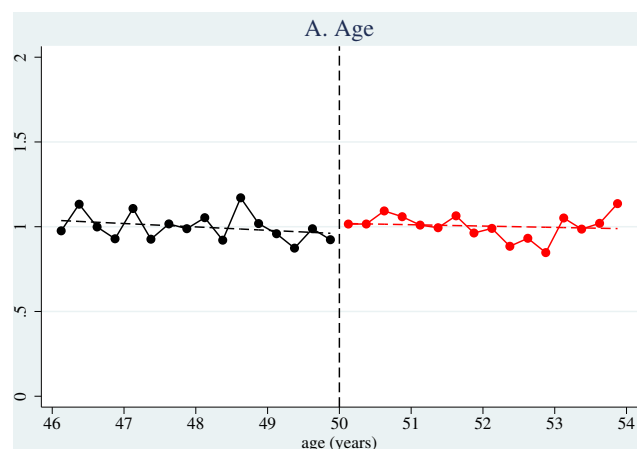
discontinuous around cut-off? McCrary (2008) or Cattaneo et al. (2018), both implemented in Stata. Are covariates balanced around cut-off?

Illustration: Lalive (2008). In some Austrian regions, duration of unemployment benefits = 30 weeks for unemployed whose spell starts when they are below 50, and 209 weeks for unemployed whose spell starts when they are above 50. What is the effect of having longer unemployment benefits on unemployment duration?

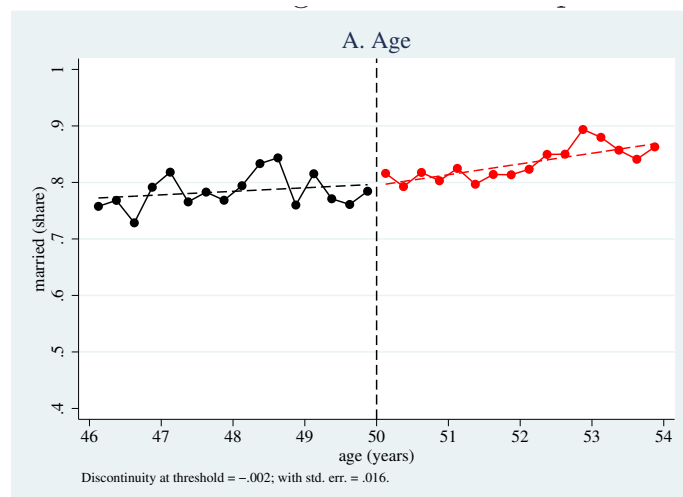
Strong positive effect on men's unemployment duration.



No discontinuity of density of running variable around threshold for men.

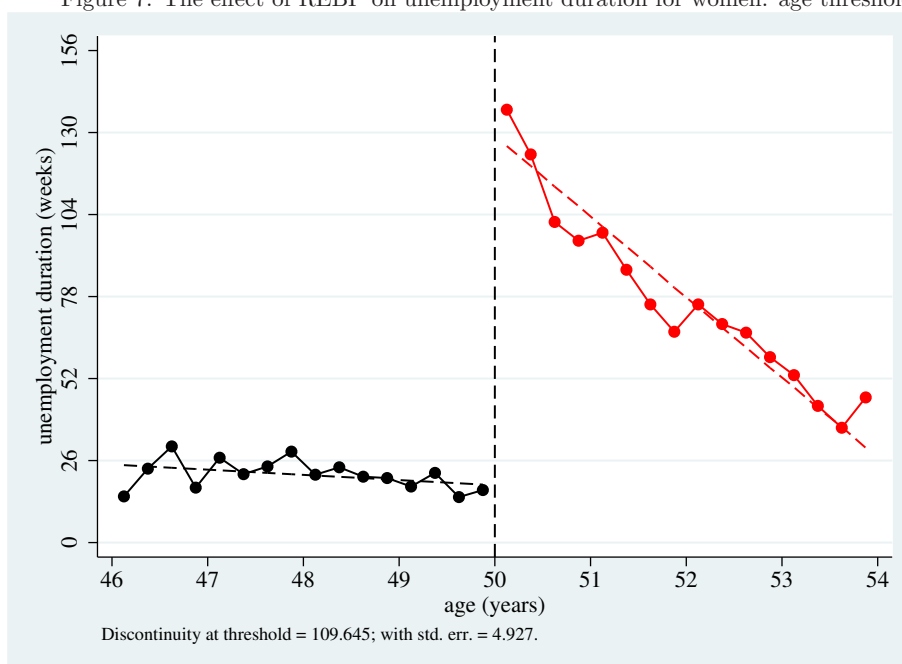


Covariates balanced around threshold for men.



Even stronger positive effect on women's unemployment duration.

Figure 7: The effect of REBP on unemployment duration for women: age threshold



But strong discontinuity of density of running variable around threshold for women.



Plots like those above are a key aspect of a RD analysis. When building those plots, the researcher needs to choose the size of the bins of the running variable in the plots. Calonico et al. (2015) propose a method to choose the size of those bins, and they have developed R and Stata packages implementing their proposed methods.

10.1.2 Fuzzy designs

$D \neq 1\{S \geq s_0\}$, but probability of getting treated jumps discontinuously at s_0 . E.g.: in principle, sharp rule, but imperfect compliance with the rule. Analyzed in IV framework. Each unit has potential treatment depending on whether she is below or above cut-off. Monotonicity assumption: no defiers, people who get treated if below cut-off, but do not get treated if above cut-off. Then, under some assumptions on continuity of CEFs of $Y(0)$, $Y(1)$, $D(0)$, and $D(1)$,

$$\frac{\lim_{s \rightarrow s_0, s > s_0} E(Y|S = s) - \lim_{s \rightarrow s_0, s < s_0} E(Y|S = s)}{\lim_{s \rightarrow s_0, s > s_0} E(D|S = s) - \lim_{s \rightarrow s_0, s < s_0} E(D|S = s)} = E(Y(1) - Y(0)|S = s_0, C).$$

$E(Y(1) - Y(0)|S = s_0, C)$: even more local effect than $E(Y(1) - Y(0)|S = s_0)$.

10.2 Estimation and inference

We focus on sharp case. How do we estimate the difference between the limits of two CEFs? **Complicated literature, brief and very simplified summary below.**

Initial idea in RD literature: regress Y on a high-order polynomial in S to the left of the cut-off and use $\hat{E}(Y(0)|S = s_0)$, predicted value of Y at $S = s_0$ according to that regression, as your estimator of $E(Y(0)|S = s_0)$. Then, regress Y on a high-order polynomial in S to the right of the cut-off and use $\hat{E}(Y(1)|S = s_0)$, predicted value of Y at $S = s_0$ according to that

regression, as your estimator of $E(Y(1)|S = s_0)$. Finally, use $\widehat{E}(Y(1)|S = s_0) - \widehat{E}(Y(0)|S = s_0)$ as your estimator of treatment effect.

Estimation strategy not used anymore. Several issues. How to pick the order of the polynomial? Big degree of freedom for researchers, and results may be sensitive to that choice. Relatedly: estimator may be inconsistent if model misspecified and true functional form of $E(Y(0)|S = s)$ and $E(Y(1)|S = s)$ is not a high order polynomial. Other argument against high-order polynomials estimation strategy laid out in Gelman & Imbens (2017). E.g.: $\widehat{E}(Y(0)|S = s_0)$ is a weighted average of the Y of all control units, where the weights can be estimated. In applications, it turns out that with high-order polynomials, $\widehat{E}(Y(0)|S = s_0)$ gives a lot of weight to units whose running variable is far from the cut-off. At odds with the RD idea, where we want to only use units just below the cut-off to estimate $E(Y(0)|S = s_0)$.

Alternative strategy: non-parametric regression. Most basic version: use $\frac{1}{n^-} \sum_{i:s_0-h_n < S_i < s_0} Y_i$ to estimate $E(Y(0)|S = s_0)$, where n^- is the number of observations with $s_0 - h_n < S_i < s_0$, and h_n is a sequence that go to 0 when the sample size goes to infinity. $\frac{1}{n^-} \sum_{i:s_0-h_n < S_i < s_0} Y_i$: **Nadaraya-Watson kernel estimator**, with a uniform Kernel (see (Nadaraya 1964) and (Watson 1964)) Then, use $\frac{1}{n^+} \sum_{i:s_0 < S_i < s_0+h_n} Y_i$ to estimate $E(Y(1)|S = s_0)$, where n^+ is the number of observations with $s_0 - h_n < S_i < s_0$. Finally, estimator is

$$\frac{1}{n^+} \sum_{i:s_0 < S_i < s_0+h_n} Y_i - \frac{1}{n^-} \sum_{i:s_0-h_n < S_i < s_0} Y_i.$$

Intuition: we compare the average outcome of units to the right of, but very close to, the cut-off, to the average outcome of units to the left of, but very close to, the cut-off. As h_n goes to 0 when n goes to infinity, when the sample size goes to infinity we compare the $Y(1)$ and the $Y(0)$ of people exactly at the cut-off.

Issues with that alternative strategy. Finite sample bias: $\frac{1}{n^-} \sum_{i:s_0-h_n < S_i < s_0} Y_i$ is biased estimator of $E(Y(0)|S = s_0)$. Conditional on n^- , expectation of $\frac{1}{n^-} \sum_{i:s_0-h_n < S_i < s_0} Y_i$ is $E(Y(0)|S \in (s_0 - h_n, s_0))$ which may differ from $E(Y(0)|S = s_0)$. To reduce the bias, instead of Nadaraya-Watson kernel estimator, better to use $\widehat{\alpha}^-(h_n)$ to estimate $E(Y(0)|S = s_0)$, where

$$(\widehat{\alpha}^-(h_n), \widehat{\beta}^-(h_n)) = \underset{(a,b) \in \mathbb{R}^2}{\operatorname{argmin}} \sum_{i:s_0-h_n < S_i < s_0} (Y_i - a - b(S_i - s_0))^2.$$

We run a regression of Y_i on a constant and $S_i - s_0$ to the left of the cut-off, among units that are less than h_n away from the cut-off. Predicted value for the Y_i of units with $S_i = s_0$ in that regression is $\widehat{\alpha}(h_n) + \widehat{\beta}(h_n)(s_0 - s_0) = \widehat{\alpha}(h_n)$. **Local linear regression** estimator proposed by Fan (1992), who shows that less biased than Nadaraya-Watson kernel estimator.

Therefore, Hahn et al. (2001) propose to use $\widehat{\alpha}^+(h_n) - \widehat{\alpha}^-(h_n)$ to estimate $E(Y(1)|S = s_0) - E(Y(0)|S = s_0)$, where

$$(\widehat{\alpha}^+(h_n), \widehat{\beta}^+(h_n)) = \underset{(a,b) \in \mathbb{R}^2}{\operatorname{argmin}} \sum_{i:s_0 < S_i < s_0+h_n} (Y_i - a - b(S_i - s_0))^2.$$

$\hat{\alpha}^+(h_n)$ is just the intercept in regression of Y_i on a constant and $S_i - s_0$ to the right of the cut-off, among units that are less than h_n away from the cut-off. Instead of running two separate regressions, to estimate $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$ one can just estimate a regression of Y_i on a constant, $S_i - s_0$, $1\{S_i \geq s_0\}$, and $1\{S_i \geq s_0\} \times (S_i - s_0)$, among units to the left and to the right of the cut-off that are less than h_n away from the cut-off. One can show that the coefficient of $1\{S_i \geq s_0\} \times (S_i - s_0)$ in that regression is equal to $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$.

Hahn et al. (2001) show that under a bunch of technical conditions, if $h_n = An^{-1/5}$ for some real number A , then

$$n^{2/5} (\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n) - E(Y(1) - Y(0)|S = s_0)) \xrightarrow{d} N(\mu, \sigma^2).$$

A few things to notice:

1. $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$ is a $n^{2/5}$ consistent estimator, not $n^{1/2}$ estimator. Intuitively, that's because only observations "close enough" to the cut-off are used in the estimation. Take-away: to get a precise estimator of the treatment effect in RD, you need a very large sample!
2. Result holds for any $h_n = An^{1/5}$ for some real number A . With $A = 10000$ or with $A = 0.0001$, $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$ may be very different. $\Rightarrow$ degree of freedom for the researcher.
3. Notice asymptotic distribution not centered at 0 but at some number μ . That's because the bias of $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$ goes to 0, but at the same speed as $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$ converges towards $E(Y(1) - Y(0)|S = s_0)$. First-order bias. Very roughly speaking,

$$\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n) \approx \frac{E(Y(1) - Y(0)|S = s_0)}{n^{2/5}} + \frac{\mu}{n^{2/5}} + \frac{N(0, 1)}{n^{2/5}}.$$

Consequence: $\frac{\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n) - E(Y(1) - Y(0)|S = s_0)}{\hat{\sigma}/n^{2/5}}$ does not follow $N(0, 1)$ distribution, so comparing $\frac{\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)}{\hat{\sigma}/n^{2/5}}$ to critical values of $N(0, 1)$ is not a valid test of $E(Y(1) - Y(0)|S = s_0) = 0$. However, applied researchers typically run a regression of Y_i on a constant, $S_i - s_0$, $1\{S_i \geq s_0\}$, and $1\{S_i \geq s_0\} \times (S_i - s_0)$, among units to the left and to the right of the cut-off that are less than h_n away from the cut-off, and compare coeff of $1\{S_i \geq s_0\} \times (S_i - s_0)$ divided by s.e. to critical values from $N(0, 1)$. Incorrect to do so!

To improve on 2., Imbens & Kalyanaraman (2012) derive a formula for an "optimal" bandwidth, that minimizes an asymptotic approximation of

$$E \left(\left(\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n) - E(Y(1) - Y(0)|S = s_0) \right)^2 \right),$$

the MSE of $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$. Intuition: the wider the bandwidth, the more biased $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$ may be. E.g.: when you increase the bandwidth, you use observations further to the right of the cut-off to estimate $E(Y(1)|S = s_0)$, but those observations have a value of the

running variable quite far from s_0 , so their expected Y may be far from $E(Y(1)|S = s_0)$. On the other hand, the wider the bandwidth the lower the variance of $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$, because the more observations are used in the estimation. MSE of an estimator = variance + squared bias, increasing bandwidth reduced first term but increases second one $\Rightarrow$ there is an optimal bandwidth.

Applied researchers are sometimes reluctant to using that optimal bandwidth. Indeed, optimal bandwidth is function, among other things, of the variance of the outcome at the cut-off. $\Rightarrow$ optimal bandwidth, and accordingly the sample included in the estimation, will change according to the outcome variable one considers. Then, applied researchers argue that if they find an effect on outcome A and not on outcome B, this could be due to the fact the effects for outcome A and B are for different subgroups, so they prefer to stick to the same, unoptimal and arbitrary, bandwidth for all the outcomes they consider. This reasoning is misguided: irrespective of the bandwidth, RD is always giving you an estimate of the treatment effect for units “at the threshold” $\Rightarrow$ even if the bandwidths for outcomes A and B are not the same, RD with the optimal bandwidth gives you an estimate of the treatment effect on outcomes A and B for units at the cut-off $\Rightarrow$ for the same population of units.

To improve on 3., Calonico et al. (2014) propose an estimator $\hat{\mu}$ of the first-order bias μ . Then, they show that

$$n^{2/5} \left(\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n) - \frac{\hat{\mu}}{n^{2/5}} - E(Y(1) - Y(0)|S = s_0) \right) \xrightarrow{d} N(0, \sigma_{BC}^2).$$

The estimation of the first-order bias changes the asymptotic variance, but Calonico et al. (2014) propose an estimator $\hat{\sigma}_{BC}^2$ of σ_{BC}^2 . Finally, one can compare $\frac{\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n) - \frac{\hat{\mu}}{n^{2/5}}}{\hat{\sigma}_{BC}/n^{2/5}}$ to critical values of $N(0, 1)$ to get a valid test of $E(Y(1) - Y(0)|S = s_0) = 0$. The computation of $\hat{\alpha}^+(h_n) - \hat{\alpha}^-(h_n)$ for the optimal bandwidth, $\hat{\mu}$, and $\hat{\sigma}_{BC}$ is automated in Stata and R (rdrobust package).

RD is still a very active area of research. Out of the most recent contributions, let me emphasize 2. Armstrong & Kolesár (2018) derive confidence intervals for $E(Y(1) - Y(0)|S = s_0)$ that may exhibit better finite sample properties than that in Calonico et al. (2014), and that are also valid if the running variable is discrete (so far, we have assumed that the running variable is continuous, which is often not the case). Imbens & Wager (2017) derive the estimator of $E(Y(1) - Y(0)|S = s_0)$ which is a linear estimator of the Y_i s and that has the lowest worst-case mean-squared-error, across all possible DGPs such that the second derivatives of $E(Y(1)|S = s)$ and $E(Y(0)|S = s)$ are bounded by a constant B . Their estimator may also exhibit better finite sample properties than that in Calonico et al. (2014), is also valid if the running variable is discrete, but their confidence intervals seem tighter than those in Armstrong & Kolesár (2018).

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